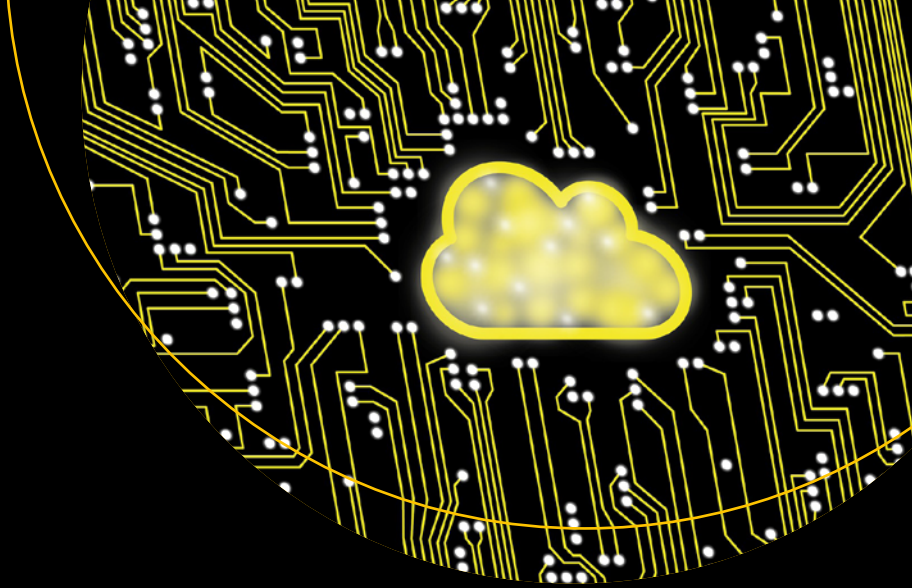




SAP HANA Cloud in a Nutshell

Design, Develop, and Deploy
Data Models using SAP HANA Cloud

Miguel Figueiredo

apress®

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Deploy Data Models using
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About the Author



Miguel Figueiredo is a passionate software professional with more than 30 years of experience in technical solution architecture. He has a degree in information systems and an MBA from Mackenzie University, as well as an international MBA in business administration from FIA Business School in partnership with Vanderbilt University.

Miguel gained his experience delivering business intelligence solutions for a number of Fortune 500 companies and multiple global corporations. As the SAP HANA Services Center of Excellence leader, he was responsible for the evangelization and best-practice adoption of data management and business intelligence in his region.

Currently, he advises companies to maximize value realization in their digital transformation journeys and move to cloud initiatives.

Miguel is dedicated to supporting his family and encouraging the development of good habits for health of body and mind.

About the Technical Reviewer



Henrique Pinto is a seasoned data and analytics architect and go-to-market expert, having had different roles in consulting, product management, presales, and business development across more than 15 years. He is currently an analytics platform specialist at Amazon Web Services, focusing on helping companies to leverage AWS data and analytics services to become more data-driven. Prior to joining AWS, he spent 12 years at SAP, where he was responsible for the go-to-market of SAP

HANA and, more recently, SAP HANA Cloud, having worked with early adopters from the inception of the product through its maturity.

Acknowledgments

Back in 2020, I was working at home, still living the impacts and restrictions brought by the COVID-19 pandemic, which was responsible for many changes in our lives. All this adverse scenario made 2021 a challenging year for most of us and for me was not different. The economic scenario in the world was one of a slow recovery, but my wish was that 2022 would be a better year. When the year started I was pleasantly surprised to receive an invitation to write a book. I felt very happy because it was something I always dreamed of doing. The challenge would be how to conciliate everything to make the book a reality.

I experienced a large part of the SAP HANA journey publicly started in 2010 and for me the technology and the business benefits brought by this solution were and still are disruptive for many companies. Since then, the desire to write a book about it has always haunted me, So when I got the invitation, I saw the opportunity to publish a concise and practical SAP HANA book that was easy to read on essential concepts and that could allow anyone to get hands-on experience through guided exercises using a SAP HANA Cloud that is provided as a cloud service.

In my life, I have known many amazing people. I learned that amazing people make others feel appreciated, and usually they know the power of an authentic “Thank you.” They know how recognizing and expressing appreciation makes others feel good.

So, before you start reading, I want to express gratitude to you, the reader. Thank you! This book is for you. It is my belief that if you can find usable, essential information in this book to support your digital transformation journey, the investment you made will have paid off.

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I'm eternally grateful to my awesome wife, Alessandra, for listening to my first ideas about this book and for always trusting and giving me all the support I've needed in all these years together. Thank you so much, dear.

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Happy reading!

Introduction

This book introduces SAP HANA Cloud and helps you develop an understanding of some of its key areas, including technology, architecture, and data modeling. *SAP HANA Cloud in a Nutshell* will help you to develop the skills needed to use the core features of the completely managed and in-memory cloud-based data foundation available in the SAP Business Technology Platform. This book covers modern modeling concepts and equips you with practical knowledge to unleash the best of SAP HANA Cloud. As you progress, you will learn how to provision your own SAP HANA Cloud instance, as well as how to work with different roles and data modeling for analytical and transactional use cases. Additionally, you will learn how to pilot SAP BTP Cockpit and work with entitlements, quotas, account structure, spaces, instances, and cloud providers. You will learn how to perform administration tasks such as stopping and starting an SAP HANA Cloud instance and making it available for use.

This book is for anyone who wants to gain knowledge about SAP HANA Cloud. To unleash the essential knowledge inside this book, you will find practical step-by-step instructions for how to establish a cloud account model and create your first SAP HANA artifacts. This book is an important prerequisite for those who want to take full advantage of SAP HANA Cloud.

Book Structure

This book contains the following chapters:

Chapter 1: Introduction to SAP HANA Cloud

This chapter helps readers to understand which new demands are being created by the digital transformation and how these changes have impacted most business organizations. Also, this chapter describes the main drivers behind the development of SAP HANA Cloud, introduces the Intelligent Enterprise concept and SAP Business Technology Platform, and highlights the SAP HANA Cloud architecture and its main capabilities.

Chapter 2: Administration of SAP HANA Cloud

This chapter introduces key concepts for managing an SAP Business Technology Platform account, describes steps necessary to create an SAP HANA instance, helps readers to understand which administration tools are available, and provides an overview of administration tasks, highlighting user and role management as well as backup and recovery operations.

Chapter 3: Analytical Processing with SAP HANA Cloud

This chapter explains the benefits of the new programming approach versus the classic approach; explores the main differences between complex and simplified applications; helps readers to know how to work with object modeling and different

structures, like calculation view, dimensions, cubes, analytical views, and how to enhance models using SQL script and procedures; and introduces advanced analytics capabilities, such as SAP HANA Cloud Spatial, predictive analytics (APL and PAL), and graph modeling.

Chapter 4: Developing Applications with SAP HANA Cloud

This chapter helps readers to understand the application development environment used with SAP HANA Cloud, which developer tools are available, and what content types can be used to start building your apps. It also presents key elements like Cloud Foundry, Cloud Application Programming Model (CAP) and Git architecture, service plans, UI5, and OData services.

Chapter 5: Working with SAP Estimator and SAP HANA

This chapter explains how to use the SAP Estimator to define an SAP HANA Cloud deployment setup; highlights capacity unit's relationship with resource allocation, like memory, CPU cores, and storage; and covers how to share such estimations.

Chapter 6: How to Activate Your SAP BTP Account

This chapter helps readers through a set of practical step-by-step instructions to configure an SAP Business Technology Platform trial account, provision an SAP HANA Cloud instance, and enable SAP Business Application Studio to start with database artifacts development.

INTRODUCTION

Chapter 7: How to Create Artifacts in SAP HANA Cloud

This chapter helps readers through a set of practical step-by-step instructions to start creating database artifacts directly in the provisioned SAP HANA Cloud instance. Readers will learn how to create tables, views with anonymized data, indexes, and calculation views, and start modeling in SAP HANA Cloud.

To get a better understanding of this book, I recommend starting with Chapter 1 and reading through the book sequentially.

Source Code

All source code used in this book can be downloaded from github.com/Apress/Apress-SAP-HANA-Cloud-in-a-Nutshell-by-Miguel-Figueiredo.

CHAPTER 1

Introduction to SAP HANA Cloud

Before we start our journey, we should take a step back and look at what motivated the development of the SAP HANA Cloud service.

In our modern, digital world, we are more connected than ever before, with huge amounts of valuable data being generated every second by organizations and individuals. More and more services and new digital businesses come online on a seemingly daily basis. This trend is set to continue growing exponentially. Forward-thinking companies have already taken steps to adapt to this ever-evolving digital world and grow their business by leveraging new technologies.

The New Digital World

Let's look at some of the opportunities for developing modern applications in the new digital world (Figure 1-1).

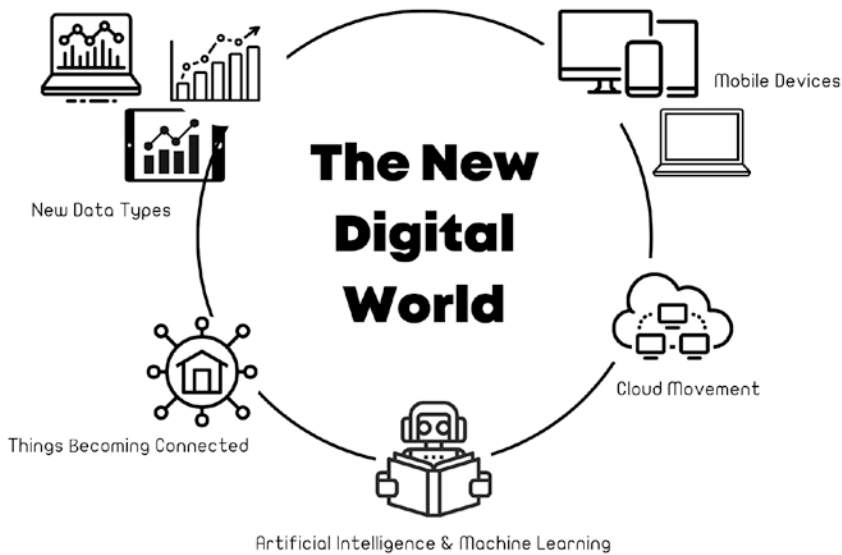


Figure 1-1. *New demands are recognized as opportunities*

The ubiquity of mobile devices and social media and the massive amounts of data they generate have changed the way we live and work. Most companies say most of their employees use mobile devices in their roles to for everything from email to project management to content creation.

But it's not just people who are networked. We are seeing a dramatic increase in the number of devices that have become "smart" with built-in CPUs and internet connections. We refer to this trend as the Internet of Things (IoT). Today, it is not uncommon for devices to be able to connect online to create intelligent and real-time responsive environments.

New types of data have become available as well. In addition to the traditional structured columns and rows that represent master data and transactions, we now have access to other types of data, such as unstructured text, spatial, and graph data. We can now develop applications that combine these data types to provide even more insights.

Cloud computing offers organizations the ability to develop new applications faster than ever before by deploying cloud platforms that provide an instant development and runtime environment. It supports flexible scalability so you can start with small pilot applications and then scale them to reach a larger audience. It also provides massive amounts of computing power on a subscription basis, so organizations only pay for what they need. Businesses can easily mix technologies by combining different cloud services. This means they can use the best tools and platforms for each application.

Until recently, data science was out of reach for most companies because they did not employ highly skilled people in this area. Today, data science is within reach of business users, who can employ simple-to-use tools to develop powerful data models that give them valuable insights into their vast amounts of data.

These are exciting times, and there are unprecedented opportunities for all companies to grow their business by leveraging the latest technologies.

For many companies, however, it is a major challenge to exploit these new opportunities with their current resources and technologies.

IT Challenges

Information technology (IT) landscapes have evolved over time into a variety of complex arrangements of purchased, acquired, and homegrown applications that are powered by multiple platforms. These platforms may be based on different types of hardware from different vendors.

Generally, existing systems run different operating systems and more than one type of runtime database, as well as multiple programming language frameworks. To make it all work together, companies have developed multiple interfaces between these systems that must be constantly maintained as the connected systems change—a difficult and extremely costly task.

The use of large data stores and solutions in a landscape as a result of increasing data and analytics requirements over time presents challenges because legacy applications were not designed to handle data stored in different online layers of varying data temperatures. The applications simply expected the data to be offloaded to archives for analysis. Not all organizations are ready or determined with their architecture to take advantage of large databases, cloud data, or data lakes.

The way we consume information has changed, and this behavior reflects how we demand IT and how we can support users (Figure 1-2).

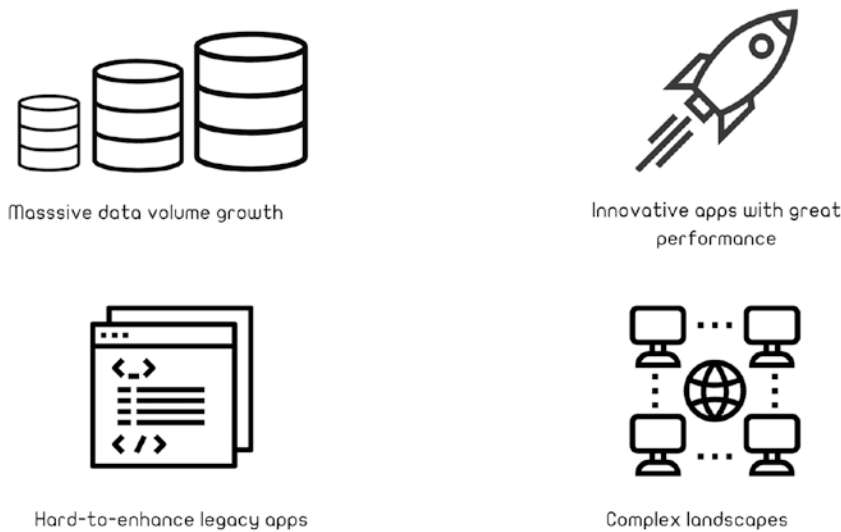


Figure 1-2. *IT challenges*

Most business users, as data consumers, are becoming more and more familiar with smart, instant-response customer apps in their personal lives. As a result, they expect a demand for innovative, real-time apps in their work as well. Clearly, businesses need deeper insights and faster decision support to help users perform their business tasks more flexibly and powerfully. People also expect to be able to go about their daily activities using their smartphones and tablets, which allow them to respond instantly even when they are away from their desks.

Companies have developed applications that have been enhanced and modified over many years. These applications are increasingly difficult to improve as the code becomes more complex and, in some cases, even outdated. A simple change request to an application can lead to a large effort in this scenario.

In addition, some business systems cannot cope with the increasing demands of the highly connected digital world, which is generating data (Figure 1-3) on an unprecedented scale. Adding more technologies and complexity to the existing IT landscapes will not ensure that enterprises keep up with existing trends.

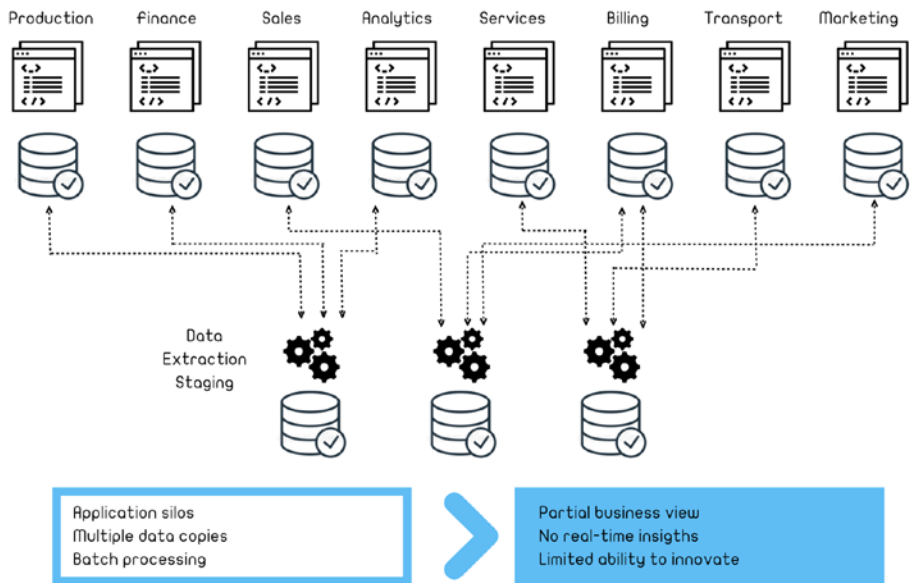


Figure 1-3. Data silos

So, in many cases, we must start with a blank slate and think about how we can review business systems with a bottom-up approach, using the latest technologies.

A simple and commonly used approach is “lift and shift,” where applications and associated data are migrated to the cloud with minimal or no changes. Applications are “lifted” out of existing environments and “shifted”—exactly as they are—to new hosting platforms (the cloud). Thus, there are often no significant changes to be made to the application’s architecture, data flow, or authentication mechanisms.

The most important considerations in a lift and shift migration are the application’s compute, storage, and networking requirements. They must be mapped from the resources currently available in the source infrastructure to the corresponding resources of the cloud provider.

With this method, significant costs can be reduced during migration. And since most cloud service providers offer dynamic upgrades, you can start with a smaller size and then scale up over time. This is a minimal risk approach for maximum ROI (return on investment).

However, there are several situations in which the lift and shift method is inefficient.

Some businesses offload substantial amounts of data from transactional applications into dedicated analytical systems on a regular basis.

Using specialized apps, businesses are able to clean and synchronize data with other sources. This method also relieves the transactional system of the burden of retaining data that is only required for analytical purposes. Multiple copies of the data, however, are produced in various locations.

Because this process frequently occurs at a different time, generating a consistent picture of data throughout the company appears to be a difficult and complex undertaking.

The Intelligent Enterprise

To really understand the role of and opportunities within SAP HANA Cloud, it is fundamental to get an introduction to what SAP has defined as an intelligent enterprise.

SAP coined the term “intelligent enterprise” (Figure 1-4) to define an extensive strategy that joins cutting-edge technology with business processes to create considerable corporate value. Data is the primary motivator in this strategy. In general, it is expected that intelligent companies adopting this strategy can definitely create and deploy data-driven applications to power their operations.

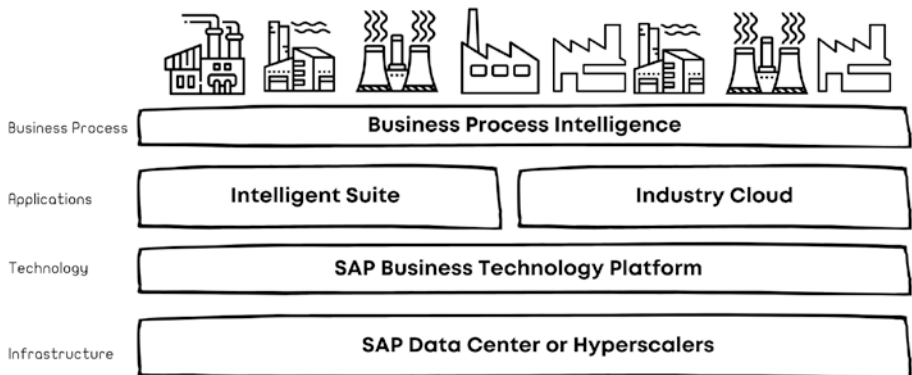


Figure 1-4. *The intelligent enterprise*

To make this strategy happen, it is necessary to consider a mix of on-premises and cloud solutions.

SAP has developed many solutions and enabled a partner ecosystem so as to deliver highly integrated software and services to support its strategy.

The intelligent enterprise is divided into four categories (Table 1-1).

Table 1-1. *Intelligent Enterprise Categories*

Category	What it does
Business Processes	Functional part of any business
Applications	Backoffice and industry solution of any business
Technology	Platform as a service foundation where all cloud services (including SAP HANA Cloud) are available to support app development
Infrastructure	Hyperscalers and SAP data centers that provide the fundamental layer to run any application

All software components are provided by the SAP Business Technology Platform (SAP BTP), a Platform-as-a-Service (PaaS) offering that has all cloud and on-premises technologies required to run applications.

Moreover, SAP BTP (Figure 1-5) has a collection of cloud services that delivers technologies to allow you to connect and expand the usage of SAP and non-SAP systems.

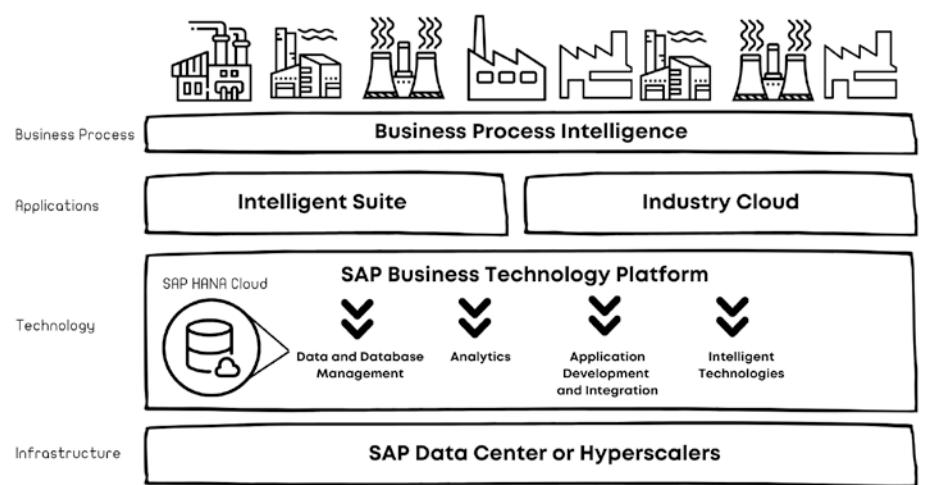


Figure 1-5. *The SAP Business Technology Platform*

SAP BTP supports different data-driven application scenarios, allowing you to take advantage of data beyond traditional usage, exploiting it in advanced scenarios by using machine learning and artificial intelligence techniques.

Users may utilize available cloud services (Figure 1-6) to swiftly convert data into insights, connect processes, and develop and enhance applications.

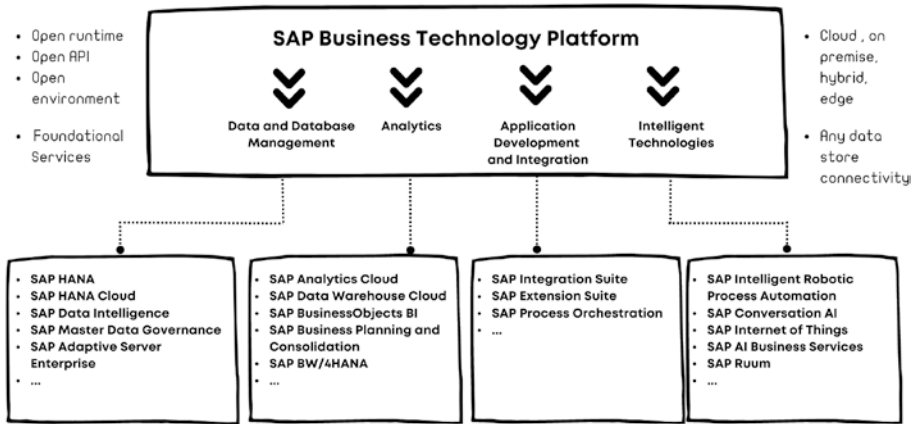


Figure 1-6. Cloud services of the SAP Business Technology Platform

Note SAP Business Technology Platform (SAP BTP) is the platform for the intelligent enterprise that helps you accelerate business outcomes with integration, data-to-value, and extensibility.

The four areas of SAP BTP are summarized as follows:

1. Database and Data Management

Database and data management products are at the core of SAP BTP. You can manage, govern, integrate, and analyze corporate data to assist your company in making smart business decisions.

Furthermore, you may run your data anywhere, decrease data redundancy, integrate data across varied geographies, and develop and manage a comprehensive master data system. SAP BTP is compatible with the following:

Database: SAP HANA, SAP HANA Cloud, SAP IQ, SAP Adaptive Server Enterprise, and SAP SQL Anywhere are the databases.

Data Management: SAP Master Data Governance, SAP Data Intelligence, SAP Information Steward, SAP Data Quality Management, SAP PowerDesigner, SAP Information Lifecycle Management, SAP Landscape Transformation Replication Server, and SAP Data Services are all part of the data management suite.

Data Management Solutions Extensions: SAP Extended Enterprise Content Management by OpenText, SAP Advanced Data Migration by Syniti, SAP Information Steward Accelerator by Syniti, SAP Master Data Governance, retail and fashion management extensions by Utopia, SAP Data Mapping and Protection by BigID, and SAP Asset Information Workbench by Utopia are examples of data management solutions extensions.

2. Analytics

SAP BTP provides you with the tools you need to get the most value from your company data. Machine learning, artificial intelligence, business intelligence, and augmented analytics provide real-time insights by analyzing past and present conditions and modeling future possibilities. SAP BTP is compatible with the following:

Business Intelligence: SAP Analytics Cloud for business intelligence, SAP BusinessObjects Business Intelligence (BI) suite, and SAP Crystal Reports.

Data Warehousing: SAP Data Warehouse Cloud, SAP HANA, and SAP BW/4HANA.

Enterprise Planning: SAP Analytics Cloud for planning and SAP Business Planning Consolidation.

3. Application Development and Integration

SAP BTP offers a variety of tools, services, and programming models for connecting and expanding your systems and solutions. There are several options and routes for connecting and expanding your SAP and non-SAP apps and systems. SAP BTP is compatible with the following:

Application Development: SAP Extension Suite and SAP Process Mining by Celonis and SAP Signavio Process Intelligence.

Application Integration: SAP Integration Suite and SAP Process Orchestration.

4. Intelligent Technologies

Using SAP BTP, you can keep up with developing technology. To supply the apps and processes that your firm needs, use artificial intelligence, robotic process automation, machine learning, IoT, and blockchain. SAP BTP supports the following:

- **Artificial Intelligence:** SAP Intelligent Robotic Process Automation, SAP AI Business Services, and SAP Conversation AI.
- **IOT:** SAP Internet of Things and SAP Edge Services.

The SAP HANA Cloud

SAP HANA Cloud is a cloud Database-as-a-Service (DBaaS) offering that is completely managed and in-memory. It is the SAP Business Technology Platform's cloud-based data foundation. SAP HANA Cloud allows you to build, operate, and innovate applications.

SAP HANA Cloud is made up of several components (Figure 1-7). The primary component is the SAP HANA database; other ones—for example, the data lake—can be added at any moment.

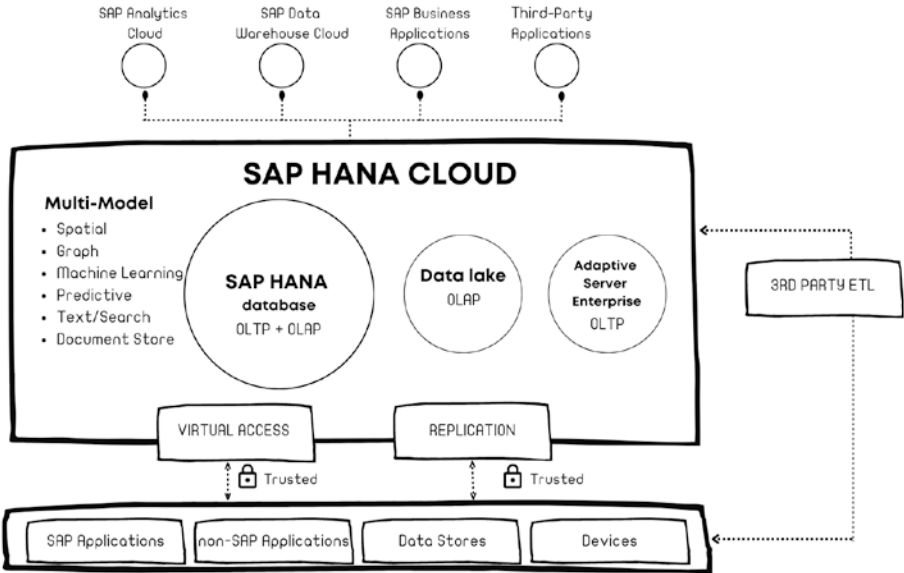


Figure 1-7. The SAP HANA Cloud architecture

Note It is important to highlight that SAP HANA Cloud is a cloud-native service that was created from the ground up, not just a cloud deployment of the previously existing on-premises SAP HANA.

SAP HANA Cloud provides most of the functionality of the on-premises edition, as well as certain extra services not available in the on-premises version.

Customers that have deployed applications based on SAP HANA on-premises may easily move them to SAP HANA Cloud.

SAP HANA Cloud may be used in different scenarios to support data warehouses, traditional and advanced analytical applications, and even custom or third-party applications.

Data from any data source may be loaded into SAP HANA Cloud on a continual or planned basis using built-in technologies. SAP HANA Cloud allows one to connect any source system remotely in real time. Data may be derived from any application, both SAP and non-SAP.

SAP HANA Cloud is appealing to clients that want to use SAP HANA without having to spend the time and money that are usually necessary to run SAP HANA on-premises. Companies don't need to buy or lease an infrastructure when using SAP HANA Cloud. It is operated by SAP and cloud hyperscalers, which handle provisioning.

There are four key components of SAP HANA Cloud (Figure 1-8).

- SAP HANA Cloud, SAP HANA database
in-memory database with built-in advanced analytics 
- SAP HANA Cloud, data lake
store and query large data sets 
- SAP HANA, adaptive server enterprise
high-performance relational database 
- SAP HANA Cloud, adaptive server enterprise replication
bi-directional real-time data replication 

Figure 1-8. *Key components of SAP HANA Cloud*

The part that delivers the in-memory capability is the SAP HANA database. In addition, there are tools to help with the creation of sophisticated data models that can be ingested by different purpose applications.

SAP HANA Cloud has a data lake that allows one to store infrequently accessed data.

This implies that very big data sets may be stored and searched straight from the SAP HANA database or be accessed in real time.

SAP HANA Cloud additionally has the SAP ASE, which is a cloud relational database to support high-performance transactional applications. In general, this type of application requires low-latency atomic inserts and in general quick response over large databases. For customers that have the on-premises version of SAP ASE it can be interesting to consider when looking for a cloud migration path, while also making it possible to link the application to an SAP HANA database for additional benefits.

Another existing feature is Adaptive Server Enterprise Replication, a log-based, real-time data-replication engine that enables on-premises and cloud databases to be synchronized in real time.

Provisioning with Flexible Scalability

To make an SAP HANA Cloud database instance available for use, we just need some minutes and a web browser. In the setup assistant, you must provide the amount of processing power and data storage that are necessary. These options are changeable at any moment, which allows the database to scale (Figure 1-9).

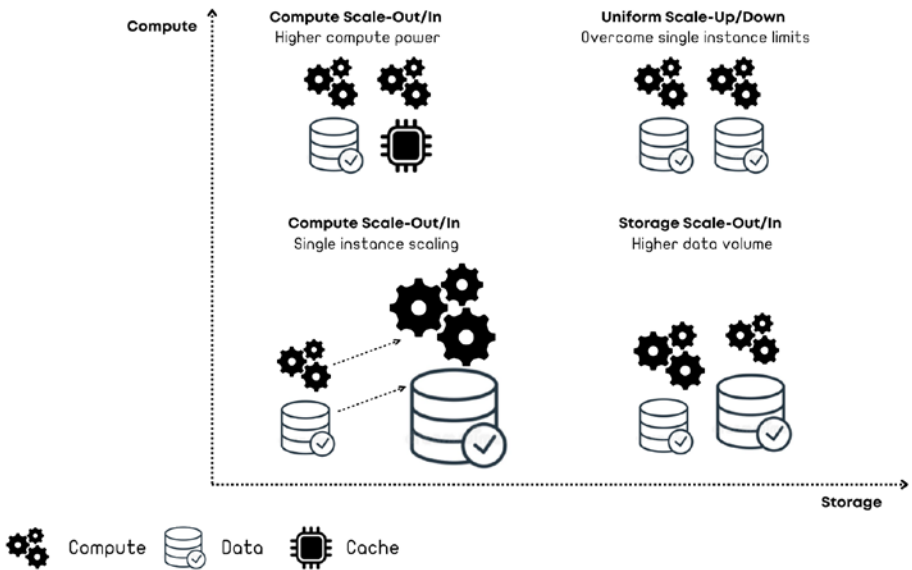


Figure 1-9. *Computing scale*

The scalability feature of SAP HANA Cloud is considerably compelling in terms of computing and data storage setup. Using a simple interface, customers can simply increase or reduce their computation and/or data storage capacity.

SAP HANA Cloud's flexibility, in particular, is the right answer for SAP HANA on-premises clients that may utilize computing and data storage to increment their on-premises setup for temporary peaks in demand or to replace existing on-premises systems permanently.

You can set the CPU's size during the deployment. This may be adjusted when compute power demand grows or shrinks.

SAP HANA Cloud supports multi-temperature data tiering (Figure 1-10).

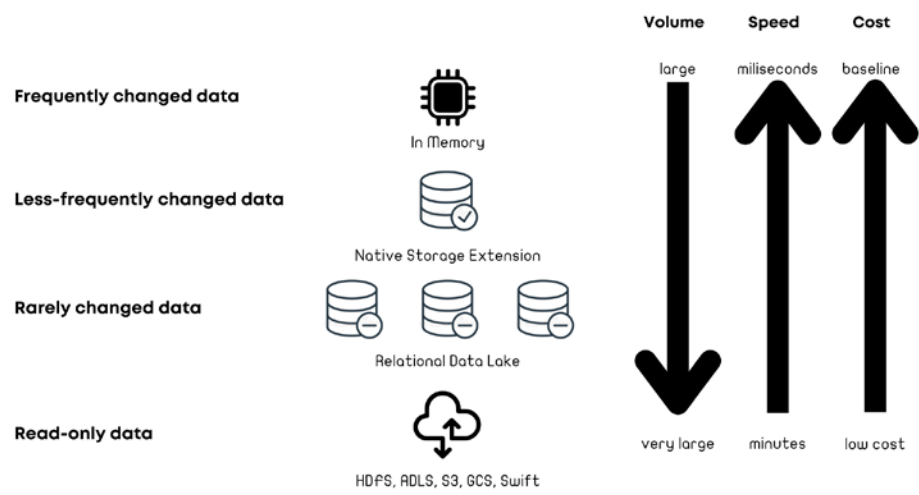


Figure 1-10. SAP HANA Cloud storage tiering

Hot data is data that is accessed often, requires optimal speed, and is kept in the SAP HANA Cloud database’s memory. Warm data is data that does not require high speed and is saved on disk, maintained via a component called Native Storage Extension (NSE) in SAP HANA Cloud.

NSE employs a smart caching method that collaborates with memory to define when to retain the frequently used data in memory, or to temporarily promote less-frequently accessed data to hot while it is being demanded, returning it to the warm tier when it is no longer necessary. When you do the setup, you may specify your SAP HANA database’s memory size (for frequent-access data) and amount of disk storage (for less-frequent-access data).

During provisioning, you may choose between hot and warm data layers, as well as an additional data lake. A data lake is an area used as a storage facility for cold data. Cold data is still saved on disk, but it manages far larger amounts of data than NSEs and at a significantly cheaper cost.

Data lakes are perfect for managing massive volumes of data when access speed does not need to be optimal.

Note The hot, warm, and cold layers permit not only reading but also writing data in real time. Additionally, Smart Data Access (SDA) is a feature of SAP HANA Cloud that provides fast, federated access to remote data sources. The SAP HANA Cloud data lake is connected to the SAP HANA Cloud database via SDA.

You may connect SAP HANA Cloud to most cloud native storage services, such as Amazon S3, Azure Storage, and Google Cloud Storage, to read data. The advantages of utilizing very large data providers include access to virtually infinite data storage capacity at a lower cost. However, the disadvantage of this strategy is that the response time of this type of read-only access is much slower than direct database access, potentially leading to performance bottlenecks if used by applications with more frequent data-access needs.

Usage Scenarios

SAP HANA Cloud plays an important role in the open Platform-as-a-Service SAP Business Technology Platform, supporting most applications' development in a pure cloud or hybrid configuration.

SAP HANA Cloud may be utilized in a wide range of IT settings. Let's look at several scenarios in which SAP HANA Cloud may be employed (Figure 1-11).

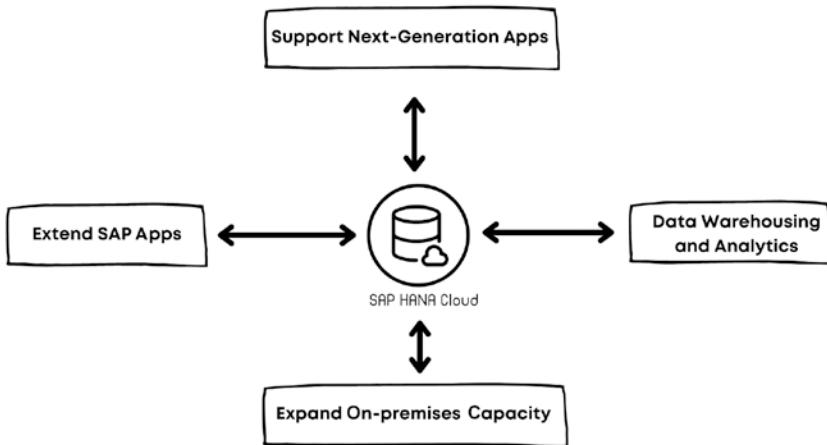


Figure 1-11. Use cases for SAP HANA Cloud

SAP HANA Cloud has several use cases and scenarios. These are some examples:

- To offer a database for modern apps that demand instantaneous access to huge amounts of data.
- Powering different analytical scenarios using solutions such as SAP Data Warehouse Cloud, for example. Gaining productivity with SAP Analytics Cloud by linking to and organizing distributed data into a single, consistent data model.
- Scaling resources like data storage or processing capacity to support on-premises applications, especially when additional users are demanded.
- Extending and adopting new features of Cloud Services to existing applications.

Now, let's review some characteristics of SAP HANA Cloud in a mixed landscape.

Working in a Mixed Landscape

SAP HANA Cloud may be quite useful in a hybrid environment. Along with moving to the cloud, customers frequently keep operating on-premises apps and related databases.

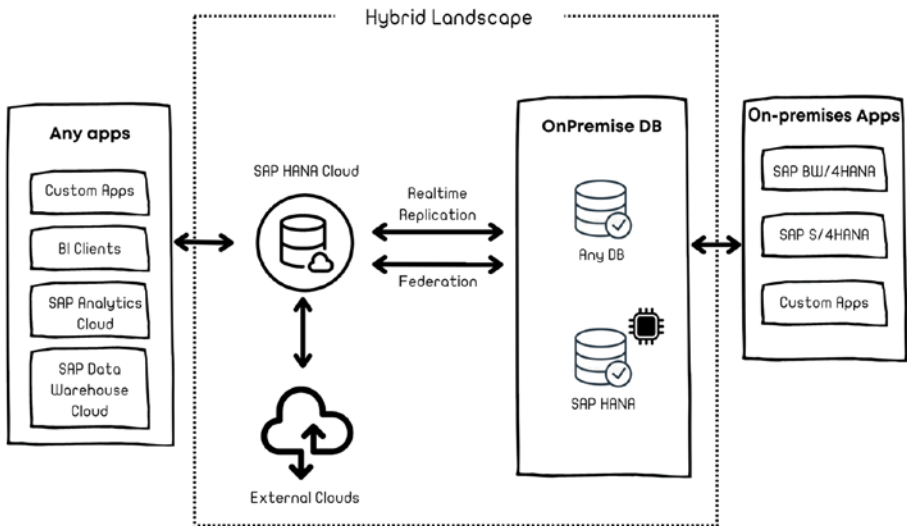


Figure 1-12. *Mixed landscape*

However, instead of operating decoupled databases, SAP HANA Cloud and SAP HANA (on-premises) may be tightly linked to create a combined set of data storage and processing capabilities (Figure 1-12).

Data may be duplicated in real-time or batched between SAP HANA Cloud and SAP HANA (on-premises). However, replication is not mandatory, and virtualization (federation) is also feasible in both directions.

Some clients use SAP HANA Cloud as a single hub through which all data from various sources is centralized. Some data sources deliver data to SAP HANA Cloud, while others give a remote link to live data. The advantage of this method is that SAP HANA Cloud becomes a centralized data point for all applications.

Some companies opt to upgrade their central mission-critical applications with SAP HANA on-premises while also utilizing SAP HANA Cloud services. This is a smart approach to extend applications with new features while keeping the core safe without impact.

SAP HANA debuted in 2010, only accessible as an on-premises deployment at that time. It was necessary for customers to provision their own hardware and have a certified professional install the SAP HANA software. The customer would oversee day-to-day operations as well as ongoing maintenance and improvements.

SAP HANA Cloud, a more easily deployable version of SAP HANA, was announced in 2020 and was the third cloud version, being SAP HANA as a Service the second time and SAP HANA Neo Service the first time. The cloud version is managed by SAP, and cloud hyperscalers are responsible for all elements of the platform's operations.

SAP HANA Cloud receives upgrades on a regular basis. When new features are introduced, a new release is made every quarter. While this may seem intriguing to certain clients who wish to create apps employing cutting-edge technology, it is not desirable to enterprises that don't need to support applications that are not ready for the latest improvements or must stay in a specific release. In general, these customers remain on SAP HANA on-premises, choosing total control over the landscape and the timeliness of upgrades (Figure 1-13).

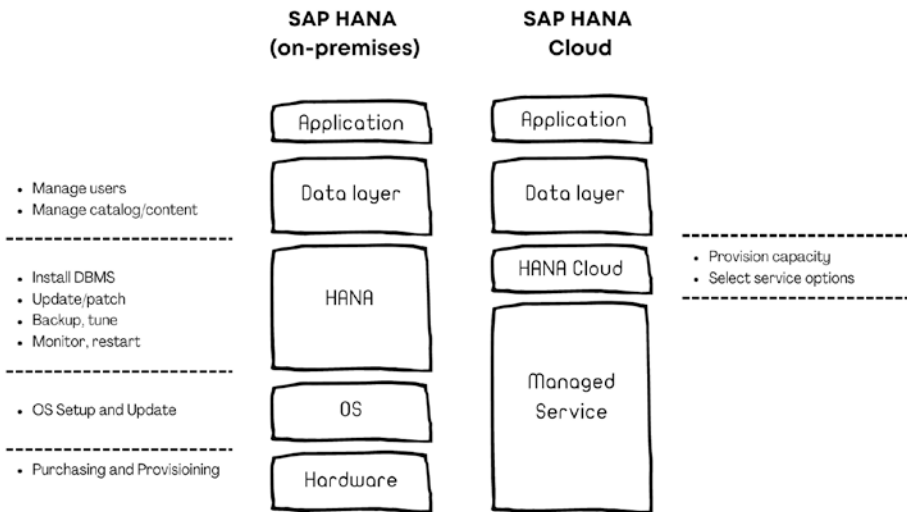


Figure 1-13. SAP HANA on-premises and Cloud customer responsibilities

Caution SAP HANA Cloud should not be confused with SAP HANA Enterprise Cloud (HEC), a private deployment that is used by just one customer who has total control over it. SAP HANA Cloud is a public cloud service that, while logically isolated, is physically run on infrastructure that can be shared by multiple customers in a multi-tenant approach.

The SAP HANA Toolbox

To start using SAP HANA Cloud, you will need to use tools (Table 1-2) accessible through the SAP BTP console.

Table 1-2. *SAP HANA Cloud Toolbox*

Tools	What It Does
SAP HANA Cockpit	Used to perform monitoring and administration jobs
SAP HANA Database Explorer	Used to explore and create runtime database objects and to perform SQL statement execution
SAP Business Application Studio	Used to perform the creation of design-time database objects

Feature Differences

In terms of features, SAP HANA Cloud is fairly similar to SAP HANA on-premises, although there are some distinctions (Figure 1-14).

SAP HANA Cloud is a fully managed service, and the majority of the system and database configuration parameters are not visible to the client, as they are in SAP HANA on-premises.

Feature	SAP HANA Cloud	SAP HANA On-premises
Row/Column in-memory Database	✓	✓
Data Modeling	✓	✓
Data Replication	✓	✓
Data Virtualization	✓	✓
Data Cleansing (SDQ)	✗	✓
Text Analysis and Mining	✓	✓
XS Classic and XS Advanced	✗	✓
Multi-tenancy	✗	✓
SAP Data Warehousing Foundation	✗	✓
SAP Streaming Analytics	✗	✓
Data Lake	✓	✗

Figure 1-14. *SAP HANA feature differences*

However, it is not a question of selecting between on-premises and cloud computing. Most businesses currently have a hybrid-multi-cloud strategy in place, which means they recognize the benefit in integrating on-premises and multi-cloud deployments. Customers frequently opt to combine SAP HANA Cloud and SAP HANA on-premises to benefit from a strong, hybrid SAP HANA architecture that combines the advantages of both techniques (Figure 1-15).

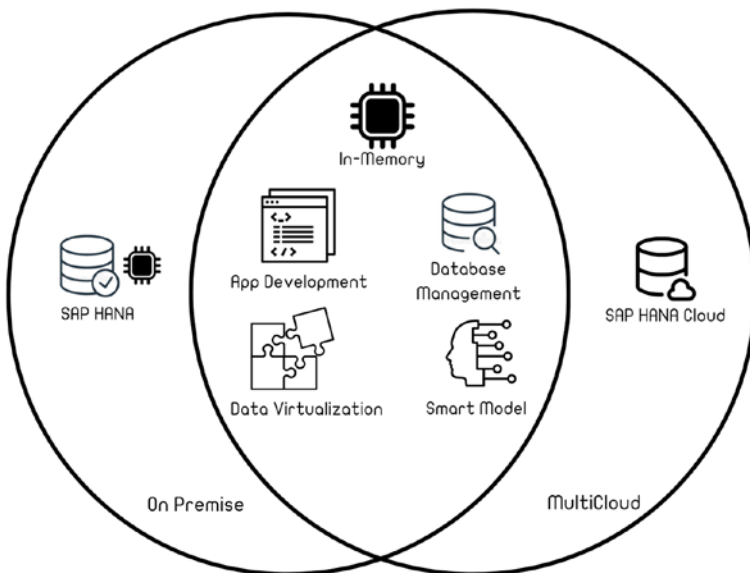


Figure 1-15. *Hybrid landscape*

Simplified Data Models

A classic data model built on legacy technology sometimes contains workaround techniques to address performance issues. Because code must be built to maintain various workaround solutions, application code maintenance and operation are more difficult. Additionally, coding used to assist application performance offers little value to core business functions.

Integration with other apps and upgrades are challenging due to the complex data model and application code. These programs are simply insufficiently agile to keep up with today’s fast-paced corporate environment.

We have a streamlined data model in SAP HANA Cloud, which makes application code simpler, to allow improvements and add new features.

Traditional Applications

A hierarchical data model was historically used to build traditional applications (Figure 1-16). To improve systems’ efficiency, granular data was summarized into aggregate layers. On top of these aggregates, we created more aggregates as well as unique versions of the database tables to enable certain applications. In addition to keeping additional copies of data, we had to write application code to manage extra database objects and keep them up to date. Because a backup of these extra database objects was also necessary, IT activities were also disrupted.

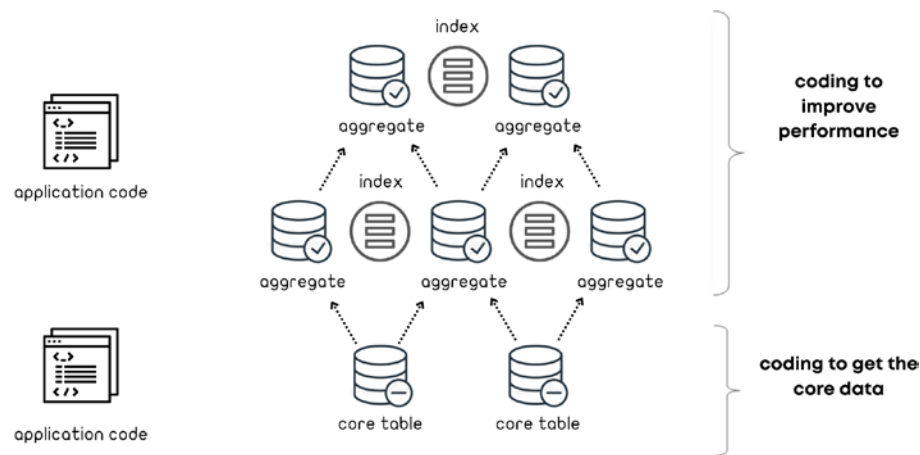


Figure 1-16. *Traditional application*

In addition to the aggregates, we must eliminate data-access inefficiency. This is managed by creating database indexes that boost access speed. Indexes must be discarded and rebuilt every time the tables are modified. As a result, additional code is required to control this operation.

Simplified Applications with SAP HANA Cloud

You may do an aggregation operation on the fly from any line-item table using the capabilities of SAP HANA Cloud. There is no need to work with prebuilt aggregation (Figure 1-17). SAP HANA Cloud may provide multiple views of the data from the basic source data at runtime.

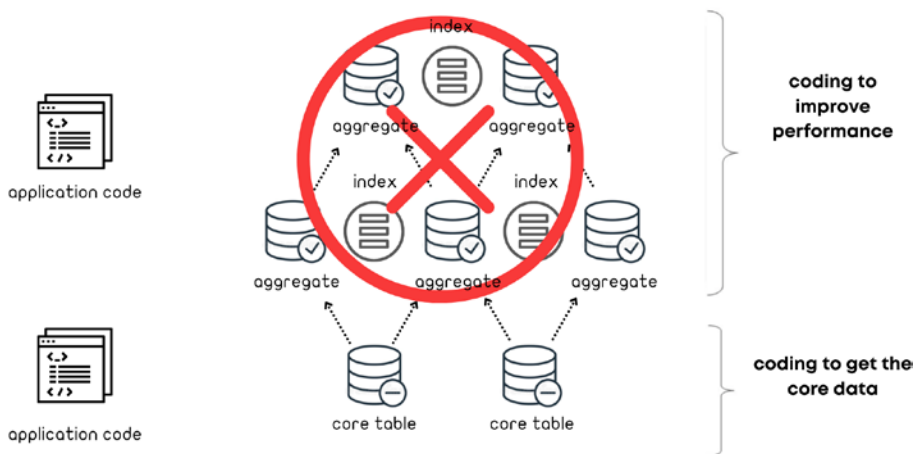


Figure 1-17. *Simplified application*

SAP HANA Cloud provides column stores that make indexes rarely required, and even when they do exist they usually offer little improvement.

In addition to eliminating aggregation and indexes, a considerable chunk of the application code can be eliminated. This results in a simpler data model as well as cleaner application code with a potentially lower maintenance resource demand.

Classic Approach

Previously, the database's duty was to handle any data-related commands given by the application. This included general demands for data creation, reads, updates, and deletions.

These requests were often simplistic, requiring little or no real data processing or calculation. In the instance of a read request, for example, the database would return the raw data to the application layer, which would do the computations. The database activities were limited to fetching huge pieces of data, writing large portions of data, and so on. However, the SAP HANA Cloud database is by far more than a simple database; it is an in-memory data-processing platform that can perform additional complex calculations on behalf of the applications when necessary.

Some of the data-processing capabilities of SAP HANA are as follows:

- Perform data aggregation and disaggregation
- Filter, sort, and rank data
- Combine multiple data sources (multi-dimensional modeling)
- Generate calculated values from existing data
- Predict and forecast values
- Perform currency or unit measure conversions
- Harmonize data
- Perform geospatial and graph processing

New Approach

Whenever possible, data instructions, whether basic or complex, should be pushed down to the SAP HANA Cloud database by an application. Only processed data results are sent back to the application, reducing the computation needs of the application layer and the traffic of unnecessarily granular data over the network.

Consider the following application request to the SAP HANA Cloud database: Calculate the net worth after discounts based on the last five years of smartphone sales by area and year.

SAP HANA Cloud evaluates the data request and returns just qualified results to the application layer. This results in a significant benefit with less data transmitted to the application. All data processing occurs in-memory, resulting in very fast speeds.

“Push-down” refers to the process of moving data-processing activities from the application layer to the database layer (Figure 1-18). When using a push-down strategy, application developers must reconsider how they code.

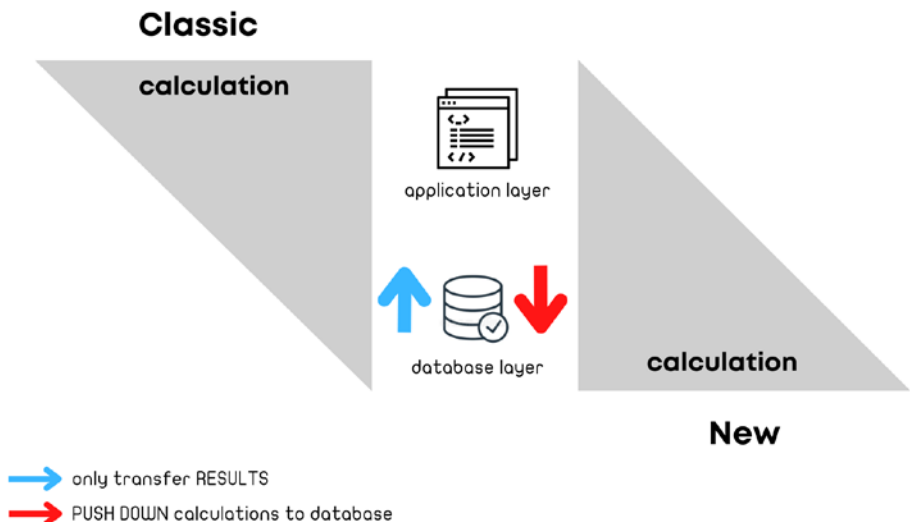


Figure 1-18. Classic versus new approach

Usually, a developer might load records from the database into internal tables and then develop a code to loop through the data set, reading and moving it into an auxiliary table. However, in an SAP HANA Cloud database, coding may now be developed directly in the database. SQLScript is an upgraded form of SQL that developers utilize in the SAP HANA Cloud database. SQLScript provides a programming language framework for developers to create sophisticated views as well as construct stored procedures and functions that can be invoked from application code while passing parameters, when necessary.

A Single Point of Truth

SAP HANA Cloud database can offer the opportunity to eliminate any data redundancy by keeping one copy of data, making it available to all apps (Figure 1-19).

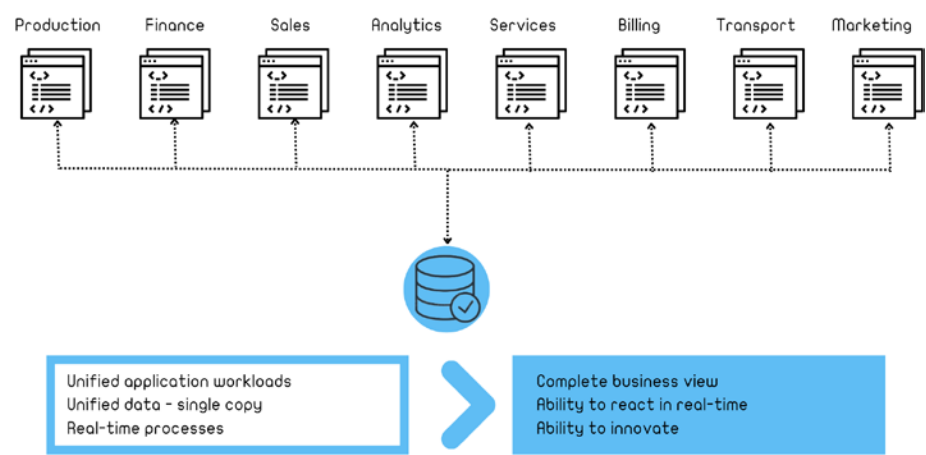


Figure 1-19. *One data set for all*

SAP HANA Cloud has many capabilities (Table 1-3) that empower companies to keep one single data set to be used among applications.

Table 1-3. *Key Capabilities of SAP HANA Cloud*

Capabilities	What It Does
Hybrid data models	Support different data models
In-memory column store database	Provides high performance for instant insights
Data connectivity	Connects to any source
Elastic storage	Supports data lake connectivity to consume historic data

Modern Applications Embedding Insights

Historically, apps were run on their own specialized platforms. To transport data between apps, data would need to be moved, which would typically require the use of complicated additional solutions to clean and rearrange data. This resulted in delays, data silos, and, of course, complexity. A preferable method would be to power all apps with a single high-speed architecture. This entails a single architecture with just one data management system, where all data is available in real time to all applications. There will be no need for wasteful data migration or administration of various data stores, as well as no need for sophisticated platform-to-platform designs with expensive maintenance interfaces (Figure 1-20).

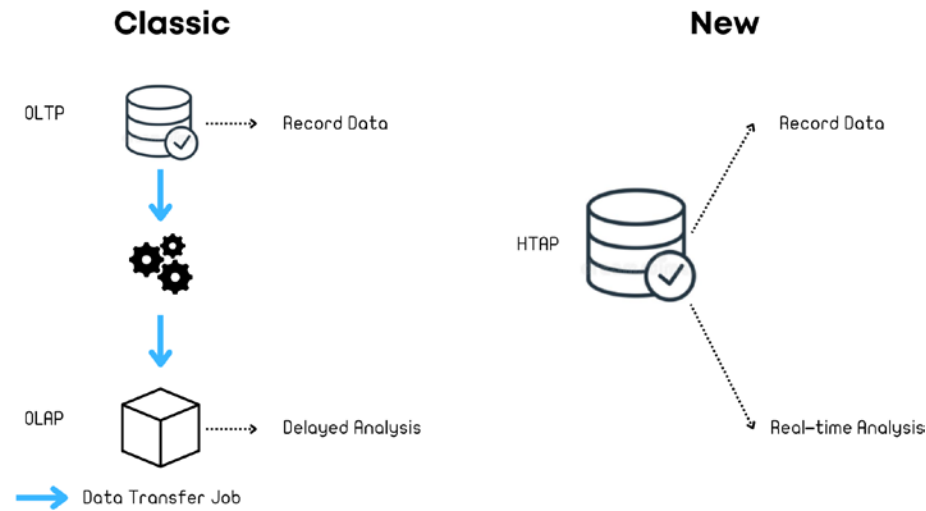


Figure 1-20. *The new approach with SAP HANA*

Let’s have a look at how SAP HANA Cloud combines transactional and analytical demands in a single architecture.

Today, most systems of record (transactional business applications) are managed in systems where the hardware, database, and data models are all structured with a focus on rapid data read/write at the record level (atomic data processing). On the other hand, systems for analytics purposes (systems of differentiation) are designed to function separately.

The hardware, database, and data models are designed with batch loading, aggregated storage, and an emphasis on very intensive reading and caching in mind. As a result, online transaction systems (OLTP) and online analytical processing systems (OLAP) are frequently configured as separate systems and linked via interfaces where data is extracted, transformed, and loaded on a regular basis.

This method causes delays in the analysis of transactional data. It is not uncommon to have data available for analysis purposes only on the next day (D+1).

SAP HANA Cloud may combine transactional and analytical requirements into a single platform. Hybrid transactional/analytical processing is the abbreviation for this sort of consolidated system (HTAP).

SAP HANA Cloud's database and data architecture are designed to support both transactional and analytical processing. Because transactional and analytical users operate on a single copy of data, there is no need for data to be moved from a record system to an analytical system. This implies we can migrate data using distinct tools and interfaces since we have live data. It also implies that new, innovative applications combining transactions and analytics can be developed.

Here's an example: When there are few vehicles available on the lot, a rental car system evaluates the projection of peak demand based on running client orders and blocks out the cars available for renting. Previously, two distinct programs on separate platforms would have been required: one for analytics purposes and another to record transactions.

The new technique has transformed the way applications are designed, making it possible to develop modern apps that not only enable transactional workloads as a system of record, but also provide embedded and instant analytical insights.

Working with Different Data Types and Sources

It is very common to see modern apps taking advantage of geolocation, networked information, and classic transactional data. SAP HANA Cloud allows one to work with different data types (Table 1-4).

Table 1-4. *New Data Types*

Data Type	What Data Describes
Spatial	Specific location, such as maps, routing, floor plans, and so on.
Graph	Highly networked things like social networks, supply chains, and so on

SAP HANA Cloud can connect to any type of data source (Figure 1-21) and make it available in different ways to support modern apps.

New Type of Data



Graph



Spatial

Any Source



Data Warehouses



Archives



Social Networks



Clouds



Remote Databases



Enterprise Systems

Figure 1-21. New data types and sources

SAP HANA Cloud can also access data from any type of source, including the following:

- **Enterprise Systems:** SAP S/4HANA is an example of an enterprise system.
- **Data Warehouses:** SAP BW is an example of a data warehouse.
- **Archives:** SAP HANA Cloud can create an online connection to remote archives.
- **Big Data:** Apache Hadoop is an example of Big Data.
- **File Format Types:** XLSX, CSV, and XML are examples of file format types.

- **Databases:** This includes any relational database (i.e., SAP ASE, Oracle, SQL Server, DB2, etc.).
- **Social Networks:** Twitter, Facebook, and LinkedIn are examples of social networks.

Virtualization has an important role in providing remote data access to a data source without having to load data into an SAP HANA Cloud database (federated access).

Now, let's see which key technologies are inside the SAP HANA Cloud.

Key Technologies

To manage complex data processing, SAP HANA uses a number of key technologies that you must know.

New Technologies and Opportunities

SAP HANA Cloud was designed to take full advantage of the latest trends and advancements in computing technology. It was created by completely redesigning and redeveloping an existing software solution. This comprehensive development was carried out to guarantee that it could deliver a next-generation platform based on cutting-edge technologies.

The high cost of memory for some time represented an important restriction on the amount of this resource available for use. Limited memory was a significant barrier to data movement from drive to CPU. It made no difference how fast the CPU was if the data from the disk couldn't reach it quickly.

However, the cost of memory has dropped in recent years and continues to decline year after year. Server manufacturers are increasingly providing massive quantities of RAM that allows scale-up to several terabytes faster and easier than before. (Figure 1-22).

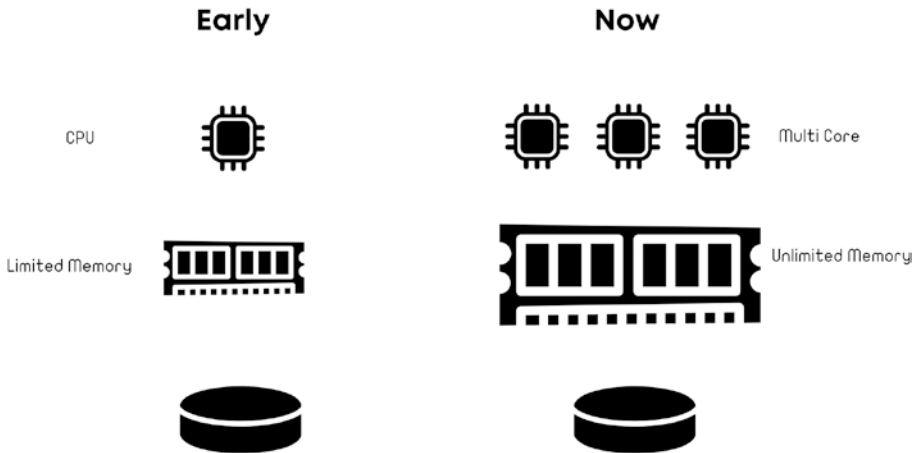


Figure 1-22. Elastic computing

Since more quantities of memory are now accessible, we can now keep corporate databases entirely in memory, rather than on disk. This provides rapid data access since there is no need to wait for data transfers from disk to memory.

We need to utilize 64-bit operating systems to access enormous quantities of memory. Traditional 32-bit operating systems are incapable of addressing the massive quantities of memory that are presently accessible.

However, having a large amount of RAM sending data to the CPU is necessary to improve the speed and capacity of CPU processing. This is where high-speed and multi-core processing can break down complicated operations and perform them simultaneously to deliver remarkable results.

The most compute-intensive analytical activities may run at any time. SAP HANA Cloud can distribute workloads to many CPU cores for maximum performance. Performance improves as the number of CPU cores increases. This is known as scaling up.

Cloud providers may now simply and rapidly add extra RAM and CPUs to their servers thanks to contemporary blade-server architecture. This enables the hardware to be quickly scaled up to accommodate larger

workloads and data volumes. Once the hardware's scale-up constraints are achieved, we can consider scale-out, where more worker nodes (servers) are added to handle the workload.

However, the majority of common database systems were not built to make use of cutting-edge technology and would not know how to operate properly with vast amounts of RAM and multi-core CPUs. As a result, SAP created the SAP HANA Cloud database from the ground up to run on cutting-edge technology, allowing applications to benefit from in-memory and massive processing techniques.

SAP designed SAP HANA Cloud to make full use of cutting-edge technologies working in collaboration with hyperscale cloud providers.

The Path to In-Memory Database

Previously, databases were entirely kept on disk, with only the data needed by applications being copied to memory and then delivered to the CPU for processing. Due to memory-size restrictions, it would quickly fill up, requiring data to be off-loaded back to disk to create room for new requests. This scenario brought intense disk swapping that usually directly affected the application-use experience.

To overcome such impacts, some application-layer workarounds were usually created in an attempt to prevent swapping, not fixing permanently the fundamental problem of limited memory. Now, you may store a whole database in memory using SAP HANA Cloud and the massive memory capacities available (Figure 1-23). This means that there is no need to load data from disk into memory; all data is always available to the applications instantaneously.

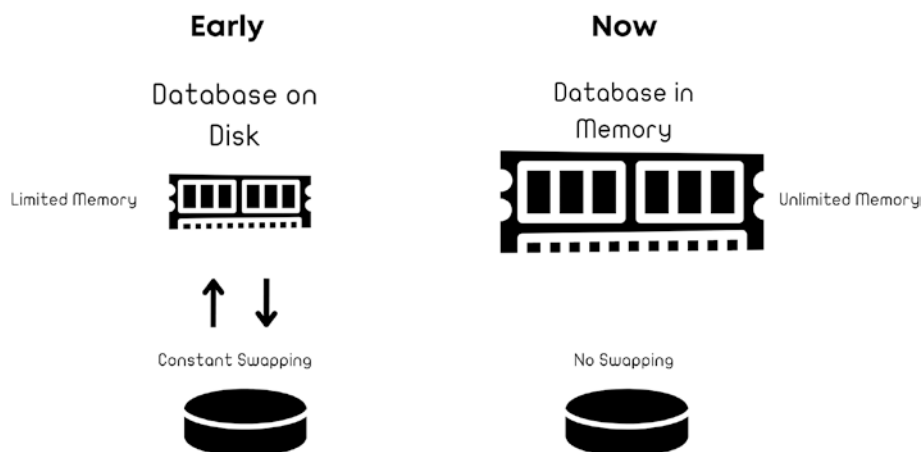


Figure 1-23. *The in-memory database*

It is possible to put an entire database into memory with the collaboration of two crucial factors:

- Extremely large quantities of memory
- Data-compression algorithms

SAP HANA Cloud automatically applies compression algorithms over the data. Usually very large databases can have their footprint reduced to a fraction of their original size, shrinking by about 90 percent.

So, There's No Need for a Disk?

We are certain that a complete database can be moved from disk to memory. So, there's no need for a disk? No, SAP HANA Cloud continues to consume disk space.

Data that resides in memory is classified as hot. Hot data regularly supports businesses processes. In general, hot data is the very recent and highly relevant data in most customer-facing application scenarios. For optimal read speed, this data should be kept as close to the CPU as possible.

Warm data is data that is infrequently accessed, implying that speed is not as relevant. Warm data is kept on disk and moved into memory only when it is required. Most businesses do not need all data in memory since they consider just a portion of it to be hot. Memory costs keeps falling but it remains relatively expensive when compared to disk.

Thus, it's recommended that customers carefully analyze the memory-size needs of hot and warm data when selecting their SAP HANA Cloud size, and select a memory/disk ratio arrangement that gives exceptional speed on the most frequently accessed data and a reasonable experience for the less frequently accessed data.

When you perform an SAP HANA Cloud tenant configuration, you select and define the sizing required.

Row and Column Store

SAP HANA Cloud is ACID-compliant.

Note “ACID” stands for Atomicity, Consistency, Isolation, and Durability (ACID). This is a criterion for a database to demonstrate that it is completely reliable for mission-critical applications. Even when numerous tables are being updated at the same time, the database must ensure data correctness and integrity. It should also be capable of data rollback and recovery.

Row store is the foundation of most traditional enterprise relational database tables. In an online transactional processing (OLTP) application, row store is thought to be the best storage solution. An OLTP application demands rapid record-level changes, with all columns that make up the record typically required for processing.

OLTP applications employing row store are fully supported by SAP HANA Cloud. The row store is thought to be the best storage solution for OLTP. OLTP applications often deal with large volumes of data that must be promptly aggregated via ad-hoc queries.

In comparison to many databases that only allow row or column store, the SAP HANA Cloud database provides both types (Figure 1-24).

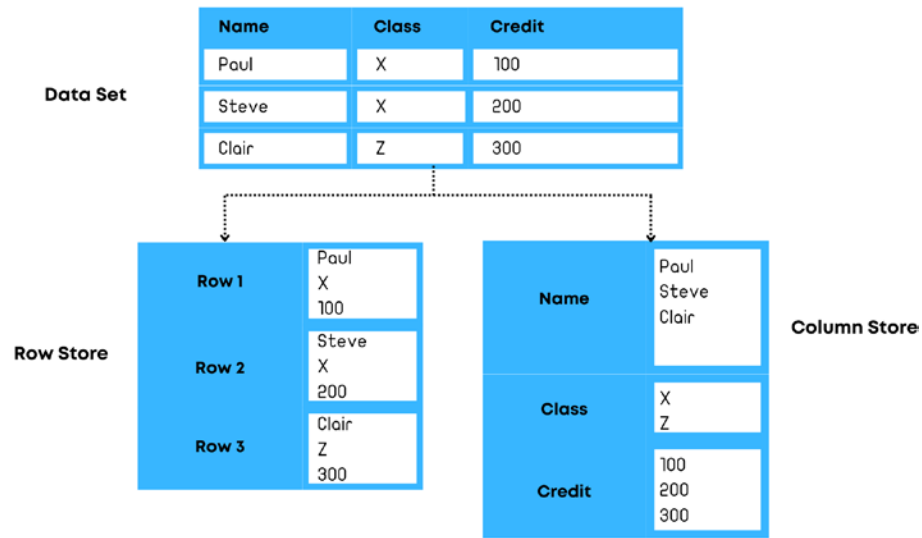


Figure 1-24. Row and column store

SAP HANA Cloud is the appropriate database on which to develop modern applications that work both as a system of record as well as a system providing analytical processing (*transalytics* applications).

Column tables are mostly efficient to support analytical use cases where data-selection demands are unpredictable. Queries made to the database from analytical applications frequently demand only a fraction of the data in the table. Typically, just a few columns from the table are necessary, as are only a few rows from those columns.

Only the essential columns are handled by columns tables, so you don't need to access columns that are unnecessary in a given query. Furthermore, the data is well organized, with all values of a given column available one after the other.

In a row store table, the data value is read from a column first, moving over to the other undesired columns until we return to the needed column to get the next qualified value that applies to your criteria. This is inefficient and detrimental to performance.

SAP HANA Cloud reads data columns in a column store table so quickly that extra indexes, while possible, are rarely needed. This makes things simpler by eliminating the need to repeatedly construct, remove, and rebuild indexes.

There is no complexity to modify column tables, and you can add or delete columns without having to drop and reload data. Column tables were designed to perform efficiently, taking advantage of parallel processing where each CPU core can handle each separate column.

In general, rebuilding whole records from the individually stored columns is the disadvantage of column tables. Reconstruction is mostly used in systems of record that demand whole records for updating, copying, or deleting. Thus, it is important to understand the purpose of a given data model, selecting carefully between row or column store according to the most relevant characteristic of that model.

However, more and more applications (for example, SAP S/4HANA) are blurring the distinction between transactional and analytical features. In this instance, you must select the storage solution that is appropriate for you. More and more applications are embedding analytical features that perform more efficiently when the data model has column store tables.

Note A table can be defined as row- or column-based.

Supporting Transactional and Analytical Applications

Column tables use compression algorithms to transform and store data to support high-performance reading and calculations. While this results in good read speed, it creates a challenge when writing operations are needed.

In a column table where the data is compressed, new data cannot be directly inserted into the compressed data. In a column store with data compression, there is a procedure that examines the whole data set and then produces a new compressed version of the data.

To directly add new data, the compressed existing data must be decompressed before the new data is inserted, and the complete data set is then compressed to generate a new version.

When many data insertion operations are necessary, more uncompressing and compressing tasks will be demanded, so it could result in a very expensive operation.

So, to address these difficulties, SAP HANA Cloud implements a smart approach that basically segments a column store table into two sections: delta store and main store. All writing operations use delta store, where the storage is not compressed. As a result, fresh data may be entered directly and quickly. The main store maintains compressed data. This data has been encoded and optimized for extremely high-performance reading. SAP HANA Cloud controls how and when data moves from delta to main store (Figure 1-25).

Delta Merge

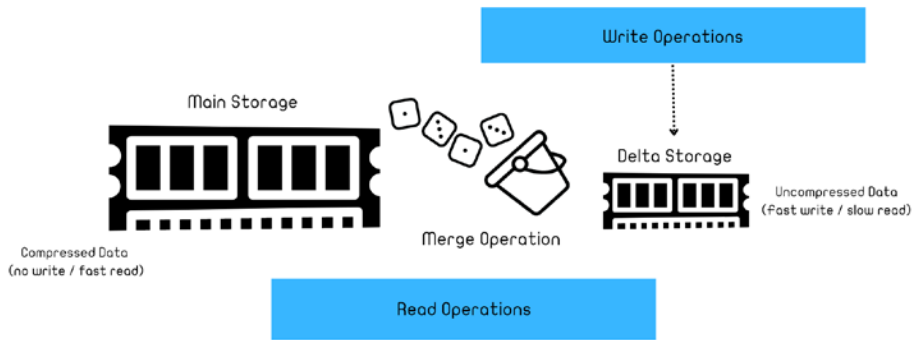


Figure 1-25. *The delta merge operation*

Reading operations merge data from the main and the delta storage, making it accessible to all applications. Developers are not required to understand the complexities behind this data movement between storages. They just request data from the table, regardless of the store where the data is located.

Naturally, as additional data is uploaded to the column store table, the delta storage will ultimately fill up. The more full the delta storage, the less efficient the read operation becomes as the computational cost of merging both storages grows. As a result, applications may begin to operate poorly. Furthermore, database memory is not optimized and fills up when delta storage data is uncompressed.

There is a threshold—which is set by a system parameter—that controls when the delta storage should be merged with the main storage. After this, the delta storage is emptied, and the cycle can begin again.

Data Footprint Reduction

The column store data table is compressed automatically, which lowers the size of the data footprint. It is not uncommon to obtain data footprint reductions of up to 90 percent.

A smaller data footprint allows the following:

- Store whole business databases in memory and eliminate disk access, resulting in faster searches and computations.
- Get more data into the CPU cache and thereby minimize main memory access to obtain even higher performance levels.
- Fit more data into your selected SAP HANA Cloud configuration, allowing you to get the most out of your investment.

The maximum data compression potential is influenced by the type of the business data. When there are redundant column values, compression results are most effective. Let's have a look at an example.

A big sales order database in an e-commerce system of record has the column COUNTRY, with distinct values Brazil, Chile, Mexico, United States, or Portugal. This would imply that those five values would appear several times in the column. This repetition is unnecessary.

1. Compression eliminates value duplication in a column by keeping each different value just once in a particular location known as a dictionary store. An integer is created for each unique dictionary value. As a result, the dictionary store would only have five records, each with its own distinct integer value representing the actual nation name.
2. Indexes are then constructed, with each integer positioned beside the original record position.
3. The original uncompressed data with its repetition is no longer required, thus it is eliminated from the table to generate a significantly smaller footprint.

Aside from decreasing storage requirements, replacing repetitious string values with integers is extremely efficient for in-memory processing.

The process just described is based on a compression method known as dictionary encoding.

Dictionary encoding is a first-level compression approach that is used to compress all columns in a column store table. However, there are other compression algorithms that may be used. Run-length encoding, cluster encoding, and prefix encoding are examples of second-level compression.

All users and developers are unaware of the compression algorithms that are in the core of SAP HANA Cloud database. The Database Explorer tool may be used to see the compression information of each column table.

During the delta merging procedure, compression is used. That is, data added into a column store database stays uncompressed until the first delta merging is executed. As a result, if you have a significant number of initial records in a table, a delta merge should be executed as soon as possible to decrease the footprint (Figure 1-25).

All delta merge operations recreate the compressed data to include any newly added and changed records.

Note Row store tables are not compressed.

Now, let's see which roles are available to support an SAP HANA Cloud implementation.

Key Roles

SAP HANA Cloud has a set of key roles that support the execution of main common tasks. Let's examine.

The Administrator Role

The SAP HANA administrator is responsible for the deployment and monitoring of the SAP HANA Cloud service.

Common Tasks:

- Analyze SAP HANA Cloud requirements for apps
- Define sizing and deployment
- Support the implementation of security settings
- Tune data models
- Analyze data growth of existing data models
- Schedule and trigger updates
- Migrate other databases into SAP HANA Cloud
- Monitor and troubleshoot SAP HANA apps

The required tool to perform these tasks is SAP HANA Cockpit.

If a migration from an on-premises database to SAP HANA Cloud is necessary, the administrator should learn how to use the available tools to automate and assure a successful conversion.

The administrator's primary task is to guarantee that SAP HANA Cloud is properly operational. This comprises creating and implementing a fail-over strategy. This might include the configuration of redundant backup servers (scale-out).

For monitoring and tracing, SAP HANA Cloud includes the SAP HANA Cockpit. Administrators should understand how to use these tools so that SAP HANA Cloud applications may be completely optimized. When compared to on-premises SAP HANA deployments, the administrator's job in SAP HANA Cloud is greatly reduced. This is because SAP HANA hardware and software do not need to be installed and maintained. SAP HANA Cloud is installed utilizing a simple interface and does not

require any IT expertise. However, the administrator will most likely be engaged in sizing the required cloud tenant, which includes the database computation, memory, disk, and, if necessary, the data lake.

They will also be required to help link on-premises and other cloud apps to SAP HANA Cloud. However, the day-to-day administration tasks of keeping SAP HANA functioning, as well as backups, are the responsibility of SAP or hosting partners, which is a significant distinction between cloud and on-premises deployments.

The Modeler Role

The virtual data layer in the SAP HANA Cloud database is built by the SAP HANA Cloud data modeler. This is mostly composed of calculation views, but it may also contain other database artifacts such as user-defined functions and procedures. Calculation-view development is the most important skill of an SAP HANA Cloud data modeler. A simple calculation view is simple to design, but understanding how to develop high-performance yet easy-to-maintain models that harness the full capabilities of the SAP HANA Cloud in-memory database takes a lot of knowledge and training.

Common Tasks:

- Develop high performance data models
 - Calculation views
 - Graph workspaces
 - Predictive models
- Analyze analytical requirements of apps
- Design and configure security user access
- Select correct infrastructure setup to support analytical models

The required tool to perform these tasks is SAP Business Application Studio.

SAP HANA Cloud data modelers should also be proficient with SQL because they will be writing custom logic to expand the conventional capabilities of the calculation views. SQL is also used to test and track the execution of the calculation views; thus a solid knowledge of SQL is required for this function.

Aside from basic modeling using calculation views, data modelers should also learn about advanced SAP HANA data-modeling topics, including spatial, predictive, and graph data processing. SAP HANA Cloud is a strong data-processing platform, and a data modeler who can create sophisticated models that utilize various forms of data is an important element of an SAP HANA Cloud development team.

Because most applications require some level of analytical processing, the data modeling role overlaps with that of an application developer. Instead of managing the data-processing logic within the application code, the analytic processing part of an application is best handled by generating and consuming data models that utilize the supplied SAP HANA Cloud database artifacts, such as calculation views and SQL-based models. Application developers are often focused on building application and UI code and are usually not experts in complex data modeling. When data modeling requirements get sophisticated, application developers frequently need to recruit the help of a specialist SAP HANA Cloud data modeler.

Security may also be a point of overlap with the team members in charge of this aspect of the implementation. Protecting data from illegal access is a task for the data modeler, who creates specific database objects called analytic privileges to give data-access rules to each user/role. However, this might also be the responsibility of the security experts on the team. But whoever takes ownership of this task, collaboration is essential between the security experts and data modelers.

The Application Developer Role

The role of the application developer is to build cloud applications using different programming languages.

Common Tasks:

- Design, develop, and maintain cloud apps
- Migrate SAP HANA on-premises apps to SAP HANA Cloud
- Analyze analytical requirements of apps
- Collaborate in the design of security user access
- Support discussion around infrastructure requirements and data growth of analytical models

The required tool to perform these tasks is SAP Business Application Studio.

SAP HANA Cloud provides the tools to support application developers. The main tool is the SAP Business Application Studio. This tool is used to develop cloud foundry applications in multiple programming languages.

The abilities required are determined by the programming languages used for application development. SAP HANA Cloud on Cloud Foundry supports the use of many languages to develop the stack's various tiers: UI, application, and database. JAVA, Node.JS, Python, SQL, and Core Data Services (CDS) are examples of different programming languages and frameworks supported by SAP HANA Cloud.

The Security Architect Role

The security role is often taken over by the administrator. But as rules tighten around data privacy and data access, organizations often employ dedicated security experts to take on this role.

Common Tasks:

- Plan and implement security settings and policies on SAP HANA Cloud
 - Authentication
 - User authorization
 - Compliance and governance
 - Trace, monitor, and audit security
- Support discussions around apps and data-access rules
- Establish data access to models and data-privacy rules

The required tool to perform these tasks is SAP HANA Cockpit.

The security role covers many different areas of SAP HANA Cloud. The main areas related to security are authentication (who has access) and authorization (what can be accessed).

Security also covers the increasingly important topic of data governance and compliance. SAP HANA Cloud includes many features and tools to ensure cloud applications are built to be secure and yet easily accessible by those who should have access.

The cloud providers secure the customer's data against unauthorized external access, but the customer still plays an important role in maintaining the policies for internal access to apps and data.

The Data Architect Role

Data management is becoming a big subject as companies collect more data than ever before and data-driven applications become the norm. SAP HANA Cloud provides various tools for data management, and this has now become a recognized function in an SAP HANA Cloud development project.

Common Tasks:

- Plan, implement, and manage a data architecture
 - Implement data tiering
 - Data integration
 - Data quality
- Map data-access requirements, data growth, and archiving of models
- Understand infrastructure alternatives to support correct data management
- Establish data access to models and data-privacy rules

The required tools to perform these tasks are SAP HANA Cockpit, SAP Business Application Studio, Smart Data Access (SDA), and Smart Data Integration (SDI). Previously, IT would serve as data architects, supplying storage capacity as needed and establishing archiving solutions. However, application data requirements have gotten more sophisticated, and we demand access to data from all sources and in real time, in addition to access to data lakes. As data and its intelligent use are increasingly acknowledged as a business's competitive advantage, new jobs in organizations such as data manager or data architect have emerged.

SAP HANA Cloud includes capabilities for connecting, integrating, and cleansing data from on-premises and cloud sources. Working with these specialized tools requires abilities that are not often available in other SAP HANA roles.

The data architect learns to deal with native SAP HANA data management tools, usually SDI, SDA, and NSE, as well as tools that can interact with SAP HANA, like SAP Cloud Integration Suite, SAP Cloud Data Integration, SAP LT (SLT), and others.

Summary

You have successfully completed this chapter, and now it's time to review what you learned and see what's next.

- The way we consume information has changed, and this behavior reflects how we demand IT and how we can support users.
- The intelligent enterprise is an extensive strategy that joins cutting-edge technology with business processes to create considerable corporate value.
- SAP Business Technology Platform (SAP BTP) is a Platform-as-a-Service offering that has all cloud and on-premises technologies required to run applications.
- SAP HANA Cloud is a cloud Database-as-a-Service (DBaaS) that is completely managed and in-memory. It is the SAP Business Technology Platform's cloud-based data foundation. SAP HANA Cloud allows you to build, operate, and innovate applications.

Now we will continue our journey with essential learning about the SAP HANA Cloud administration.

CHAPTER 2

Administration of SAP HANA Cloud

The SAP Business Technology Platform (SAP BTP) offers a wide range of services. Every SAP BTP service offers one or more service plans. A service plan is a subset of a service. The SAP HANA Cloud database service, for example, is available in a variety of configurations and sizes, as shown in Figure 2-1.

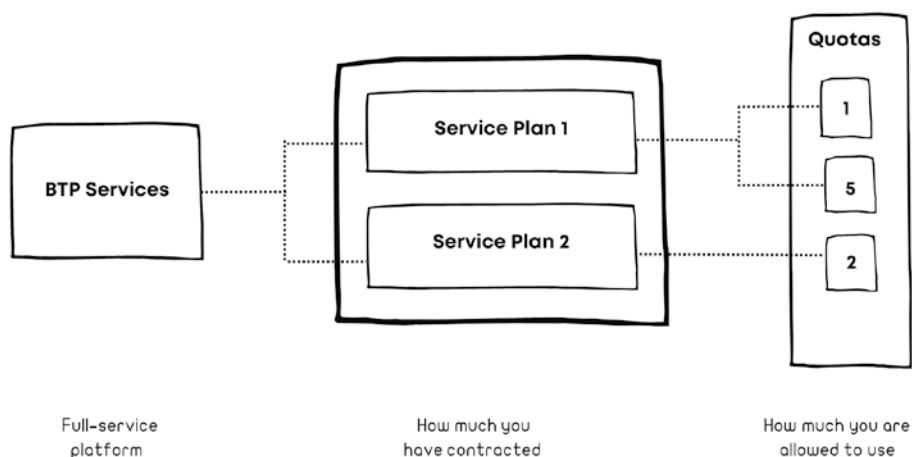


Figure 2-1. *Entitlements and quotas*

Entitlements and Quotas

SAP BTP provides different services that have at least one service plan. A **service plan** offers a different configuration for a given service. The service plans you select are referred to as **entitlements**. There is another important concept known as **quotas**. A quota indicates how much of a service plan you are allowed to utilize.

Entitlements and quotas are allocated at the global account level and utilized by subaccounts that need them.

When quotas or entitlements are removed from a subaccount, they become free again and can be used in other subaccounts.

To see how services are provisioned and consumed at the global account level, you can navigate through the Usage Analytics panel available in SAP BTP Cockpit.

Global Accounts

To begin using SAP HANA Cloud, you must first have an SAP BTP global account. In BTP, this is the highest level of account administration. Consider the global account to be the “contract” level.

SAP BTP supports two types of global accounts: enterprise accounts and trial accounts.

A trial account enables you to test the SAP BTP services for free. Access is available to everyone. Trial accounts are designed for personal use rather than production use or team application development. They grant limited access to the platform’s resources and services. The trial duration varies according to current trial rules defined by SAP.

After you’ve created your global account, you may establish one or more subaccounts that use the global account’s features. You have to select the cloud provider as well as the region when creating a subaccount.

For instance, Azure in North America and AWS in South America. The cloud provider region does not have to be the same as the location of consumption. The list of supported cloud providers and regions will grow as new cloud providers and regions become accessible.

Subaccounts are logically isolated from one another. This implies that users, apps, authorizations, and so on are not automatically shared between subaccounts, as shown in Figure 2-2.

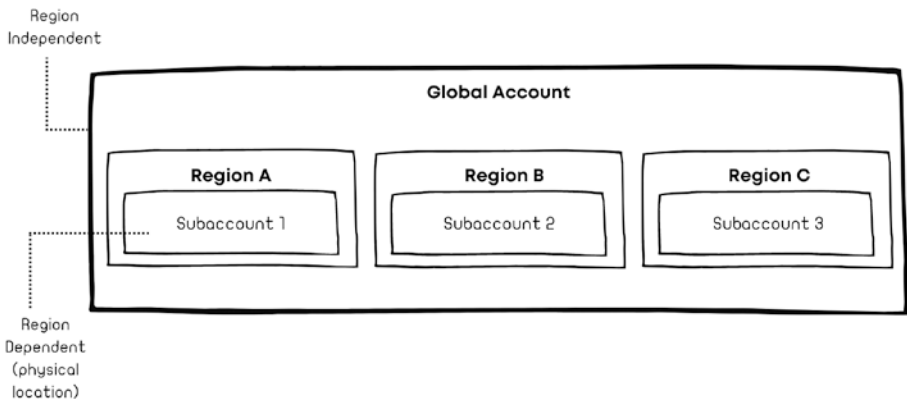


Figure 2-2. *Global and subaccounts*

Directories can be used to manage subaccounts if desired. Directories enable you to organize subaccounts for easier management if you have a large number of them. If you don't want to set quotas directly to subaccounts, you may instead allocate these quotas to directories (Figure 2-3).

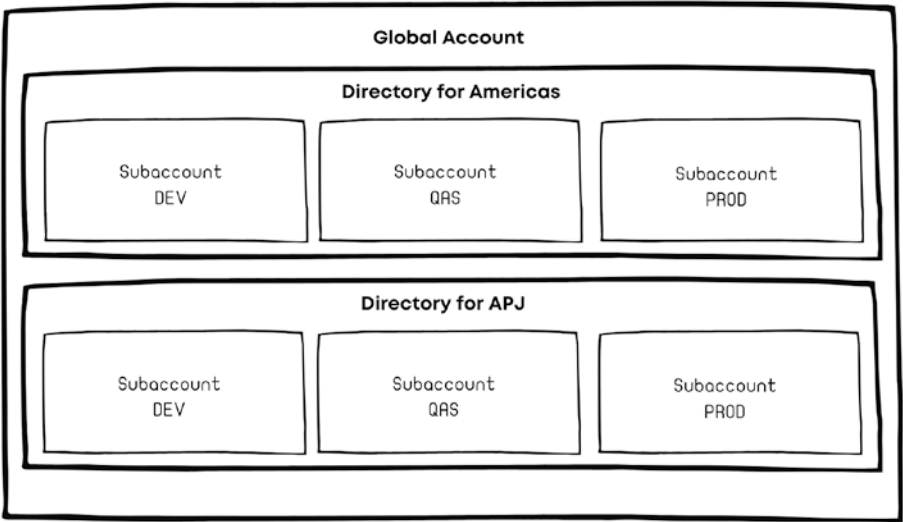


Figure 2-3. Account structure

To establish a directory hierarchy, a directory can be subdivided into several directories. Any subaccount may be mapped to any level of the directory hierarchy. Now, let’s see how space and instance concepts support the usage of SAP HANA Cloud.

Space and Instance

When you sign up for the SAP HANA Cloud service, the SAP BTP walks you through the process of generating a copy of SAP HANA Cloud’s SAP HANA database component in your SAP BTP account. This replica is known as an *instance*, and the process of producing an instance is known as *provisioning*. Using basic questions, a SAP BTP wizard walks you through the process of establishing an instance.

However, before you can build your first instance, you must first establish a Cloud Foundry space. A space offers total separation from other applications, allowing them to operate independently. A space can hold several database instances (Figure 2-4).

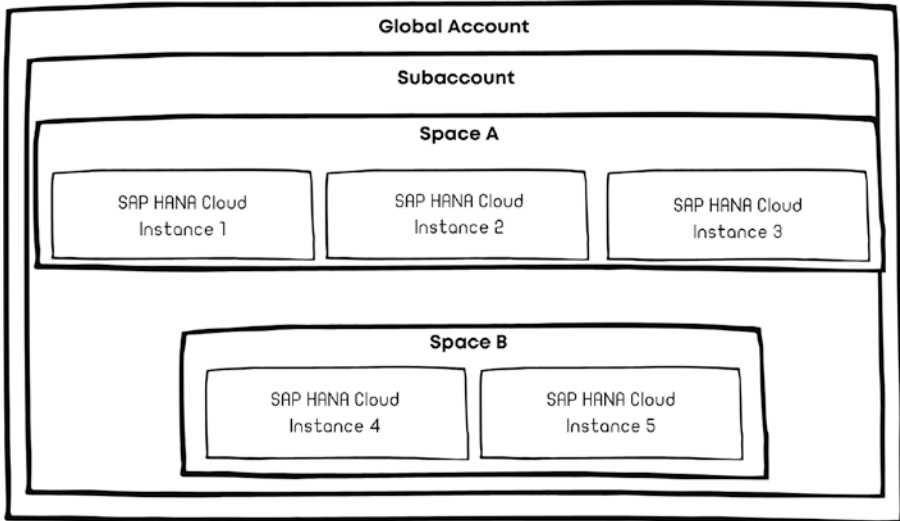


Figure 2-4. Account structure

You may start constructing your BTP application after you have a SAP HANA Cloud database instance. You may select which database instance to bind to during application deployment.

Note In the context of SAP BTP, a service is a software product you use on a subscription basis through your SAP BTP account. SAP HANA Cloud is an example of an SAP BTP service you subscribe to.

To move ahead with the process, it is necessary to define where SAP HANA Cloud service will run, so you must select the hyperscaler cloud provider that will provide that service.

Hyperscalers

Choose your cloud provider based on the region from which you would like your service to be provisioned. Also, consider that not all cloud providers offer the same services, such as high availability. They also may not offer all application development and runtime environments.

Here is a list of current hyperscalers:

- SAP
- Amazon Web Services (AWS)
- Google Cloud Platform (GCP)
- Alibaba Cloud
- Microsoft Azure

Common tasks owned by SAP are as follows:

- Installing, configuring, and upgrading the operating system
- Backing up, restoring, and recovering the database software
- Tuning the database to run optimally on underlying operating system and hardware

Common tasks that are not owned by hyperscalers are as follows:

- Sizing and provisioning the SAP HANA Cloud database
- Monitoring the SAP HANA Cloud database
- Managing users, roles, and permissions in the SAP HANA Cloud database

The customer is in charge of these tasks. So, let's check some important details to keep in mind when setting up an SAP HANA Cloud database instance.

Setting Up a SAP HANA Cloud Database Instance

Before you can build a SAP HANA Cloud database instance, you must first complete the following prerequisites:

- Create a subaccount.
- Enable Cloud Foundry for this subaccount.
- Assign a quota to this subaccount.
- Create a space in this subaccount.

Using the provisioning interface of SAP HANA Cloud Central, you can create a database instance (Figure 2-5).



Figure 2-5. *Create a database instance*

Using the wizard, you are guided through the following steps to create the database instance:

1. **Location and Basics:** Choose a Cloud Foundry organization and space. Choose a name and description for the instance. Choose a username and password to log on to the instance.
2. **SAP HANA Database:** Choose how much memory, CPU, and disk storage you require.

3. **SAP HANA Database Availability Zone:** Choose a cloud provider and geographic region where your instance will be hosted.
4. **SAP HANA Database Advanced:** Choose the type of connectivity that you will permit for your instance. For example, can only other SAP BTP apps connect to the instance, or can this include other external cloud apps? Do you allow connection to on-premises apps?
5. **Data Lake:** Choose to set up an optional data lake to work with your database instance for storing huge amounts of warm data.

Working with Administration Tools

To support administration tasks of an SAP HANA Cloud instance, you can use different tools that provide alternatives ways via an easy-to-use graphical interface or via command line.

SAP HANA Cloud Central

Use SAP HANA Cloud Central to provision and manage SAP HANA Cloud instances (Figure 2-6).

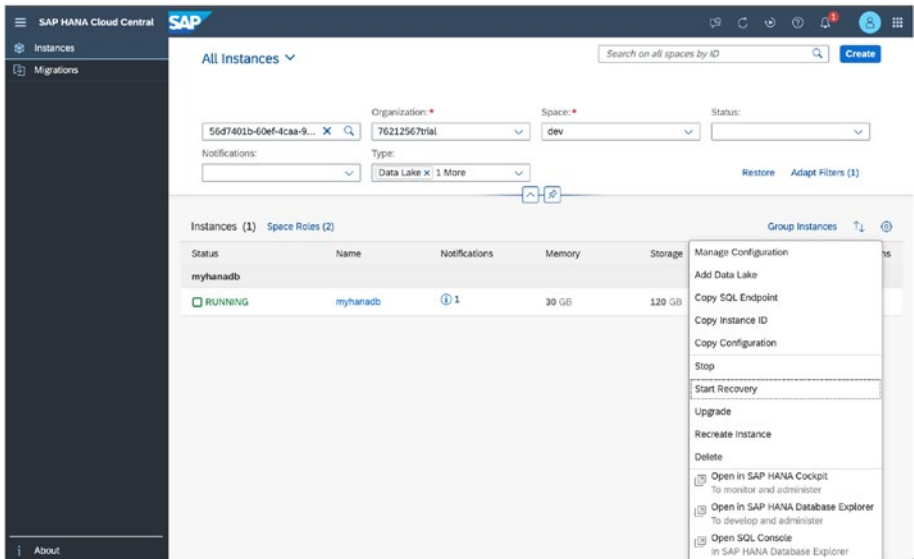


Figure 2-6. SAP HANA Cloud Central

In addition to creating, updating, or deleting instances, you may utilize SAP HANA Cloud Central to search for and display all specific information about each instance, including the following:

- Type of instance (data lake, database)
- Status of each instance (stopped, starting, creating, etc.)
- Notification/alerts
- Memory, storage, and CPU assigned to each instance

Using the filter tool, you may look for instances based on type, alert, status, or other parameters. You can add or remove columns of the instance list by clicking the Settings icon.

SAP HANA Cockpit

A key tool used by the administrator of SAP HANA Cloud is the SAP HANA Cockpit (Figure 2-7).

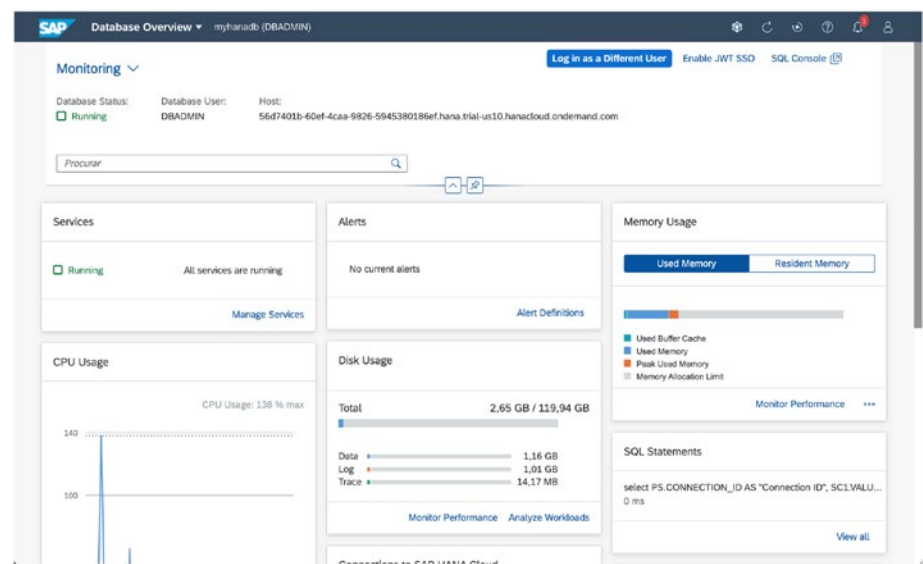


Figure 2-7. SAP HANA Cockpit

The SAP HANA Cockpit gives you a single point of access to a variety of tools for managing and monitoring your SAP HANA Cloud databases. Using customizable cards, the tool displays essential database information. Each card contains only the most important information. However, each card may be drilled down into to access more extensive information. The cards are organized into views. SAP provides four standard views, but administrators can define their own. A custom view can include only the cards that the user is interested in, as well as the settings and filters that apply to them.

The key areas of SAP HANA Cloud that can be managed and monitored using the SAP HANA Cockpit include the following:

- **Services** – database services such as indexserver, nameserver
- **Memory** – monitor memory usage and check out-of-memory issues
- **Alerts** – be warned of critical situations, such as disk becoming full
- **Workload** – organize jobs into workloads for better system utilization
- **Table Usage** – ensure tables are optimally designed for best performance
- **Database Configuration** – manage configuration (*.ini) files that determine database behavior

The SAP HANA Cockpit for your SAP HANA instance may be accessed via SAP HANA Cloud Central, SAP BTP Cockpit, or the direct URL.

The SAP HANA Cockpit cloud version is built on the same design as the SAP HANA Cockpit used in SAP HANA on-premises deployments.

The SAP HANA Cockpit has a different release cycle from SAP HANA Cloud. New features are instantly made available to all clients. There is no need to upgrade; this is handled by the cloud provider.

Note Take care not to mix up SAP HANA Cockpit with SAP BTP Cockpit. The latter is used to administer all SAP cloud applications, including SAP HANA Cloud, among many others. SAP HANA Cockpit solely manages SAP HANA Cloud databases.

Cockpits give an easy-to-use interface for performing important management activities. Now, let's explore command-line interfaces that allow a more direct operation.

Command-Line Interfaces

There are two command-line interfaces (CLIs) provided for use with SAP HANA Cloud setup and management. Administrators who wish to write scripts to automate processes or copy/paste frequently used commands generally choose command-line interfaces.

SAP BTP Command-Line Interface (btp CLI)

For customers who prefer to operate with a command line, the SAP Business Technology Platform (BTP) Command Line Interface (CLI) offers an alternative to the BTP Cockpit. This interface is commonly referred to as btp CLI (Figure 2-8).

```
C:\Users\>btp
SAP BTP command line interface (client v2.0.0)

Usage: btp [OPTIONS] ACTION [GROUP/OBJECT] [PARAMS]

CLI server URL:      https://cpcli.cf.eu10.hana.ondemand.com
User:                not set
Configuration:       C:\Users\\AppData\Local\SAP\btp\config.json

You are currently not logged in.

Tips:
  To log in to a global account of SAP BTP, use 'btp login'. For help on login, use 'btp --help login'.
  To display general help, use 'btp --help'.

OK
```

Figure 2-8. *btp CLI*

These are some of the tasks that you perform with the `btcli` CLI (Table 2-1):

- Creating subaccounts and directories
- Managing entitlements of global accounts and subaccounts
- Managing users and their authorizations in global accounts and subaccounts
- Subscribing to applications

Table 2-1. *Examples of Tasks Using `btcli` CLI*

Task	Command
List subaccounts	<code>btcli list accounts/subaccount</code>
Get details of the global accounts	<code>btcli get account/global-account</code>
List role collections	<code>btcli list security/role-collection</code>

A trial account can be used to test the BTP CLI.

Cloud Foundry Command-Line Interface (cf CLI)

For managing subaccounts in the Cloud Foundry environment, such as creating orgs and spaces or maintaining quotas, use the Cloud Foundry Command Line Interface (cf CLI), as shown in Figure 2-9.

```
~ % cf
cf version 7.1.0+4c3168f9a.2020-09-09, Cloud Foundry command line tool
Usage: cf [global options] command [arguments...] [command options]

Before getting started:
  config      login,l      target,t
  help,h      logout,lo

Application lifecycle:
  apps,a      run-task,rt   events
  push,p      logs         set-env,se
  start,st    ssh          create-app-manifest
  stop,sp     app          delete,d
  restart,rs  env,e        apply-manifest
  restage,rg  scale         revisions

Services integration:
  marketplace,m      create-user-provided-service,cups
  services,s         update-user-provided-service,uups
  create-service,cs  create-service-key,csk
  update-service    delete-service-key,dsk
  delete-service,ds service-keys,sk
  service          service-key
  bind-service,bs  bind-route-service,brs
  unbind-service,us unbind-route-service,urs
```

Figure 2-9. *cf CLI*

These are some of the tasks that you perform with the cf CLI (Table 2-2):

- Create spaces
- Add ORGANIZATION members
- Add SPACE members
- Create SPACE quota plans
- Assign quota plans to SPACES

Table 2-2. *Examples of Tasks Using cf CLI*

Task	Command
Create a space	cf create-space SPACE [-o ORG] [-q SPACE_QUOTA]
Set a space role	cf set-space-role USERNAME ORG SPACE ROLE
Set a space quota	cf set-space-quota SPACE_NAME SPACE_QUOTA_NAME

Now that you are a bit familiar with some tools that allow you to interact with the SAP HANA Cloud, let's explore the SAP HANA database content.

SAP HANA Database Explorer

The SAP HANA Database Explorer is used to query database information and display information about your SAP HANA database's catalog objects (Figure 2-10).

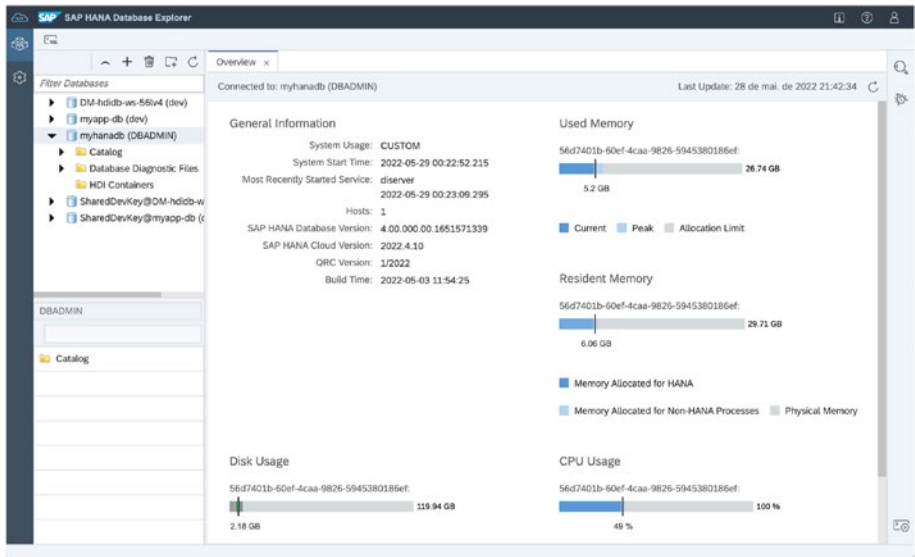


Figure 2-10. SAP HANA Database Explorer

The built-in SQL Console is a significant component of the SAP HANA Database Explorer. This is where you'll write SQL and SQLScript statements. It has code correction and autocomplete features. To identify errors, you may run the SQL statements, inspect the results, and review the SQL plan and run-time data. Some additional features are:

- Database items can be located using search tools.
- The Database Explorer works with both HDI containers and classic schemas.
- The SAP HANA Database Explorer is accessed via the SAP HANA Cockpit or the Business Application Studio.

Next, let's see how to check HANA Deployment Infrastructure (HDI) information from an administration perspective.

HDI Cockpit

Containers are important in database development because they provide an effective way of segregating database runtime artifacts and encourage modularization. Users and roles are granted container rights so that they may access the resources contained inside them.

A SAP HANA Cloud database will often have a large number of containers. Each container will have a unique set of user and role rights allocated to it. This can quickly become tough to handle.

The HANA Deployment Infrastructure (HDI) Cockpit is a tool that allows administrators to quickly explore containers, organize them, and display the users and roles that have access to them (Figure 2-11).

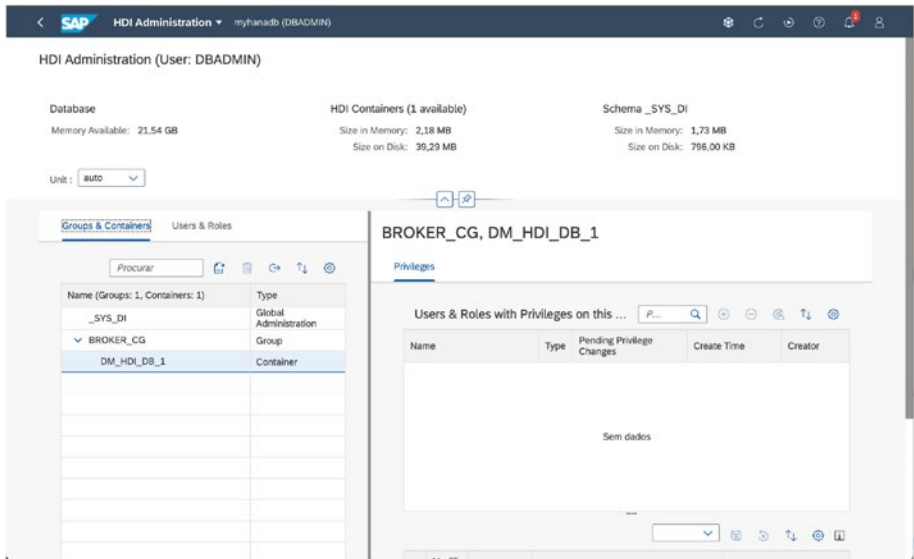


Figure 2-11. *HDI Cockpit*

Administration Tasks

Now that you are a bit familiar with administration tools, let's look at the most common tasks supported.

Launching and Shutting Down

The SAP HANA Cloud administrator is in charge of launching and shutting down SAP HANA Cloud database instances. SAP HANA Cloud Central is the avenue by which to accomplish this.

When operating a database there are some circumstances when you may need to stop and restart it. Some examples are as follows:

- Adding a data lake
- Scaling up the database to add more RAM
- Maintaining the application that is running on the database
- Restarting after troubleshooting, at suggestion of cloud provider support team

Note Database instances created with a trial BTP account will be stopped automatically after a period of inactivity. If you want to use them again, you must restart them.

Working with New Releases

A new update of the SAP HANA Cloud database is usually released every three months. It is the SAP HANA Cloud administrator's responsibility to upgrade the SAP HANA Cloud database for each database instance that has been provisioned. However, if the administrator doesn't update the instance manually, SAP will automatically update it when the threshold for a given version is reached. Bear in mind that in a maintenance window the customer is notified in advance.

Different than with on-premises systems, the upgrading of an SAP HANA Cloud database is extremely rapid and totally automated.

The SAP HANA Cloud Central tool is used to initiate the upgrade. The database will be stopped and restarted throughout the update, thus this activity should be carefully planned so that it does not disrupt the company.

Cycle of SAP HANA Cloud Releases

One of the primary advantages of adopting a cloud solution is that new features are accessible much more quickly than with on-premises systems. Whereas an on-premises system's new release occurs once a year or twice a year, cloud-solution upgrades occur more regularly—every few weeks, on average.

For SAP HANA Cloud, currently, a new version is released every three months. This is known as the quarterly release cycle (QRC). The release code format is QRC QQ/YYYY (Figure 2-12).

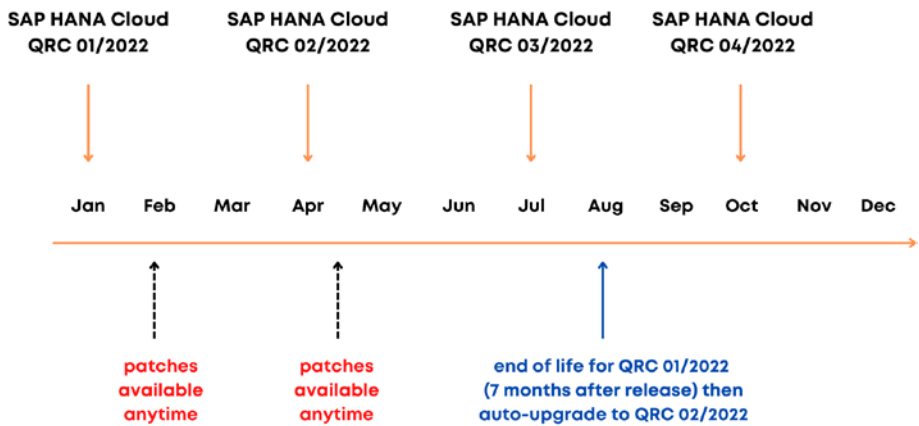


Figure 2-12. SAP HANA Cloud release cycle

Customers can opt to update as soon as the new release is ready, or they can postpone the upgrade for up to seven months. If a consumer does not choose to upgrade after seven months, the upgrade is completed automatically. The automated upgrade takes the user to the release that comes after the one they were using, rather than the most recent available.

SAP releases some updates that address security and other high-priority areas of SAP HANA Cloud on a regular basis, and these patches do not wait for a QRC release.

The SAP HANA Cloud update demonstrates one of the most significant distinctions between SAP HANA Cloud and SAP HANA on-premises. Whereas an on-premises SAP HANA Cloud update demands a large investment in resources and time, and comprises several processes, a SAP HANA Cloud upgrade is automated, relatively straightforward, and begins with a single menu selection chosen by the client.

Backup and Recovery of the SAP HANA Cloud Database

As an administrator of an SAP HANA Cloud, sometimes you will be required to back up and recover a database. Next, we will cover these topics.

Back Up the SAP HANA Cloud Database

Backups of your HANA Cloud instance occur automatically to guarantee that it can be restored to its most recent consistent state. Multiple backups are saved in order to restore it not just to the most recent backup, but to any earlier backup that is accessible (Figure 2-13). This is highly handy since it allows you to pick a secure state of the database, even if it is not the most recent backup.

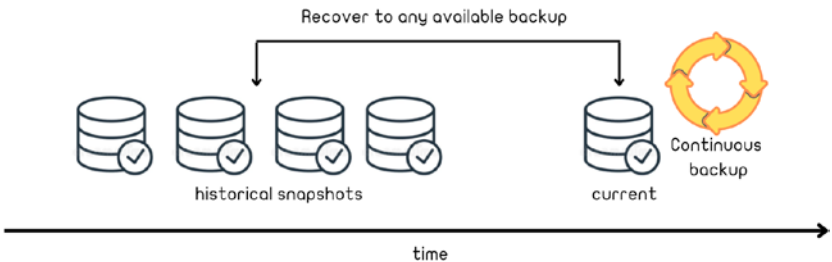


Figure 2-13. SAP HANA Cloud release cycle

Backups are replicated across several availability zones within the same region. This provides additional protection against data loss in the event of a single data center's encountering a disaster. SAP HANA database instances are backed up on a regular basis.

- The recovery point objective (RPO) is 15 minutes or less. This implies that any potential data loss is restricted to a maximum of 15 minutes of data updates.
- Snapshot backups can be made and saved to establish historical recovery points. Up to 15 backups are kept.
- In the SAP HANA Cockpit, you may view information about available database backups.
- The SAP HANA Cloud administrator is in charge of backup management.

Recover the SAP HANA Cloud Database

SAP HANA Cloud Central is the tool used by the administrator to start the recovery process. With this tool, the administrator is able to recover the database up to a selected point in time for which a backup is available.

Recovery of the SAP HANA Cloud database is the responsibility of the SAP HANA Cloud administrator.

Manage and Monitor Performance

The SAP HANA Cloud database administrator is responsible for monitoring the availability and resource consumption of the SAP HANA Cloud database services.

Although SAP HANA Cloud considerably decreases and streamlines the work for administrators, the administrator still has several responsibilities, including the following:

- **Workload management** – balance consumption of database resources across applications
- **Managing tables** – manage partitions, define placement of tables for scale-out, monitor replication of data, check table consistency, monitor compression ratios, load/unload to memory
- **Schedule administrative tasks** – write scripts to automate repetitive tasks
- **Set up alerts** – define alerts to warn of potential issues
- **Utilize support resources to fix issues** – look up SAP notes, schedule chat with support team, collect log info for support to analyze, use SAP Community to support
- **Manage data tiers** – configure tables to use NSE and monitor the performance, monitor the data lake

Secure the SAP HANA Cloud Database

Security administration is a subset of general administration that is often administered by security specialists, who must evaluate the full IT landscape rather than simply the SAP HANA Cloud database. They must guarantee that all solutions are compliant.

In a SAP HANA Cloud environment, the security administrator is responsible for a variety of activities. Let's look at some examples:

- **Monitoring critical security settings** – SAP offers security advice such as password expiration for users, how system rights should be divided, and how to establish the database activity trace level. These should be reviewed and checked by the security administrator.
- **Data encryption** – To safeguard data and logs, the SAP HANA Cloud database employs a number of encryption services. Although SAP handles encryption, the security administrator should use the SAP HANA Cockpit to monitor the state of encryption.
- **Auditing activities** – monitor and record specific actions executed in the SAP HANA database, enabling visibility into who did what in the database (or attempted to do what) and when. Changes to user authorization, the creation or deletion of database items, user authentication, modifications to system configuration, and access to or changes to sensitive information are some examples.
- **Manage certificates and keys required for trust validation** – SAP HANA Cloud offers a variety of authentication mechanisms, many of which make use of keys and certificates.
- **Monitor data anonymization** – SAP HANA Cloud database offers anonymization services to preserve data privacy. This is accomplished by establishing SQL views with an anonymization option. The security administrator generates reports that show which views are anonymized (and which are not) as well as numerous KPIs that show how data is safeguarded.

Manage Users, Roles, and Permissions

The security administrator is in charge of user management. “Users” is a broad term that may relate to both developers and app users. Typical administration activities include the following:

- Creating users and user groups and assigning roles and privileges
- Investigating authorization of authentication issues
- Deactivating users

The SAP HANA Cloud database base user is created by a security administrator. The user is set to one or more roles.

The role includes one or more privileges that grant role-based access to database objects like tables and views, as well as the data they expose.

Users can be allocated to user groups, and administrators can be set to manage all users in a certain group (Figure 2-14).

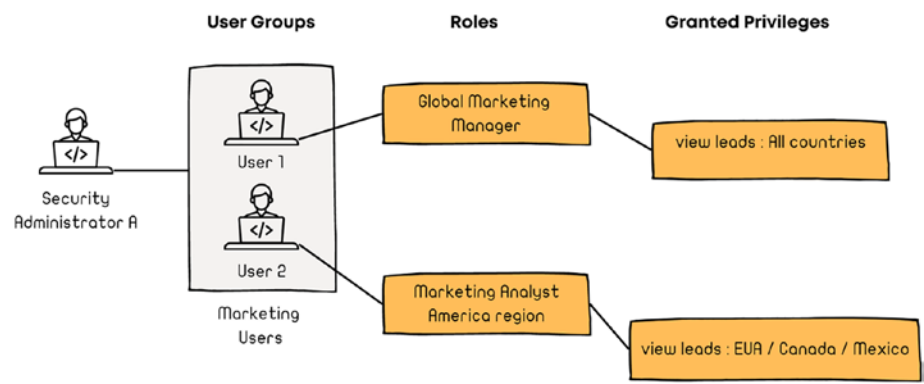


Figure 2-14. *Users and roles*

The setup sequence is usually as follows:

1. Create privileges (many standard privileges are supplied by SAP).
2. Create roles.
3. Assign privileges to roles.
4. Create users.
5. Create user groups.
6. Assign users to user groups.
7. Assign users to roles.

Note Different than SAP HANA on-premises, in SAP HANA Cloud there is no system-wide “user administrator” user; that is, a user with the system permission USER ADMIN. This is due to the fact that the user SYSTEM in SAP HANA database instances is reserved for SAP’s usage. DBADMIN is the default administration user in SAP HANA database management. As the default user group’s administrator, DBADMIN may be used to create additional user groups and grants administrative rights to users belonging to these groups.

Summary

You have successfully completed this chapter, and now it’s time to review what you learned and see what’s next. We covered the following:

- Understand key concepts of SAP BTP
- Get an introduction of account structure, entitlement set, and global and subaccounts

CHAPTER 2 ADMINISTRATION OF SAP HANA CLOUD

- Map the steps necessary to create an SAP HANA Cloud instance
- Identify how SAP HANA supports backup and recovery operations
- Get an overview of how SAP HANA Cloud supports users and roles

Now we will continue our journey with essential learning about SAP HANA analytical processing capabilities.

CHAPTER 3

Analytical Processing with SAP HANA Cloud

In this chapter, you will learn key features of data object modeling as well as different structures, such as calculation views, dimensions, cubes, and analytical views. You'll also be introduced to how models can be enhanced using security modeling, SQLScript, table functions, and procedures, as well as advanced data processing capabilities with SAP HANA Cloud spatial, predictive analysis, and graph modeling.

Modeling in SAP HANA Cloud

With SAP HANA Cloud, the main type of modeling object is called a calculation view. A calculation view mixes data from several tables or views, applies filters, computes new data, and aggregates the data to provide summarizations at any level. Calculation views are created in SAP HANA Cloud with simple graphical modeling tools and are deployed to the SAP HANA Cloud database. When a calculation view is generated in a run-time object, it produces a column view that can be queried with SQL.

Instead of embedding sophisticated code in the application to handle the data, the application now calls the needed calculation views, and the processing is pushed down to the SAP HANA Cloud database. This is advantageous in the following ways:

- Because it does not have to deal with as many data-processing duties, the application code is simpler. These tasks are sent to the SAP HANA Cloud database for in-memory processing.
- Data processing occurs where the data lives, eliminating the need to transport raw data from the database to the application. We solely send the data-processing results to the application.
- We avoid redundancy by reusing calculation views across various apps.

Classic Approach: Data-Modeling Code in the Application

The database's part in a classic application is to facilitate the storage and provision of raw data. The data in the database is generally processed relatively little, if at all.

The raw data is transferred straight from the database to the application. The program then starts processing the data by merging, aggregating, and conducting calculations to create something relevant.

We may have to move a large amount of raw data between the database and the application. When we bring raw data to the application layer, we complicate the application code. This is due to the fact that it must handle data-processing and modeling activities in addition to handling all of the other process-flow control, business logic, user interface (UI) operations, integration of data from different sources, and so on (Figure 3-1).

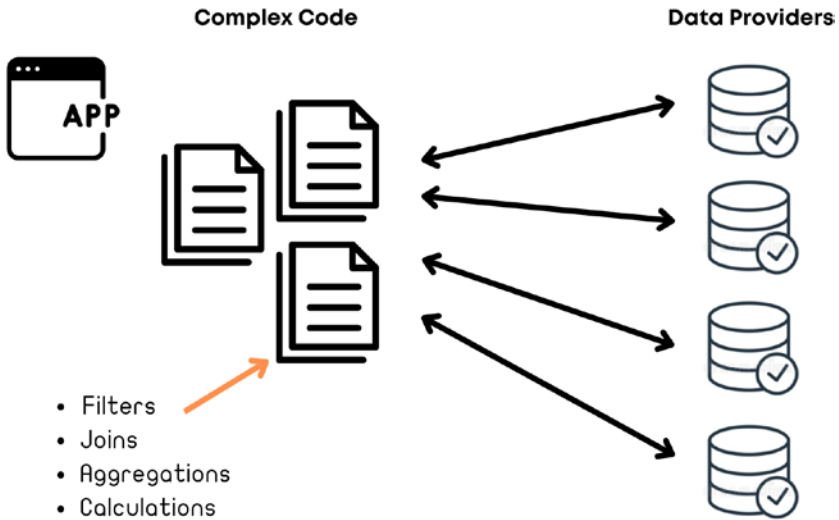


Figure 3-1. *Complex application*

New Approach: Data Modeling in the Database

We can develop a comprehensive modeling layer on top of the database tables, right inside the database, using SAP HANA Cloud. This means that we may first process the raw data in the database and convert it into something relevant before giving it to the application for consumption (Figure 3-2).

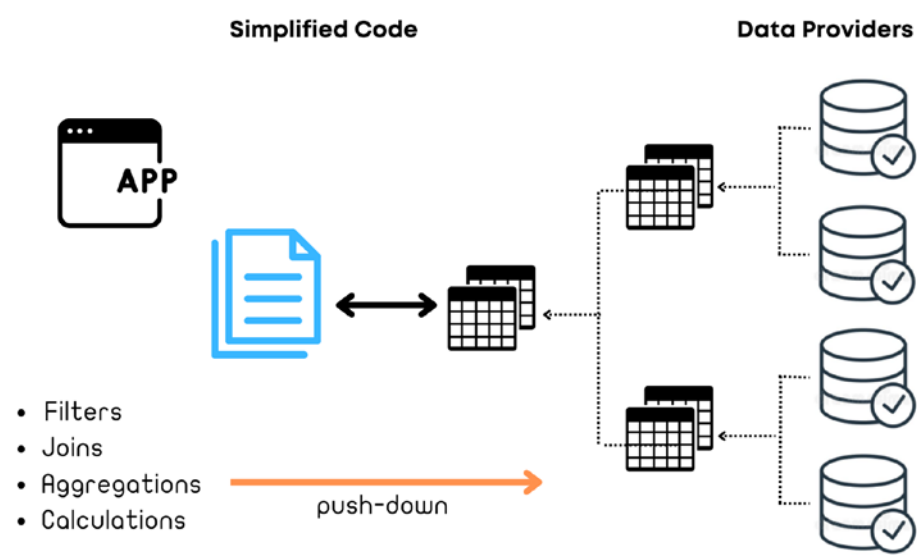


Figure 3-2. *Simplified application*

By using this approach, you will make application code simpler, with the benefit of reusing what you pushed down in other applications. Let’s learn how in the next section.

Reuse of Models

The application code in classical programs has a significant degree of redundancy. Developers are always writing the same code to process data.

When dealing with highly normalized database models, such as those used by SAP Business Suite, numerous independent tables are called and each query result is joined in the application code. In most databases, these joins may be pushed down. However, the SAP HANA Cloud database extends beyond assisting with table joins. All data-processing tasks from the application may be handled by the SAP HANA Cloud database. The SAP HANA Cloud database handles complicated calculations and data-flow logic, such as aggregation and disaggregation.

Dynamic placeholders may be used in SAP HANA Cloud calculation views. This implies that the application can send variables from a response to a user prompt down to the calculation views. Many calculation views can also invoke processes with input parameters.

Other calculation views' results can be consumed by calculation views. This promotes a high level of modularization and reusability of calculation views (Figure 3-3). A calculation view is typically part of a stack of calculation views that form a whole data model. It is a best practice to create a calculation view with future usage in mind so that it may be readily reused. This reduces model redundancy and facilitates simpler maintenance.

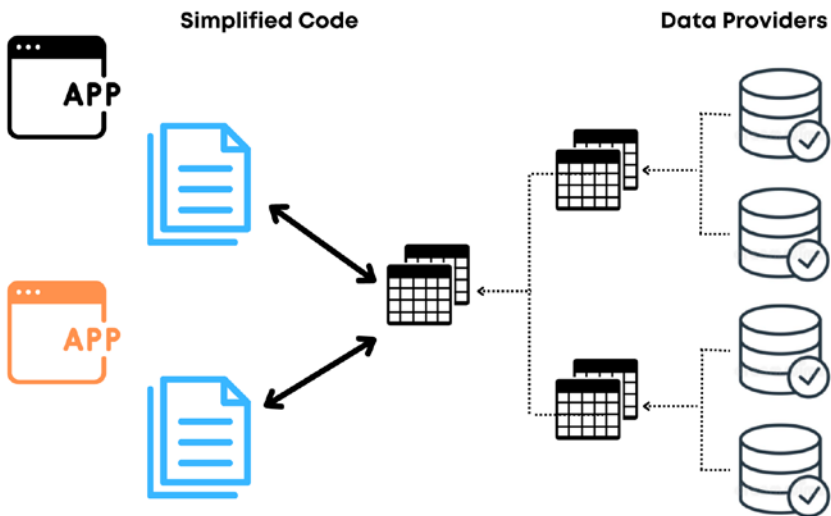


Figure 3-3. *Modeling reuse*

Now, let's get more details about the advanced data-processing capabilities of SAP HANA Cloud.

Advanced Data Processing

SAP HANA Cloud comes with powerful data processing capabilities. Spatial, graph and predictive modeling are examples of these (Figure 3-4). Although sophisticated modeling functions can be utilized independently, they can also be conveniently accessed through calculation views. This means that it is feasible to create extremely sophisticated data models capable of handling any form of data processing.

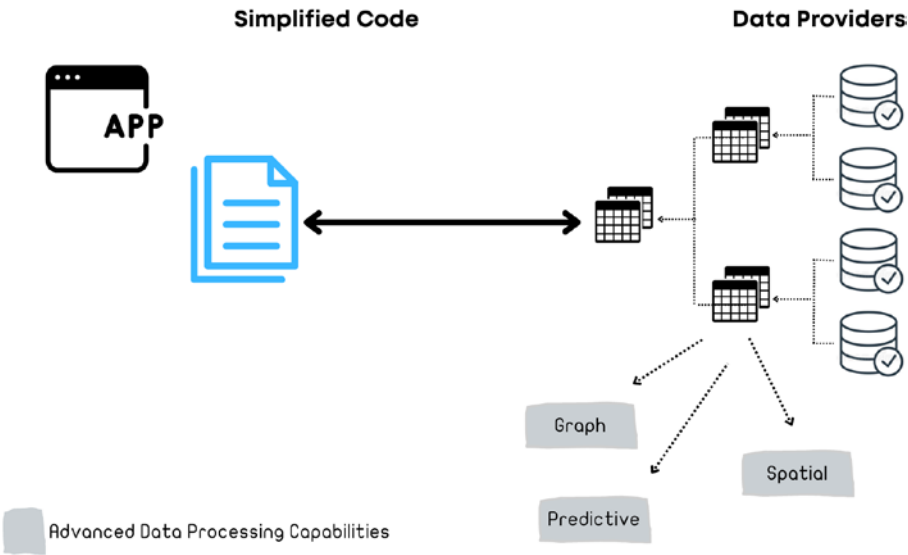


Figure 3-4. Advanced data processing

Types of Calculation Views

In SAP HANA Cloud, the calculation view is the primary modeling object. There are three types of calculation views:

- Dimension
- Cube without star schema
- Cube with star schema

Calculation views are developed with the Business Application Studio graphical editor. You should be familiar with fundamental data-modeling techniques in order to create a calculation view.

Modeling Dimensions

A dimension type calculation view's goal is to establish a set of related characteristics. A calendar dimension, for example, might have features such as its date, month, and year (Figure 3-5). Although this sort of calculation view may be consumed directly by an application, it is more likely to be found as a consumable view in a cube with star schema calculation view to supply the cube's dimensions (Figure 3-5).

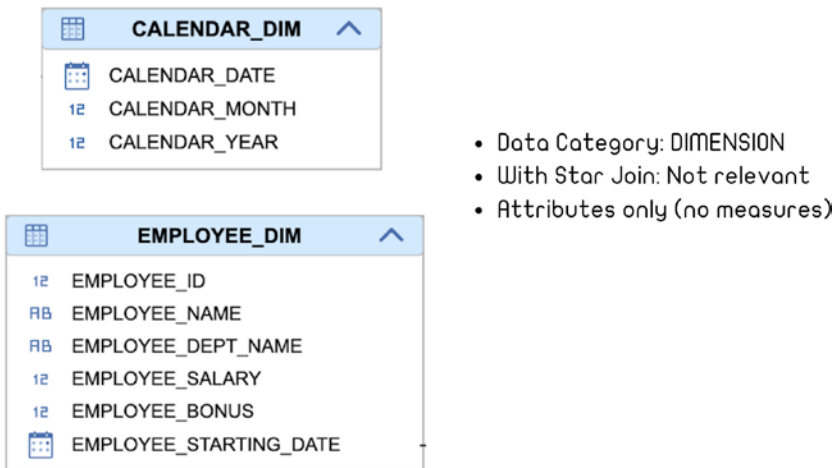


Figure 3-5. *Dimensions*

Calculation views of type DIMENSION can be thought of as core master data views. You define them once and then reuse them several times.

To begin using calculation views of type DIMENSION, change the data category to DIMENSION.

The next step is to specify the source tables and, if necessary, joins. Construct filters and then specify which output columns should be accessible by the view. Additional derived properties can also be defined. A new column might be created to construct a Measure category based on a range of measures, as an example. You might verify each measure and assign it to a category using an IF statement. You can also have a more generic statement by creating calculated columns based on multiple string, numeric, and other types of functions.

You may rename any source column in the calculation view to make it more understandable to the business user. Keep in mind that column names are derived from database tables, and these names are frequently technical and unintelligible to business users.

Other dimension calculation views can be consumed by a dimension calculation view. For instance, you might add a new dimension called Partners that joins the Customer dimension and Vendor dimension.

Modeling Cubes

The cube calculation view is the next type of calculation view (Figure 3-6). This calculation view is used to construct a data set made up of characteristics and measurements. As a result, they are perfect for transactional data modeling. You may set the aggregation behavior for each measure, allowing business users to slice and dice data as needed, and all measures are aggregated based on their behavior. A cube calculation view can consume other cube calculation views or dimension calculation views. A cube calculation view is a relational model, not a dimensional model.

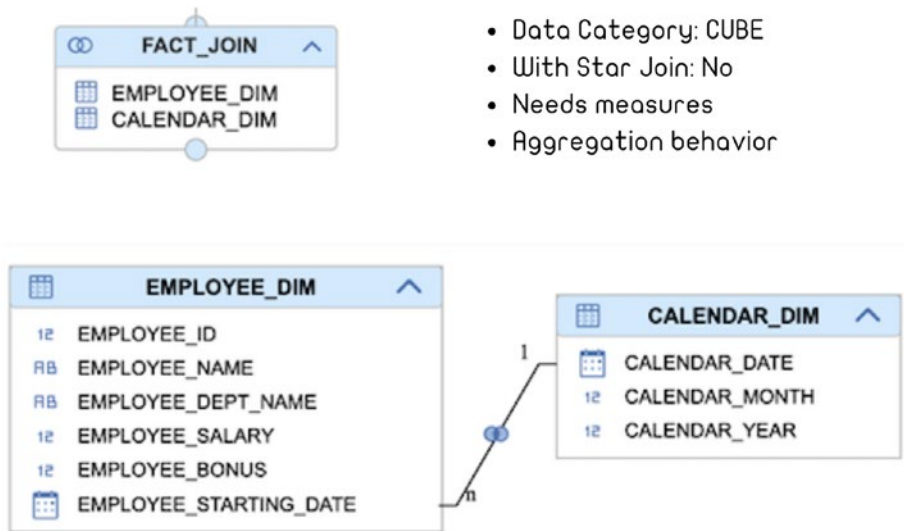


Figure 3-6. Cubes

This type of calculation view may be accessed directly by applications. SQL statements can also be used to access them.

To begin, change the data category to CUBE.

Do not select the Star Join flag.

You then pick a table or a list of tables that will be included in the model. Typically, you pick a transactional table with columns from which to create characteristics and metrics. It is possible to incorporate more than one transactional table (e.g., when you have Sales Orders and Sales Order Items on separate tables). To complete the image of an employee hiring transaction, for example, you may need to include further employee attributes. In this situation, you just join the tables. You may also integrate transactional tables by using a union.

Then, choose which columns from the tables should be exposed. Filters and extra calculated columns can be optionally defined as options.

Finally, rename any columns to give the business user appropriate names.

Modeling Star Schemas

Now we arrive at the most powerful of all calculating views: the cube type, but with a new setting—star join (Figure 3-7). This is how we model a star schema. This model is based on a cube calculation view. However, it is connected to one or more dimension calculation views. This results in a very strong and versatile dimensional data architecture that allows for drill-down, slice, and dice analysis across any combination of variables and measurements. When compared to cube-type calculation views, adding several dimensions greatly expands the analytical options.

Non-star join calculation views can also support multiple “dimensional” tables being joined to one central table. The difference is the performance optimizations a star join calculation view allow when you have a true star or snowflake schema.

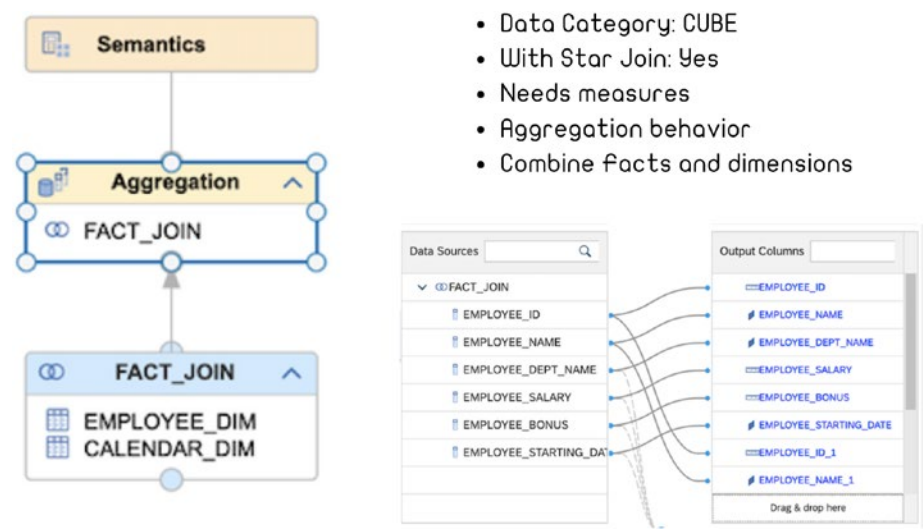


Figure 3-7. Cube with star join

To begin, ensure that the data category is changed to CUBE and that the Start Join flag is selected.

Then, choose the transactional tables and, if necessary, establish joins to integrate the transaction tables. Then, choose the columns to be exposed, apply any filters, and construct any calculated columns. Up to this point, you've been constructing a fact table that serves as the core hub of the star schema.

The last step is to create the star schema by connecting appropriate calculation views of type DIMENSION to the fact table.

Finally, utilize the rename method in the Semantic node to enhance the names of any columns so that it is entirely business-user facing.

Cubes with star join calculation views may consume cube with star join calculation views as well as ordinary cube calculation views.

In the next section you will see how to take advantage of advanced data-processing capabilities.

Modeling with SAP Business Application Studio

The SAP Business Application Studio is used to construct calculation views. It is a multi-purpose integrated development environment (IDE) tool that is used to produce all artifacts needed in a full-stack application (Figure 3-8). This contains the user interface (UI), application code, and database objects.

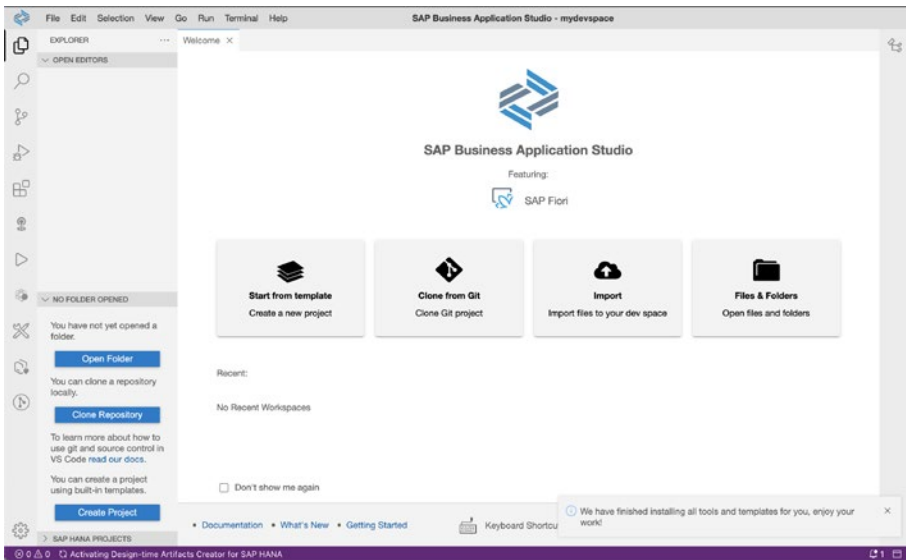


Figure 3-8. *SAP Business Application Studio*

The SAP Business Application Studio is more feature rich than its antecessor tool, Web IDE for SAP HANA (released in 2016).

Instead of utilizing a GUI, the SAP Business Application Studio features a terminal console where instructions may be typed directly via a command-line interface (CLI).

You may use SAP Business Application Studio to construct several data-modeling artifacts, such as the following:

- Calculation views
- Procedures
- Table and scalar functions
- Flowgraphs
- Analytic privileges

- Local and virtual tables
- Replication tasks

Artifacts are generated with either graphical or text editors. You may switch between several types of editors. The artifacts are saved in simple text files, making them extremely portable. Aside from SQL, the files are normally in JSON or XML format.

Some artifacts have complicated syntax. For example, the graphical editor should be used for calculation views. However, the text editor is also handy for browsing through these sorts of files or copying and pasting code snippets.

The SAP Business Application Studio's primary features include an explorer on the left side, which is divided into two sections. You may open a workspace in the top section to examine your project(s) and all the files they include. Many of these files will be SAP HANA database objects like tables and views. These tiles may be seen in the lower section, where you can also observe the deployment status.

The interface's right side is made up of editors and a log section. You may easily reposition them; for example, you might have numerous editors open side by side or stacked on top of each other.

The open-source Git framework is used for version management of database artifact files. Business Application Studio completely integrates with Git-compatible repositories. This implies that when you create artifacts, you can save them to the team repository. You can also return to an earlier version of any artifact. Git allows developers to clone a project and work on pieces of it collaboratively with other developers without impacting others. When the development is complete, Git can be used to merge artifacts from different branches of the project.

Now, let's see how SQLScript can be used along data-object modeling.

Extending Calculation Views with SQLScript

Calculation views are created graphically, with no scripting necessary. However, the graphical method may not always give all the functions and data-flow logic needed for a complex calculation view. This is where functions come into play. SQLScript, SAP HANA Cloud’s native database language, is used to create functions (Figure 3-9).

Note There are two types of functions: scalar functions (which return a single value, such as the current date) and table functions (which return a data grid). In this part, we’ll look at table functions.

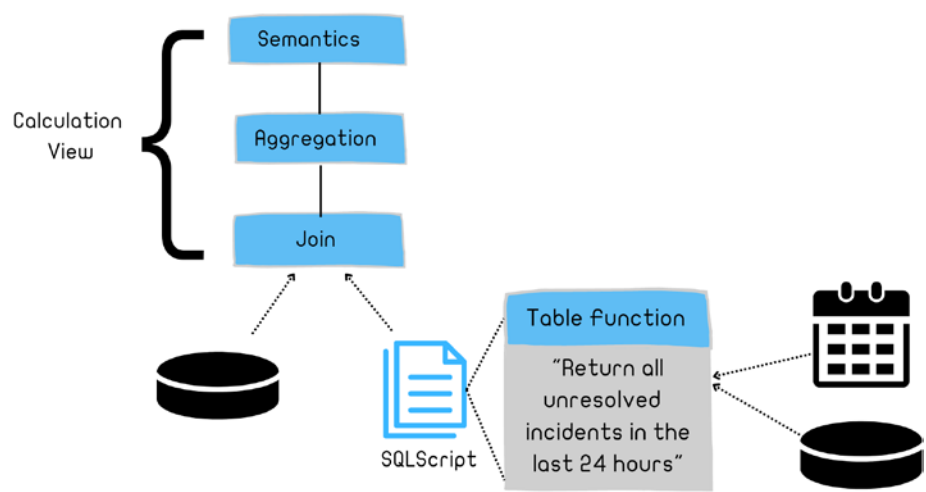


Figure 3-9. Table functions

Table functions are commonly used in SAP HANA Cloud modeling to produce a tabular data collection that is then utilized as a data source in a calculation view. Table functions can be utilized in SQLScript directly in the FROM clause of a SQL SELECT statement like a standard table. A table function encapsulates the logic in a reusable form that may be reused in other objects.

Table functions can take one or more parameters as input. Table functions are read-only, which means they cannot be used to modify data. Table functions generate precisely one tabular result. They have the ability to call other functions.

Procedures specify reusable data-processing logic that may be used to improve a calculation view (Figure 3-10). Procedures are comparable to functions in that they are written in SQLScript, have one or more inputs, and always have a result. Procedures, however, can generate several output data sets with varied structures, while a table function can only provide one tabular output structure. Procedures play just a minor part in data modeling. Procedures cannot be used as data sources in calculation views, but they can be used in other places, such as to automatically derive values for an input parameter.

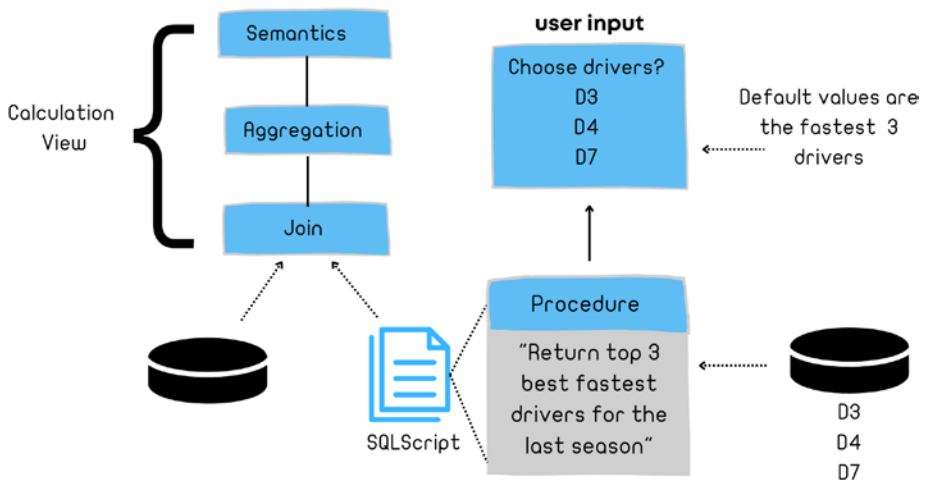


Figure 3-10. Procedures

A procedure may be called from SQLScript directly, which implies it can be called from a function or even another procedure.

In general, procedures are called stateless (side-effect free) when they do not affect any data or metadata in the database. However, procedures may be stateful when are responsible for updating, inserting, and deleting data as needed. Stateful procedures are not permitted to be called from calculation views.

Now, let's look at important concepts about managing access levels in your data modeling.

Securing Data Models

Calculation views are used to develop data models in SAP HANA Cloud. Calculation views have several security safeguards to guarantee proper data-access permissions (Figure 3-11).

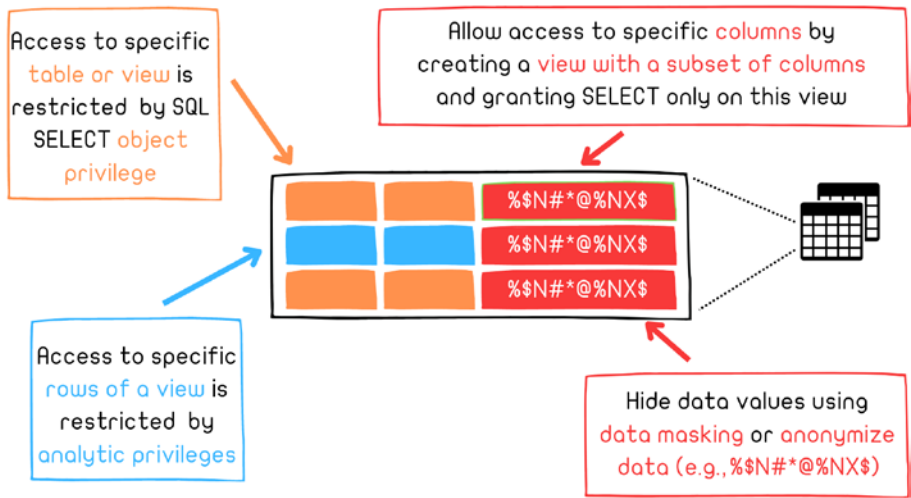


Figure 3-11. Security in modeling

There are three levels of access to consider, as follows:

1. The initial level of access is to the data model's real database objects. This generally refers to the source tables, procedures, and functions that the calculating view consumes. This is accomplished using conventional SQL database permissions. In most analytical cases, we simply require the SELECT permission on the database objects. Permissions can be provided to individual users directly, but it is more probable that roles will be developed first to organize the numerous users, and permissions will be issued to the roles such that they are inherited by the assigned users. This method is more efficient in terms of maintenance.
2. If there were no analytical privileges in place, a user would be able to access all data from a table they have SELECT permission to. Only after analytical privileges are created and assigned to a user or a role to which the user has been allocated is row-level access enforced. An analytic privilege is an object that is built to explain the rules that enable data access for each calculation view.

The conditions might be as basic as Peter's being able to examine any data but only for the startup Bricks. The conditions can also be more complicated, such as for Mary's being able to examine data for the startup Bricks but only between 2021 and 2022, and just for product groups Beverage or Snacks. A user/role can have various analytic rights granted to them. Analytic permission logic may be constructed using basic form-based rules or more complicated SQLScript rules.

3. We can now view all data within our permitted rights; however, certain users may not want to see the complete value of a field. By adding a masking rule to each column, we may hide whole or partial data values for sensitive data. Instead of displaying a phone number, for example, we only show XXX-XXX-8942. We may specify which users or roles should be subject to the masking rule.

Now that you are a bit familiar with most data-modeling capabilities, let's check some of advanced capabilities available in SAP HANA Cloud.

Advanced Analytics with SAP HANA Cloud

In this section, you will learn how to take advantage of some advanced capabilities available in SAP HANA Cloud to support a modern application development.

SAP HANA Cloud Spatial

You may create unique apps that give greater insight to uncover possibilities or limit risk by merging standard business data with geographical data. You can allow business users to have a better experience through modern applications' adopting a user-friendly interface, as shown in Figure [3-12](#).

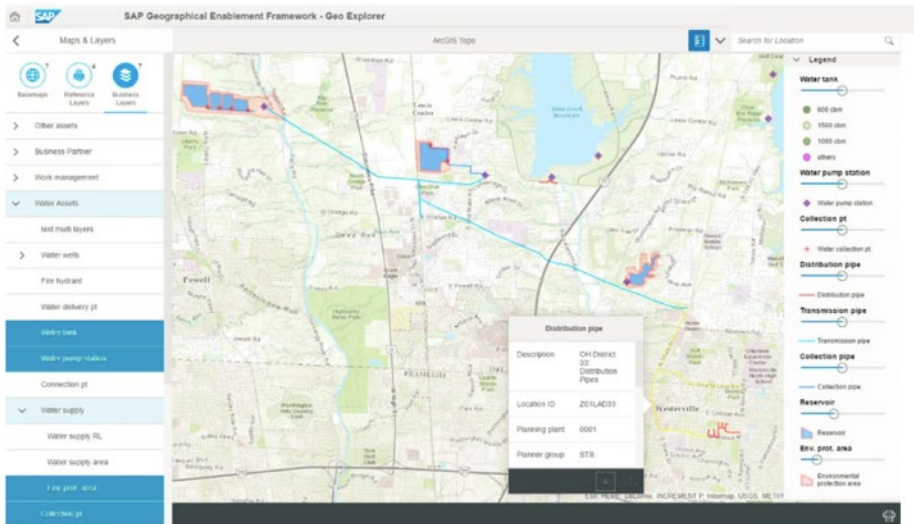


Figure 3-12. SAP HANA Cloud Spatial

Many companies currently rely on geographic data processing and utilize specialized applications in addition to, but separate from, business-process applications. For example, if a power station reports a failure, a coordinator will create an emergency repair order in the SAP ERP system. The coordinator will then use a second GIS (geographic information system) based application to manually enter the tank's asset number. The tank then will appear on a map, and the coordinator will use another program to locate the nearest competent engineer, who then will be deployed. This business process demands the use of many disconnected apps.

It would be more efficient if the ERP program could find exactly the place in the station where the failure occurred while producing the repair order. It could then locate and send the nearest competent engineer with adequate working hours to finish the repair, as well as provide valuable geographic information to the engineer to describe how to get to the tank

as fast as possible. The application might then offer information on other nearby equipment that is due for an inspection shortly. This eliminates the need for many visits. This possibility may be realized if the underlying business process were combined with geographical data analysis.

Aside from business applications, there are fascinating use cases for spatial analysis in sports. SAP has created a suite of programs that offer in-depth analysis of player performance. In soccer, for example, by attaching a sensor to each player, we may construct a graphical history that shows a player's stats and how they can improve their performance. Major sports organizations all around the world are already utilizing this sort of software.

Many applications might benefit significantly from the use of geographical data.

SAP Spatial adds additional data types for storing geometrical data, including points, lines, and polygons. These can be used to describe specific locations, highways, and areas. SAP HANA Cloud Spatial adheres to open standards and so integrates seamlessly with well-known, top geospatial providers such as ESRI, OGC, OpenStreetMap, and Google Maps.

SAP HANA Cloud not only stores geographical data, but also includes spatial query methods that can be readily integrated into SQLScript. Here are a few functions to get you started:

- **Within** – Which customers are in my region?
- **Distance** – What is the longest distance a high-value customer must travel to reach the outlet?
- **Crosses** – Where does truck route A cross truck route B?

SAP HANA Spatial additionally has techniques for determining clusters. This enables a company to pinpoint particular places that may be profitable based on income data and other intriguing qualities linked with customers.

Predictive Analysis

SAP HANA Cloud allows you to create predictive models (Figure 3-13). SAP HANA Cloud has two libraries that allow for either a manual or an automated approach to predictive modeling, as follows:

- Predictive Analysis Library (PAL)
- Automated Predictive Library (APL)

Predictive Analysis

PAL
Predictive Analysis
Library

- SAP library of algorithms
- Requires knowledge of algorithms
- Machine learning

APL
Automated
Predictive Library

- fully automated predictive and scoring model creation
- Requires no specialist knowledge of algorithms
- Machine learning

Figure 3-13. *Predictive analysis*

SAP HANA Predictive Analysis Library (PAL)

The Predictive Analysis Library (PAL) offers a vast number of algorithms for developing predictive and machine learning models. Some of these algorithms are employed for pre-processing tasks in data mining, such as the following:

- Sampling – select a few records from large data sets (for example, we need 1,000 people from each country)
- Binning – grouping records into basic categories (for example, age ranges)

- Partitioning – creating sets of data for training, testing, and validation used to train models and check their predictive accuracy

The vast majority of algorithms are employed in scoring or predictive modeling. Algorithms are available for all major data-mining categories, including the following:

- Association
- Classification
- Clustering
- Regression
- Time series
- Neural networks

PAL algorithms can be accessed directly from SQLScript operations, or they can be incorporated into an SAP HANA flowgraph created with a graphical editor in the SAP Business Application Studio (SAP BAS). The data inputs, data processing and outputs, and parameters utilized in the predictive model are all defined by a flowgraph. Using PAL requires an understanding of statistical concepts and data-mining techniques. This is due to the fact that the developer must decide which algorithm to utilize.

As a result, it is critical that the developer first recognize the distinctions between the algorithms used for data preparation, scoring, and prediction. They must, however, be able to fine-tune the algorithms to achieve the desired result. A developer, for example, would need to select when to utilize the time-series method for double exponential smoothing versus triple exponential smoothing, as well as how to alter the parameters to account for trends or to eliminate outliers. Developers that deal with SAP HANA PAL are usually involved in predictive analytic projects or have a good knowledge of the subject.

Automated Predictive Library (APL)

The Automated Predictive Library (APL) is designed for developers who do not have (or do not want to have) an in-depth understanding of the algorithms and mathematics underlying the models. The methods used for data preparation, scoring, and prediction are totally automated (hence the name). All the developer needs to do is submit the data, and APL will evaluate it to find the optimum data preparation (cleaning up the data) and prediction models.

Graph Modeling

Graphs are used to describe data that would be better represented by a network. Supply-chain networks, transportation networks, utility networks, and social networks are a few examples (Figure 3-14). The core concept behind graph modeling is that it allows a modeler to specify a sequence of items (nodes) and connect them with lines (edges) to form a network. This network illustrates how each node is connected to all other nodes.

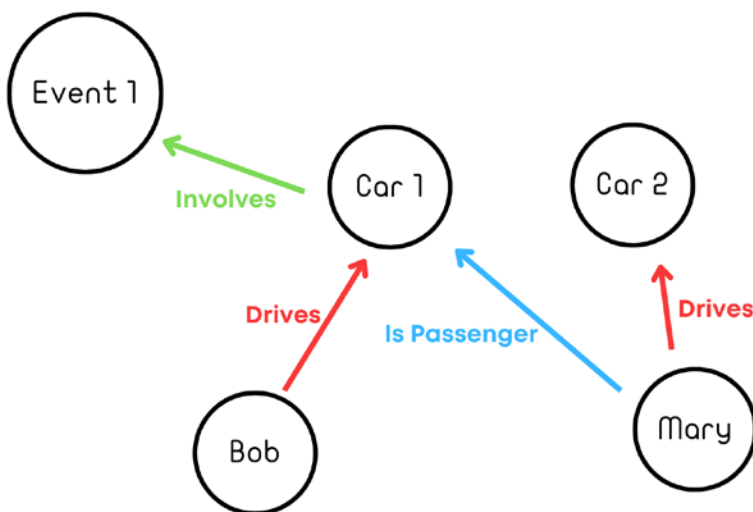


Figure 3-14. Graph model example

Graph models can also represent the direction of flow between entities, and the nodes or the lines that link them can have any number of properties. This implies that more meaning may be added to the network and queries can be conducted to ask network-related questions.

Consider a complicated supply chain represented as a graph, with information kept along the links for all manufacturers, suppliers, distributors, customers, and consumers. The advantage of this type of modeling is that it makes it simple to create applications that can explore large graphs at high speeds. Therefore, you can ask things like the following:

- How many hours has the package traveled between two specified points in the network?
- Where are all the possible points of sourcing of this package?
- Describe the entire journey of a package by listing all the stop-off points in its path.
- What is the shortest or most cost-efficient path between point A and point B?

Graph analysis enables you to uncover hidden patterns and correlations in your network data in real time.

Graph Model: Example

There are several applications for SAP HANA Graphs, including the following:

- **Medical**
Create a network of patients, ailments, treatments, and results that may be reused in the diagnostic and treatment planning of additional patients.

- Social Network

Find your clients and their friends, friends of friends, likes, and dislikes using data offered by major social media networks to develop marketing initiatives.

To develop and process an equivalent model, regular SQL tables with standard data definitions and query code may be used. However, defining such a model in simple SQL and querying the model would be computationally inefficient. In addition, processing times may be onerous. SAP HANA Graph includes graph-definition tools as well as extra language for graph queries to make model building more natural and simpler. It also ensures that the processing is flexible and, of course, optimized for in-memory processing by utilizing a specialized in-memory graph engine located directly within the database.

Summary

You have successfully completed this chapter, and now it's time to review what you learned and see what's next:

- Understand differences between classic and new programming approaches
- Map the main characteristics of complex and simplified applications
- Get an introduction to SAP HANA Cloud data-modeling concepts
- Learn about advanced data-processing capabilities provided by SAP HANA Cloud

Now we will continue our journey by getting the foundation that will allow us to start a development with SAP HANA Cloud.

CHAPTER 4

Developing Applications on SAP HANA Cloud

In this chapter, you will learn some key concepts required before you can start using the SAP Business Technology Platform (SAP BTP) for application and database development. As an open-source platform, SAP BTP supports different frameworks, languages, and models to handle myriad scenarios. Now, let's begin with the SAP BTP environment concept.

Choosing an Environment on Business Technology Platform

One of the most essential decisions to make when developing a new application is which BTP environment to use (Figure 4-1).

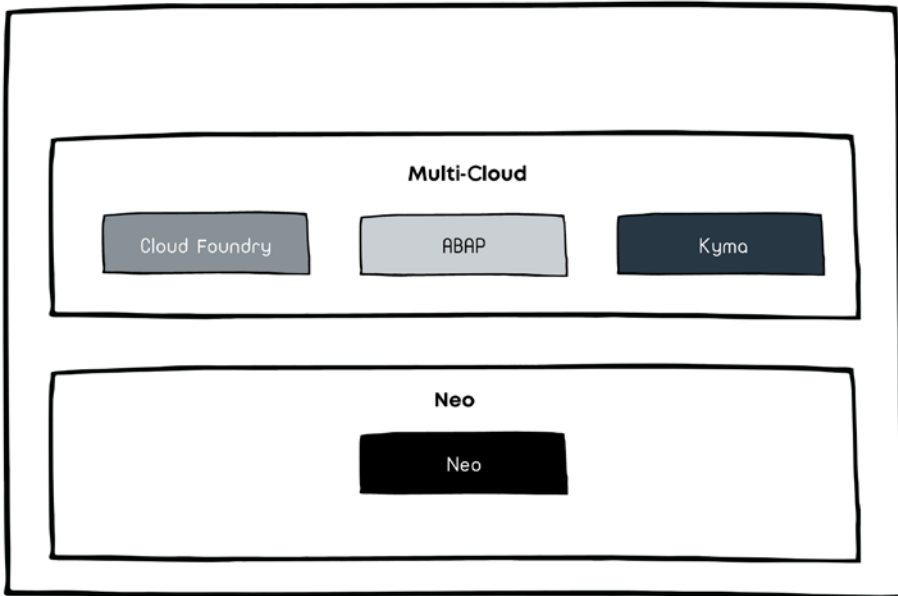


Figure 4-1. *SAP BTP environments*

Each environment has a unique set of tools, technologies, and run-times for developing applications. SAP BTP is offered in a variety of contexts, allowing for better development freedom. Customers may even mix and match environments. SAP BTP facilitates application development and deployment across numerous environments.

SAP BTP makes available the following environments:

- **Cloud Foundry**

The Cloud Foundry environment allows you to create new business applications by supporting a wide range of run-times, programming languages, libraries, and services. The Cloud Foundry environment is appealing to enterprises looking to create new multi-cloud applications using a

variety of cloud technologies and languages. They can also create apps with the Cloud Application Programming framework, which facilitates the creation of cloud applications at a greater degree of abstraction than in other settings.

- **ABAP**

The ABAP environment enables the creation of extensions for ABAP-based products, such as SAP S/4HANA Cloud, as well as the development of new cloud apps in ABAP. Existing on-premises ABAP custom code or extensions can be migrated to the cloud. The ABAP environment is interesting to enterprises who have a big, experienced ABAP development staff and seek to enhance SAP ABAP-based products.

- **Kyma**

Based on the open-source project Kyma, the Kyma environment provides a fully managed Kubernetes run-time. The Kyma environment enables developers to add serverless functionality to their SAP applications and integrate them with containerized microservices. The Kyma environment is aimed at enterprises with cloud-native developers that are skilled in many cloud technologies and wish to build multi-cloud applications with an emphasis on microservices and distributed systems, with the goal of high scalability and operational automation.

- **Neo**

The Neo environment is recognized as the very first cloud environment that SAP built, a proprietary run-time that supports just a few programming languages. You may create apps in HTML5, Java, and SAP HANA Extended Application Services (SAP HANA XS) and also create user interfaces using the UI development toolkit for HTML5 (SAPUI5). Many early adopters of Neo are now transitioning their apps to alternative multi-cloud environments like Kyma or Cloud Foundry to take advantage of their capabilities.

Now, let's take a closer look at Cloud Foundry environment.

Introducing Cloud Foundry

The SAP BTP Cloud Foundry environment is an open Platform as a Service (PaaS) designed to support microservice application building.

The microservice architecture allows developers to build programs in extremely small sections, each of which delivers a unique "service." Each service may be combined with others to form a comprehensive application. This allows for more agile development and maintenance.

The flexibility to construct apps in many programming languages is one of the Cloud Foundry environment's features. Traditionally, programs were written in a single general-purpose language, even if that language was not the best for the job at hand. We may now select the best language for each component of the program.

Cloud Foundry accelerates and simplifies the development, testing, deployment, and scaling of applications by offering a selection of underlying infrastructure (including Kubernetes), developer frameworks, and application services.

Cloud Foundry enables your business to concentrate on building and running apps rather than on the platform. It eliminates the expense and complexity of establishing, administering, and protecting infrastructure for your applications.

Cloud Foundry provides several advantages for developers and organizations, including the following:

- **Language independent**
Select any run-time language and add additional ones as needed.
- **Administration separation**
Isolating tasks enables both administrators and developers to work simultaneously
- **Standard command-line interface (CLI)**
Use one CLI against any Cloud Foundry provider.
No need to learn multiple tools.
- **Open source**
Many organizations provide continuous updates and new features.

Introducing SAP Cloud Application Programming Model (CAP)

SAP Cloud Application Programming Model (SAP CAP) is a framework of tools, languages, libraries, and APIs that combines open-source and SAP-supplied tools and technologies.

One of the most essential features of SAP CAP is that it delivers SAP best practices to assist developers in reducing their coding efforts. SAP CAP offers code that may be reused to address common application

functionalities. The idea behind CAP is that it encourages developers to focus on the added value component of the program (the domain) rather than on the low-value recurring technical characteristics of an application that can easily be given by the model and may vary across different settings. This implies that in a world of rapidly changing cloud technologies, the most valuable components of applications remain protected (Figure 4-2).

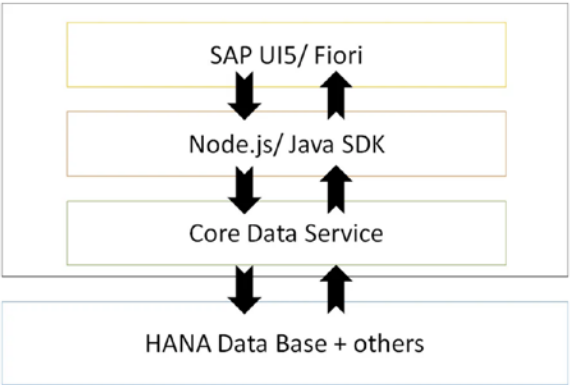


Figure 4-2. CAP model

SAP CAP is used to create new applications as well as to extend current SAP applications.

CAP is used to create apps for the SAP BTP Cloud Foundry environment.

The foundation of CAP is Core Data Services (CDS). CDS is a data description language with a high level of semantic richness. CDS is used to specify the physical data layer (for example, data types and tables). It is used to define the entities as well as the relationships between these entities. It can also be used to enforce filters, drop columns (as in a view), or define full star-schema multi-dimensional models.

CDS also provides a sophisticated querying language known as CDS Query Language (CQL). It extends ordinary SQL with components that allow for easy querying of deeply nested object graphs and document structures.

In CAP, everything is a service. Services in essence expose a standard set of APIs that may be consumed by other services or front ends. CDS services define what should happen to a dataset, and when (Figure 4-3).

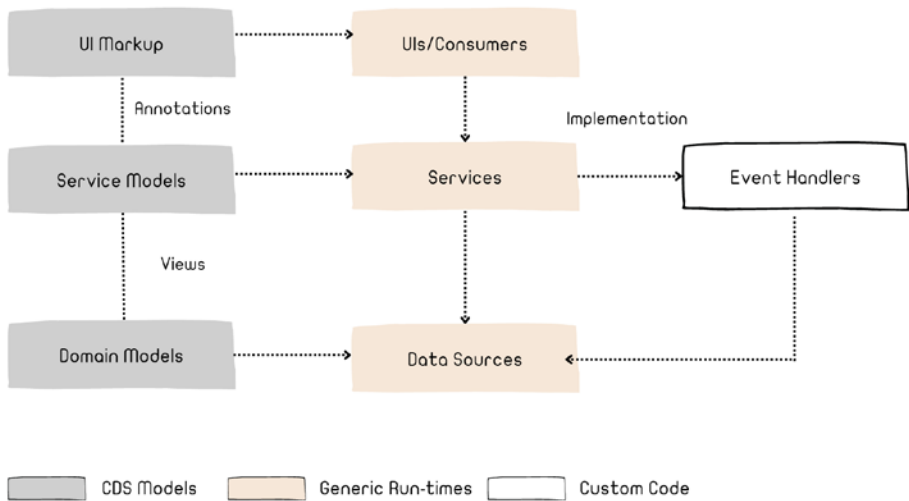


Figure 4-3. SAP CAP basic architecture

Everything in CAP happens as a result of events. CAP-based applications can send events that can be picked up by other services or front ends through subscriptions. In addition, applications can receive inbound requests and trigger services. Event handlers are used to manage events. The CAP models have generic handlers for common actions (such as CRUD operations and validations), and these handlers actually control the majority of the tasks and use cases. You may also use Java or Node.js to provide custom code to augment the generic handlers.

You can also use CAP to build native HANA Cloud database artifacts, like tables, views, and functions, and combine them with CDS components.

CAP leads developers on a route that adheres to SAP best practices for building cloud apps on the Cloud Foundry platform.

SAP suggests using CDS extensively to design models and services while using the CAP strategy, so that you are working at the abstraction layer rather than directly constructing technical objects. This implies that your work is protected even if the underlying technologies change (Figure 4-4).

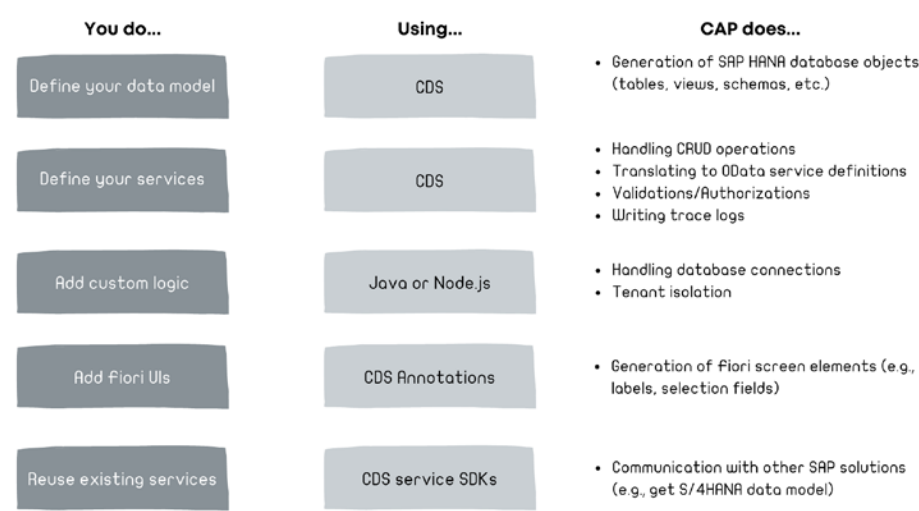


Figure 4-4. CAP best practices

Custom logic, on the other hand, is not supported by CDS; a sophisticated validation for a combination of additional input fields that are not part of a conventional data model, for example. That’s where Java and Node.js come into play to give bespoke logic.

SAP Business Application Studio is the perfect tool for designing CAP applications since it has SAP plugins and capabilities that are compatible with the CAP strategy.

Hierarchical Structure of Cloud Foundry Applications

Organizations and spaces form a stable two-level hierarchical framework within which applications execute (Figure 4-5).

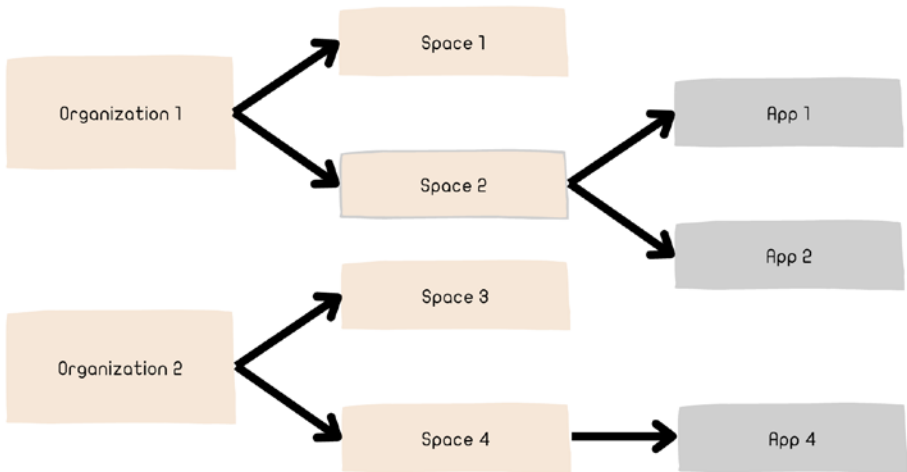


Figure 4-5. Organizations and spaces

Now, let's review the service plans available to you.

Service Plans

You select a service plan to manage the database artifacts when you provision an SAP HANA Cloud database (Figure 4-6).

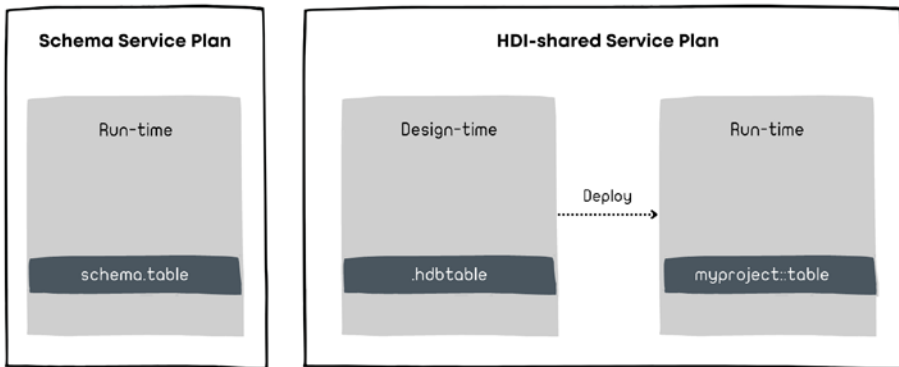


Figure 4-6. *Service plan*

SAP HANA Cloud offers two service plans for maintaining database artifacts.

1. **Schema service plan** – Using SQL, create all database objects straight to schemas. This method has been used for many years with traditional databases and appeals to businesses that merely need a database to power their applications and do not require design-time tooling to describe the database artifacts as source files. You will create schemas by hand to handle your database's real-time items.
2. **HANA Deployment Infrastructure (HDI) shared service plan** – Provides a database design-time and run-time environment in one. Using this plan, database developers specify all database objects in source files. The source files are used to create and deploy run-time objects to containers. This approach includes tools for developing and deploying database items over their entire life cycle. You don't have to worry about schemas with this plan, because they are invisible to the developer and

controlled by the HDI framework. Because database artifacts are not hardwired to schemas, this is known as schema-less development, and it allows a flexible redeployment model.

A service plan is usually added as a service entitlement at the Business Technology Platform (BTP) global account level. Thus, you may select the service plan when creating a new instance.

You are not restricted to a single service plan and can select various service plans.

Note HDI should not be confused with XS Advanced. This is a run-time that is not included in SAP HANA Cloud. SAP HANA Cloud makes use of Cloud Foundry, which provides comparable application run-time ideas to XS Advanced but is built on open standards that allow for multi-cloud development.

Now, let's review which tools are available to support application and database development.

Key Tools Used by Developers

SAP HANA Cloud comes with a toolset to cover different roles and activities.

SAP Business Application Studio

SAP Business Application Studio is an SAP Business Technology Platform (BTP) service that provides a new development environment optimized for the efficient creation of SAP intelligent enterprise business applications. Because it is a web browser-based application, there is no software to download and install, and all upgrades and new features are supplied by SAP (Figure 4-7).

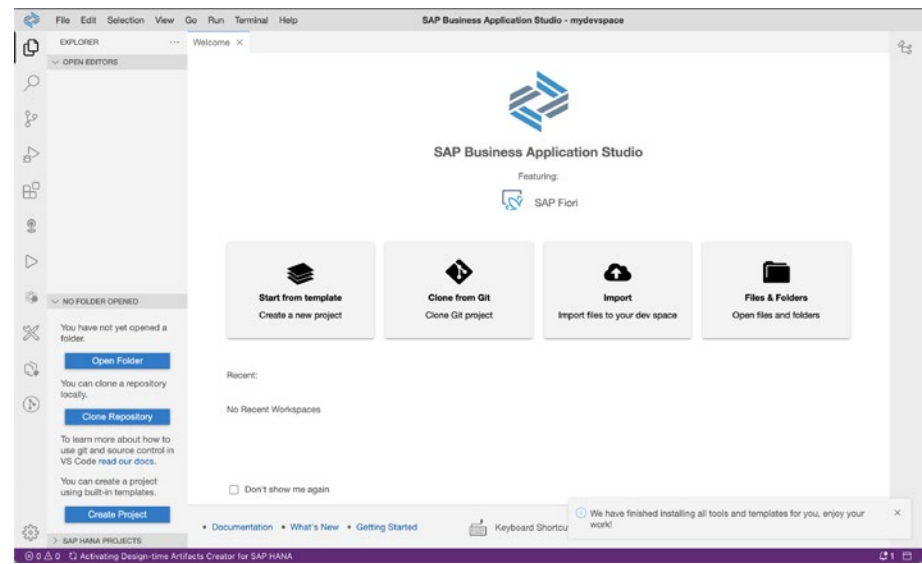


Figure 4-7. *SAP Business Application Studio*

You have access to one or more development spaces in SAP Business Application Studio.

A development space, often known as a dev space, is a development environment that includes all the tools, capabilities, and resources required to construct an application. Each application requires a particular development environment, and SAP provides several dev spaces that already have the tooling needed to get started immediately. You can subsequently add other tools to help with development. If you wish to build and consume SAP HANA calculation views from your application code, for example, you may add the Graphical Calculation View Editor to your dev area.

Caution Do not confuse development spaces with Cloud Foundry spaces.

SAP Business Application Studio resembles Microsoft Visual Studio in many ways. They have a similar look and feel and are powerful integrated development environments (IDEs). Either may be used to create BTP apps. However, SAP Business Application Studio comes with a variety of tools and templates designed specifically for designing business applications in the SAP environment.

The SAP Business Application Studio contains graphical editors for different development artifacts, such as Fiori components and calculation views, as well as text editors for coding. There is even a Cloud Foundry command-line interface (CLI) for directly executing instructions.

Tools like code assist, static code analysis, and code formatting may be added to the SAP Business Application Studio to help you build more quickly and efficiently.

Note BAS is a common abbreviation for SAP Business Application Studio; however, it is not an official SAP acronym.

Git for Source Code Management

Git is a source code version-control system. It is used in SAP HANA Cloud development projects to allow the following features:

- Collaboration during the design time
- Backup of source code
- Complete change history of source code
- Branching and merging capabilities
- Traceability and audit (e.g., linking change request to a code change)

One of the problems of ongoing application development is figuring out how to create new versions of solutions while still deploying critical updates.

Consider the following case. App V2.1.0 is currently being updated with new features, which results in the creation of new files and the modification of old files. App V2.2.0 will be the final version. However, an urgent demand for a security patch arises during development, and this must be implemented promptly to the present solution, requiring the creation of a patch version App V2.1.1. Of course, when we ultimately go live with V2.2.0, the patch will be removed. So, how can we include the patch in the final product? This is when Git comes in handy (Figure 4-8).

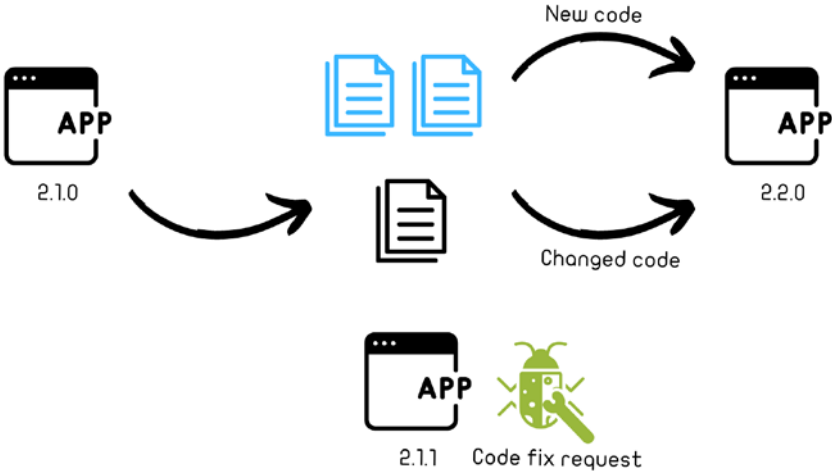


Figure 4-8. *Continuous development*

Typically, more than one developer works on an application. So, how does Git help a group of developers working on various aspects of the same project?

Each developer works in their own private Git repository and then commits their development material to a central Git repository when they are finished. Any conflicts are detected by Git and presented to the developers as an opportunity for code-comparison resolution.

It is also possible to form subteams, each with its own Git repository. This adds another staging location for local teams to collaborate before committing their progress to the central Git repository (Figure 4-9).

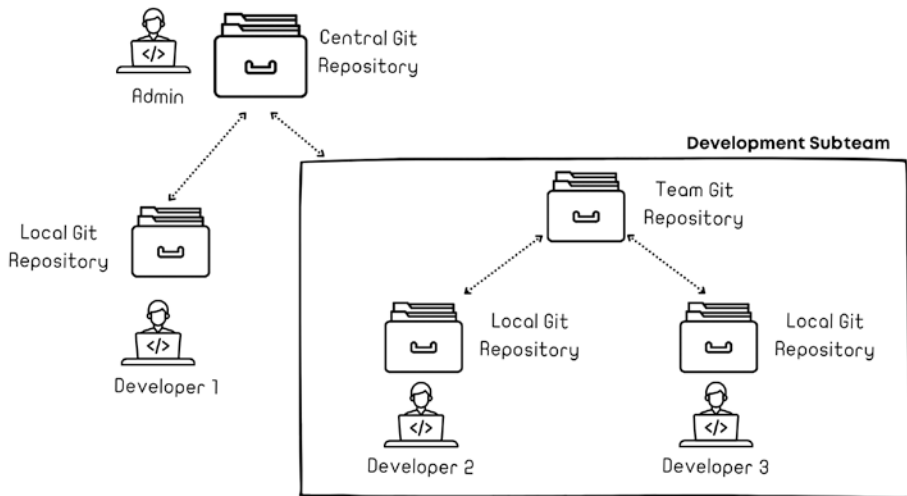


Figure 4-9. Team development with Git

Git keeps a comprehensive history of all changes made to source files and allows you to roll back to any point in time. “Commit” is a crucial phrase in Git. When a developer is confident that their material is complete, they commit one or more associated source files to the Git repository. Each commit refers to the previous commit, creating a commit history. The timeline of all commits may be seen, and any version of the source code can be returned to at any point along the commit history (Figure 4-10).

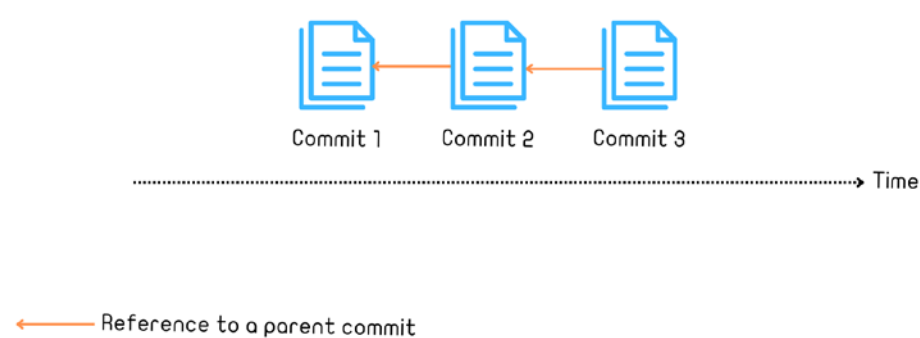


Figure 4-10. *Git commit history*

Previously, application development was linear. As a result, each developer could not begin work until the preceding development stage was done. Developers, on the other hand, may work in parallel with Git. Branching is a powerful feature of Git. A developer can establish a new branch so that their commits go on a different path from those of other developers. At some time in the future, the branches can be merged to incorporate all developments, and Git will alert developers of any conflicts that can be handled (Figure 4-11).

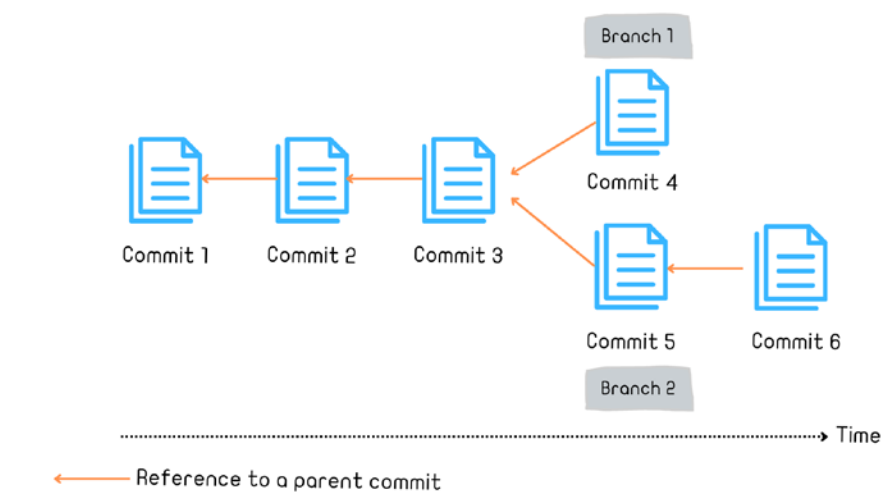


Figure 4-11. *Git branch concept*

Git is embedded straight into the SAP Business Application Studio, allowing developers to obtain and commit source files from Git repositories while working on code development.

The SAP Business Application Studio is the tool to support you all along the development cycle, but to explore database objects in the SAP HANA Cloud catalog, you need another tool, which we will now cover.

SAP HANA Database Explorer

The SAP HANA Database Explorer allows you to query your SAP HANA Cloud database and view catalog object details (Figure 4-12).

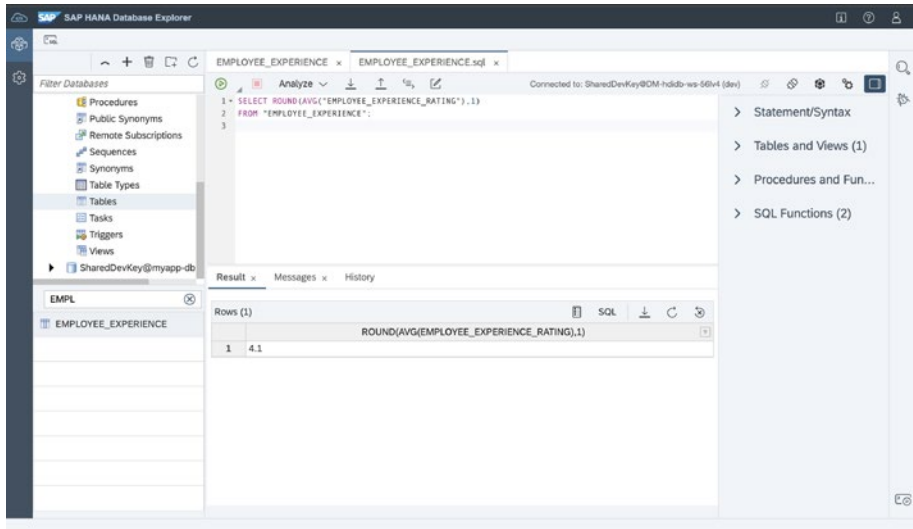


Figure 4-12. SAP HANA Database Explorer

The SAP HANA Database Explorer offers the following capabilities:

Catalog browser – View definitions for all catalog objects, such as tables, views, stored procedures, functions, and synonyms. View the content (data) of your tables and views as well.

SQL console – Create SQLScript procedures and queries, then use the SQL analyzer to run or examine their performance.

SQL analyzer – View extensive information about your queries and assess potential bottlenecks and improvements.

SQL debugger – View and evaluate expressions and variables, as well as the call stack.

The SQL HANA Database Explorer may be started from a variety of locations, including the following:

- SAP Business Application Studio
- SAP HANA Cockpit
- SAP HANA Cloud Central
- SAP BTP Cockpit

The Database Explorer can connect to both the SAP HANA Cloud database and the data lake.

Now, let's review which content types are available to support application and database development.

Choosing Content Types

As an open-source platform, SAP Business Technology Platform supports different standards and content types to allow a more flexible and wider application development. Let's review some of these standards.

UI Development

UI5 is a user interface development toolkit designed and developed by SAP that allows developers to create cross-platform, responsive, enterprise-ready front ends. It is built using HTML5 and JavaScript and runs in a web browser (Figure 4-13).



Figure 4-13. *UI5 for fevelopment*

There are two versions of UI5:

1. **OpenUI5** – This is the open-source version, which is free to use and distributed under the Apache 2.0 license. In addition to SAPUI5 libraries, OpenUI5 makes extensive use of open-source libraries. As a result, SAP has decided to make this version freely available to anybody. OpenUI5 is available for free download from Github.
2. **SAPUI5** – Similar to OpenUI5, but with many more libraries that handle UI rendering for some specific SAP elements. SAPUI5 is already compatible with SAP HANA Cloud.

The CAP mode includes out-of-the-box support for SAP Fiori Elements front ends, eliminating the need for developers to construct interface code from absolute zero.

The majority of the UI content is defined in Core Data Services (CDS) files, and it is possible to customize the display behavior with particular UI annotations. The CAP model is responsible for creating screen components based on the CDS definitions.

In addition to CDS, i18N files are used to offer localization information such as label and formatting language variations.

Note You may technically switch between OpenUI5 and SAPUI5 as long as they both utilize the same core version.

Services

In this section, you will learn the role of OData Services, Core Data Services (CDS), SQLScript, and the calculation view to support application and database development.

OData Services

The Open Data Protocol (OData) is a commonly used protocol for performing database-style create, read, update, and delete actions on resources over HTTP (Figure 4-14).

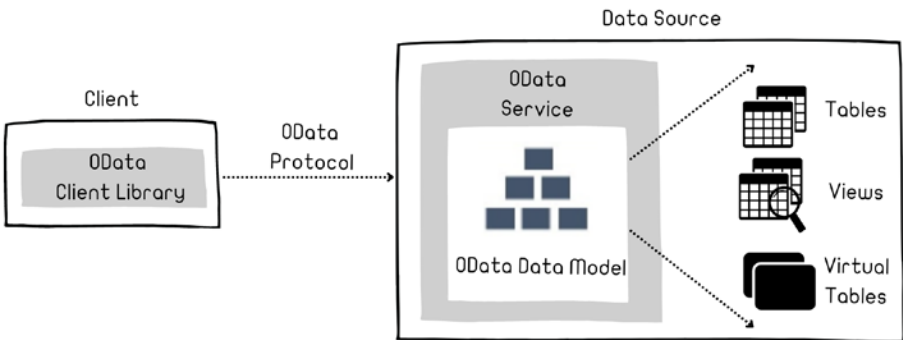


Figure 4-14. OData Services

Using the CAP model, all OData services are automatically generated from CDS files that describe the OData data model. It is possible to define specific service settings using special OData annotations.

Each programming language, such as JavaScript or Java, has prebuilt libraries for performing various sorts of OData queries and retrieving responses, removing the need for developers to build code from absolute zero.

Core Data Services

Core Data Services (CDS) is the foundation of the SAP Cloud Application Programming model (CAP). CDS is used to define all three levels of the application stack, which comprise the user interface, services, and database (Figure 4-15).

```
namespace scp.cloud;

using {
  cuid,
  sap.common
} from '@sap/cds/common';

entity Product : cuid {
  title          : String(50)           @title : 'Title';
  description    : String(1000)        @title : 'Description';
}
```

Figure 4-15. CDS example

The specification of the data model is generally the first step in the creation of a new application. This comprises views and tables. These sorts of items are defined using the CDS language. CDS is written declaratively; therefore, your CDS description does not relate to the

underlying technology on which objects are deployed. This implies you may redeploy CDS-based artifacts to newer platforms, safeguarding your development investment.

One of the most important characteristics of CDS is the ability to include extensive semantics in the CDS specification. This implies that when you construct a table, you may provide extra business information that explains the meaning and processing rules for each column in addition to giving the fundamental SQL type information, such as table type, column data types, and lengths. As an example, if the column contains a date, you may specify how it should be presented, for example, mm/yyyy. You might refer to another column that has the currency code if the column is a currency. You may also choose how the metrics should be aggregated (sum, average, count, etc.).

This instructs users on how to treat the column for formatting processing. Annotations are the name given to this supplementary information. Annotations are easily identified inside CDS definitions since they all begin with the @ character. One of the aspects that distinguishes CDS from conventional SQL for specifying data in your application is the usage of annotations. By providing processing rules and meanings directly inside the data specifications, the application code does not have to read raw data but may instead take up the rich semantics that give data meaning directly from the data definition.

SQLScript

SQLScript is an SAP HANA database development language that adds capabilities to standard SQL in order to fully use the in-memory SAP HANA database (Figure 4-16).

In your application development, you would utilize SQLScript to create modular, reusable SAP HANA Cloud database functions and processes that could be consumed by any application.

Additional data types:	text, tinyint, st_geometry
Declare variables:	USA Revenue = select sum(revenue) from balance where country = 'USA'
Conditional logic:	if-then-else, do_while...
Query parameters:	Expenses = Expense * :Local_Currency

Figure 4-16. *SQLScript*

SQLScript's purpose is to incorporate data-intensive application logic within the database. Traditionally, programs offloaded just a small portion of their processing to the database, and most of the data-processing logic was handled on the application server. That meant that data would have to be transferred from the database into the application server, processed on the application server, and the results written back to the database. The application's performance would suffer as a result of this round trip. We may skip the round trip by processing the data directly in the in-memory database with the SAP HANA Cloud database.

When executing data-intensive logic, this copying of large amounts of data can be very expensive in terms of computational capacity and data-transfer time. In addition to the data transfer, when using an imperative language like ABAP or JAVA, developers tend to write code that follows a one-tuple-at-a-time operation (for example, looping over all rows in a table). However, this approach is hard to optimize and parallelize compared to declarative set-oriented languages like SQL.

Note SQLScript contains strong capabilities that expand standard SQL and increase database efficiency. The traditional ANSI-compliant SQL is also supported.

Calculation Views

You should develop calculation views if you want to develop complex data models with the highest speed. Calculation views are generated in the SAP Business Application Studio utilizing a graphical editor (Figure 4-17).

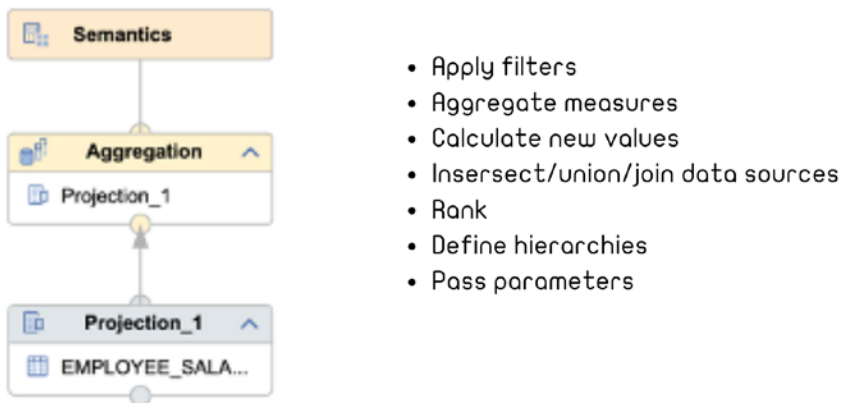


Figure 4-17. Calculation view modeling

Calculation views allow developers to indicate their intention rather than declaring how the data should be handled with SQL code. A data flow may be constructed using nodes to route data through many processing steps. This makes reading the flow much easier than attempting to understand extensive SQL code.

To execute a calculation view query, the SAP HANA optimizer guarantees that the most efficient and optimized calculation run-time object is created and considers several performance aids, such as the usage of cache or the pruning of data sources and columns.

Summary

You have successfully completed this chapter, and now it's time to review what you learned and see what's next. You've learned the following:

- Which application development environments can be used in SAP BTP
- Which tools support application development with SAP BTP and SAP HANA Cloud
- Key elements like Cloud Foundry, CAP model, and Git source code management
- How to provision database artifacts using available service plans
- How to identify main characteristics of UI5 development toolkits
- How OData services provides access to data sources

Now we will continue our journey with essential learning to start using the SAP Business Technology Platform and understand how to size the SAP HANA Cloud service.

CHAPTER 5

Working with SAP BTP Estimator and SAP HANA Cloud

In this chapter, you will learn how to work with the SAP Discovery Center and use SAP BTP Estimator to calculate capacity units and determine the sizing for your SAP HANA Cloud implementations.

Adopting SAP Business Technology Platform

One of the most essential knowledge areas you must review when getting started with SAP BTP is to explore everything inside the SAP Discovery Center (Figure 5-1).

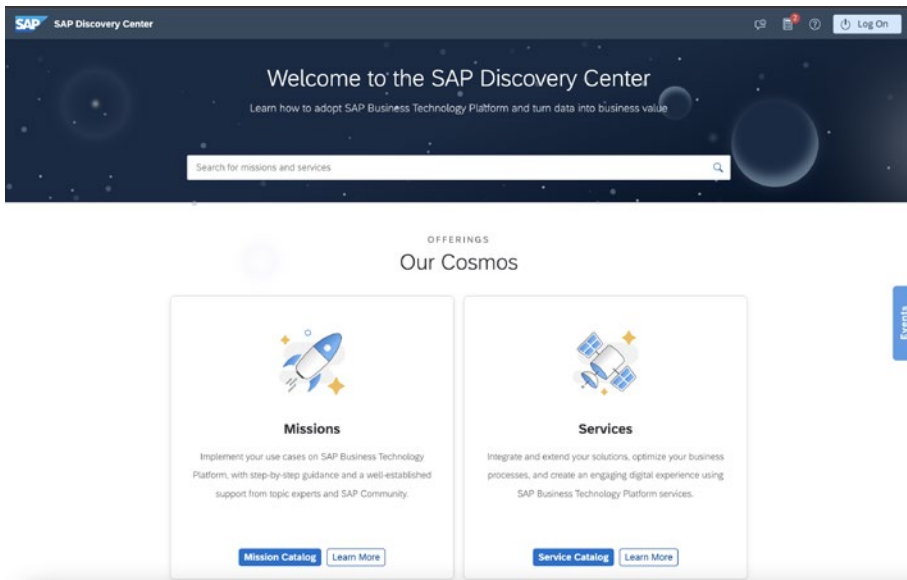


Figure 5-1. Welcome page

This area was created to centralize important content that is segmented into specific areas.

Our Cosmos – This is a space where you may find mission and service catalogs.

A *mission* is a practical set of tasks that helps you to activate SAP BTP services under a guided approach developed to support you in your journey with the SAP BTP. It is the best way to implement your use cases on SAP Business Technology Platform— with the right guidance and well-established support from topic experts and the SAP community.

A *service* is a feature-rich cloud service available in the SAP BTP, usually adopted to integrate and extend your solutions, optimize your business processes, and create an engaging digital experience.

Feature Content – This is an area where you can find a list of spotlight missions that features most applicable use cases.

Customer Stories – This is an area where you can learn with many customer use cases that are already adopting the SAP Business Technology Platform to solve different challenges.

Federated Repositories – This is the space reserved to work as a catalog of other useful content, as described in Table 5-1.

***Table 5-1.** SAP Discovery Repository*

Repository	Resource
Integration Content	API Business Hub
Customer References	Customer References Catalog
Tutorials	SAP Tutorials
Communities	SAP Community
Roadmap	Road Map Explorer
Partner Application	SAP Store

Missions

When exploring the mission catalog area in the SAP Discovery Center, it is possible to get started with a category of services by looking at practical step-by-step tutorials that carry you through the steps to implement a use case.

At the bottom of each mission, it is possible to check the number of companies that already have started that mission (Figure 5-2).

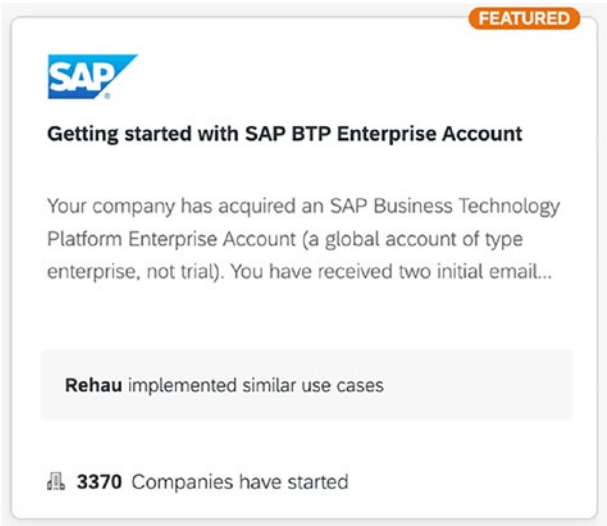


Figure 5-2. Mission tile

You can also define how the catalog sorting will be done (Figure 5-3).

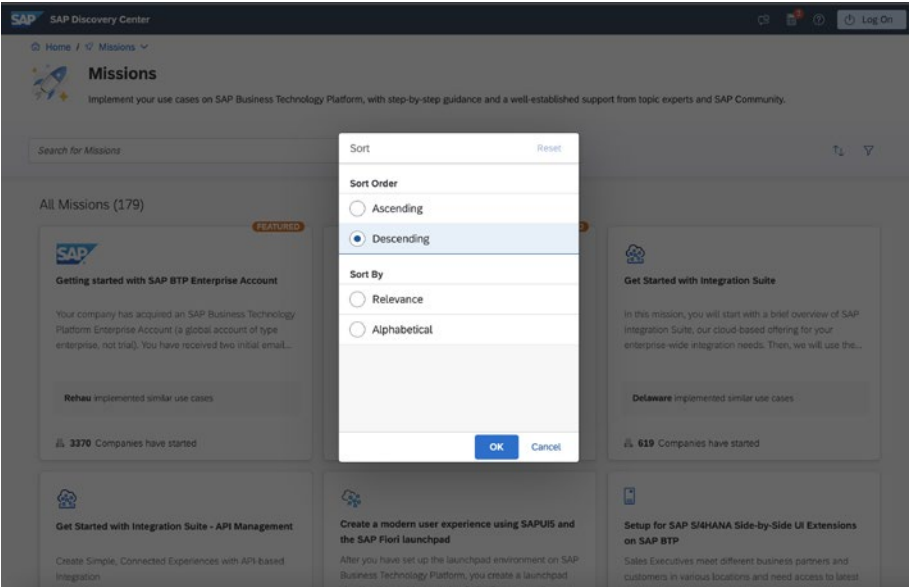


Figure 5-3. Catalog sorting

Additionally, you can define filtering on the catalog view, so you can go directly to the group of specific services you are looking for (Figure 5-4).

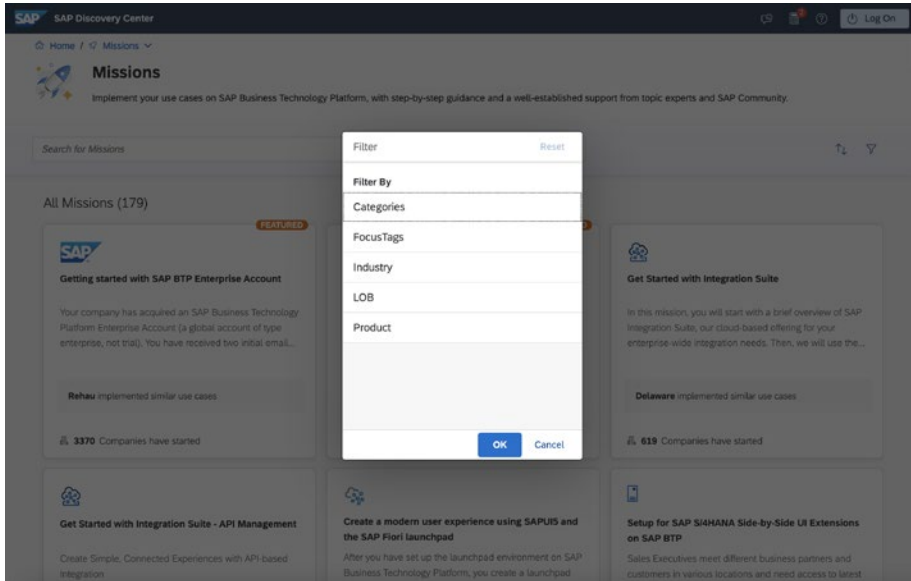


Figure 5-4. *Catalog filter*

Services

When exploring the service catalog area in SAP Discovery Center, it is possible to choose any service that you wish. It is possible to identify services using geolocation filtering by clicking the globe icon (Figure 5-5).

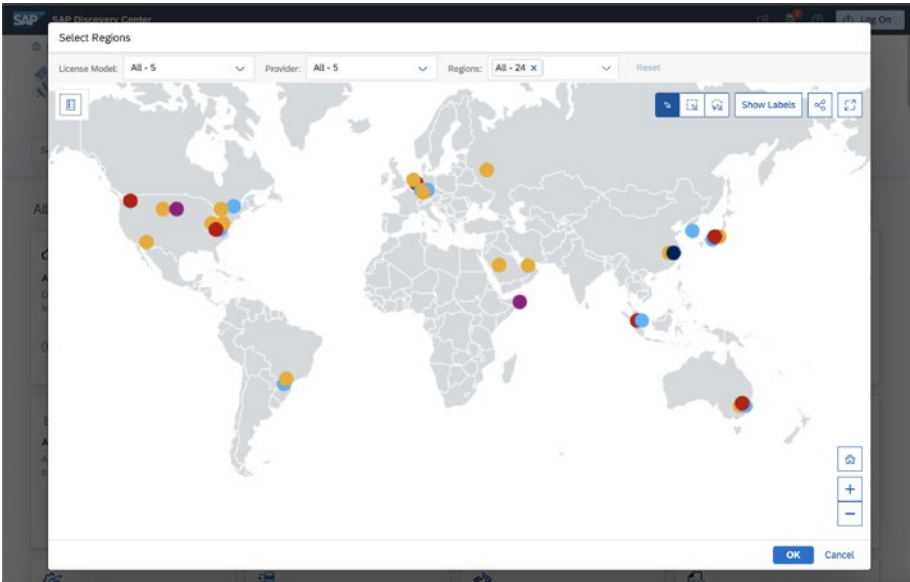


Figure 5-5. *Geolocation filter*

When navigating in the map, you can toggle on the legend to see hyperscalers’ names to support your discovery (Figure 5-6).

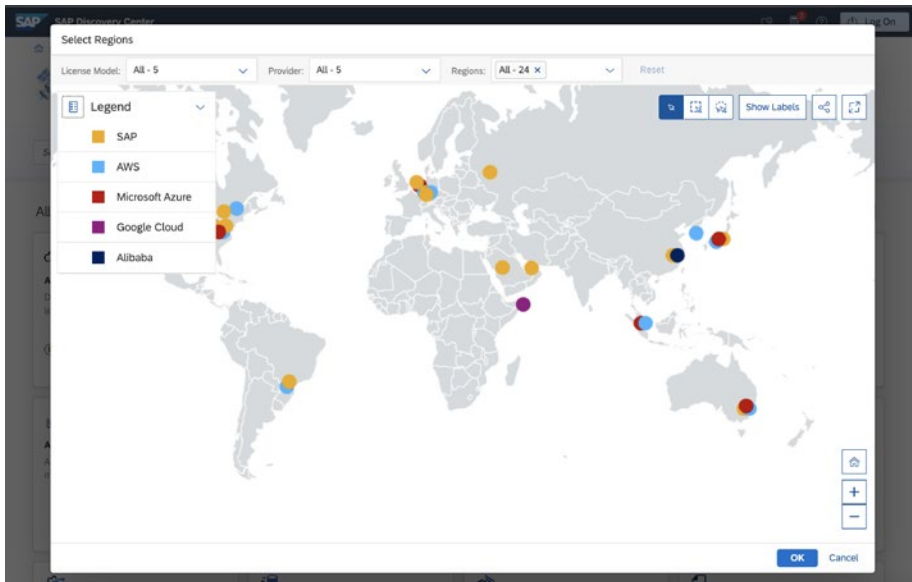


Figure 5-6. *Toggle label*

For better viewing, it is possible to toggle labels and zoom in on the map to check more closely which hyperscalers are available in a specific location (Figure 5-7).

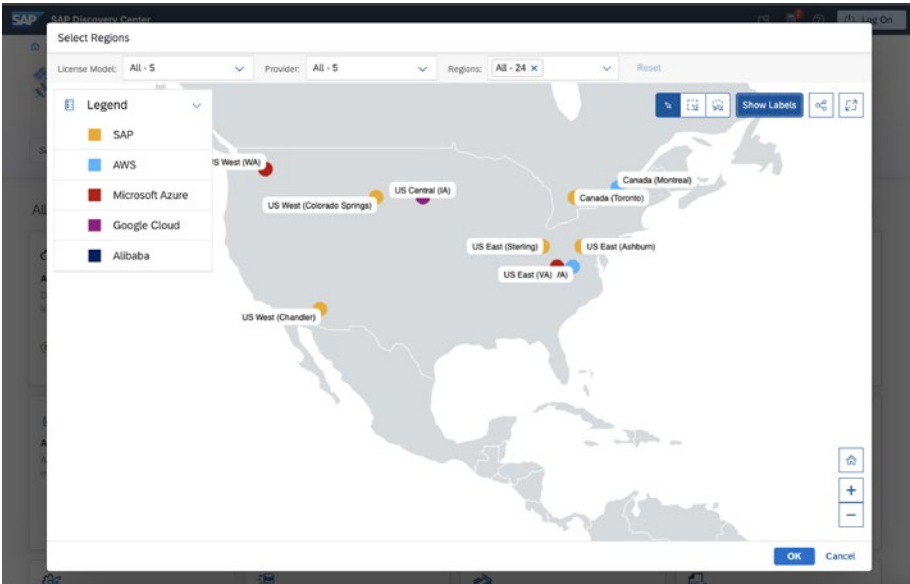


Figure 5-7. *Zooming in on region map*

The service catalog allows you to sort the list according to your needs, as shown in Figure 5-8.

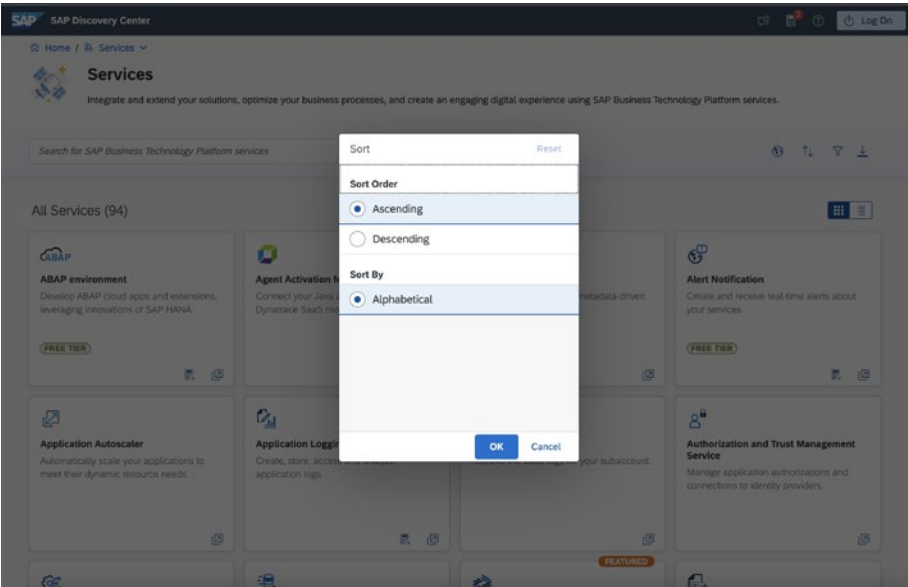


Figure 5-8. *Sorting feature*

Along with the region selection and sorting, you can also filter the service catalog by selecting the proper criteria, as shown in Figure 5-9.

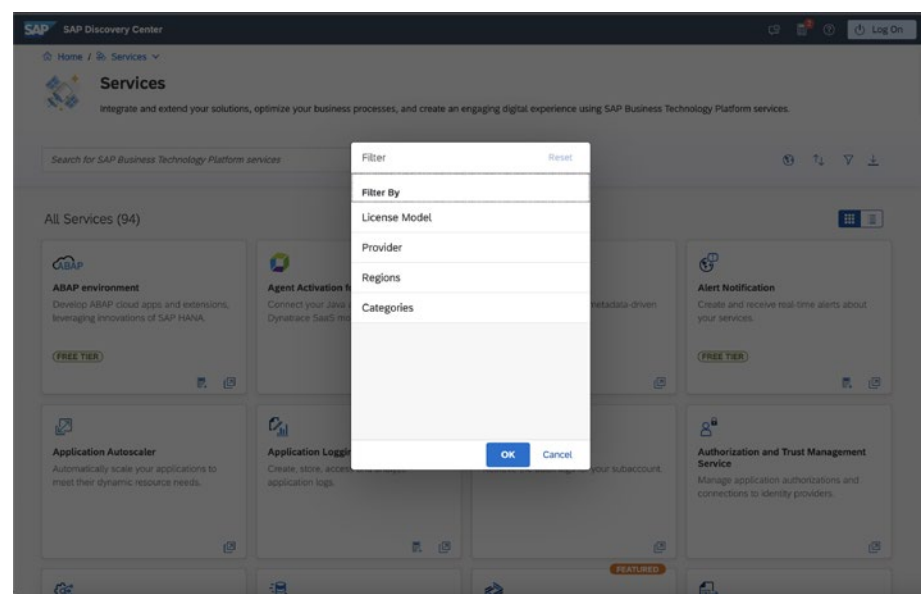


Figure 5-9. Service catalog filtering

To try out some of the capabilities of the filtering feature, we can use it to filter, for example, only services that have a free plan available, as shown in Figure 5-10.

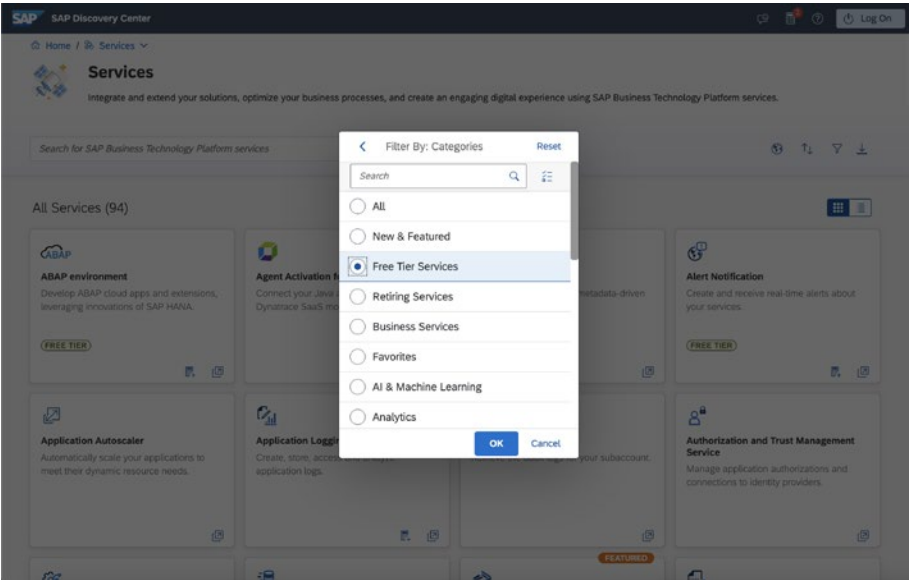


Figure 5-10. *Free Tier Services*

The interface of SAP Discovery Center allows you to perform free text searches for missions and services, so you can go directly to a specific one. In Figure 5-11, you can see an example of a search for the SAP HANA Cloud service.

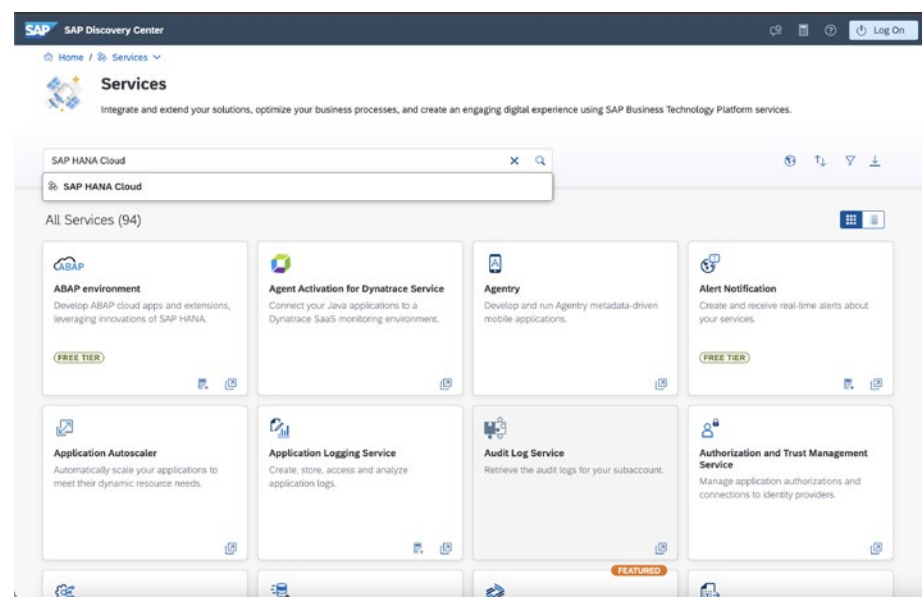


Figure 5-11. *Service search feature*

After clicking on SAP HANA Cloud, you go into the service, where you can identify related information (Figure 5-12).

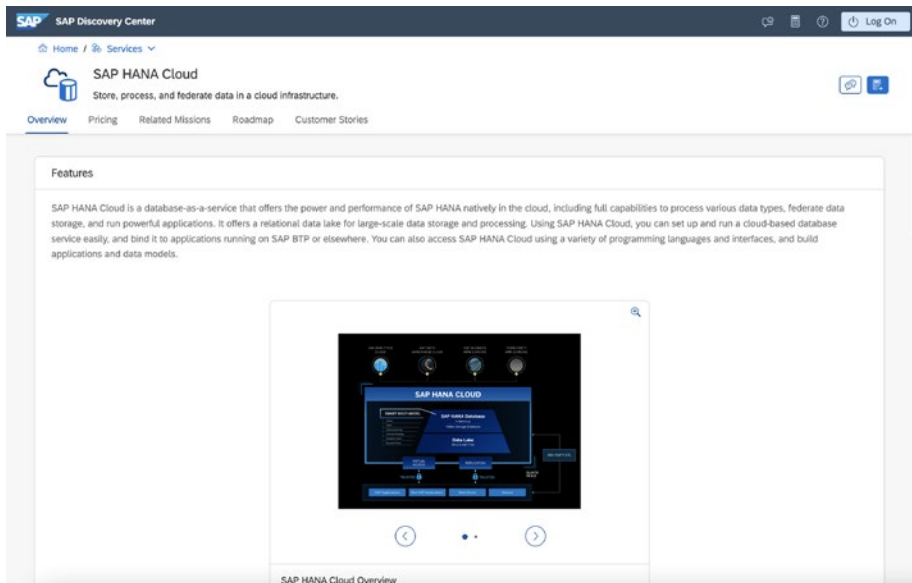


Figure 5-12. *SAP HANA Cloud page*

At the top of this page, you can locate the calculator icon, which represents the SAP Estimator tool that we will address in the next section.

Estimator

The SAP Estimator tool is an easy-to-use, web-based user interface available to anyone who wants to create a cost estimation based on a list of cloud services intended to be used in a given use case.

The starting point may be the cloud service page of SAP HANA Cloud, from which you can open the estimator tool (Figure 5-13).

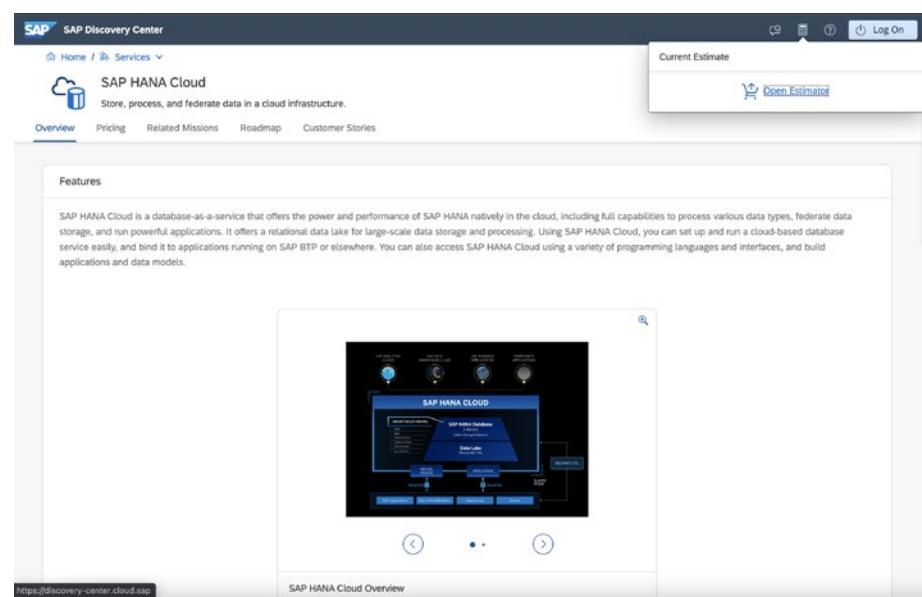


Figure 5-13. *Open Estimator feature*

Once you have the estimator tool opened, you can start setting up your scenario, but first there are some steps to be done.

Customize Estimate

Before selecting and sizing any cloud services, it is recommended that you provide some information to personalize your estimate and make it more readable (Figure 5-14).

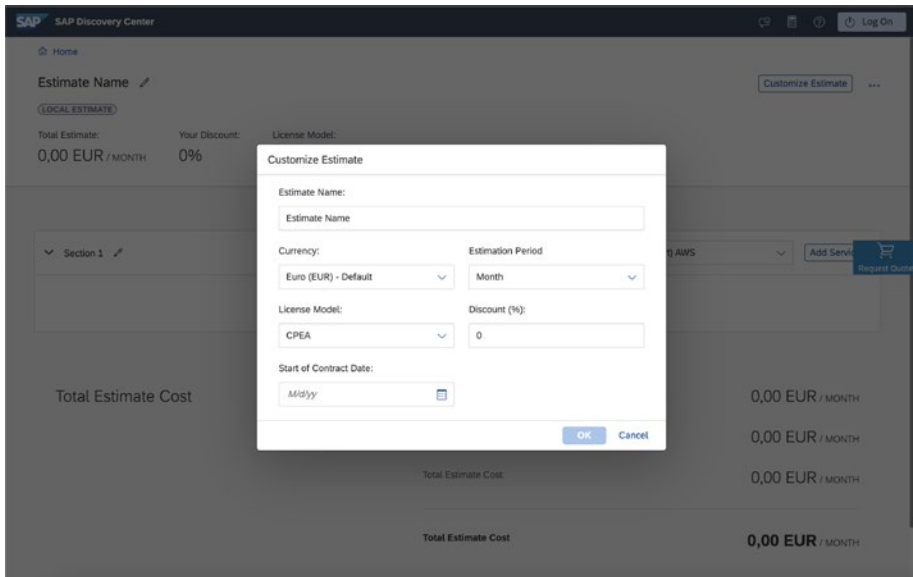


Figure 5-14. *Customize estimate*

In the Customize Estimate window, it is possible to set the following parameters (Figure 5-15):

Estimate Name – Use this field to provide a name that better describes what your estimate is being done for (e.g., MVP Cost App Analysis – wave 1, Data Lake for Mkt Campaign X, Predictive Project ABC).

Currency – Set the preferred currency to see estimate costs in your local currency. You can change this parameter any time to see it in different currencies.

License Model – Set the license model according to your needs. The possible values are CPEA (Cloud Platform Enterprise Agreement) and PAYG (Pay-As-You-Go).

Start of Contract Date – Set the desired starting date for provisioning purposes.

Estimation Period – Set this parameter to define how costs will be summarized. The possible values are Month, Quarter, Year, and 3 Years.

Discount (0%) – Set this parameter to simulate a desired discount. Bear in mind the value you input here is just to simulate an estimated value, and an official discount is defined only when you purchase a valid global account.

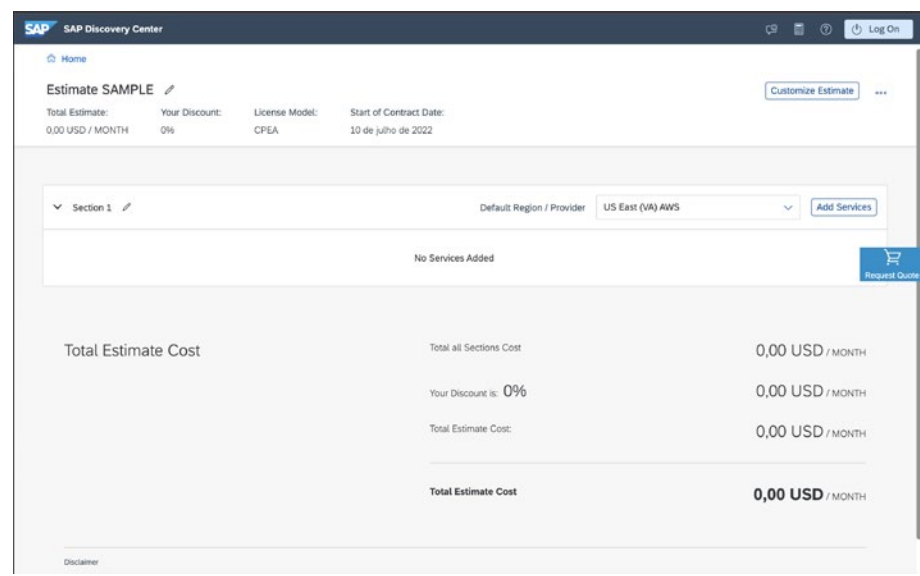


Figure 5-15. *New demands recognized as opportunities*

Once you have set general parameters you can start adding services to your estimate.

The first point of attention is that you can define the region for which you wish the cloud service to be estimated (Figure 5-16).

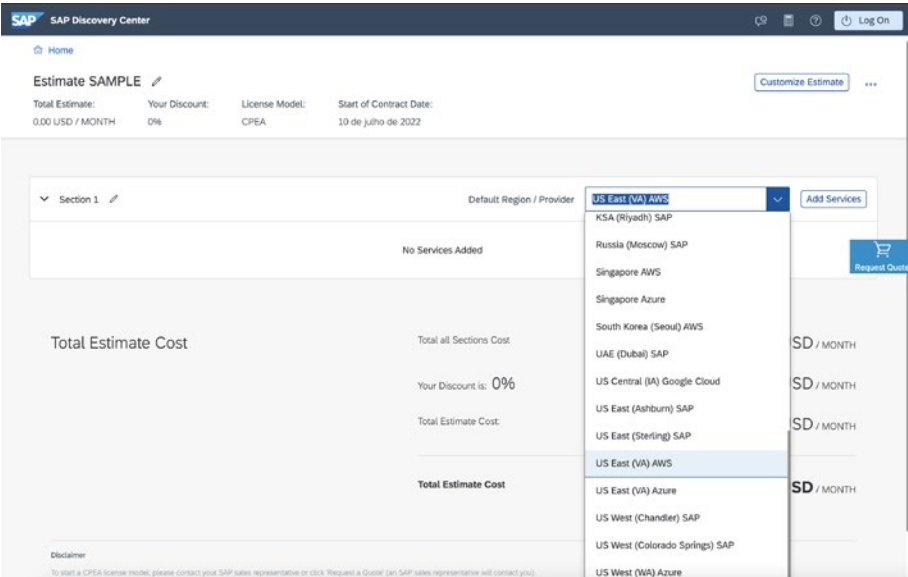


Figure 5-16. Region selection

Next, you can finally add the cloud service (Figure 5-17).

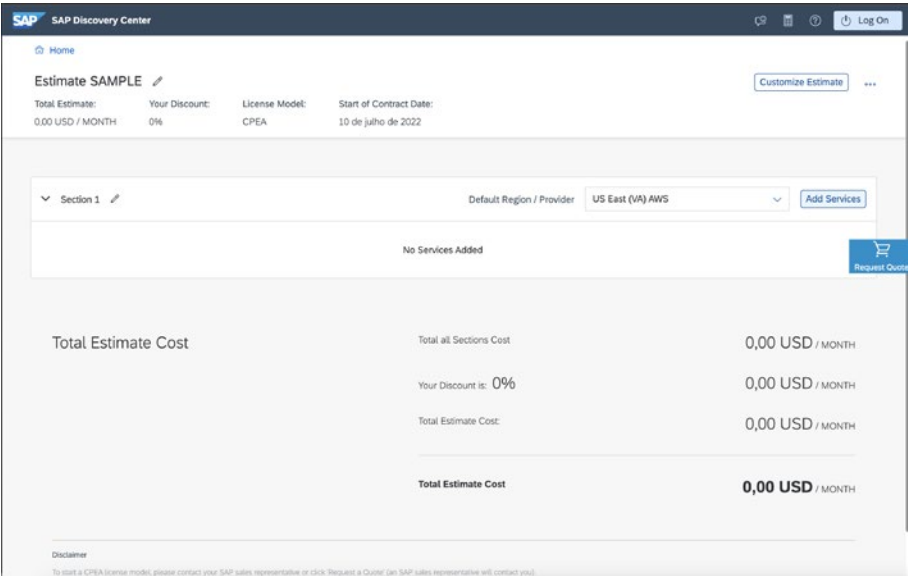


Figure 5-17. Add service feature

After clicking on the Add Services button, a popup window with all cloud services is shown. You can then select all services necessary to support your use case (Figure 5-18).

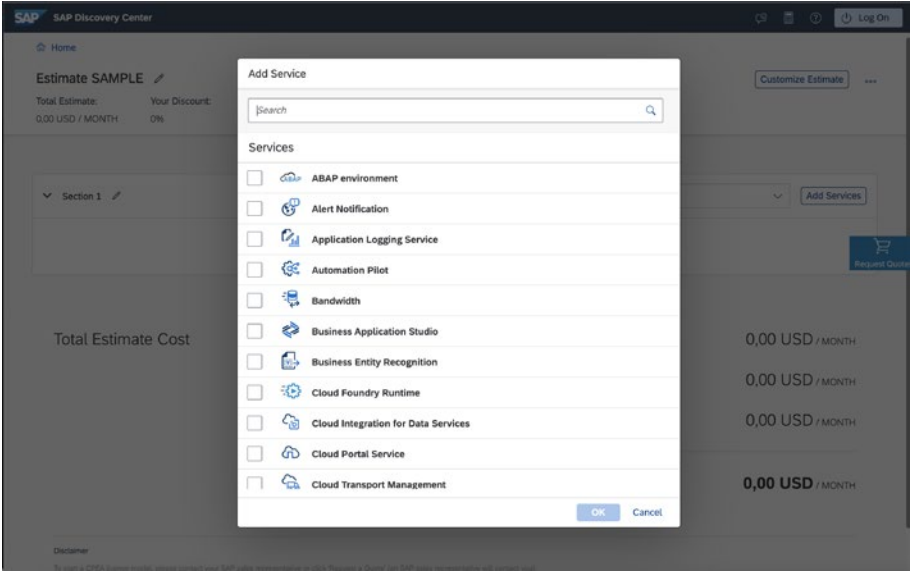


Figure 5-18. *The service catalog*

In the search field on the service list, you can input the full or partial name of a cloud service, so the list will be automatically filtered (Figure 5-19).

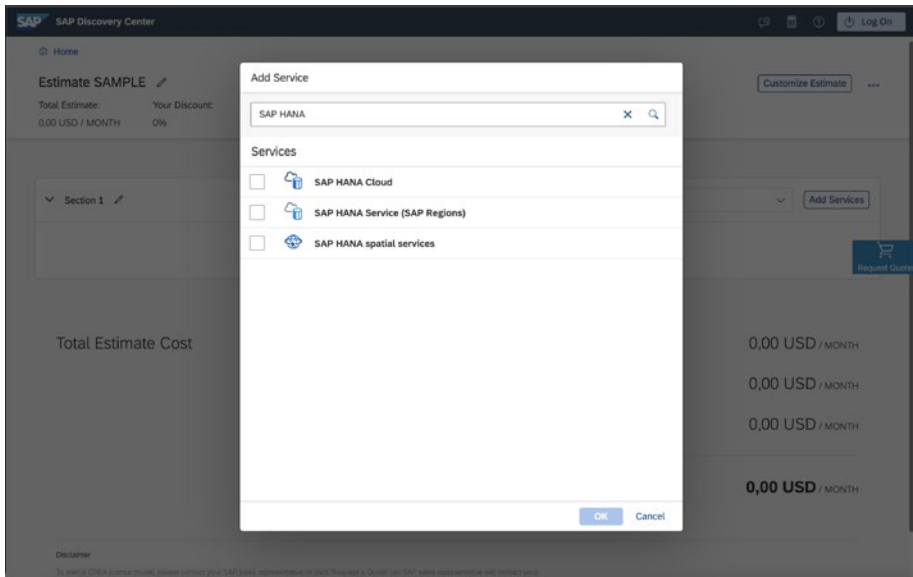


Figure 5-19. *Input field for service selection*

Next, you should click on the checkbox to select the SAP HANA Cloud service and click the OK button (Figure 5-20).

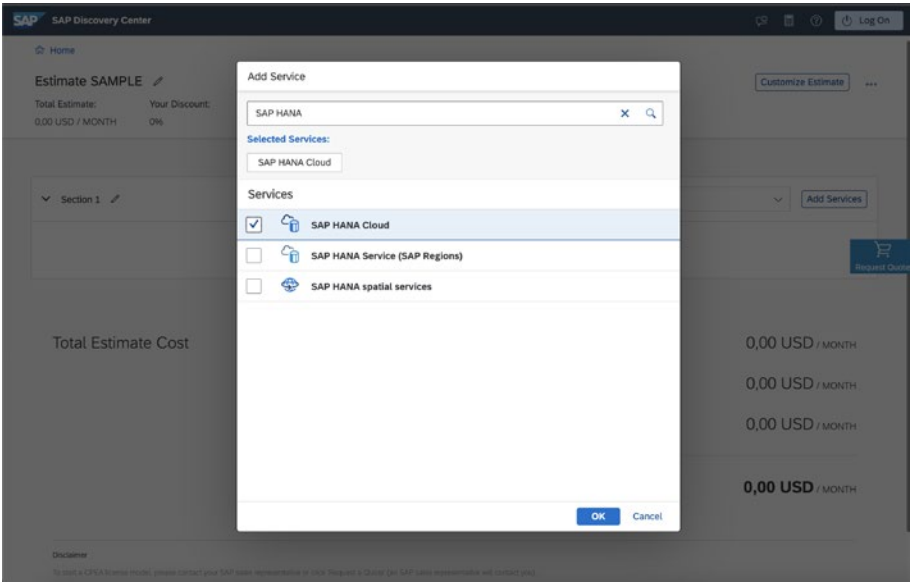


Figure 5-20. *Selecting SAP HANA Cloud service*

The next step is necessary to provide the right sizing to estimate the cost of the service. It is done by setting the service parameters (Figure 5-21).

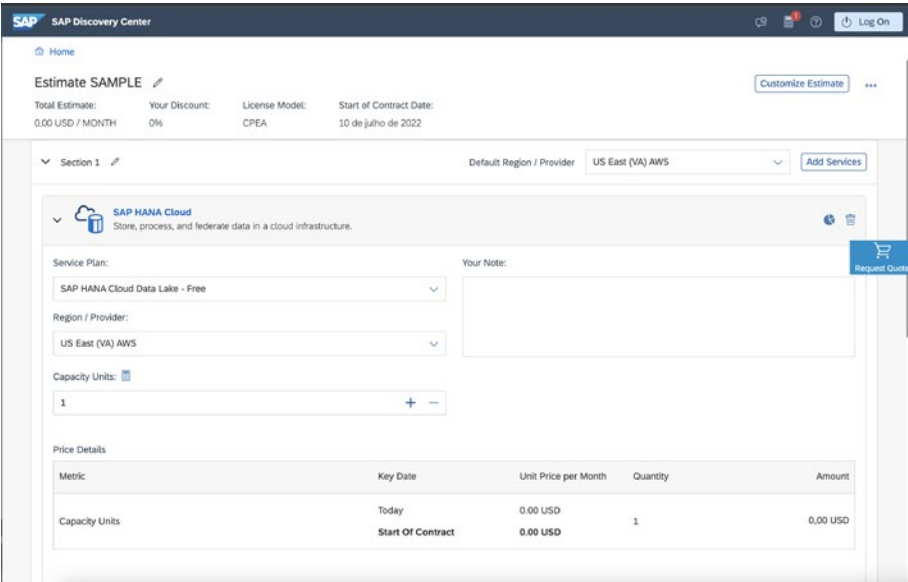


Figure 5-21. Service parameters

In the SAP HANA Cloud service, the parameters that need to be filled in are as follows:

Service Plan – This field presents all possible plans to use in the SAP HANA Cloud service, as follows:

- **HANA** – This option enables an in-memory database tenant, commonly used to support hot data models.
- **Data Lake** – This option enables a data lake tenant, commonly used to support warm data models.
- **Adaptive Server Enterprise** – This option enables a relational cloud database instance.
- **Adaptive Server Enterprise Replication** – This option enables a cloud replication server.

- **SAP HANA Cloud Data Lake-Free** – This option enables hot and warm tenants in the free tier.

Region / Provider – This field presents all available regions and cloud providers (hyperscalers) for provisioning purposes.

Capacity Units – This fields shows the amount of capacity units necessary to support a given sizing. This value is automatically generated by using the SAP HANA Cloud Capacity Estimator available via the calculator icon.

SAP HANA Cloud Capacity Unit Estimator

The capacity unit estimator tool is an easy-to-use interface that helps you to define how an SAP HANA instance will be set up (Figure 5-22).

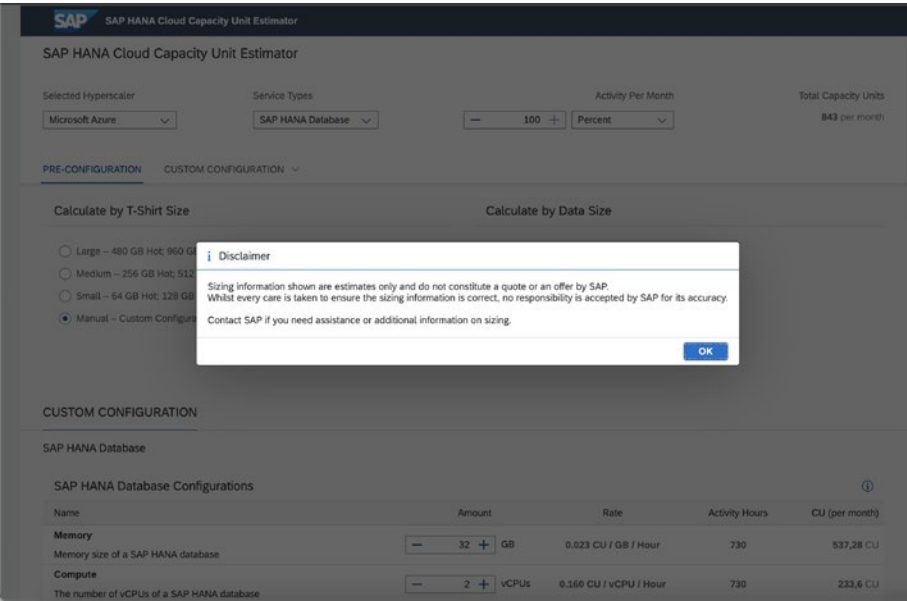


Figure 5-22. SAP HANA Cloud Capacity Unit Estimator

At the beginning, the SAP HANA Cloud Capacity Unit Estimator shows a popup window with a disclaimer to let you know that it is only an estimate and does not constitute a quote.

The estimator tool allows you to select three main parameters, as follows:

1. **Hyperscaler** – This list allows you to select in which cloud provider the estimate will be generated.
2. **Service Types** – This list shows the two data tier options for the SAP HANA Cloud service: an in-memory database tenant (hot) or a relational data lake (cold).
3. **Active per Month** – This field is used to define the expected working time for your SAP HANA Cloud instance. It is possible to define a percentage or number of hours per month.

Selecting any T-Shirt size automatically sets the database parameters for each predefined configuration of the SAP HANA database instance (Figure 5-23).

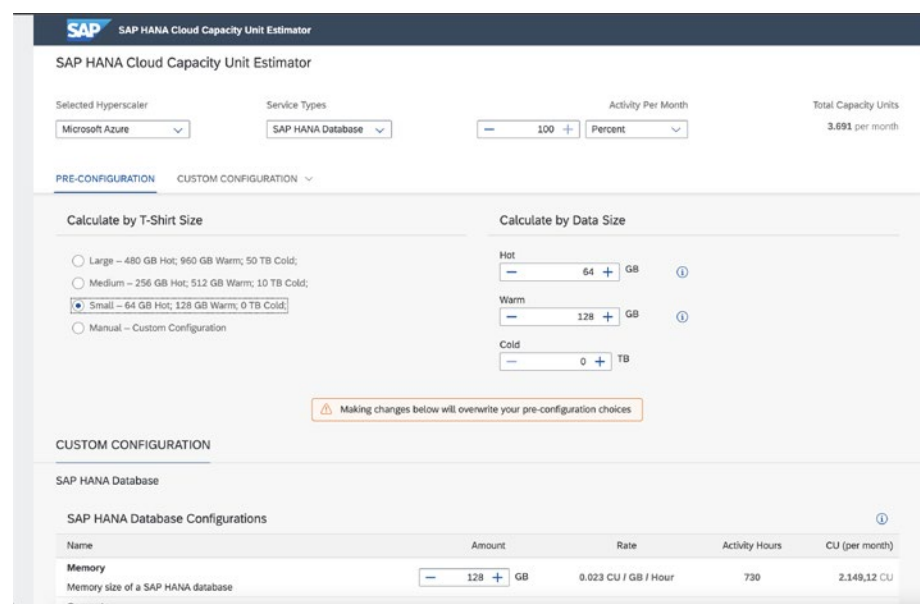


Figure 5-23. Calculating T-Shirt size with SAP HANA Cloud Capacity Unit Estimator

The available T-shirt sizes are shown in Table 5-2.

Table 5-2. T-Shirt Sizes

T-Shirt Size	Hot Storage	Warm Storage	Cold Storage
Large	480 GB	960 GB	50 TB
Medium	256 GB	512 GB	10 TB
Small	64 GB	128 GB	0 TB
Manual	Custom Configuration		

In general, you can play with T-shirt sizes and generate an estimate that will support your cost analysis and anticipate a view of a given cloud setup.

For more-specific configurations, you can still customize your configuration by setting every parameter manually. In Tables 5-3 and 5-4, you can find all parameters that can be modified in the estimate.

Table 5-3. *Custom Configuration Parameters List – SAP HANA Database*

Name	Amount	Metric	Rate	Activity Hour
Memory	Memory size of the SAP HANA Cloud database	GB	0.023 CU / GB / Hour	730
Compute	The number of vCPUS of the SAP HANA Cloud database	vCPUs	0.160 CU / vCPU / Hour	730
Storage	The disk storage space of the SAP HANA Cloud database	GB	0.013 CU / 16 GB / Hour	730
Backup Storage	The backup storage space of the SAP HANA Cloud database. It depends on the size of your hot and warm storages. It is calculated as a fixed amount of 14 times the size of the hot and warm storage combined.	GB	0.011 CU / 64 GB / Hour	730
Network Data Transfer	The amount of network traffic generated by reading from the system (egress).	GB	0.438 CU / GB	N/A
Cloud Connector	Connect to an on-premises SAP HANA database	Toggle ON/OFF	0.172 CU / Hour	730

(continued)

Table 5-3. *(continued)*

Name	Amount	Metric	Rate	Activity Hour
Synchronous Replicas	Number of synchronous replicas of the SAP HANA Cloud database	Replicas	The cost of the memory, compute, and storage of the replica is charged.	730
Asynchronous Replicas	Number of asynchronous replicas of the SAP HANA Cloud database	Replicas	The cost of the memory, compute, storage, and backup storage of the replica is charged.	730

** The Cloud Connector costs 1 vCPU and 0.5 GB of memory, which is equivalent to 0.1715 CU Per Hour.*

Table 5-4. Custom Configuration Parameters List – Data Lake

Name	Amount	Metric	Rate	Activity Hour
Compute	The number of vCPUs dedicated to data lake relational engine (memory included)	Relational Engine Compute (vCPU)	0.539 CU / vCPU / Hour	730
	The amount of file data scanned by queries per month	SQL on Files File Scan (GB)	0.539 CU / 33.06 GB	N/A
	The number of read, write, and metadata requests made against data lake file storage during a specific month. A starting estimate is 4,000,000 API calls for each TB of file data read or written.	File Access API Calls (x1000)	0.539 CU / 22825 API Calls	N/A
Storage	Data stored in data lake relational engine format to facilitate OLAP analysis	Relational Engine Storage (TB)	0.272 CU / TB / Hour	730
	Files stored in data lake file storage	File Storage (TB)	0.272 CU / 2.905 TB / Hour	730
Backup Storage	The backup storage space of a data lake relational engine	GB	0.011 CU / 64 GB / Hour	730
Network Data Transfer	The amount of network traffic generated by reading from the system (egress).	GB	0.438 CU / GB	N/A

Once you have configured the capacity unit estimator with the desired T-shirt size or custom configuration parameters, you can check what the required number of capacity units is in the top right corner (Figure 5-24).

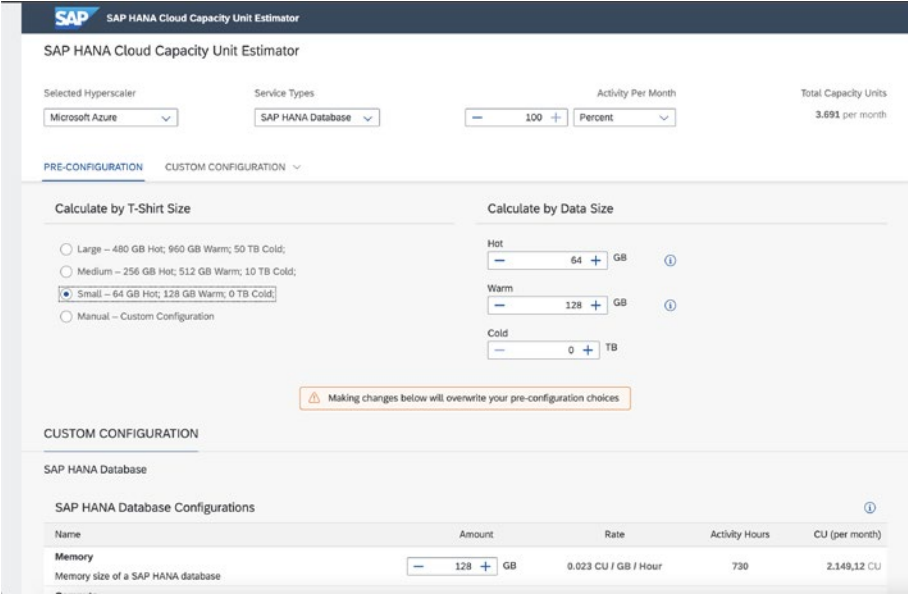


Figure 5-24. Checking the total number of capacity units

Now, you must go to the SAP BTP Estimator window and input this value in the capacity units field. After inputting it, the amount in USD will be changed as expected (Figure 5-25).

SAP

SAP Discovery Center

Log On

Home

Estimate SAMPLE

Customize Estimate

...

Total Estimate:

Your Discount:

License Model:

Start of Contract Date:

3,321.90 USD / MONTH

0%

CPEA

10 de julho de 2022

Section 1

Default Region / Provider

US East (VA) AWS

Add Services

SAP HANA Cloud

Store, process, and federate data in a cloud infrastructure.

Service Plan:

HANA

Region / Provider:

US East (VA) AWS

Capacity Units:

3691

+

-

Your Note:

Request Quote

Price Details

Metric	Key Date	Unit Price per Month	Quantity	Amount
Capacity Units	Today	0.90 USD	3691	3,321.90 USD
	Start Of Contract	0.90 USD		

Figure 5-25. *Filling the capacity unit field*

To see a summary of your estimate, you can scroll down the page and check Total Estimate Cost (Figure 5-26).

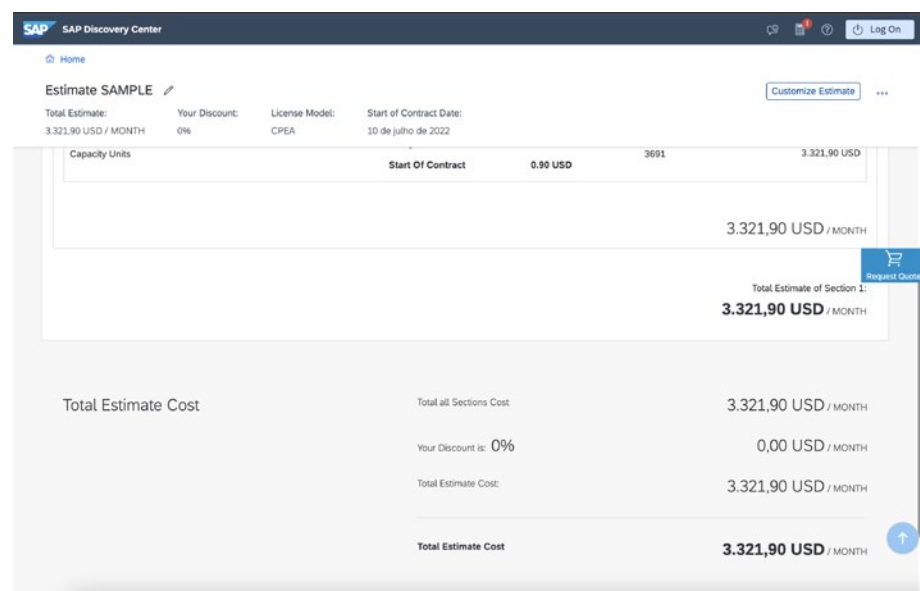


Figure 5-26. *Checking Total Estimate Cost*

Additionally, you can export, download, or share this estimate with other people in your organization. These features can be seen in Figure 5-27.

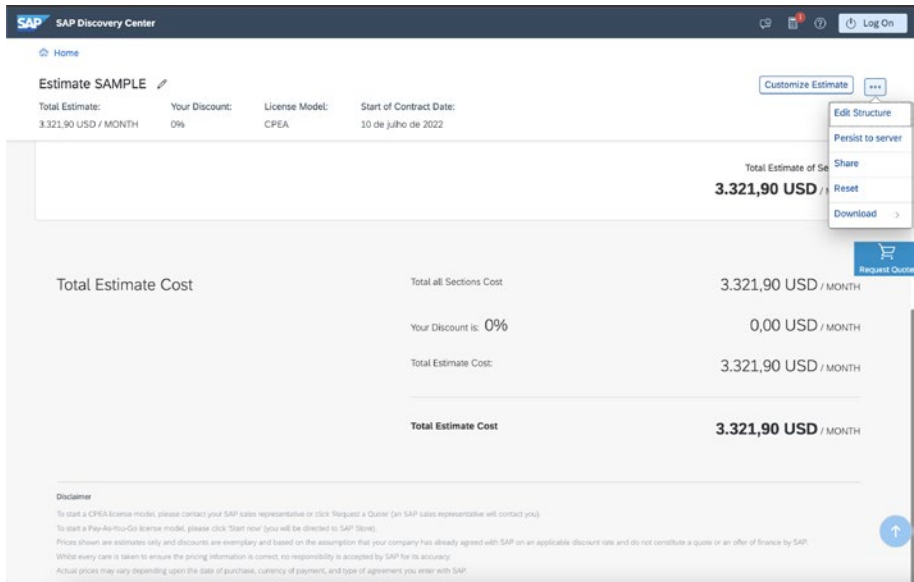


Figure 5-27. Additional features of the SAP BTP Estimator

The estimator tool is an easy way to support IT teams with a cost estimation of a use case based on required cloud services.

Summary

You have successfully completed this chapter, and now it's time to review what you learned and see what's next. We covered the following:

- How to use SAP Discovery Center
- How to calculate sizing using SAP BTP Estimator
- How to estimate SAP HANA Cloud capacity units

Now we will continue our journey with essential learning to start using the SAP Business Technology Platform and create an SAP HANA Cloud instance and artifacts.

CHAPTER 6

How to Activate Your SAP BTP Account

In this chapter, you will learn how to set up an SAP BTP trial account and create your own SAP HANA Cloud instance, and also enable the SAP Business Application Studio to start working with new artifacts.

Your Journey Starts Here

1. Navigate to <https://www.sap.com>.
2. Click the Log On icon in the upper-right corner (Figure 6-1).

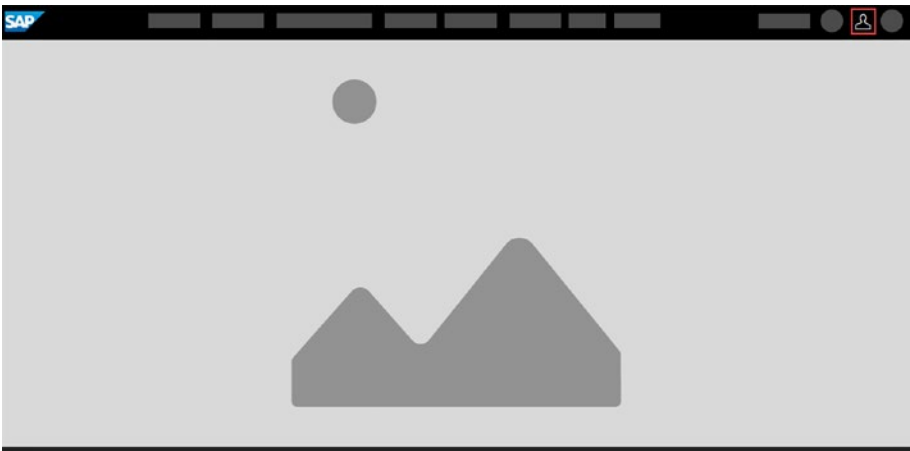


Figure 6-1. *Logon window*

You can provide your P, S, C, D, or I number, as well as your password. If you don't already have an account, use one of the other login options or click Register to create a new one (Figure 6-2).

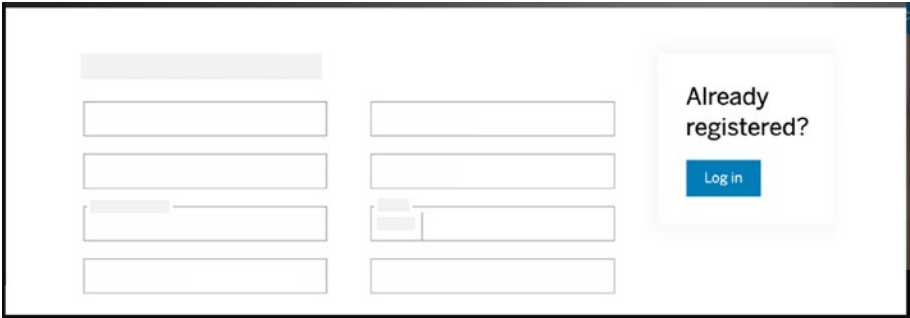
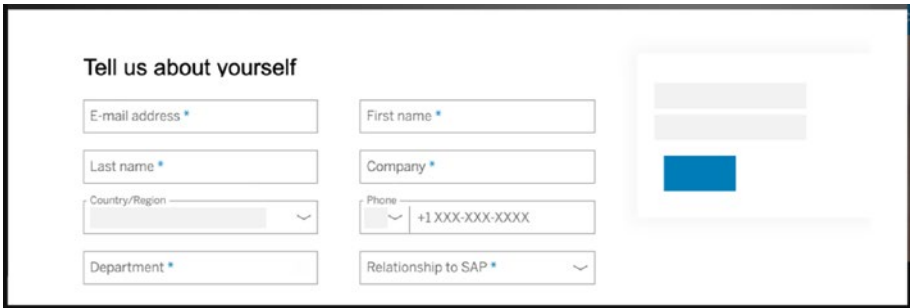


Figure 6-2. *Registration window*

Registering Your Account

You will see a registration dialog in case you do not have an account. Fill in the relevant fields and submit the form (Figure 6-3).



Tell us about yourself

E-mail address *	First name *
Last name *	Company *
Country/Region 	Phone  +1 XXX-XXX-XXXX
Department *	Relationship to SAP * 

A blue button is located on the right side of the form.

Figure 6-3. *Registration form*

An activation link will be emailed to you, as described in the message.

Activating Your Account

Look for the activation link in your inbox. To activate your account, simply click the button shown in Figure 6-4.

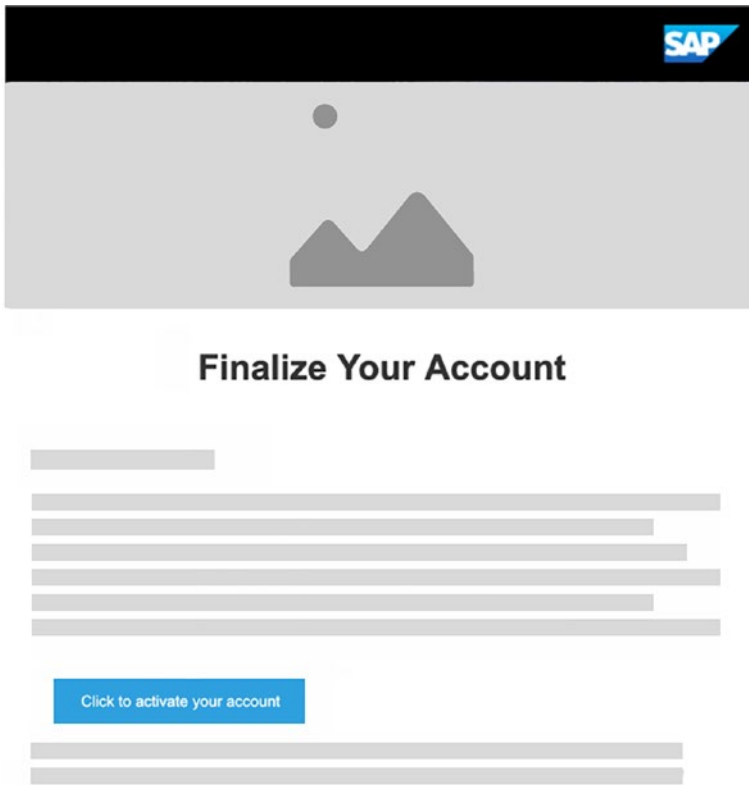


Figure 6-4. *Activating your account*

After you've activated your account, you'll get a success message.

Accessing the SAP BTP Trial

Go to the SAP BTP Trial page and click Log On after activation, or if you already have an SAP account.

A dialog will appear asking you to confirm the SAP BTP Trial terms and conditions. Click Accept after checking the boxes. This simply adds your SAP.com registration to your SAP Cloud Identity login account.

Check Your Account

1. A notice will appear stating that you must authenticate your account with a phone number. To start the flow, click [Verify Your Phone Number](#) (Figure 6-5).

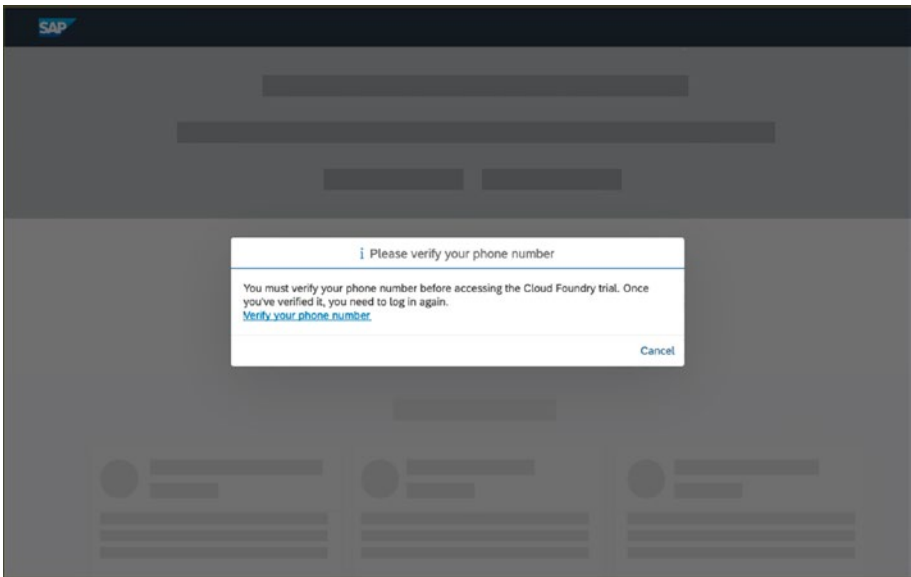


Figure 6-5. *Verify your phone number message alert*

2. To get a code, enter your phone number and select New Code.
3. To validate your account, please also input this code and click Continue. You will be immediately logged off after the verification.

Go to the Welcome Page

To access your verified account, choose Log On.

On the welcome screen of SAP BTP Cockpit, click Go To Your Trial Account to access your BTP account (Figure 6-6).

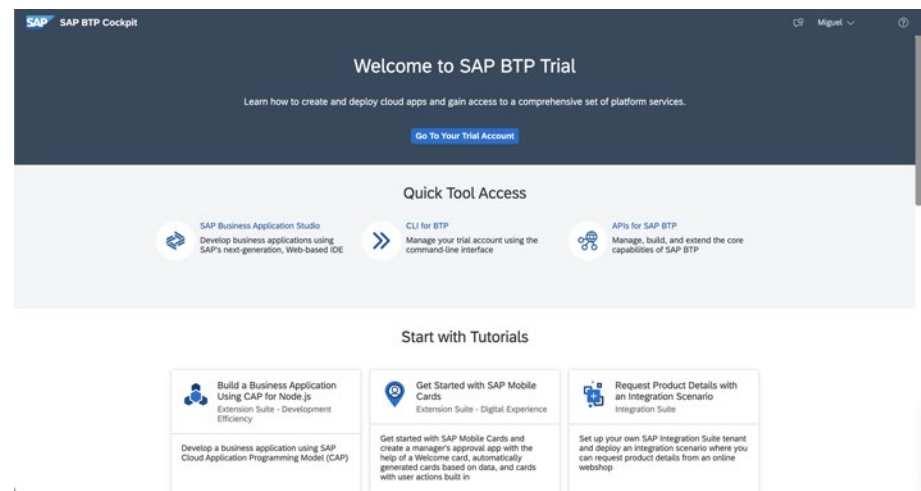


Figure 6-6. *Welcome to SAP BTP Trial*

Note Save this URL to your favorites.

Working with Subaccounts

1. Now you can create a subaccount by selecting a region and a cloud provider. To start the provisioning, click the button Create Account (Figure 6-7).

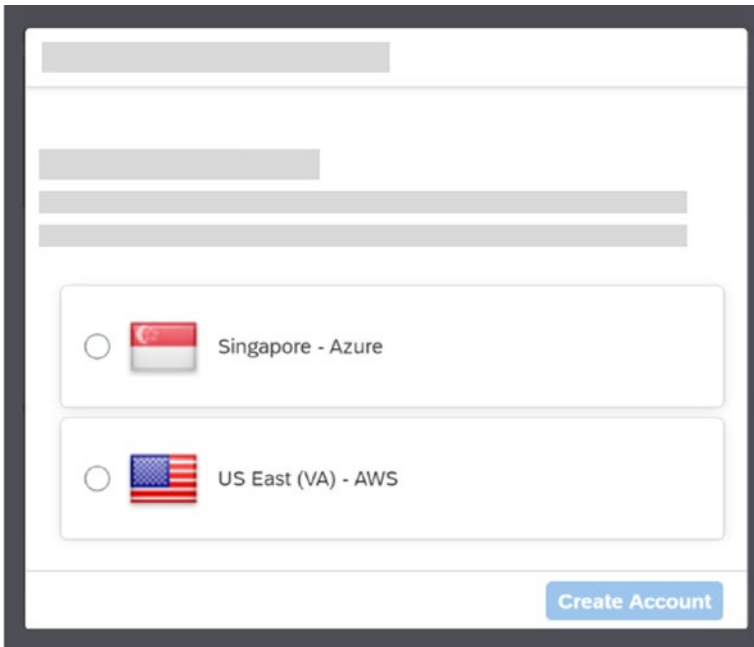


Figure 6-7. *Selecting a region and a cloud provider*

2. The process takes some time, and when finished you can get access to the new subaccount (Figure 6-8).

Note Trial accounts contain one subaccount and space.

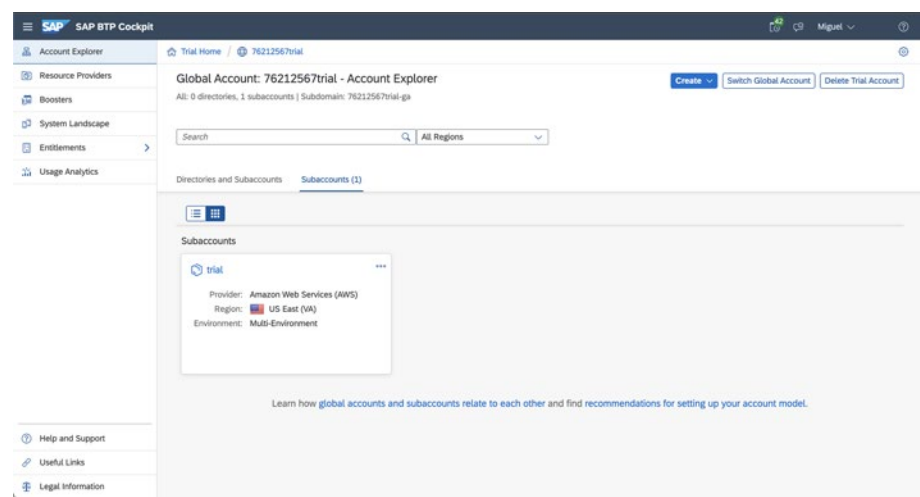


Figure 6-8. Subaccount created

3. All basic information of the subaccount is displayed on its definition page. You can explore the run-time environment and configured spaces. The API endpoint is also exposed (Figure 6-9).

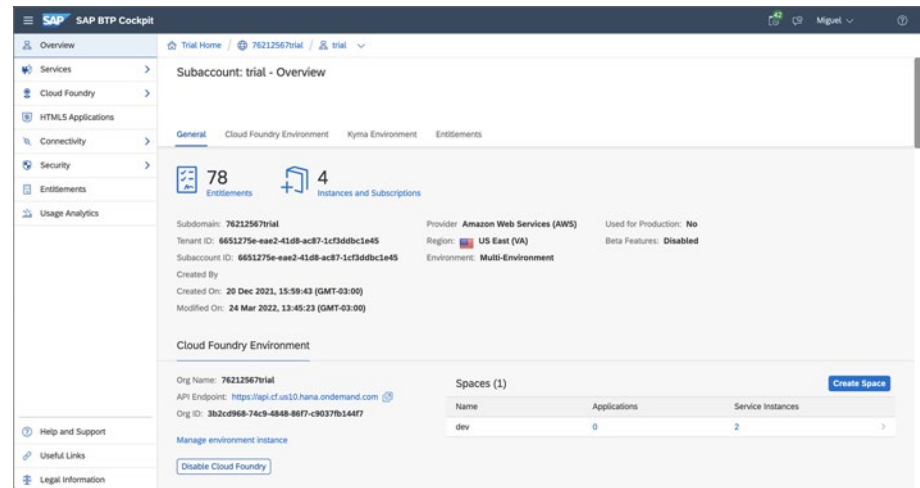


Figure 6-9. Subaccount overview

Exploring the Service Marketplace

All services that are offered on any of the settings are listed in the Service Marketplace. On the left, select Service Marketplace to see all available services, and a filter bar to find the one you want. You may create a service plan to provision service instances from here (Figure 6-10).

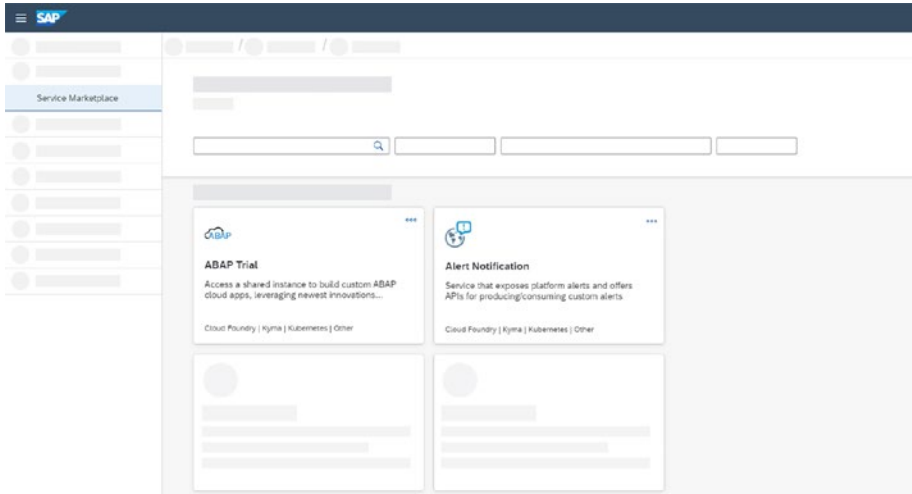


Figure 6-10. *Service Marketplace*

Creating an SAP HANA Cloud Instance

Set Up SAP HANA Cloud

SAP HANA Cloud provides database and advanced data-processing capabilities. It allows you to extract insights from various forms of data, such as text, spatial, predictive, and more.

Data operations are conducted faster in an in-memory database with a columnar store than in a typical database running with disk-oriented architecture. SAP HANA fits the category of *translytical*.¹ There is a unique benefit that you can leverage when processing transactional and analytical requests in real time, without the need to create additional data models and data loading tasks to materialize them.

Creating an SAP HANA Cloud Instance

1. Select the subaccount from the SAP BTP Cockpit (Figure 6-11).

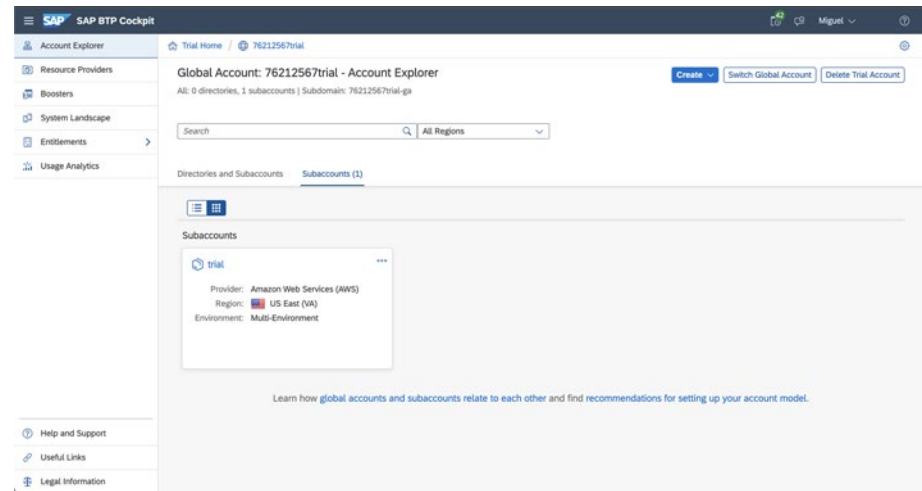


Figure 6-11. Selecting the subaccount

¹ “Translytical” is the category name used by software providers to identify solutions that offer transactional characteristics with the ability to run near real-time analytics. This is achieved using in-memory databases together with other modern database techniques.

2. Select the space in which you want to set up an SAP HANA Cloud instance (Figure 6-12).

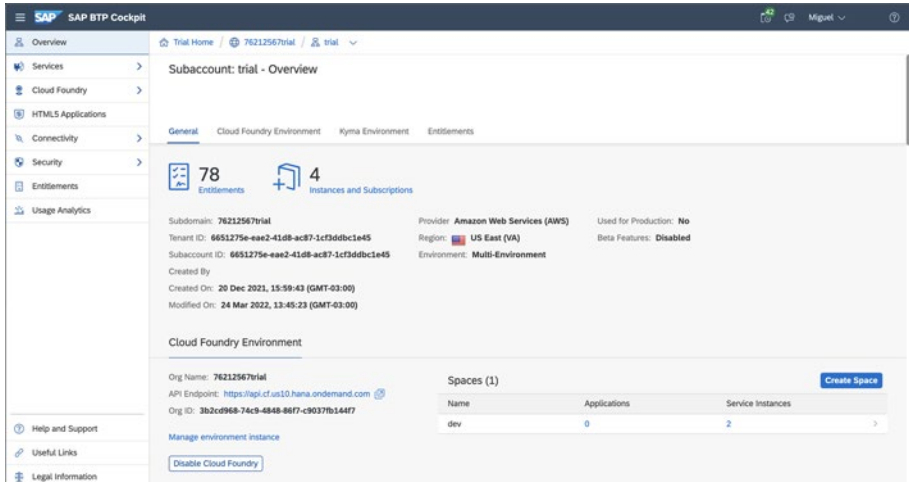


Figure 6-12. *Selecting the space*

3. Click the SAP HANA Cloud button and select SAP HANA Database to start defining your SAP HANA Cloud instance (Figure 6-13).

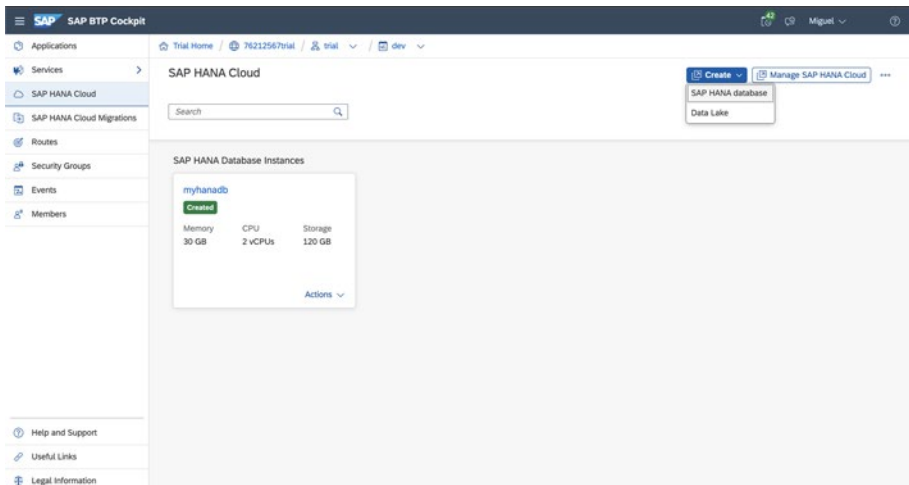


Figure 6-13. *Selecting the SAP HANA database*

Create Database

1. Complete the steps following the wizard to create an SAP HANA Cloud instance (Figure 6-14).

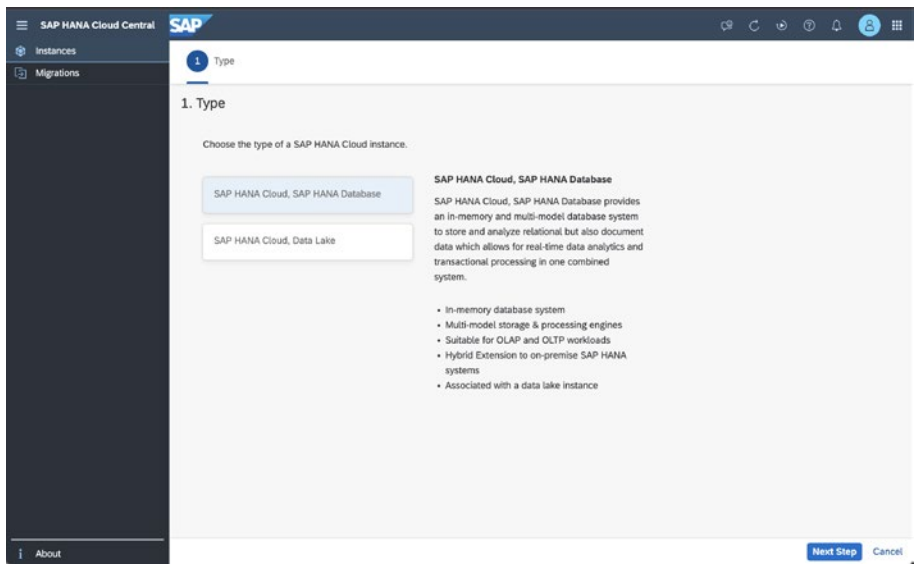


Figure 6-14. SAP HANA Cloud instance wizard

2. Provide the instance name and password in the General window (Figure 6-15).

Figure 6-15. Providing instance name and password

3. Now, define the size of its memory (Figure 6-16).

Figure 6-16. Defining the size of its memory

4. Observe the database availability zone (Figure 6-17).

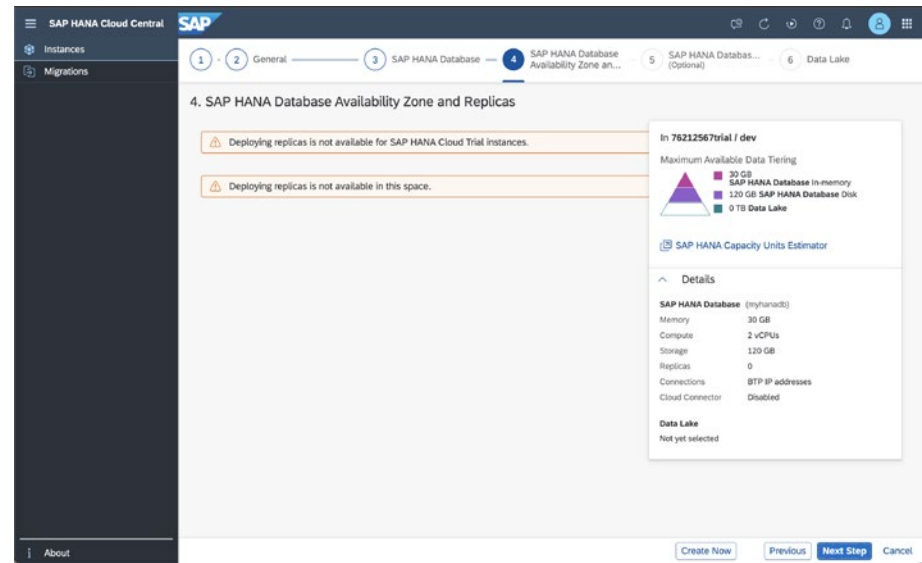


Figure 6-17. Observing the database availability zone

Note In trial accounts, this feature is not available.

5. Change the allowed connections to “Allow all IP addresses” and click the **Next Step** button (Figure 6-18).

Note The use of “Allow All IP addresses” is not recommended for production environments.

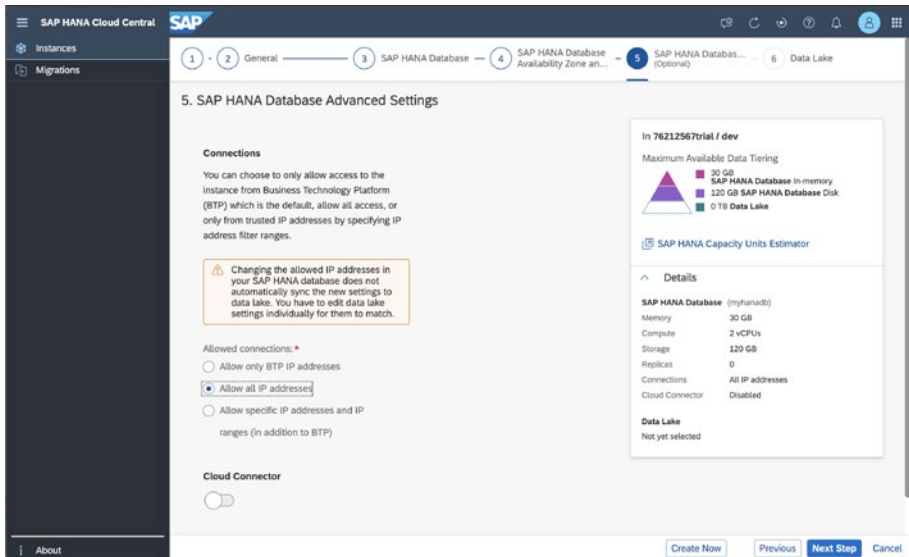


Figure 6-18. *Changing the Allowed Connections setting*

6. Click on the Create Instance button to conclude the process.
7. Now you should see your instance being created (Figure 6-19).

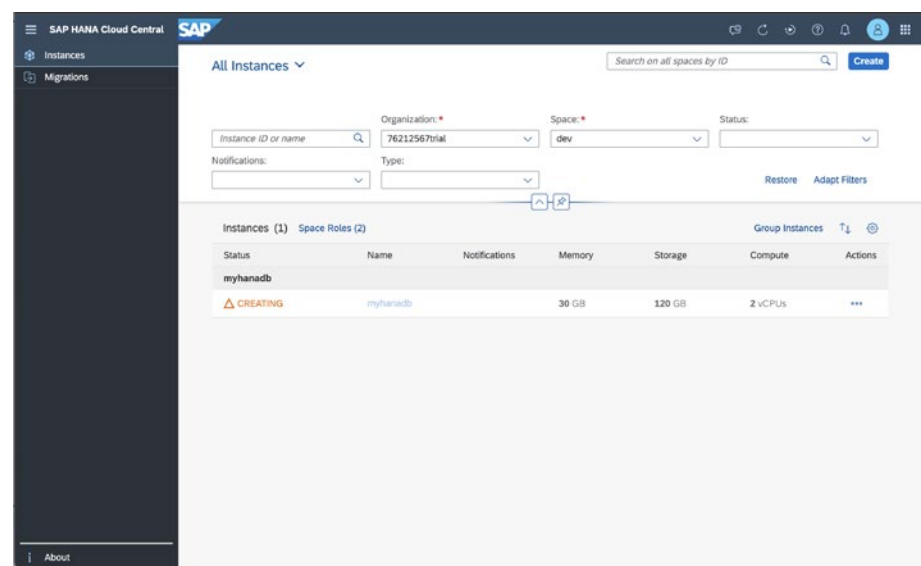


Figure 6-19. SAP HANA Cloud instance being created

- 8. Once this process is concluded, you should see an SAP HANA Cloud instance with status RUNNING (Figure 6-20).

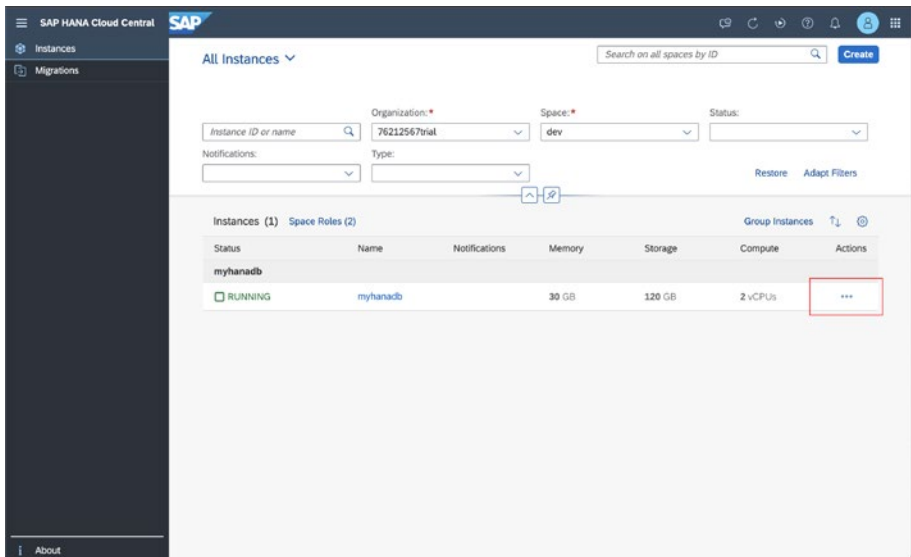


Figure 6-20. SAP HANA Cloud instance running

Note Remember that trial accounts behave differently and are stopped by the end of day, so it is necessary to check the instance status before initiating your day. If necessary, just restart it.

Managing Instances

1. To manage an SAP HANA Cloud instance, you should click on the three dots icon available in the instance line. A context menu is shown. In the next steps, we will show how to stop and start an instance (Figure 6-21).

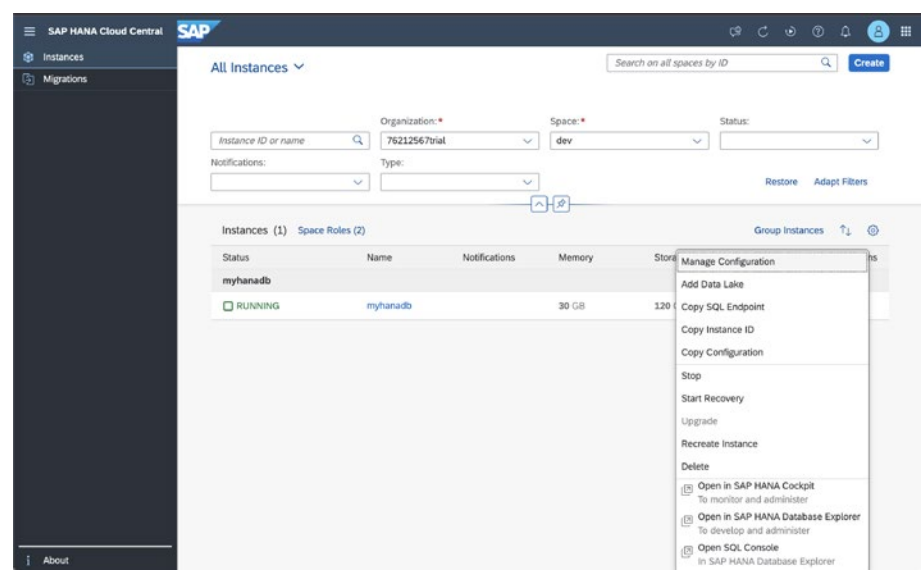


Figure 6-21. Managing the SAP HANA Cloud instance

- 2. Click on Stop to shut down the instance (Figure 6-22).

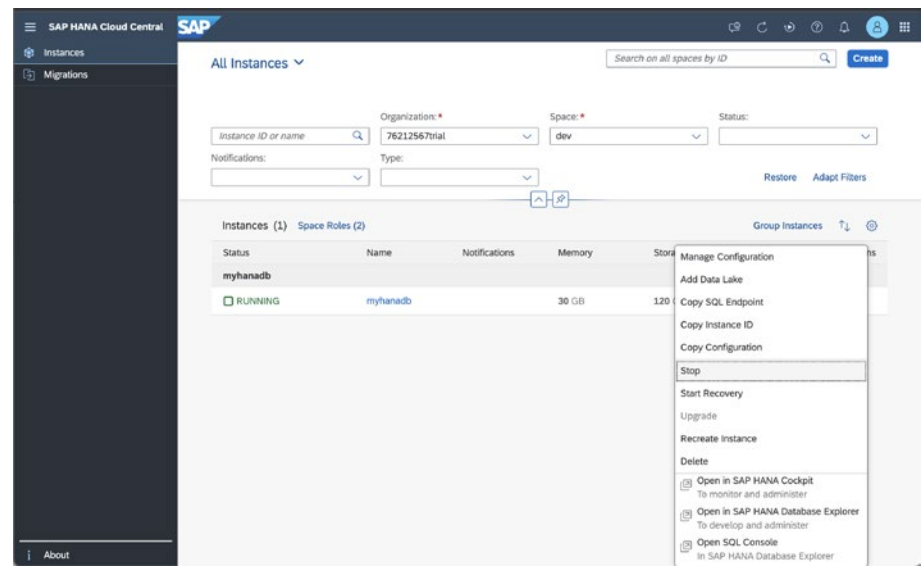


Figure 6-22. Requesting stop of the SAP HANA Cloud instance

3. An alert window is shown asking you to confirm the stop (Figure 6-23).

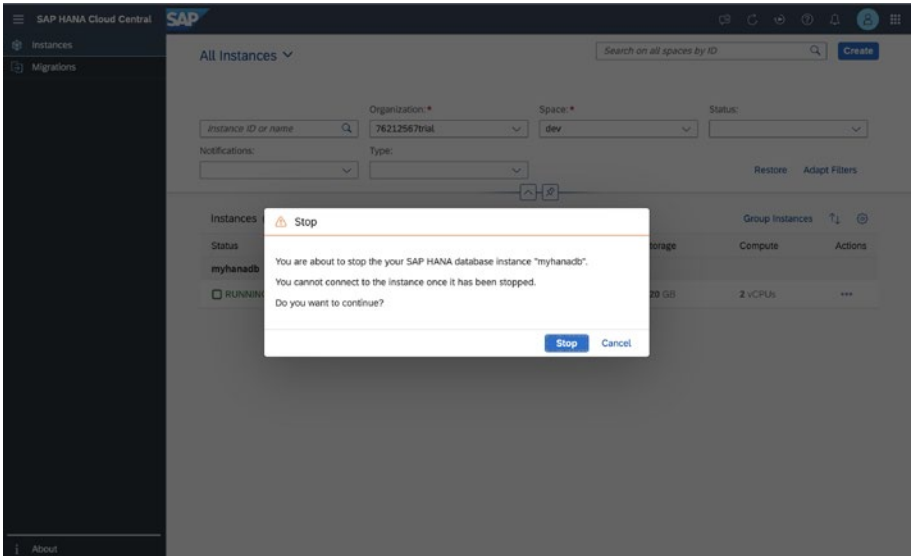


Figure 6-23. Alert window to confirm the stop

4. The shutdown process is initiated, and the instance status changes to STOPPING (Figure 6-24).

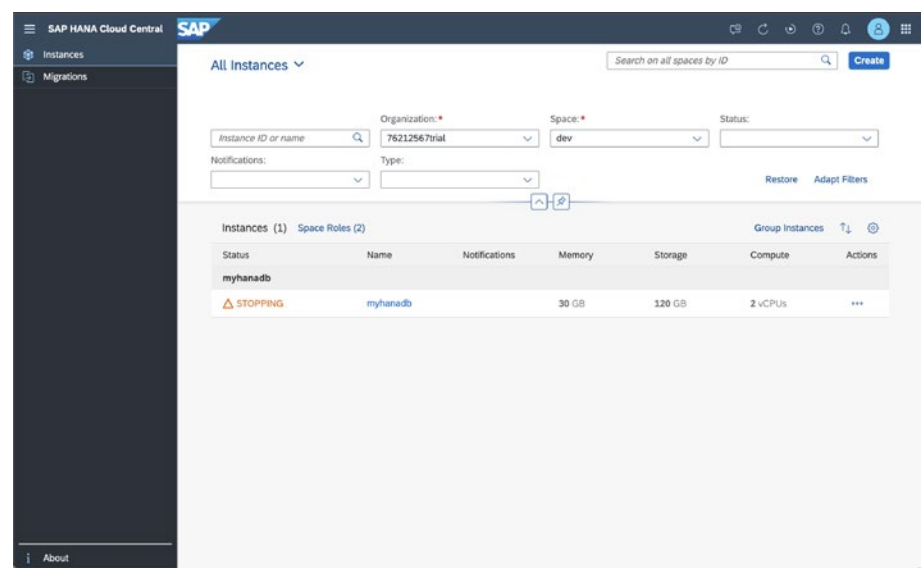


Figure 6-24. Stopping the SAP HANA Cloud instance

- 5. Once the shutdown process is concluded, the instance status changes to STOPPED (Figure 6-25).

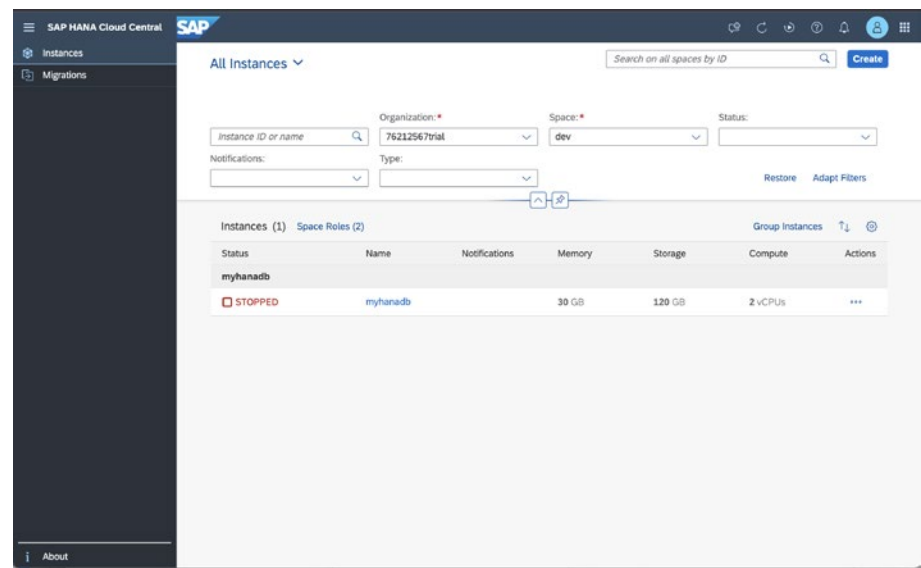


Figure 6-25. SAP HANA Cloud instance stopped

- Now you can click on the three dots icon in Actions column to open the context menu and select to start the instance (Figure 6-26).

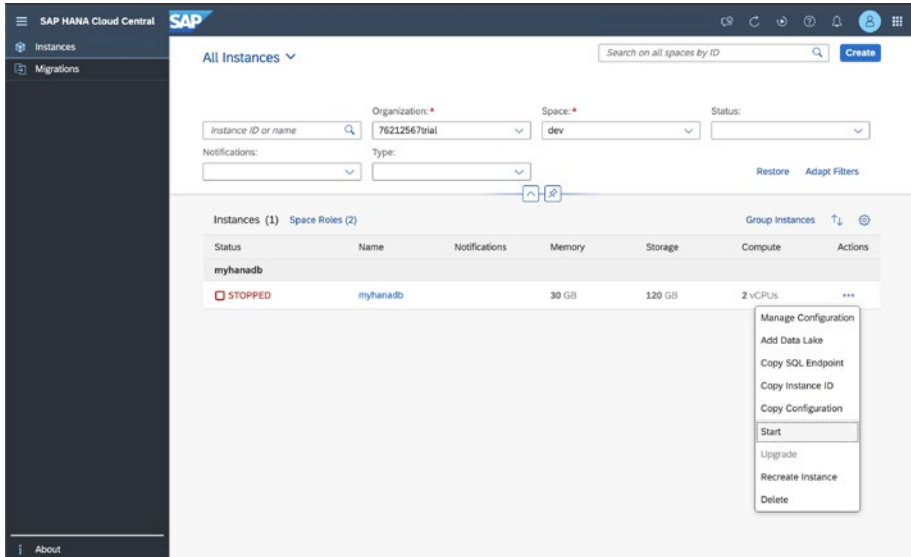


Figure 6-26. Requesting the start of the SAP HANA Cloud instance

- Once the starting process initiates, the instance status changes to STARTING (Figure 6-27).

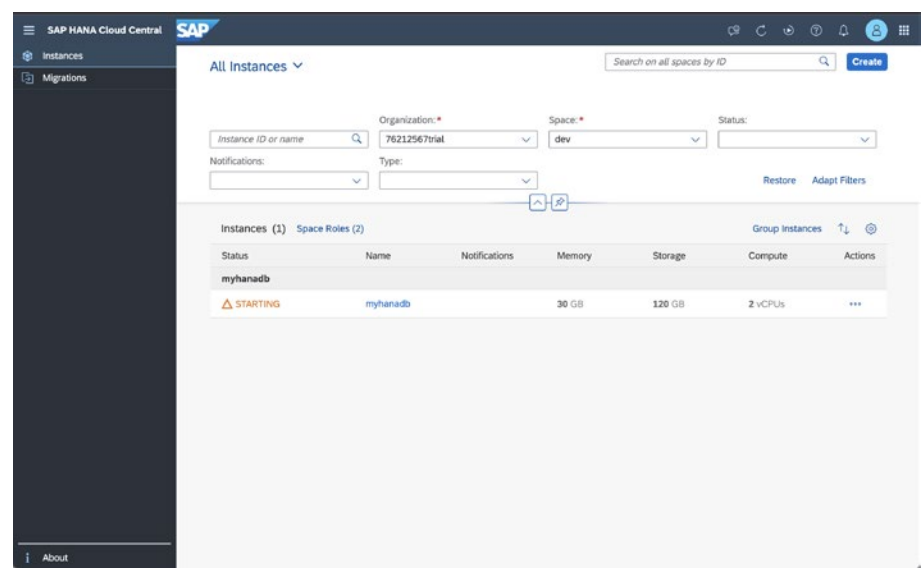


Figure 6-27. Starting the SAP HANA Cloud instance

8. After the instance has started, the instance status changes to RUNNING (Figure 6-28).

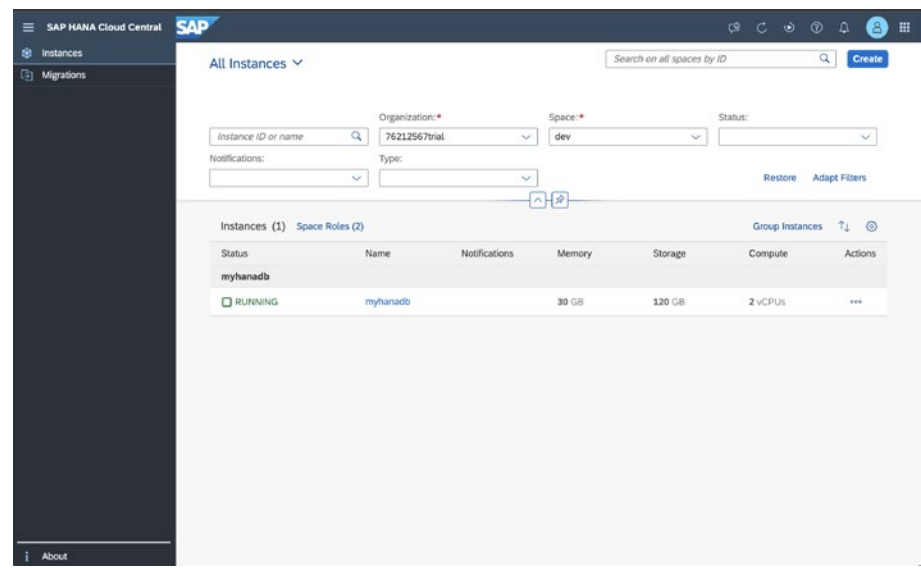


Figure 6-28. SAP HANA Cloud instance started

Set Up SAP Business Application Studio

SAP Business Application Studio is the cloud service provided in the SAP Business Technology Platform (BTP) that we use to support a cloud-based application development.

Follow the next steps to enable SAP Business Application Studio.

Starting with SAP BTP Trial Login

1. Navigate to <https://account.hanatrial.ondemand.com/> and start accessing the SAP BTP Cockpit.
2. In case you see this message, just select the checkbox and click the Accept button (Figure 6-29).

Legal Disclaimers for SAP BTP Trial

We have updated our legal documents. To access and use SAP BTP Trial, please read and accept the documents below:

Terms and Conditions

☐ I have read and understood the Terms and Conditions of SAP BTP Trial. *

*Required

Accept

SAP® ID Service

Figure 6-29. Legal disclaimers window

Accessing the SAP Business Application Studio

- 1. Click on SAP Business Application Studio to launch it (Figure 6-30).

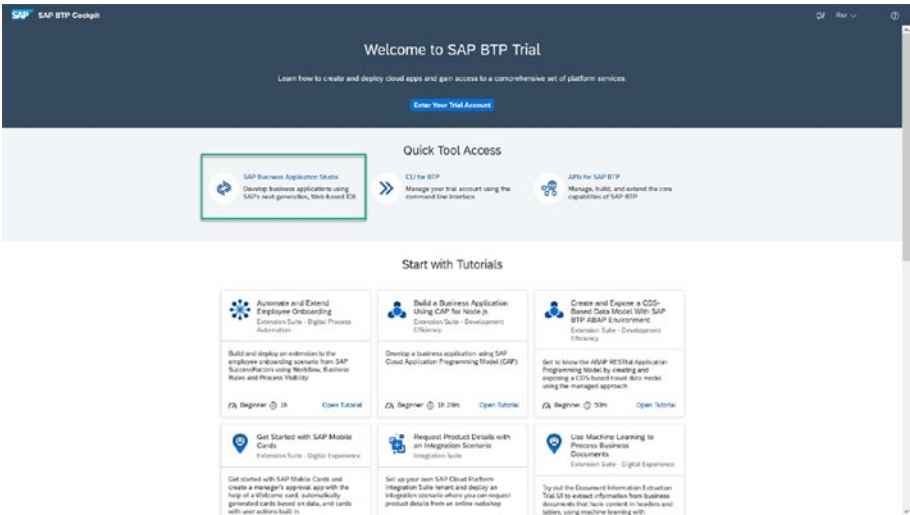


Figure 6-30. Launching the SAP Business Application Studio

- 2. Please accept the terms by clicking the OK button and checking the box to not see this message again (Figure 6-31).

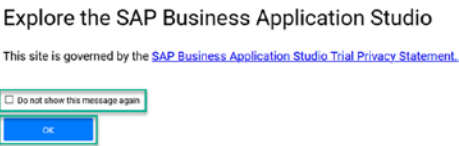


Figure 6-31. SAP Business Application Studio trial privacy statement

3. When you start using SAP Business Application Studio, a welcome page is shown (Figure 6-32).

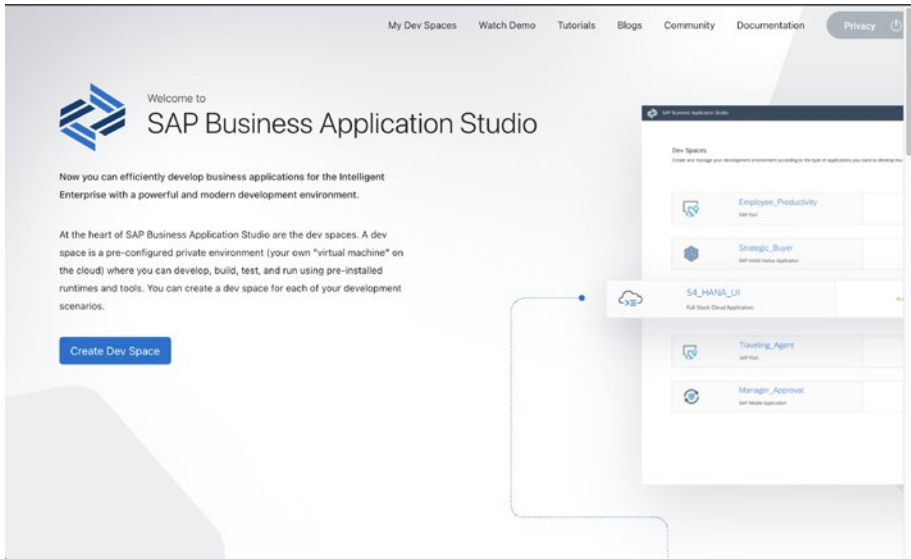


Figure 6-32. SAP Business Application Studio welcome page

4. You can manage your development spaces by clicking on the Dev Spaces link on the upper-right side of the page. If you do not have a development space created, just click the Create Dev Space button (Figure 6-33).

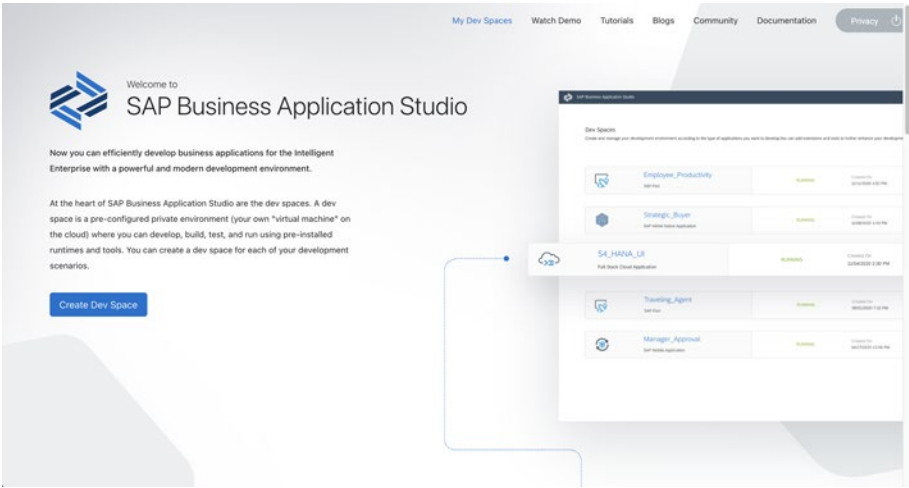


Figure 6-33. *Create Dev Space button*

Note Save this URL to save time and get back quickly.

Subscribing to SAP Business Application Studio

Go through these steps if you don't have the SAP Business Application Studio service enabled in a subaccount.

1. Click the [Go To Your Trial Account](#) button (Figure 6-34).

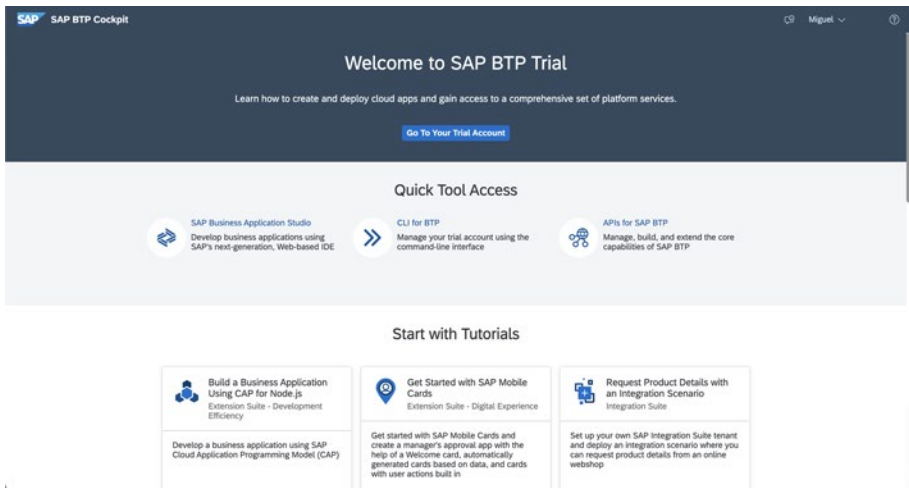


Figure 6-34. *Go To Your Trial Account button*

2. Select the subaccount where you will enable the SAP Business Application Studio subscription by clicking on the subaccount tile (Figure 6-35).

Note Remember that the SAP Business Application Studio service provided in an SAP BTP Trial account is only available in specific regions. Usually, we select the region closer to the location we are in to reduce latency, but you are free to decide.

An SAP BTP Trial account provides a trial subaccount that is automatically **created**.

CHAPTER 6 HOW TO ACTIVATE YOUR SAP BTP ACCOUNT

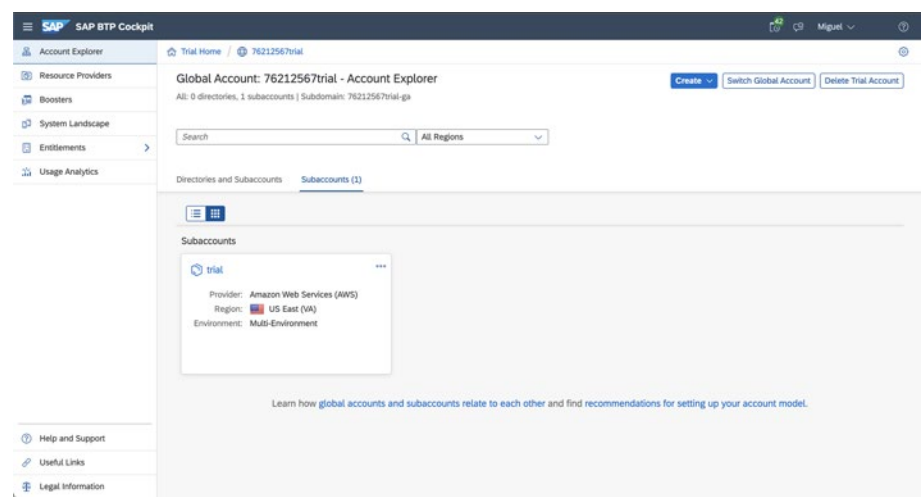


Figure 6-35. Selecting the subaccount

3. Click on Service Marketplace (Figure 6-36).

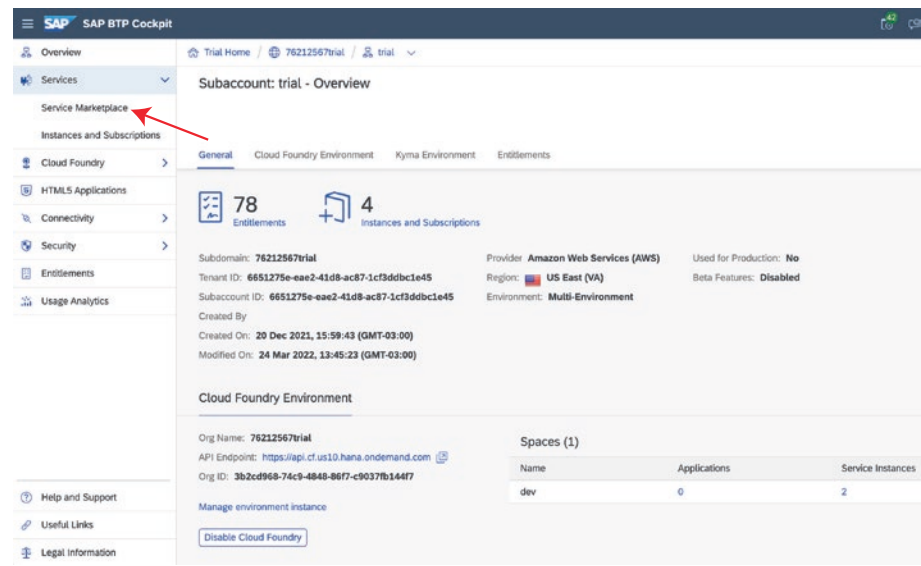


Figure 6-36. Clicking on Service Marketplace

4. Enter the word “studio” to see SAP Business Application Studio (Figure 6-37).

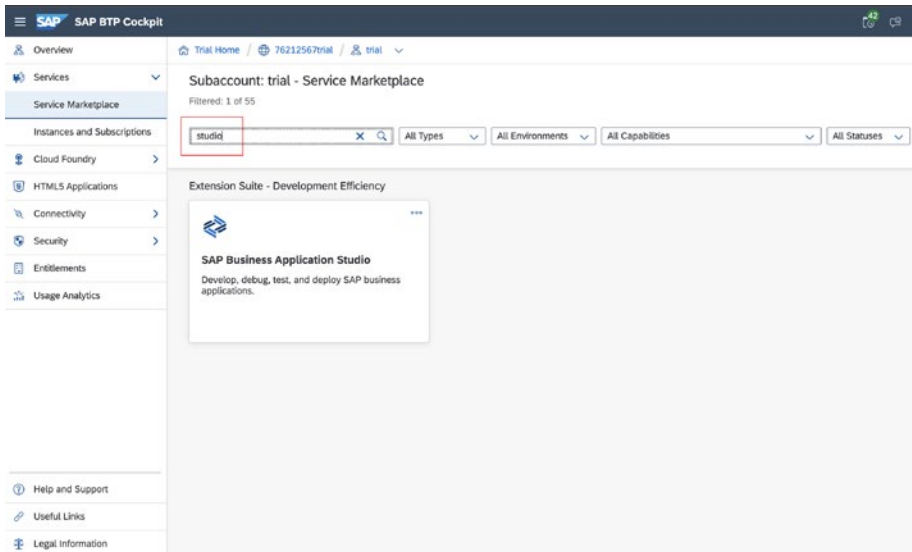


Figure 6-37. Searching for SAP Business Application Studio

5. Click on the three dots icon and click on Create to enable it (Figure 6-38).

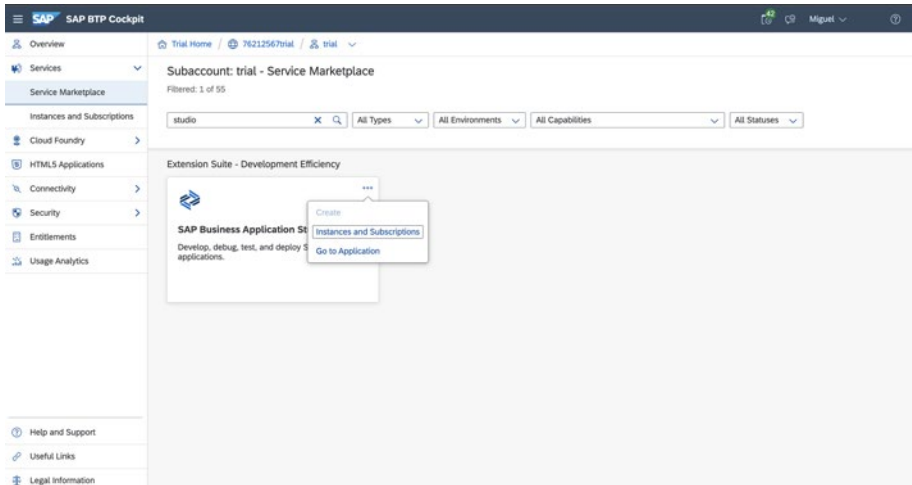


Figure 6-38. Creating the SAP Business Application Studio service instance

Note You will see the Create option in the context menu is disabled if you already have a subscription for SAP Business Application Studio enabled in this subaccount.

- 6. Keep “SAP Business Application Studio” and “trial” selected, as shown in Figure 6-39.

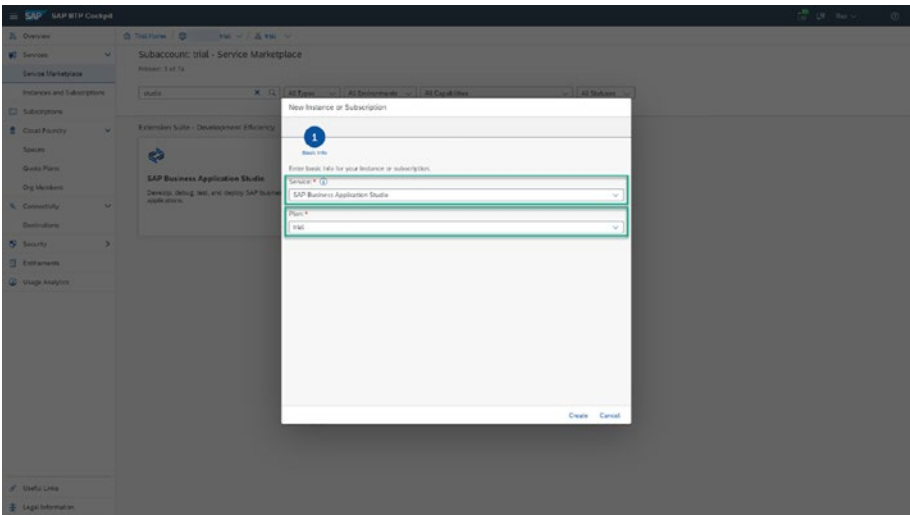


Figure 6-39. *Selecting the service plan*

- 7. Click the Create button to start the SAP Business Application Studio enabling process (Figure 6-40).

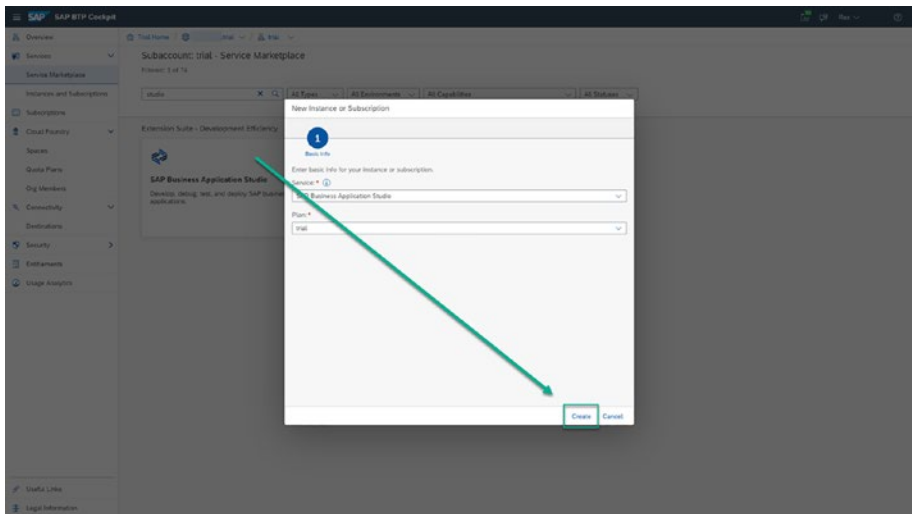


Figure 6-40. *Creating the SAP Business Application Studio service instance*

8. A Creation in Progress window is shown (Figure 6-41).

Note Later, you can check the SAP Business Application Studio subscription in the Instances and Subscriptions area.

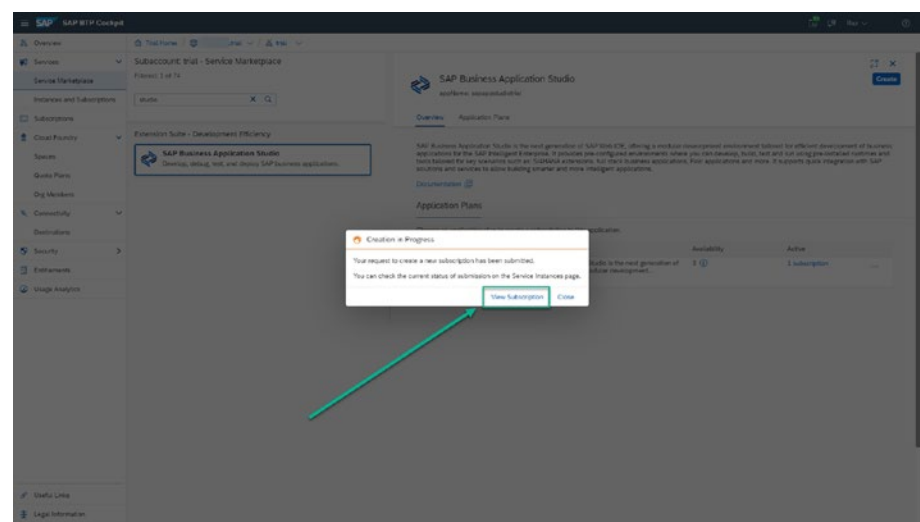


Figure 6-41. Creation in progress

Launching SAP Business Application Studio

1. Once the service is enabled, you can click the Go to Application icon to launch it (Figure 6-42).

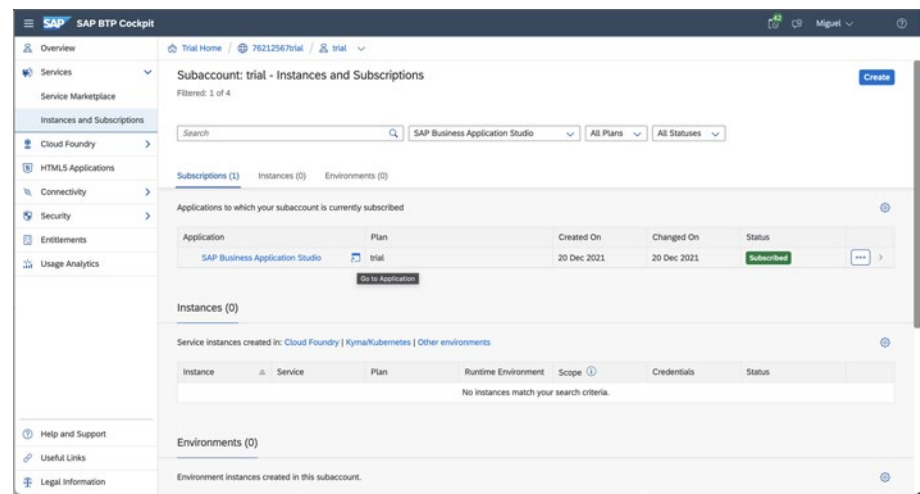


Figure 6-42. Go to application

2. If you see this message, please accept the terms by clicking the OK button and check the box to not see it again (Figure 6-43).

Explore the SAP Business Application Studio

This site is governed by the [SAP Business Application Studio Trial Privacy Statement](#).

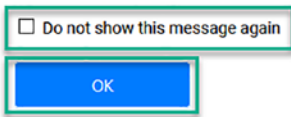


Figure 6-43. SAP Business Application Studio trial privacy statement

3. If you get an Access Denied error (Figure 6-44)...

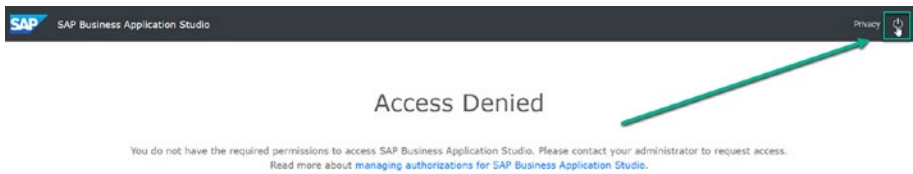


Figure 6-44. Access denied

4. ... Just log out and log back in to SAP Business Application Studio (Figure 6-45).

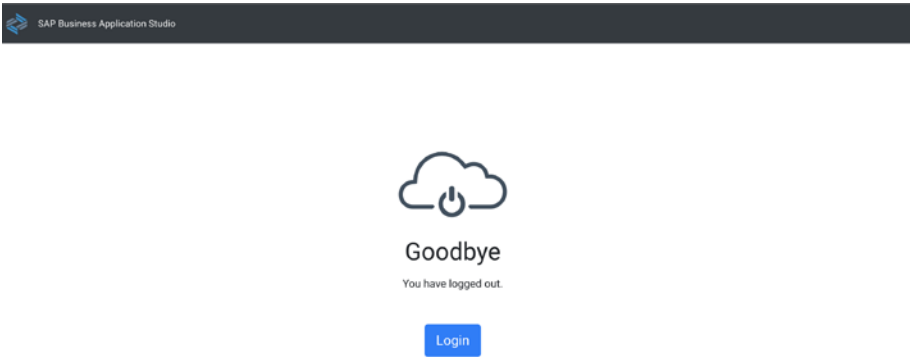


Figure 6-45. *SAP Business Application Studio login page*

5. If you see this message, please accept the terms by clicking the OK button and check the box to not see it again (Figure 6-46).

Explore the SAP Business Application Studio

This site is governed by the [SAP Business Application Studio Trial Privacy Statement](#).

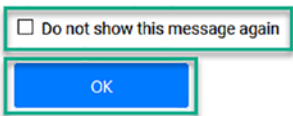


Figure 6-46. *SAP Business Application Studio trial privacy statement*

6. When you start using the SAP Business Application Studio, a welcome page is shown (Figure 6-47).

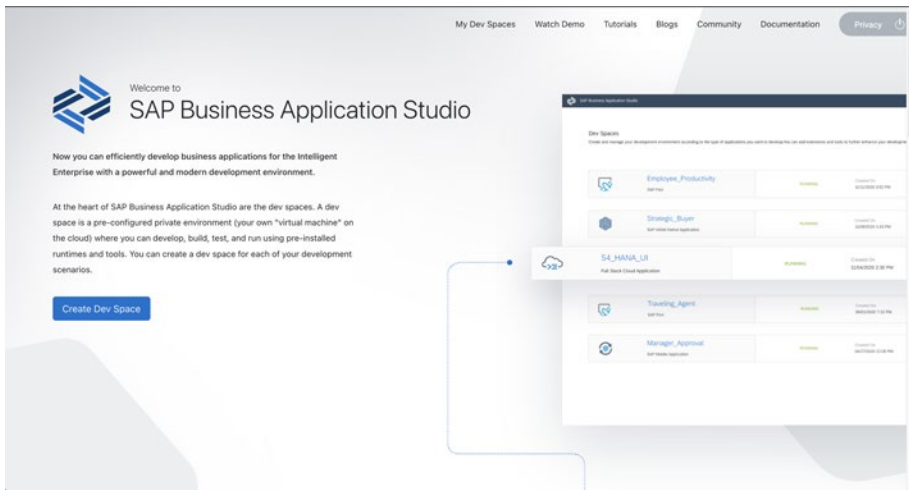


Figure 6-47. SAP Business Application Studio welcome page

7. You can manage your development spaces by clicking on the Dev Spaces link in the upper-right corner of the page. If you do not have a development space created, just click on the Create Dev Space button (Figure 6-48).

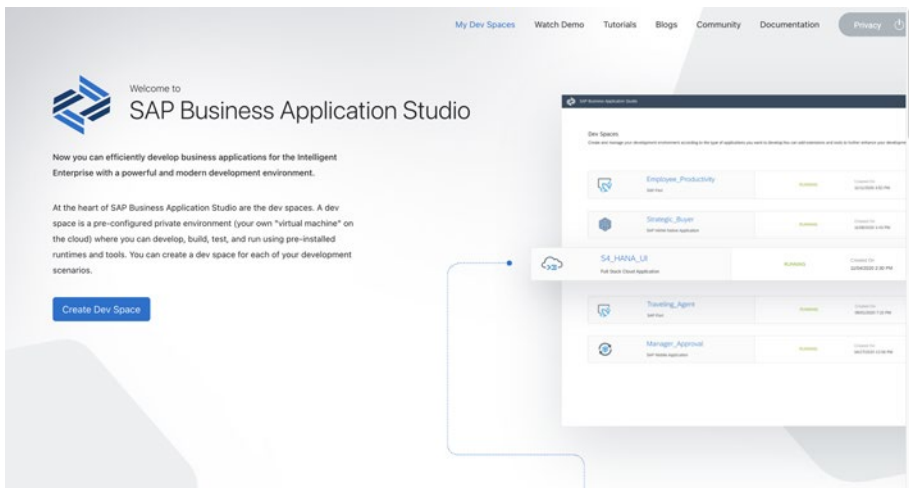


Figure 6-48. Create Dev Space button

Starting with a New Dev Space

1. Click on the Create Dev Space button to open the setup window as shown in Figure 6-49.

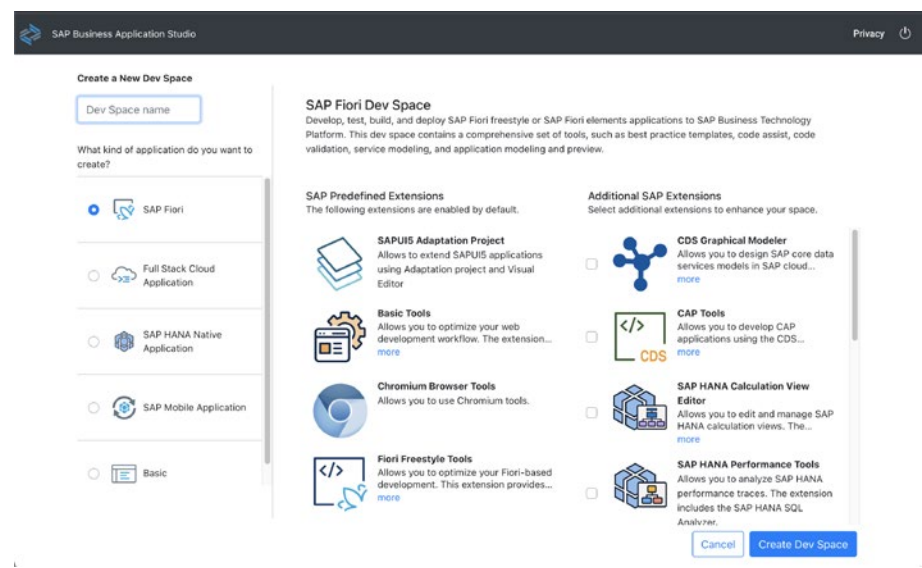


Figure 6-49. *New Dev Space setup*

2. Fill in the Dev Space Name field with a name and select the following SAP extensions (SAP HANA Calculation View, SAP HANA Database Explorer, SAP HANA Tools).
3. Click the Create Dev Space button (Figure 6-50).

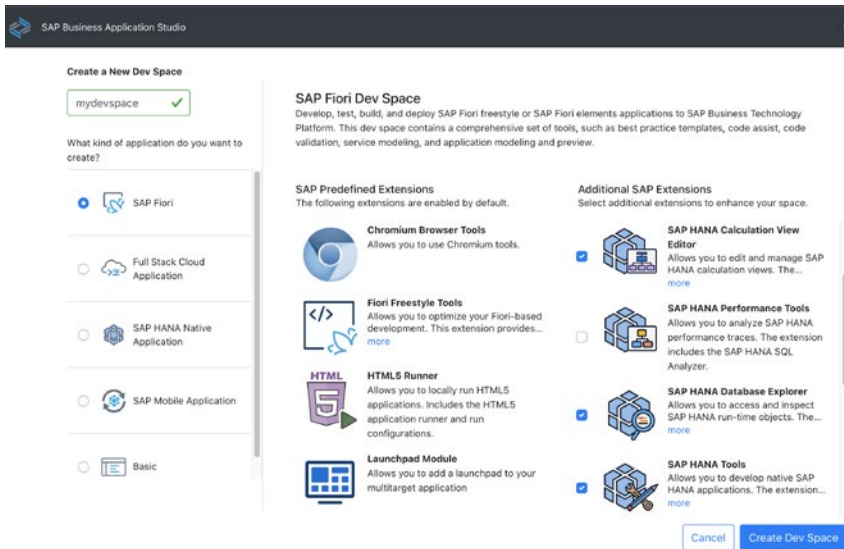


Figure 6-50. Create Dev Space button

- Once the dev space is created, you will see it starting (Figure 6-51).

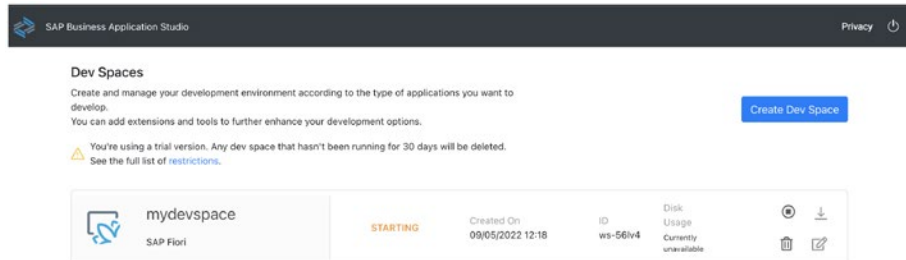


Figure 6-51. Dev space starting

5. Once it has the **RUNNING** status, click over the dev space name to open it (Figure 6-52).

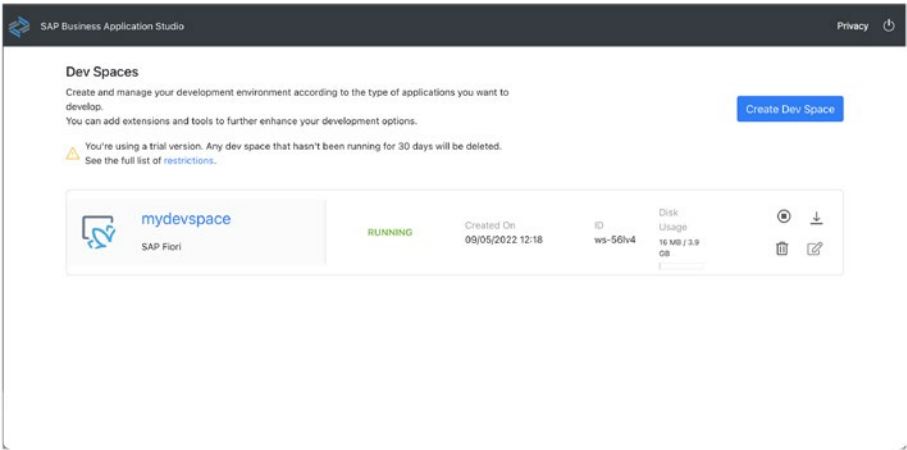


Figure 6-52. Dev space running

6. A progress window should be presented (Figure 6-53).

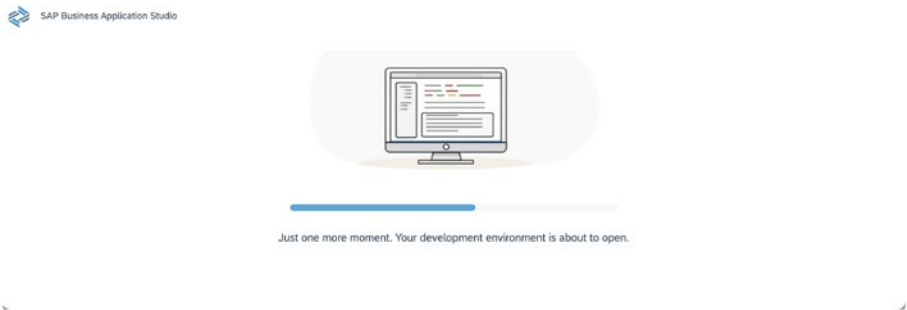


Figure 6-53. Opening the dev space created

7. That's it. You have concluded all steps to set up your dev space (Figure 6-54).

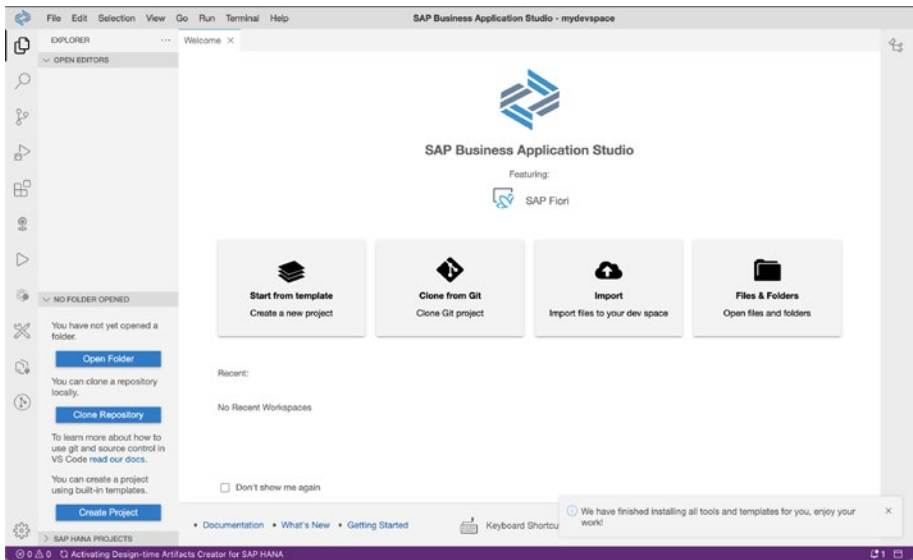


Figure 6-54. *Dev space opened*

Now you can start creating SAP HANA artifacts using the SAP Business Application Studio and explore all available capabilities.

Summary

You have successfully completed this chapter, and now it's time to review what you learned and see what's next. The following were covered:

- How to create your own SAP HANA Cloud instance
- How to set up an SAP BTP trial account
- How to enable the SAP Business Application Studio

Now we will continue our journey with essential learning to start creating SAP HANA Cloud artifacts.

CHAPTER 7

How to Create Artifacts in SAP HANA Cloud

In this chapter, you will learn how to create artifacts in SAP HANA Cloud, like tables, indexes, calculation views, and anonymized views, using the toolset you learned in the previous chapters, which will provide you practical hands-on experience with the SAP Business Technology Platform (SAP BTP).

Creating a New Project

Now that you will be creating data artifacts, it is necessary to first create a project.

1. Go to SAP BTP Cockpit trial:

<https://account.hanatrial.ondemand.com/trial/#/home/trial>

2. Click on the SAP Business Application Studio link (Figure 7-1).

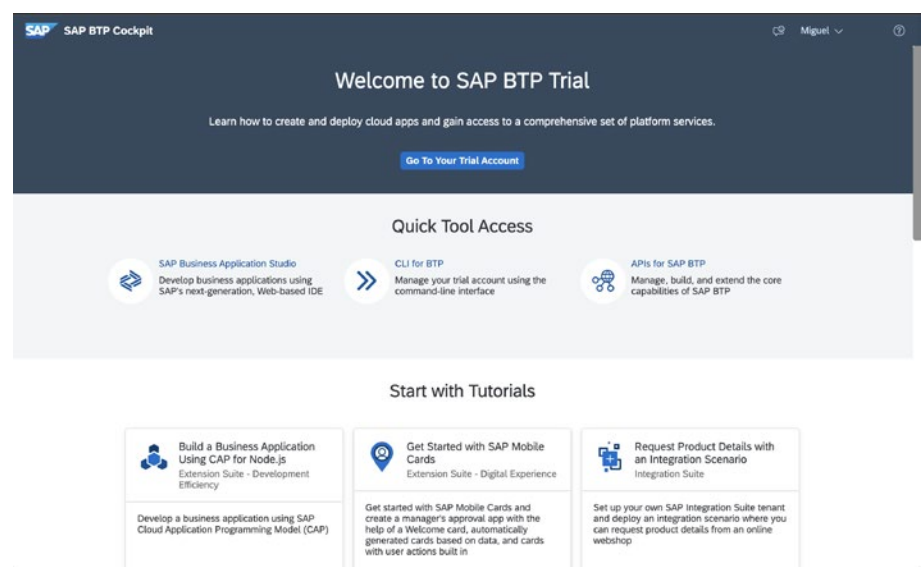


Figure 7-1. SAP BTP Cockpit page

3. Click on your Dev Space to open it (Figure 7-2).

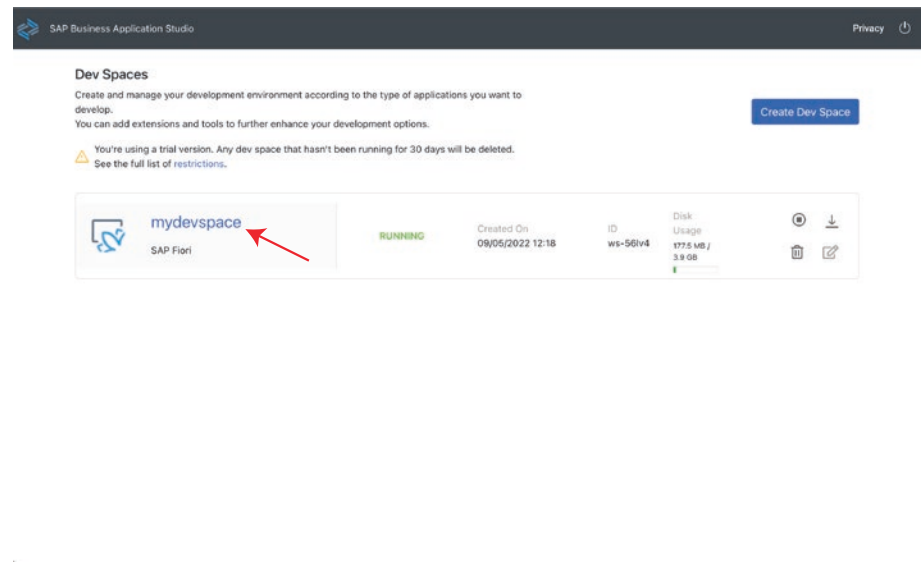


Figure 7-2. Dev Spaces page

4. Click on the “Start from template” tile to begin (Figure 7-3).

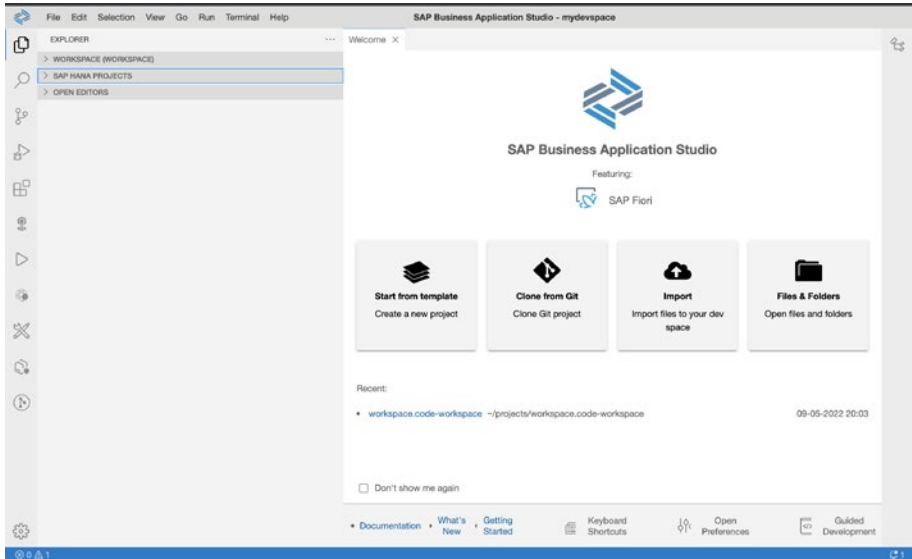


Figure 7-3. *SAP Business Application Studio workspace*

5. Select the highlighted “SAP HANA Database Project” tile and click on the Start button (Figure 7-4).

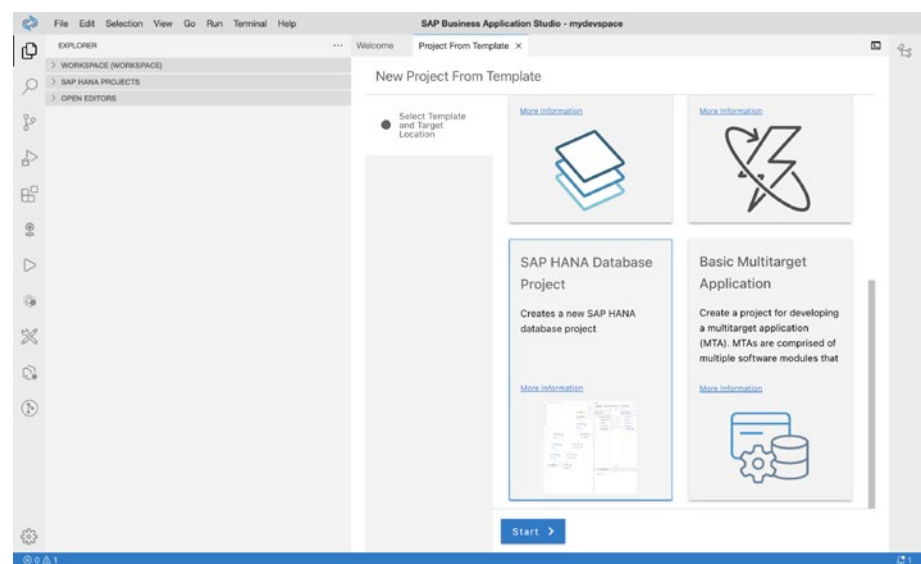


Figure 7-4. Dev Spaces page

6. Provide a name for the project (e.g., DM) as shown in Figure 7-5.

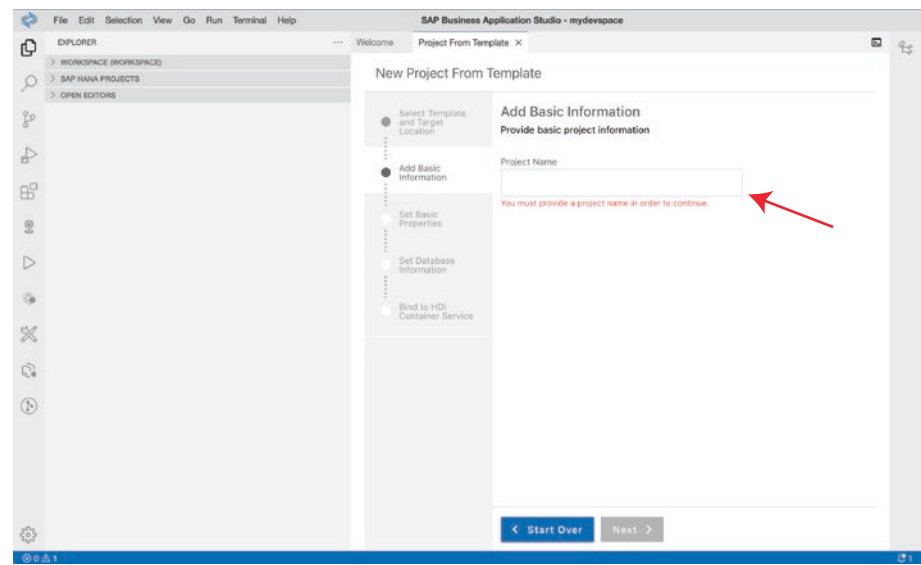


Figure 7-5. Providing basic project information

7. Click the Next button (Figure 7-6).

The screenshot shows the 'New Project From Template' wizard in SAP HANA Cloud. The interface has a top bar with 'Welcome' and 'Project From Template x'. The main area is titled 'New Project From Template'. On the left, a vertical list of steps is shown: 'Select Template and Target Location', 'Add Basic Information' (which is the current step and has a black dot), 'Set Basic Properties', 'Set Database Information', and 'Bind to HDI Container service'. The right side of the wizard is titled 'Add Basic Information' with the subtitle 'Provide basic project information'. It contains a 'Project Name' input field with the text 'DM' entered. At the bottom right, there are two buttons: '< Start Over' and 'Next >'. The 'Next >' button is highlighted with a red box, and a mouse cursor is pointing at it.

Figure 7-6. *Providing the project name*

8. Don't change the module name, and just click the Next button (Figure 7-7).

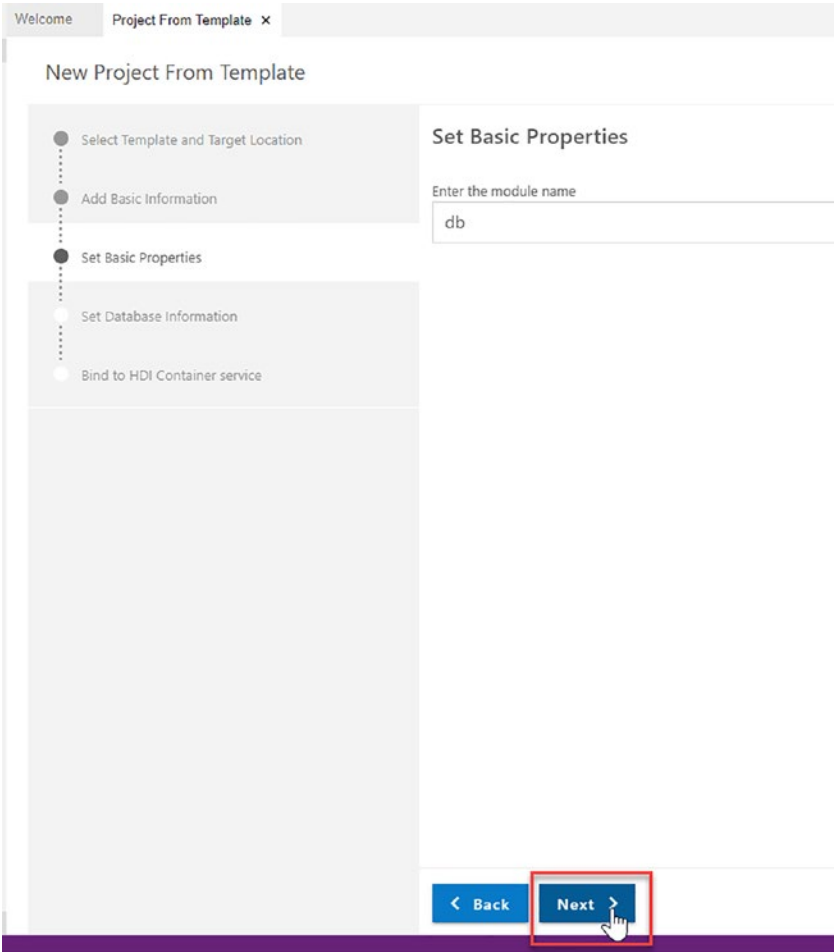


Figure 7-7. *Setting the module name*

9. Select the database version as HANA Cloud and keep the remaining fields blank. Click the Next button (Figure 7-8).

The screenshot shows the 'New Project From Template' wizard in SAP HANA Cloud. The wizard has five steps: 1. Select Template and Target Location, 2. Add Basic Information, 3. Set Basic Properties, 4. Set Database Information, and 5. Bind to HDI Container service. The 'Set Database Information' step is currently active. It contains the following fields and options:

- Namespace:** A text input field with a help icon.
- Schema Name:** A text input field.
- SAP HANA Database Version *:** A dropdown menu with 'HANA Cloud' selected. This field is highlighted with a red rectangle.
- Bind the database module to a Cloud Foundry service instance?:** Radio buttons for 'Yes' (selected) and 'No'.

At the bottom of the wizard, there are two buttons: '< Back' and 'Next >'. The 'Next >' button is highlighted with a red rectangle.

Figure 7-8. *Selecting SAP HANA database version*

10. Type the email and password used to create this trial account and click the User Login button located next to the password field (Figure 7-9).

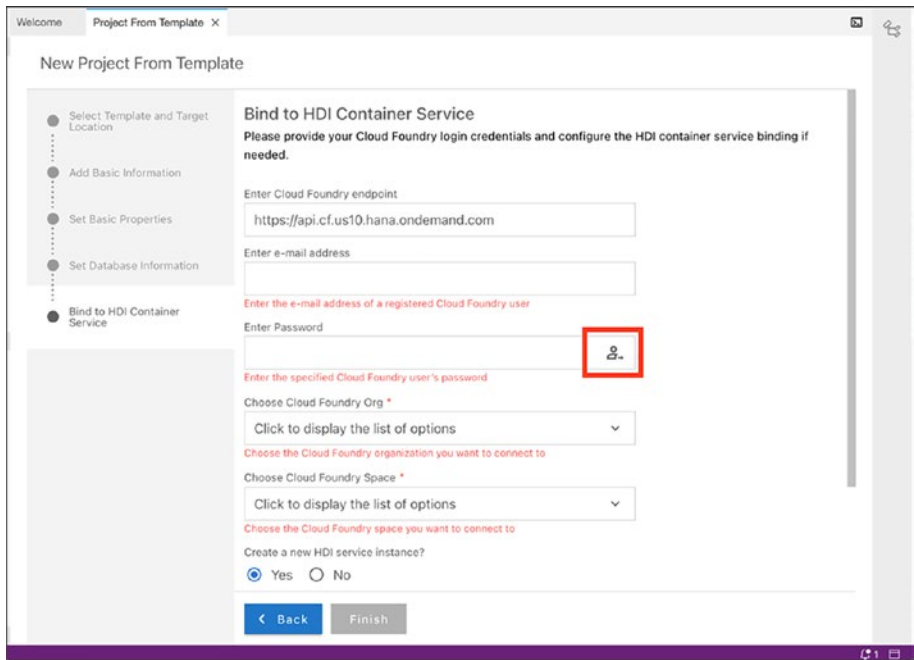


Figure 7-9. *Entering email and password*

- 11. Notice that all that other mandatory fields were automatically filled in (Figure 7-10).

The screenshot shows the 'New Project From Template' wizard in SAP HANA Cloud. The left sidebar contains a progress indicator with five steps: 'Select Template and Target Location', 'Add Basic Information', 'Set Basic Properties', 'Set Database Information', and 'Bind to HDI Container Service' (the current step). The main area is titled 'Bind to HDI Container Service' and includes the instruction: 'Please provide your Cloud Foundry login credentials and configure the HDI container service binding if needed.' The form contains the following fields and options:

- 'Enter Cloud Foundry endpoint': A text box containing 'https://api.cf.us10.hana.ondemand.com'.
- 'Enter e-mail address': A text box containing 'miguel.figueiredo@sap.com'.
- 'Enter Password': A password field with a masked input and a toggle icon.
- 'Choose Cloud Foundry Org *': A dropdown menu showing '76212567trial'.
- 'Choose Cloud Foundry Space *': A dropdown menu showing 'dev'.
- 'Create a new HDI service instance?': Radio buttons for 'Yes' (selected) and 'No'.
- 'Please enter a unique and non-existing service instance name': A text box containing 'DM-hdidb-ws-56lv4'.

At the bottom of the form are two buttons: '< Back' and 'Finish'.

Figure 7-10. *Binding to HDI Container Service*

12. Don't change other values, and click the Finish button (Figure 7-11).

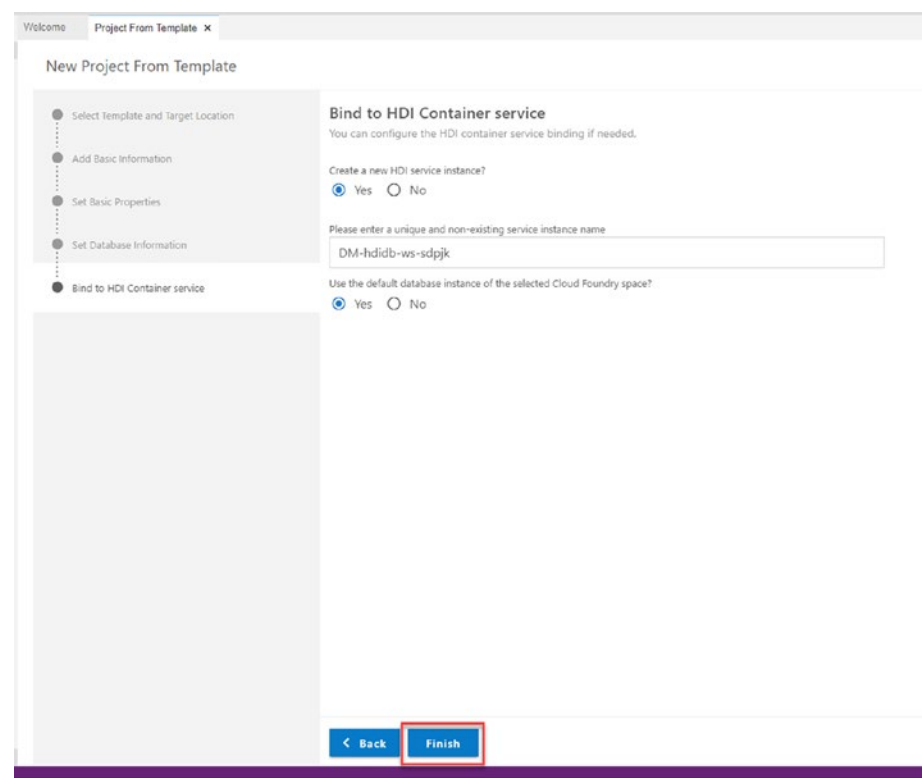


Figure 7-11. *Finishing the project creation*

- 13. The process to generate your project begins (Figure 7-12).

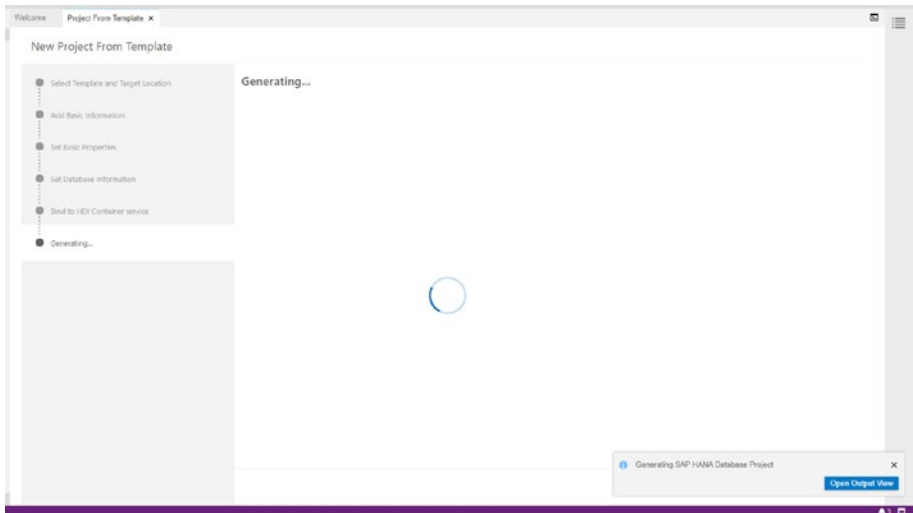


Figure 7-12. *Project generating*

14. Once the project is created, click on the Open in New Workspace button to open the new project in the workspace (Figure 7-13).

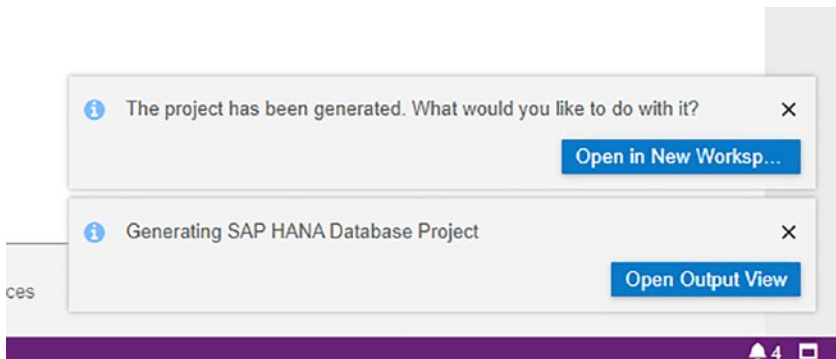


Figure 7-13. *Opening the project*

Creating Database Artifacts

The business context used in the following steps is based on a human resources data model with the purpose of tracking employee performance data, which is very common for most organizations.

The first activity is to create a table in which to store employee experience scoring and tracking ratings.

- 1. Right-click on the src folder, then choose New File (Figure 7-14).

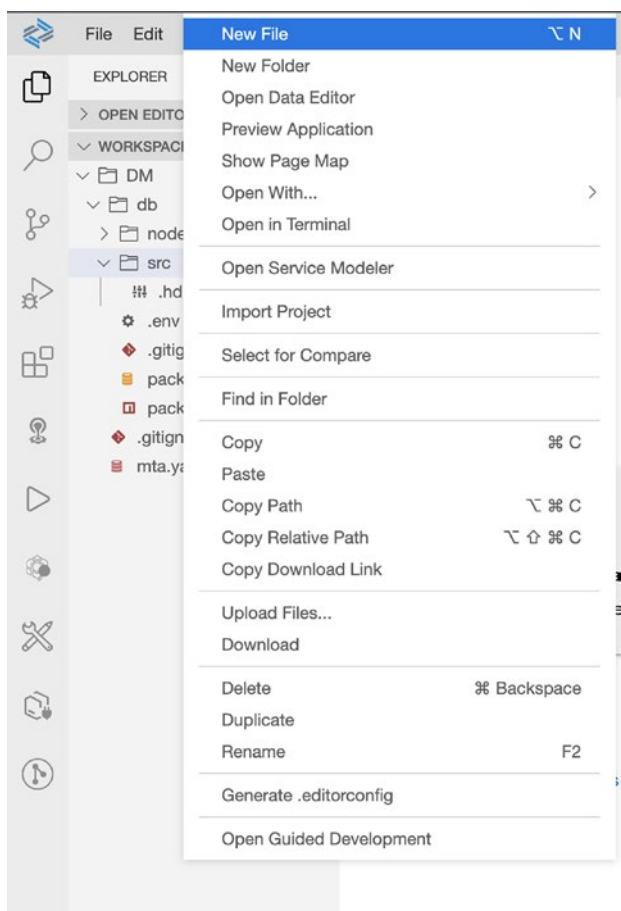


Figure 7-14. *Creating a new file*

2. Use the name `data/employee_experience.hdbtable` as in Figure 7-15.

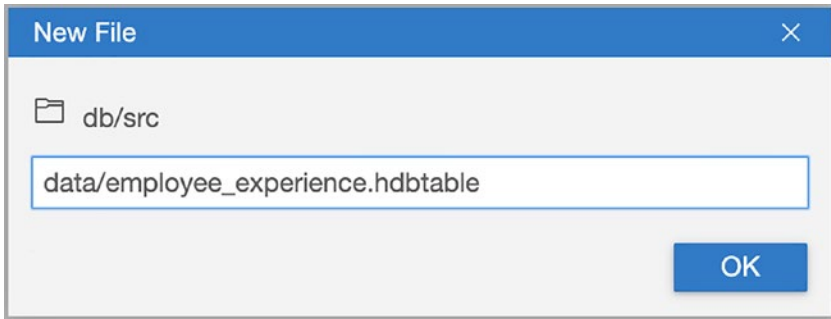


Figure 7-15. Naming a new file

3. This will create both a folder and a file. Enter the following content into the `.hdbtable` file to define a new table:

```
column table "EMPLOYEE_EXPERIENCE" (
  "EMPLOYEE_ID" INTEGER unique not null COMMENT
  'Employee Id',
  "EMPLOYEE_SCORE" DOUBLE COMMENT 'Employee Score',
  "MANAGER_ID" INTEGER not null COMMENT 'Manager Id',
  "EMPLOYEE_FEEDBACK" NVARCHAR(512) COMMENT 'Employee
feedback',
  "EMPLOYEE_EXPERIENCE_RATING" DOUBLE COMMENT
  'Employee experience rating',
  PRIMARY KEY ("EMPLOYEE_ID")
)
COMMENT 'Employee experience results'
```

4. Save the file (Figure 7-16).

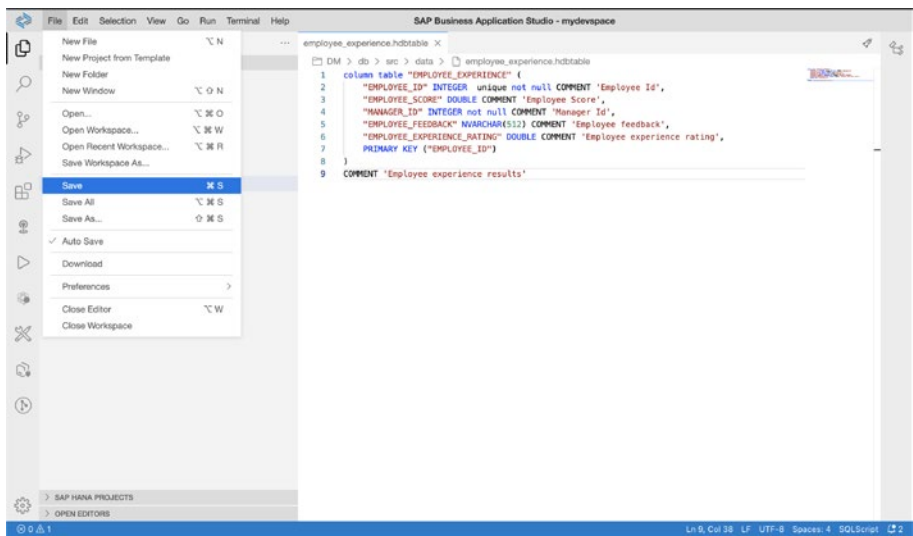


Figure 7-16. Saving the new file

Note Once you have a database module, you can later deploy the design-time artifact and generate a run-time object in the schema linked to the HDI container.

Creating a Text Index

The table you just defined will have a large text field with comments from the employees. You want to enable a fulltext index to enhance text search.

1. Create a new database artifact within the data folder.
Go to the View menu and select Find Command (Figure 7-17).

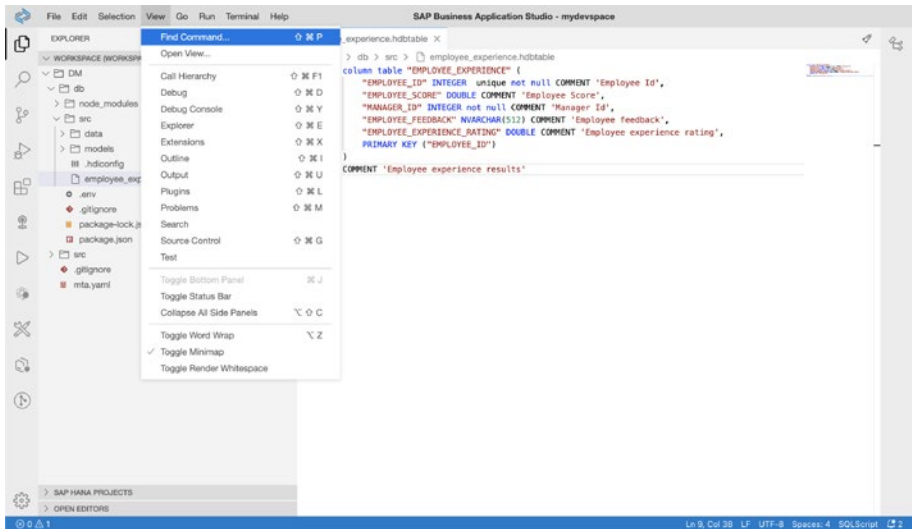


Figure 7-17. Find Command

2. Type the word “HANA” (Figure 7-18).

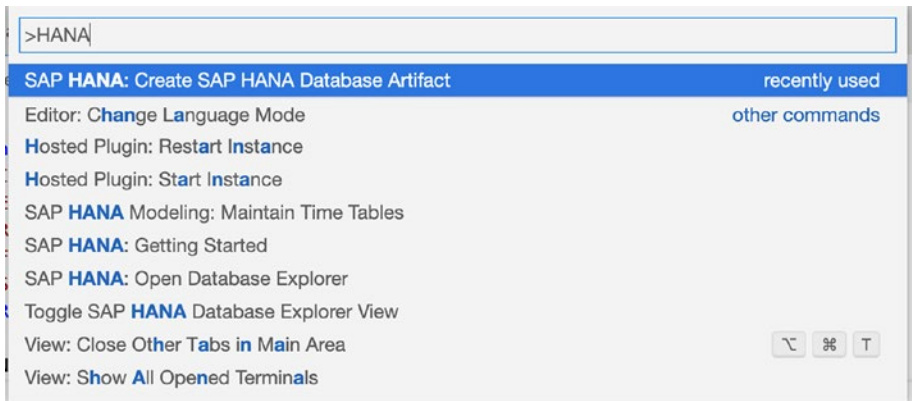


Figure 7-18. Selecting an SAP HANA database artifact type

3. Use the name `employee_feedback_text` in the Artifact Name field and choose an Artifact Type of `.hdbindex`; click the Create button (Figure 7-19).

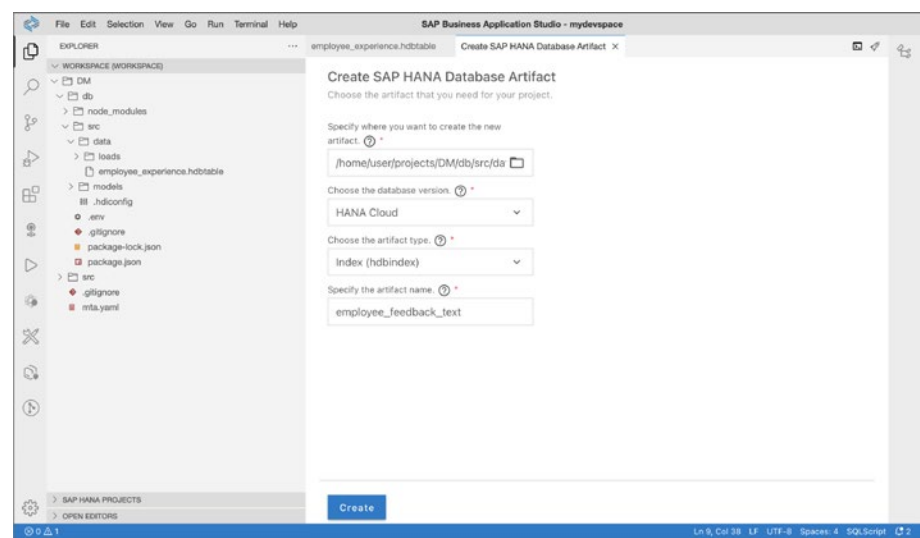


Figure 7-19. Creating an SAP HANA database artifact

- 4. Click on the index name and a tab will open the index definition (Figure 7-20).

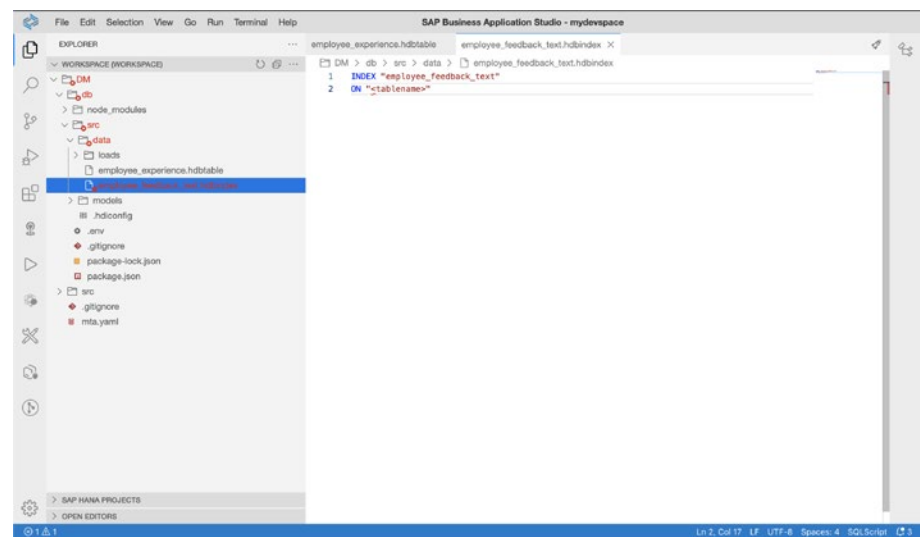


Figure 7-20. Viewing the SAP HANA database artifact created

5. Enter the following code as the index definition:

```
INDEX "EMPLOYEE_FEEDBACK_TEXT"
on "EMPLOYEE_EXPERIENCE" ("EMPLOYEE_FEEDBACK")
```

6. The index definition should look like Figure 7-21.

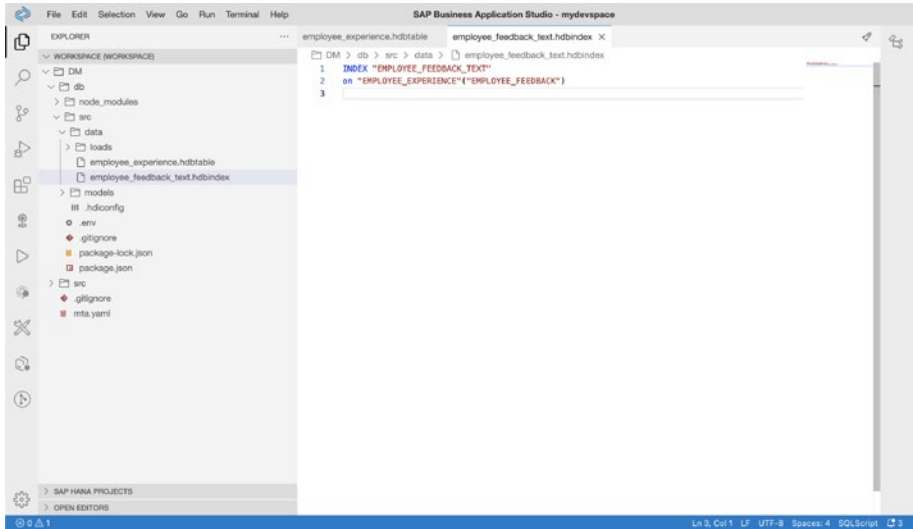


Figure 7-21. SAP HANA index definition

7. Save the file.

Deploy the Database

Now it is time to deploy your database.

1. Expand the SAP HANA Projects view (Figure 7-22).

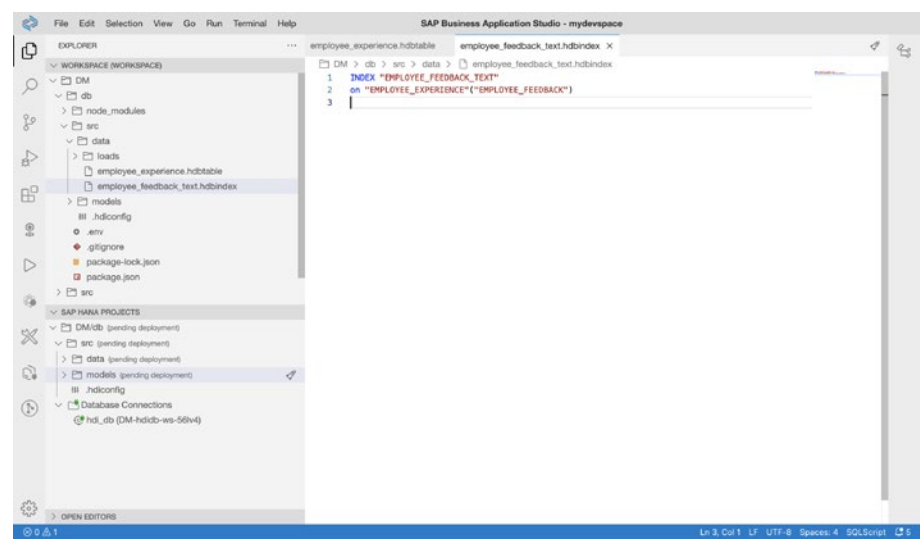


Figure 7-22. SAP HANA Projects view expanded

- 2. Click on the rocket button to deploy (Figure 7-23).

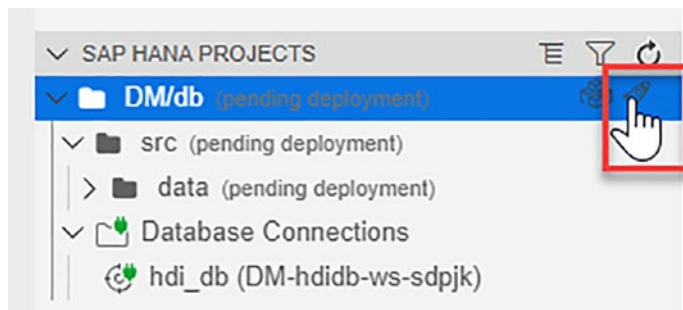


Figure 7-23. SAP HANA Projects deploy tool

The console on the bottom will show the progress. First, an HDI container will be created and bound. Scroll up the log to see it in your screen (Figure 7-24).

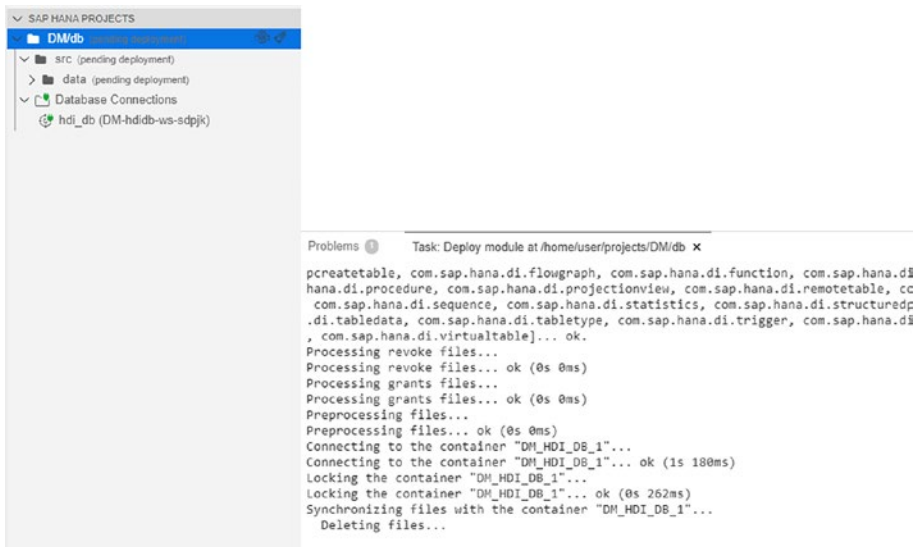


Figure 7-24. SAP HANA Projects deployment running

The hdi_db is the logical name of the database as defined in the deployment configuration file mta.yaml. The log of the process shows how table and text indexes are prepared using design-time artifacts. The DM_HDI_DB_1 is the physical schema created (Figure 7-25).

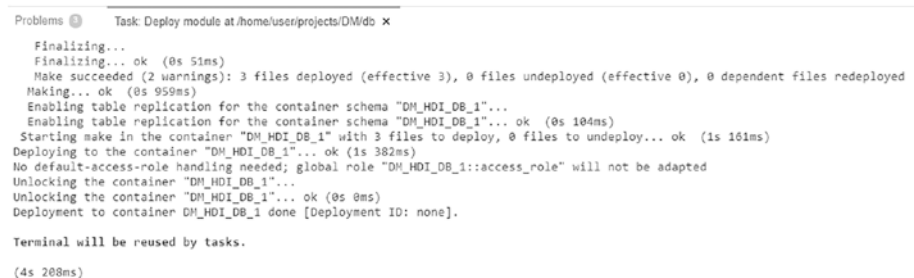


Figure 7-25. SAP HANA Projects deployment concluded

Add Data from a Local File

To make this hands-on exercise easier and more objective, we will use .csv local files to load data into our SAP HANA Cloud.

1. Create the configuration for the upload first. In the data folder, create a file called `loads/employee_experience.hdbtabledata` (Figure 7-26).

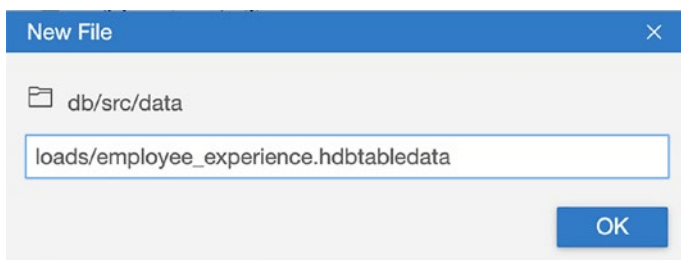


Figure 7-26. *Creating a configuration file*

2. Enter the following content into it:

```
{
  "format_version": 1,
  "imports": [{
    "target_table": "EMPLOYEE_EXPERIENCE",
    "source_data": {
      "data_type": "CSV",
      "file_name": "employee_experience.csv",
      "has_header": true,
      "dialect": "HANA",
      "type_config": {
        "delimiter": ",",
      }
    }
  ]
}
```

3. Download the .csv file https://github.com/Apress/Apress-SAP-HANA-Cloud-in-a-Nutshell-by-Miguel-Figueiredo/blob/main/employee_experience.csv to your computer.
4. Click on the Upload Files option in the File menu and select the recently downloaded employee_experience.csv file (Figure 7-27).

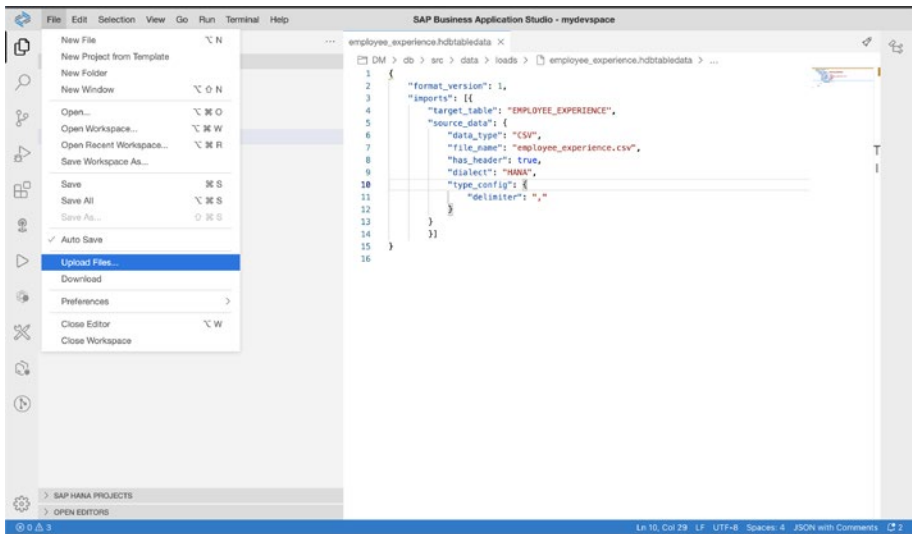


Figure 7-27. Click on Upload Files

5. Verify that the .csv file was uploaded (Figure 7-28).

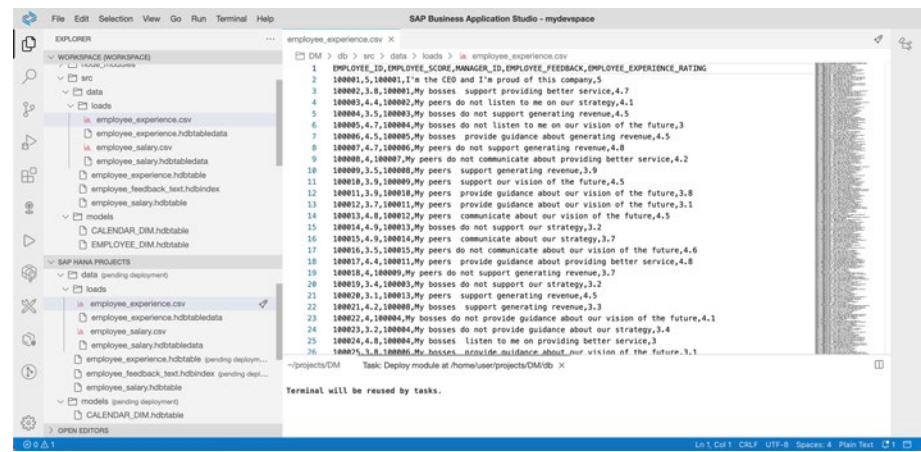


Figure 7-28. Viewing the uploaded file

6. Deploy the loads folder by clicking on the rocket button in the SAP HANA Projects view (Figure 7-29).

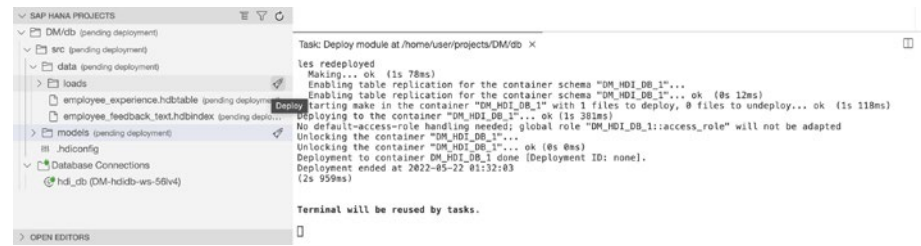


Figure 7-29. Deploying loads folder

Check the Data

Now that you have created the database table and uploaded the data into it, let's check if the content is there.

1. Click on the Open HDI Container button (Figure 7-30).

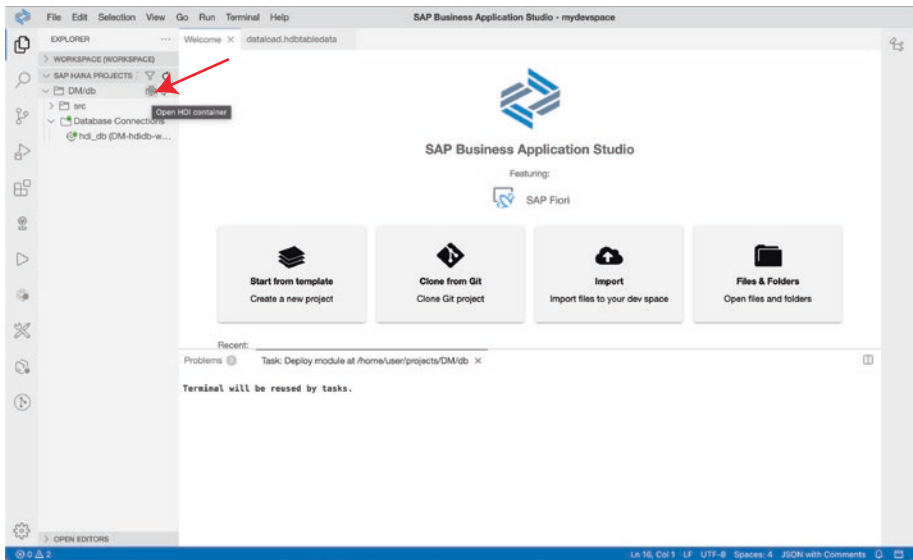


Figure 7-30. Clicking on Open HDI Container button

2. Wait until Database Explorer is fully loaded to go through with the next steps (Figure 7-31).

SAP HANA Database Explorer for SAP HANA Cockpit

Loading

Figure 7-31. SAP HANA Database Explorer loading

- 3. Select Tables in your HDI container (Figure 7-32).

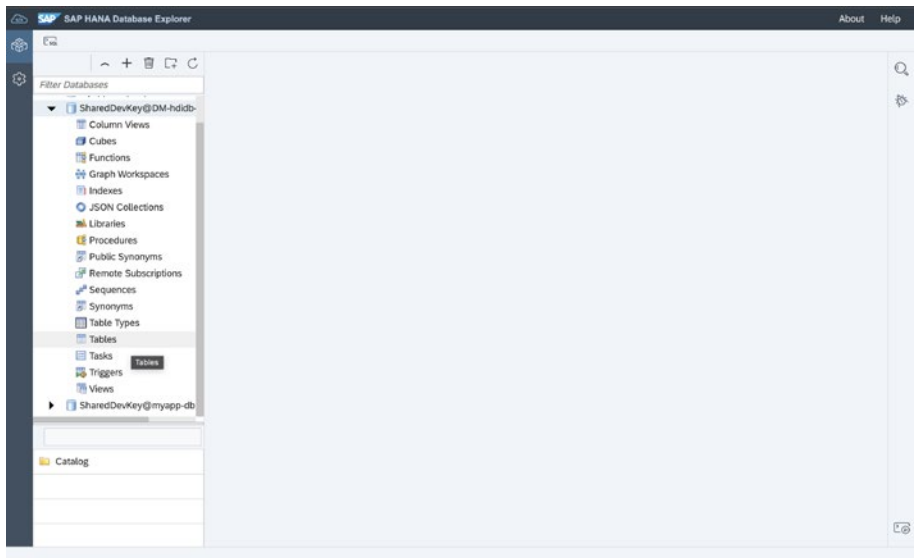


Figure 7-32. SAP HANA Database Explorer

- 4. Right-click on the EMPLOYEE_EXPERIENCE table (shown in the list in Figure 7-33) and select Open Data.

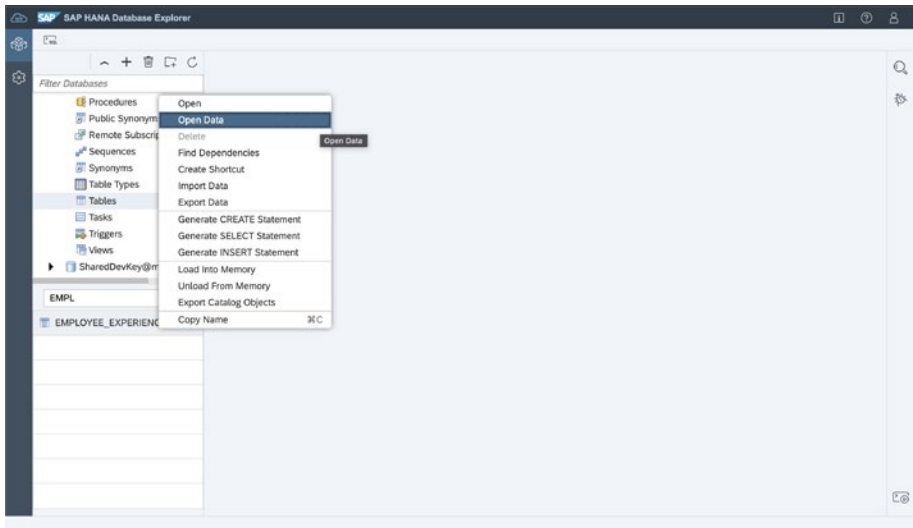


Figure 7-33. Opening data viewing for *EMPLOYEE_EXPERIENCE*

5. Now table data is shown on the canvas (Figure 7-34).

SAP HANA Database Explorer

Filter Databases

- Procedures
- Public Synonyms
- Remote Subscriptions
- Sequences
- Synonyms
- Table Types
- Tables
- Tasks
- Triggers
- Views
- SharedDevKey@myapp-db

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EMPLOYEE_EXPERIENCE

EMPLOYEE_EXPERIENCE

Raw Data

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Figure 7-34. Viewing *EMPLOYEE_EXPERIENCE*'s data

6. Open the SQL editor to execute an SQL query by clicking on the SQL button (Figure 7-35).

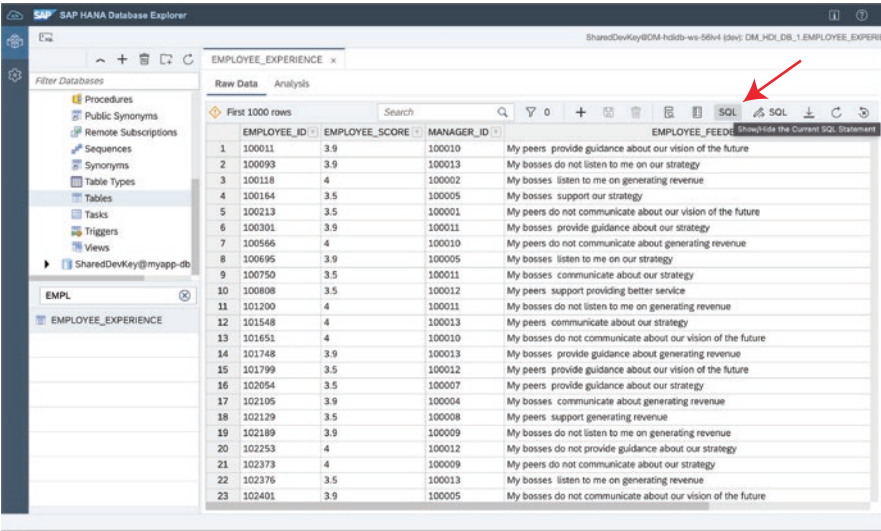


Figure 7-35. Opening the SQL editor

7. Enter and run the following SQL statement to complete the validation:

```
SELECT ROUND(AVG("EMPLOYEE_EXPERIENCE_RATING"),1)
FROM "EMPLOYEE_EXPERIENCE";
```

8. The SQL statement should display like in Figure 7-36.

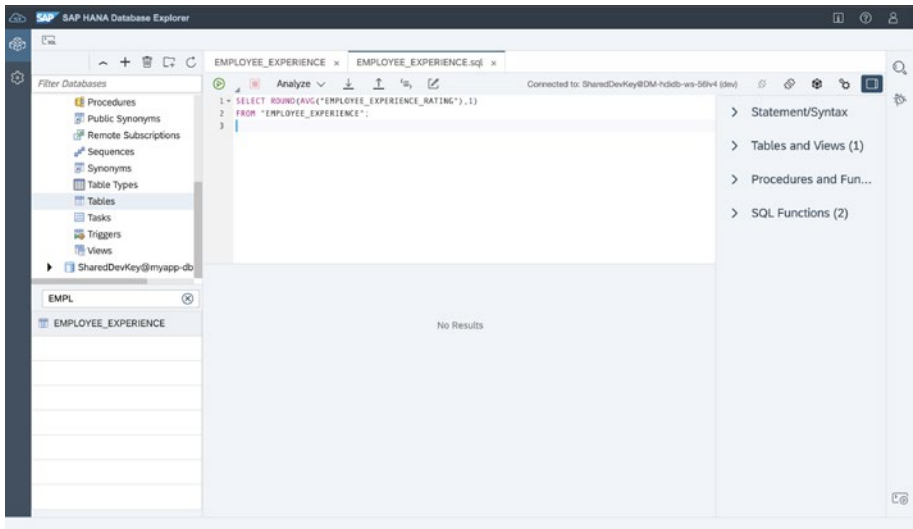


Figure 7-36. SQL statement for validation

9. Check the result in the window below (Figure 7-37).

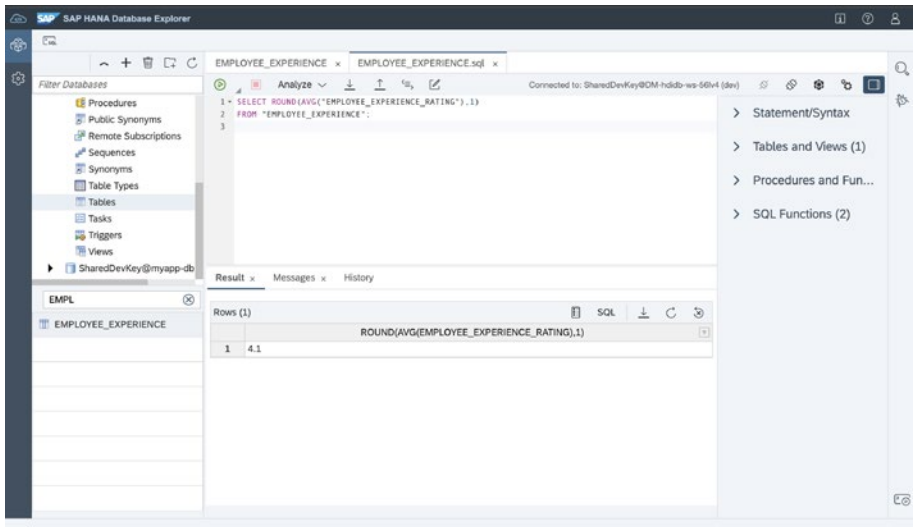


Figure 7-37. Checking results

Creating Calculations Views

Now that you're familiar with SAP Business Application Studio, you will create a calculation view using differential privacy to anonymize confidential data in SAP HANA Cloud.

Note SAP HANA Cloud supports differential privacy implementation. You may find more about differential privacy system and further characteristics visiting https://en.wikipedia.org/wiki/Differential_privacy.

You will create another table and load sample data into it. You can review a similar task you did earlier to successfully complete this activity.

1. Create a new database table named EMPLOYEE_SALARY using the .hdbtable artifact type in the data folder. Use the following table definition:

```
Column table "EMPLOYEE_SALARY" (
  EMPLOYEE_ID INTEGER unique not null COMMENT
    'Employee ID',
  EMPLOYEE_SALARY DOUBLE COMMENT 'Employee Salary',
  EMPLOYEE_STARTING_YEAR INTEGER not null COMMENT
    'Employee Starting Year',
  EMPLOYEE_GENDER NVARCHAR(1) COMMENT 'Employee
  Gender',
  EMPLOYEE_REGION NVARCHAR(4) COMMENT 'Employee
  Region',
  EMPLOYEE_CAREER_LEVEL NVARCHAR(200) COMMENT
    'Employee Career Level',
  PRIMARY KEY ("EMPLOYEE_ID")
)
COMMENT 'Employee Salary'
```

2. Create the configuration for the .csv upload. In the data/loads folder, create a file called employee_salary.hdbtabledata with the following content:

```
{
  "format_version": 1,
  "imports": [{
    "target_table": "EMPLOYEE_SALARY",
    "source_data": {
      "data_type": "CSV",
      "file_name": "employee_salary.csv",
      "has_header": true,
      "dialect": "HANA",
      "type_config": {
        "delimiter": ",",
      }
    }
  ]
}
```

3. Download this .csv file https://github.com/Apress/Apress-SAP-HANA-Cloud-in-a-Nutshell-by-Miguel-Figueiredo/blob/main/employee_salary.csv to your computer.
4. Upload the .csv file to the data/loads folder using the File ► Upload Files option.
5. Save and deploy these new artifacts (Figure 7-38).

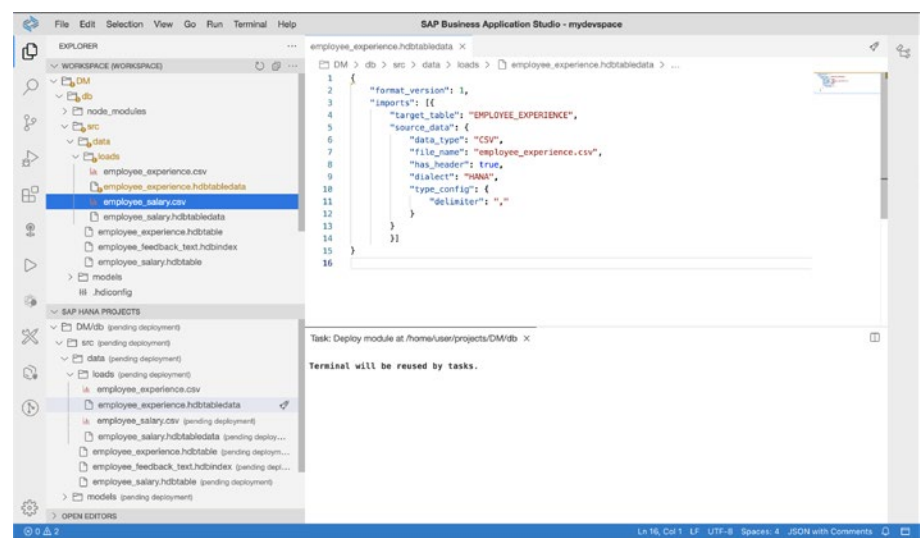


Figure 7-38. Deploying new artifacts

Now, you will create a calculation view.

1. Create a new folder called models under db/src.
Create a new calculation view via the SAP HANA:
Create SAP HANA Database Artifact command
menu entry (Figure 7-39).

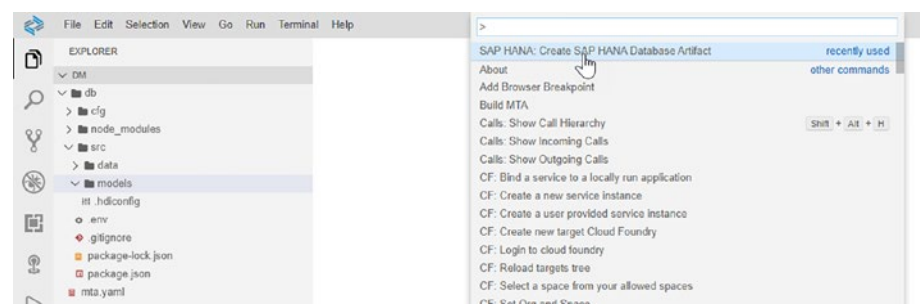


Figure 7-39. Creating a new calculation view

2. Name it `EMPLOYEE_SALARY_ANONYMIZED` and make it a CUBE. Click the Create button (Figure 7-40).

Create SAP HANA Database Artifact ×

Create SAP HANA Database Artifact

Choose the artifact that you need for your project.

Specify where you want to create the new artifact. ? *

/home/user/projects/DM/db/src/mc

Choose the database version. ? *

HANA Cloud

Choose the artifact type. ? *

Calculation View (hdbcalculationvi)

Specify the artifact name. ? *

EMPLOYEE_SALARY_ANONYMIZED

Specify the label. ?

EMPLOYEE_SALARY_ANONYMIZED

Choose a data category. ? *

CUBE

Use a star join? ?

☐ Yes ☒ No

Create

Figure 7-40. *Providing the calculation view definition*

3. Select the new calculation view by clicking on its name in the SAP HANA Projects section. The calculation view will be opened in the graphical calculation view editor at the right side (Figure 7-41).

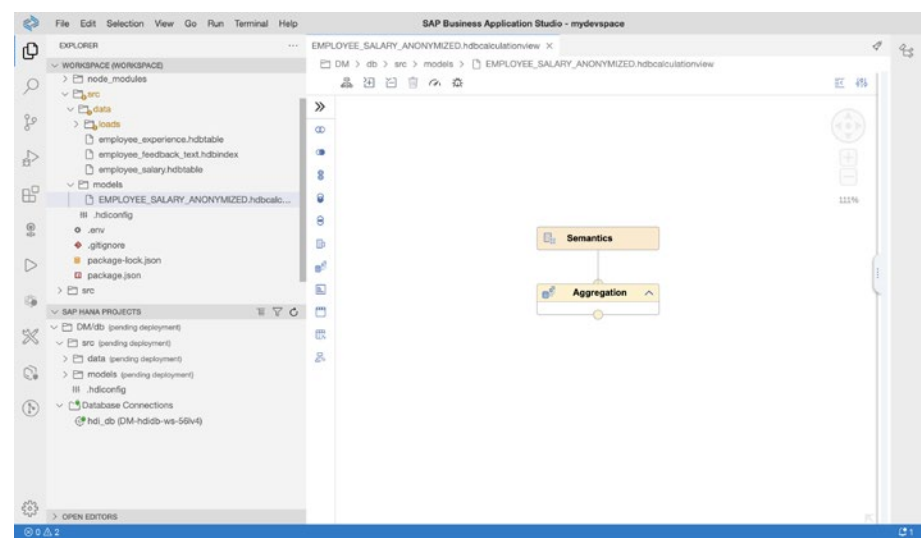


Figure 7-41. *Viewing the calculation view*

4. Click on the two-arrows icon to expand the left-side menu to show the names of the nodes.
5. Click on the Projection node and then click on the white canvas to drop it (Figure 7-42).

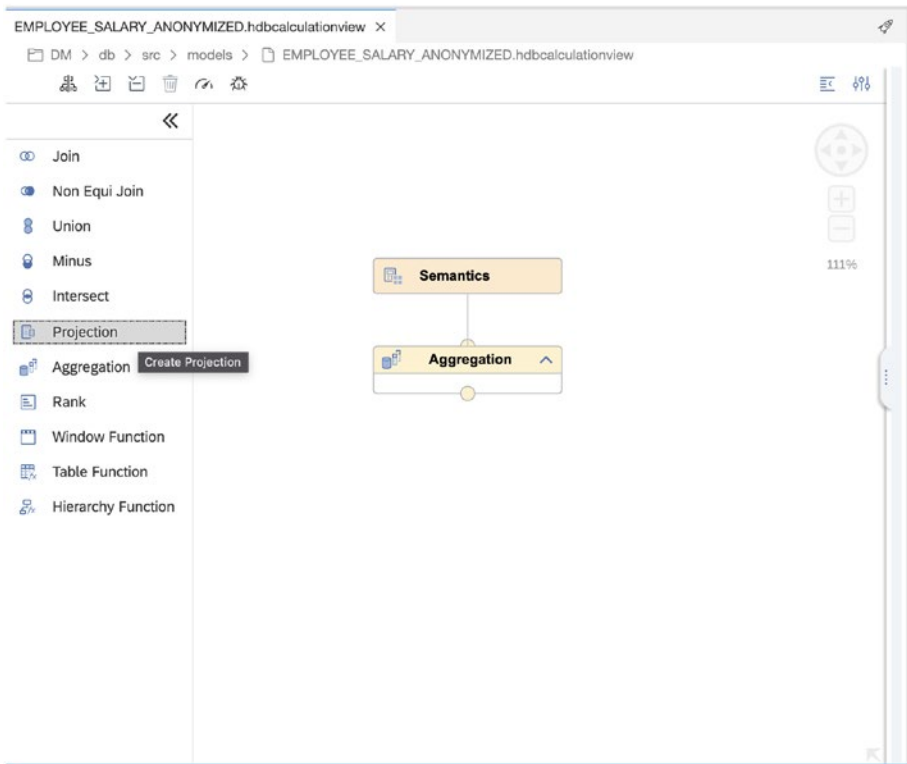


Figure 7-42. *Dropping a projection node*

6. With the Projection node selected, click on the plus button to add a table as a data source (Figure 7-43).

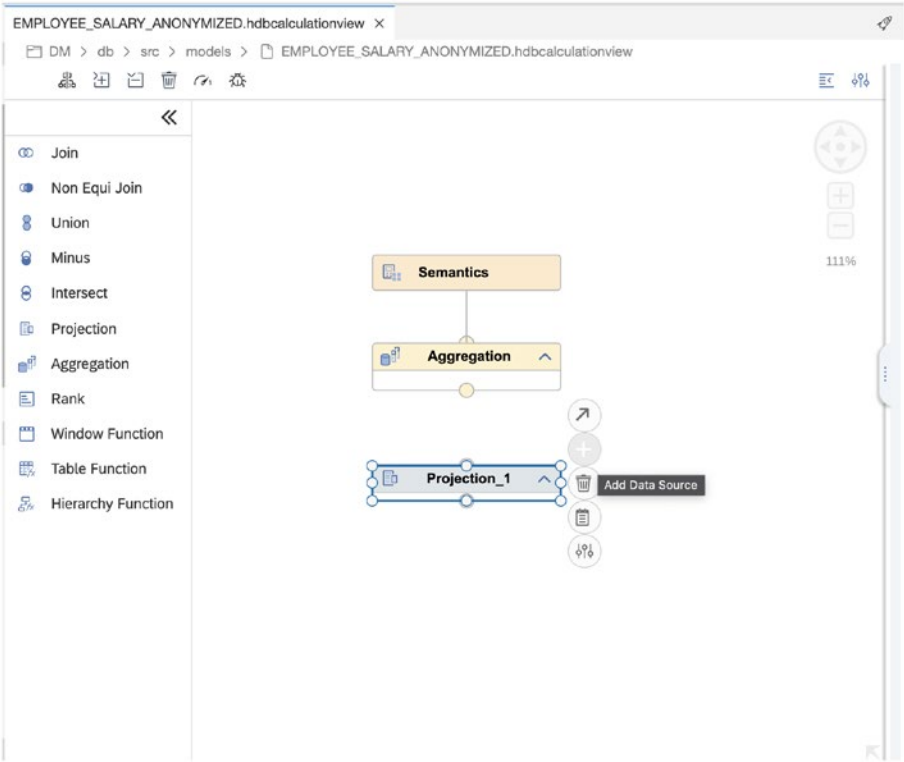


Figure 7-43. *Selecting the plus button*

- 7. Enter EMPLOYEE_SALARY as the search criteria (Figure 7-44).

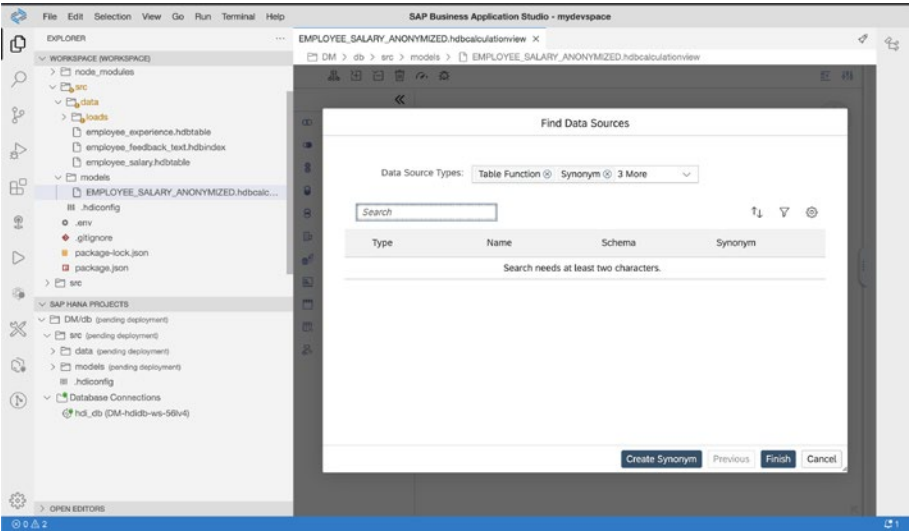


Figure 7-44. Finding data sources

8. Click on the radio button located at the left side of the table name to select it and click the **Finish** button (Figure 7-45).

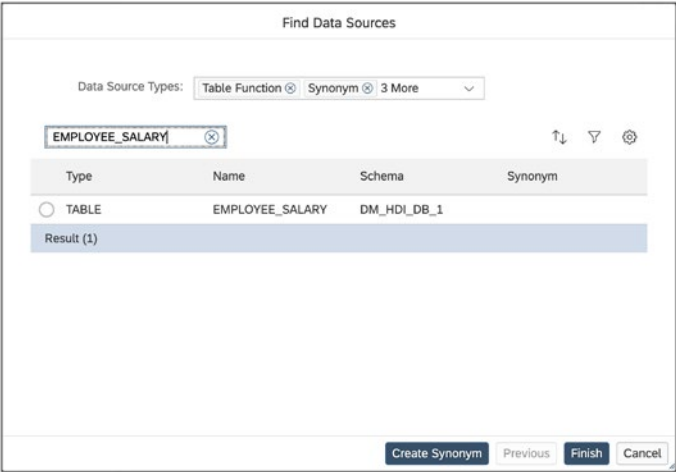


Figure 7-45. Finishing the data sources selection

9. You should see the calculation view like in Figure 7-46.

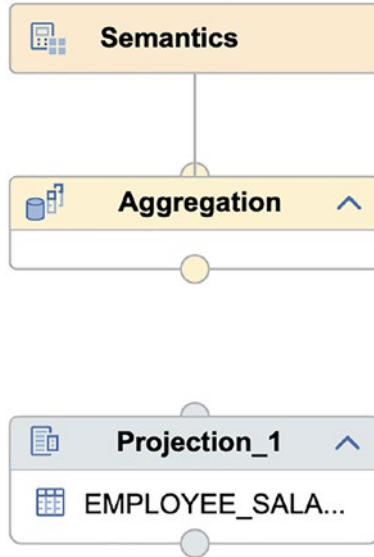


Figure 7-46. Calculation view with data source mapped

10. Double-click on the Projection_1 node. This will open the mapping view.
11. Double-click on EMPLOYEE_SALARY to add all the columns to the output. The mapping should look like the image in Figure 7-47.

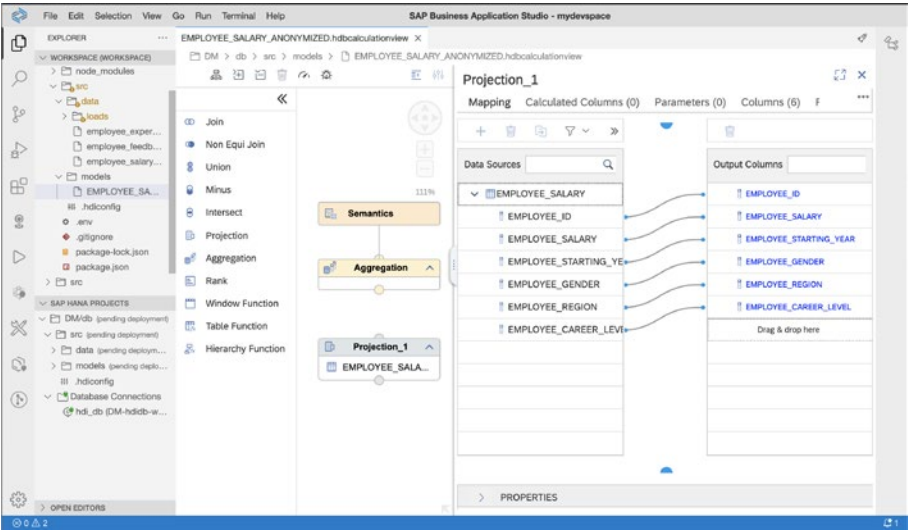


Figure 7-47. Mapping columns on the projection

- 12. Connect the Projection_1 node to the Aggregation node (Figure 7-48).

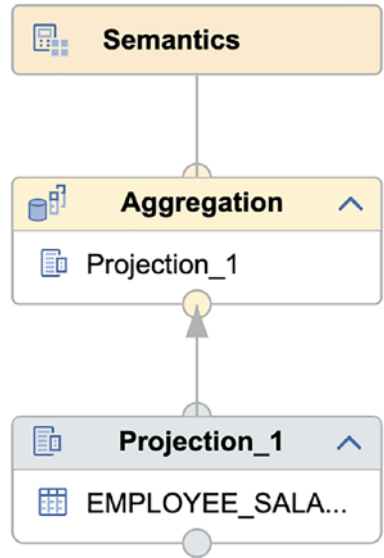


Figure 7-48. Calculation view with projection defined

13. Double-click on the Aggregation node to open the mapping view and again double-click on the projection name to add all the fields as output columns (Figure 7-49).

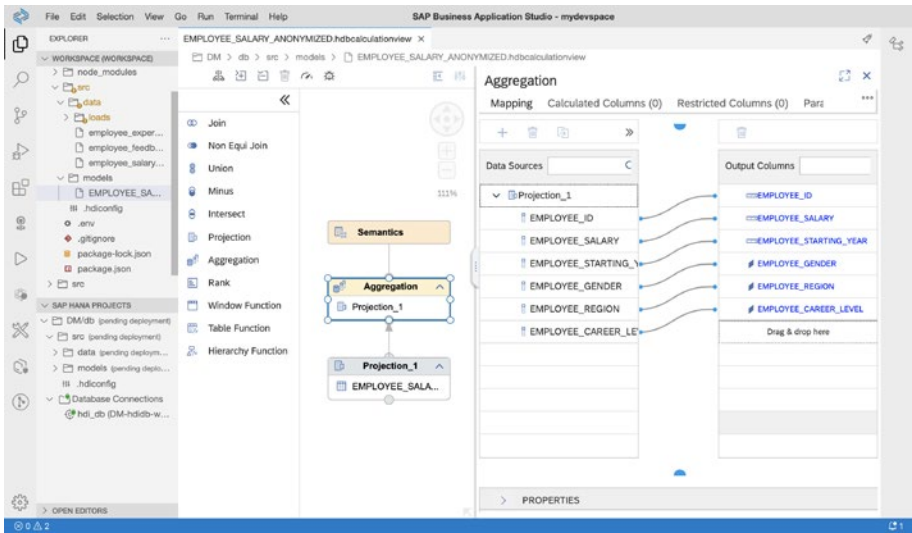


Figure 7-49. Mapping columns on the Aggregation node

14. Double-click the Semantics node and click on the checkboxes for EMPLOYEE_ID and EMPLOYEE_STARTING_YEAR. Then click the Mark as Attribute button to change these columns' type to Attribute, as shown in the image in Figure 7-50.

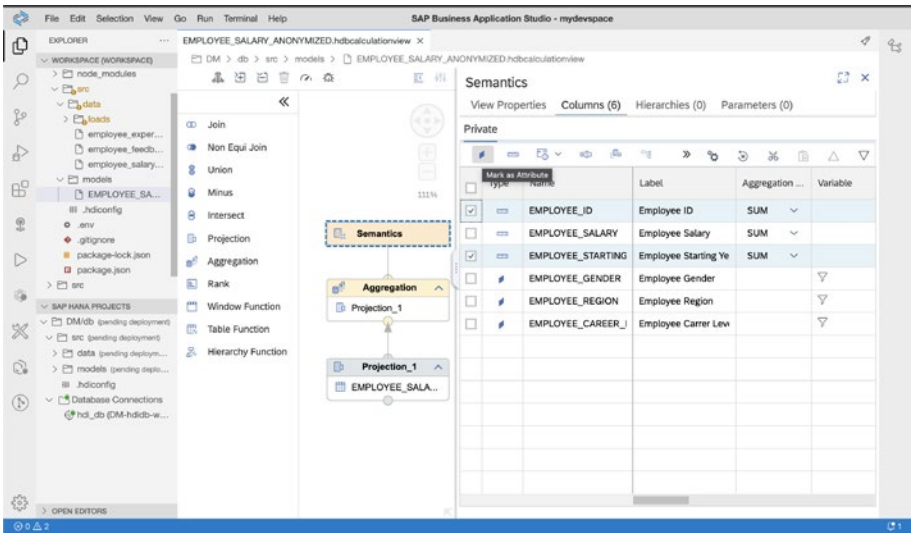


Figure 7-50. Setting specific columns as attributes

15. Save the file (Figure 7-51).

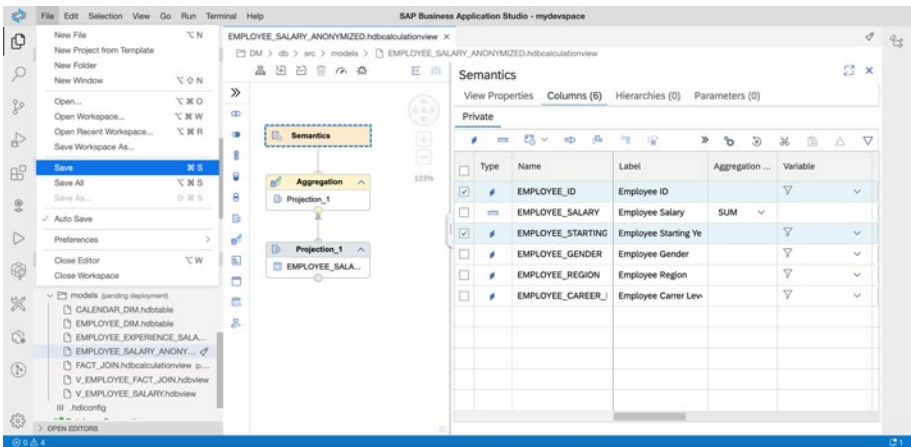


Figure 7-51. Saving

16. Click the Deploy button at the top right corner (Figure 7-52).

CHAPTER 7 HOW TO CREATE ARTIFACTS IN SAP HANA CLOUD

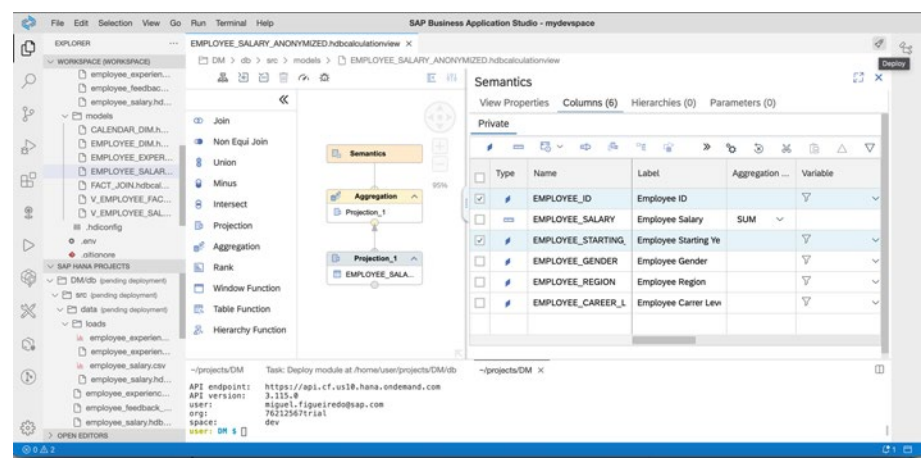


Figure 7-52. Deploying the calculation view

- 17. Test your deployment by right-clicking on the calculation view and selecting the Data Preview option from the context menu (Figure 7-53).

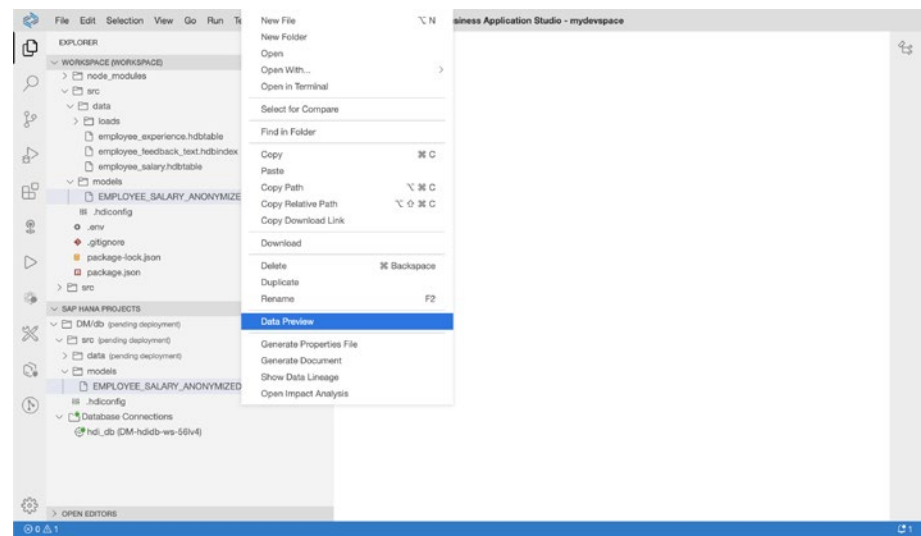


Figure 7-53. Data previewing the calculation view

18. If your deployment was successful, you will be able to open the calculation view data preview, as shown in Figure 7-54.

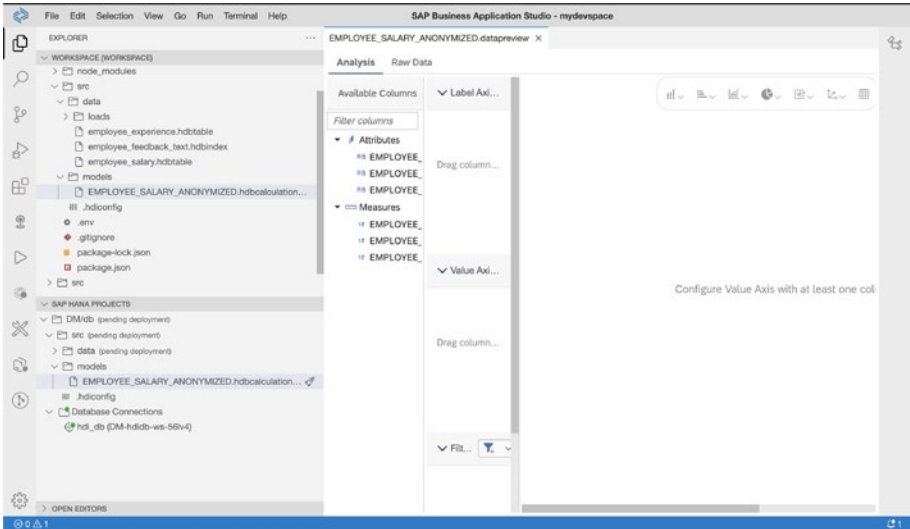


Figure 7-54. Data Preview pane opened

Configure Differential Privacy via SQL View

1. Create an SQL view (artifact type .hdbview) in the src/data/models folder named V_EMPLOYEE_SALARY (Figure 7-55).

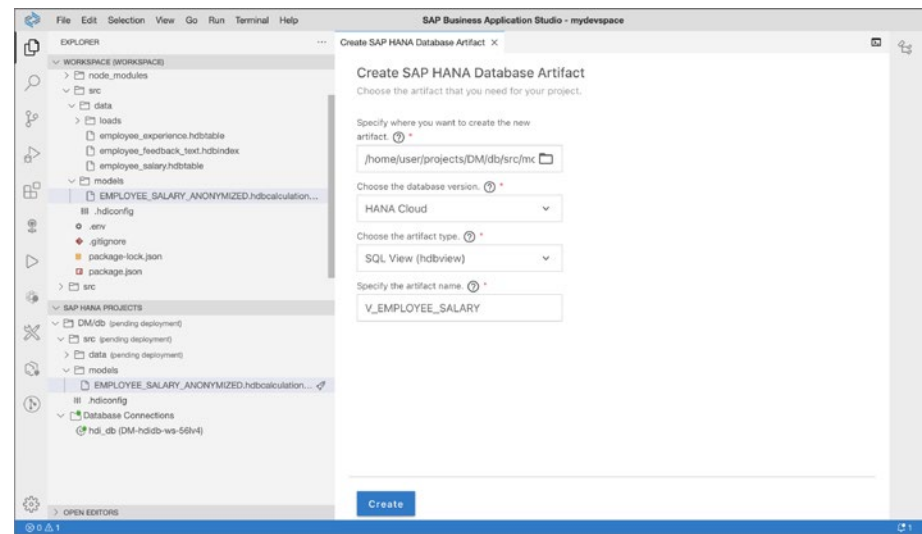


Figure 7-55. *Creating a new SAP HANA database artifact*

Note An SQL view represents an ANSI-compliant view and works like a virtual table based on the result set of an SQL statement using other views or tables. A calculation view is simpler and powerful because it mirrors the functionality found in both attribute views and analytic views, adding more analytic capabilities, and is used to define more advanced slices of the data.

You can add SQL statements and functions to a view and present the data as if the data were coming from one single table.

2. Once the view is created, click on its name on the Workspace view, and you will see the window with SQL code to be completed (Figure 7-56).

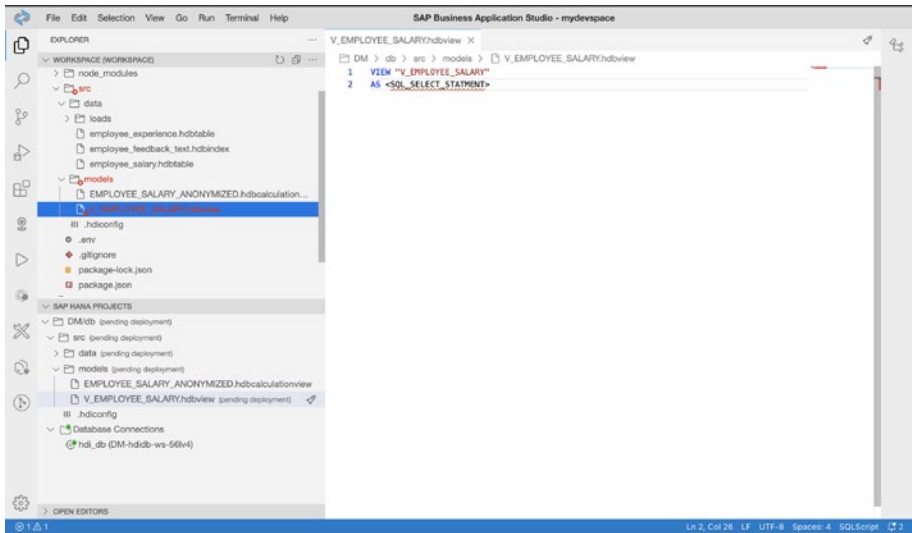


Figure 7-56. SQL code definition

3. Use the following syntax to create an SQL view that uses the Differential Privacy Anonymization approach for the EMPLOYEE_SALARY column:

```

VIEW V_EMPLOYEE_SALARY (
    EMPLOYEE_ID,
    EMPLOYEE_SALARY,
    EMPLOYEE_STARTING_YEAR,
    EMPLOYEE_GENDER,
    EMPLOYEE_REGION,
    EMPLOYEE_CAREER_LEVEL
) AS
SELECT EMPLOYEE_ID,
       EMPLOYEE_SALARY,
       EMPLOYEE_STARTING_YEAR,
       EMPLOYEE_GENDER,
       EMPLOYEE_REGION,
  
```

```
EMPLOYEE_CAREER_LEVEL
FROM "EMPLOYEE_SALARY_ANONYMIZED"
WITH READ ONLY
WITH ANONYMIZATION (ALGORITHM 'DIFFERENTIAL_PRIVACY'
  PARAMETERS '{"data_change_strategy": "qualified"}'
  COLUMN EMPLOYEE_ID PARAMETERS '{"is_sequence": true}'
  COLUMN EMPLOYEE_SALARY PARAMETERS '{"is_
sensitive":true, "epsilon":0.1, "sensitivity":15000}')
```

4. The SQL code should look like Figure 7-57.

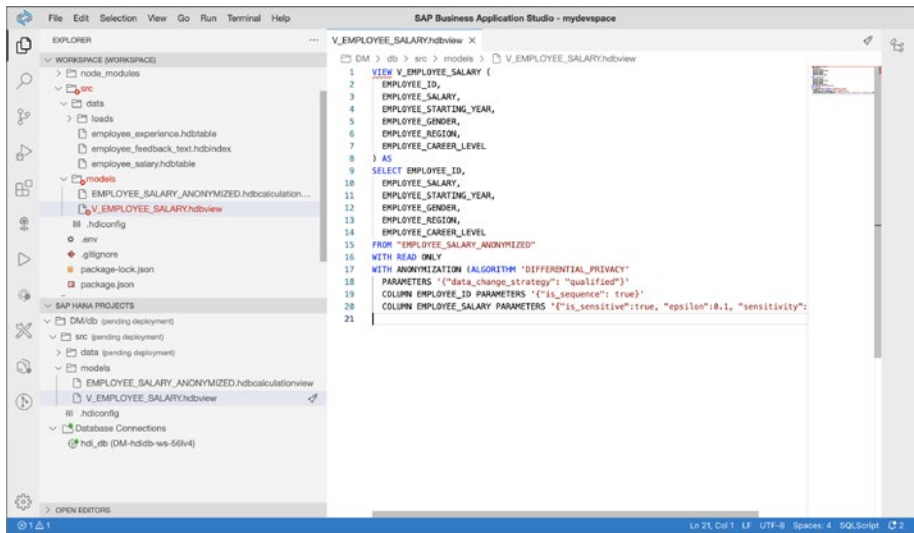


Figure 7-57. SQL code definition changed

5. If you get a “feature not supported” syntax error message, just ignore it (Figure 7-58).

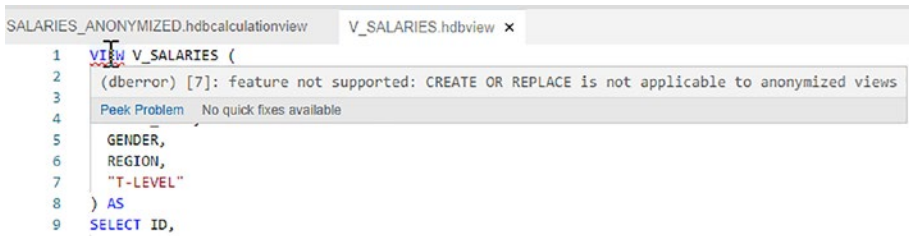


Figure 7-58. “Feature not supported” syntax error message

6. Save and deploy.
7. Open the Database Explorer for your project (Figure 7-59).

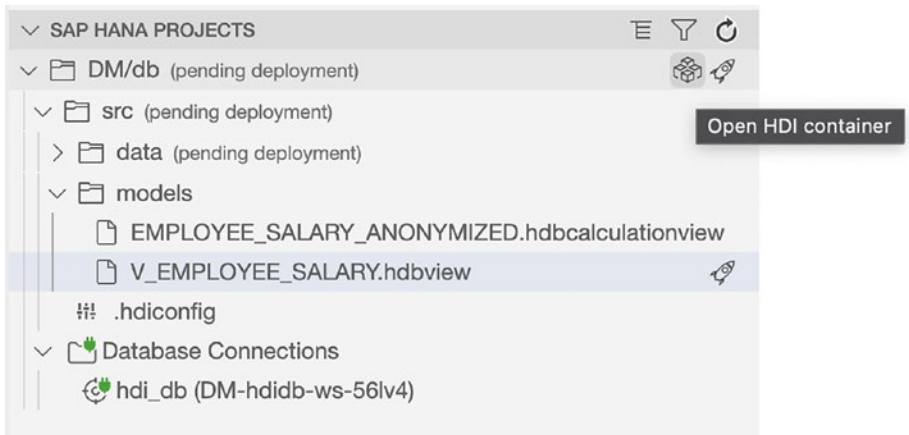
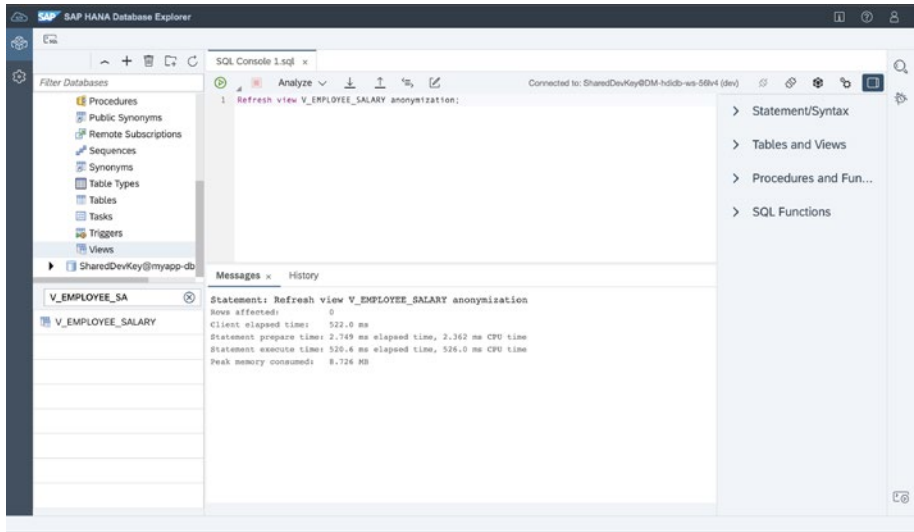


Figure 7-59. Opening HDI container

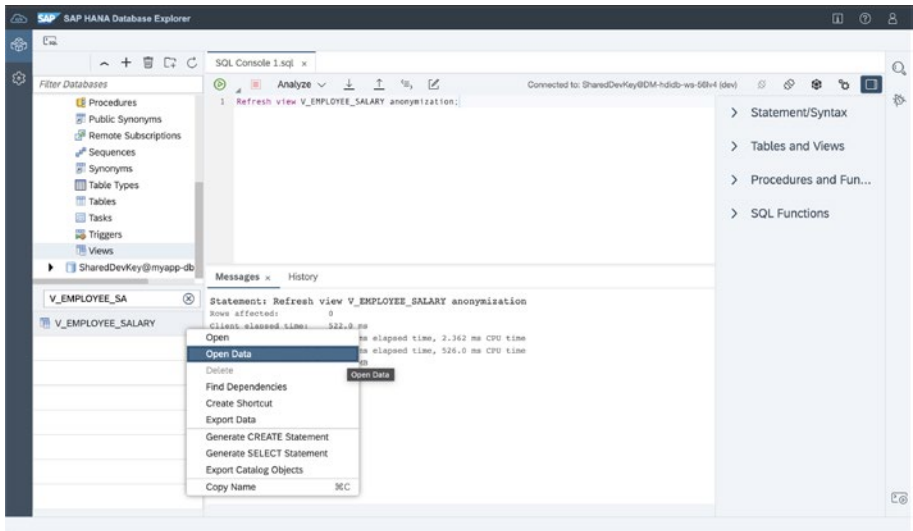
8. Open an SQL console by clicking on the SQL button on the top left corner, and then enter the following statement:
- ```
refresh view V_EMPLOYEE_SALARY anonymization;
```

9. Execute the statement in the SQL console, and you should see the output shown in Figure 7-60.



**Figure 7-60.** SQL execution results

10. You can now use the Open Data option from the context menu on the view to preview the raw data. Don't forget to select the Views folder on the upper section (Figure 7-61).



**Figure 7-61.** *Previewing data*

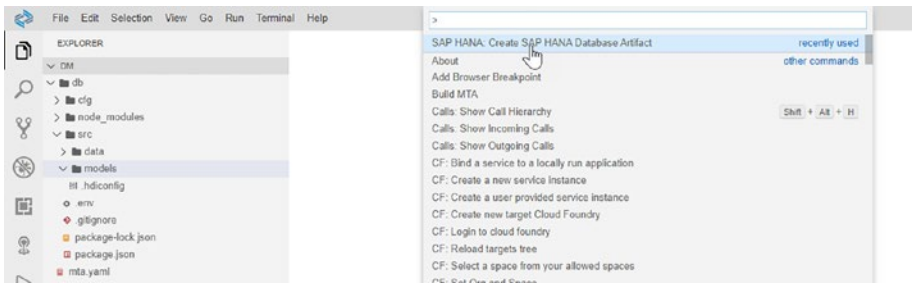
11. Now view data is shown on the canvas (Figure 7-62).

<

**Figure 7-62.** *Viewing the result set*

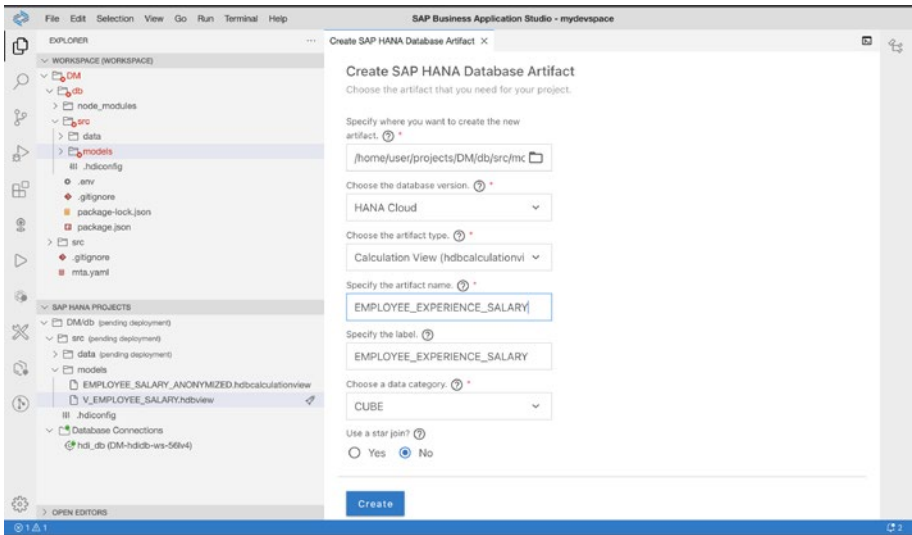
# Create a Calculation View of Type Cube

1. Create a new calculation view via the SAP HANA: Create SAP HANA Database Artifact command menu entry (Figure 7-63).



**Figure 7-63.** Creating a new calculation view

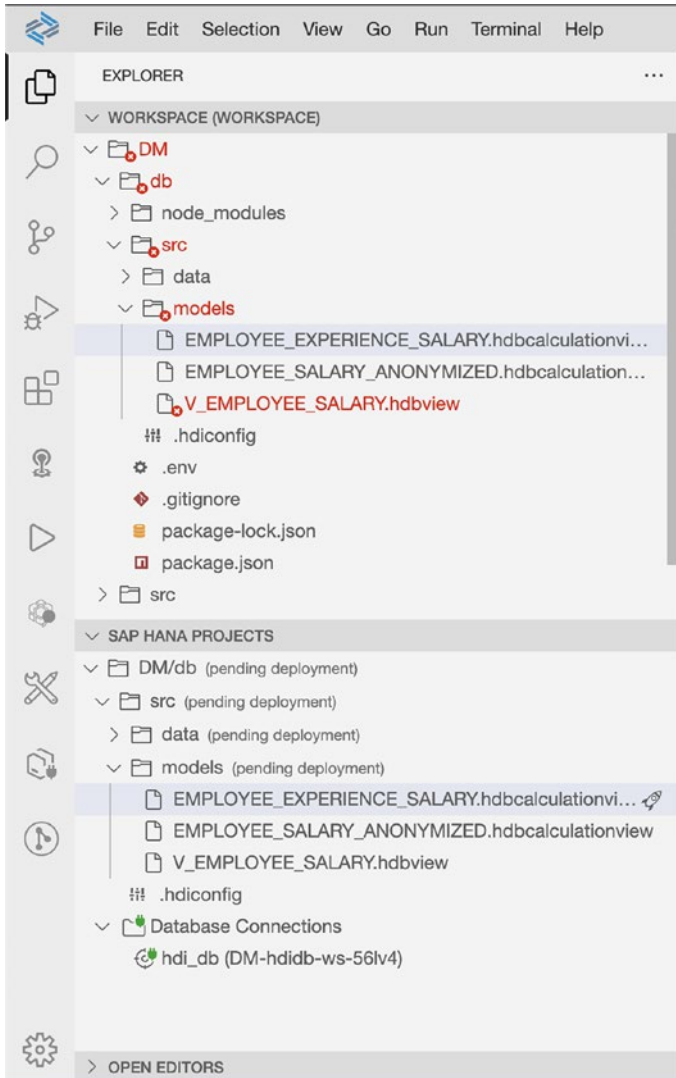
2. Create a calculation view called `EMPLOYEE_EXPERIENCE_SALARY` with `CUBE` as the data category. Click the Create button (Figure 7-64).



**Figure 7-64.** Providing calculation view definition

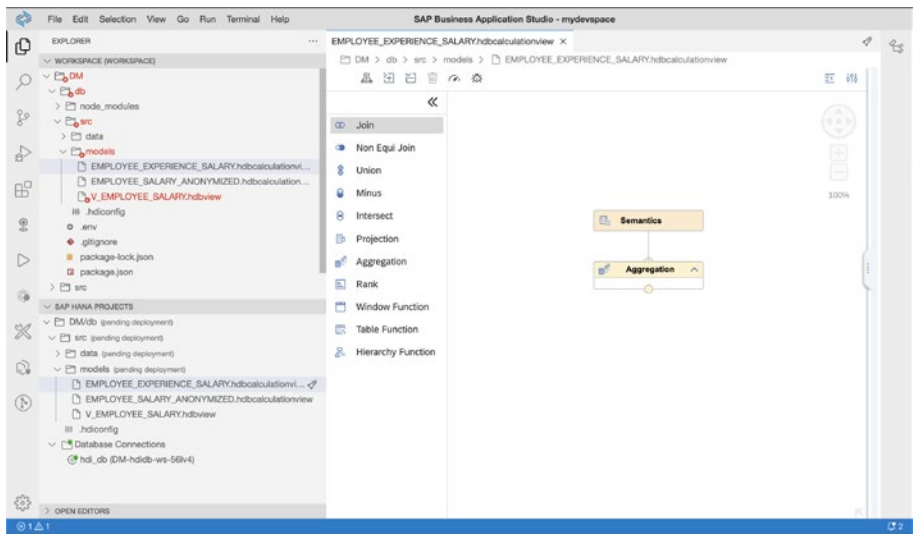
## Join the SQL View and Table

1. Double-click on the `EMPLOYEE_EXPERIENCE_SALARY` calculation view (Figure 7-65).



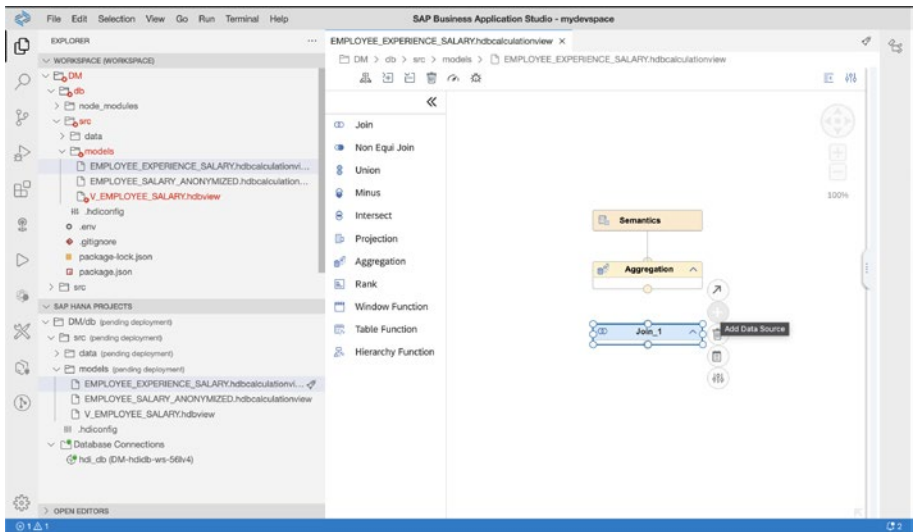
**Figure 7-65.** *EMPLOYEE\_EXPERIENCE\_SALARY* calculation view

2. Expand the left menu by clicking on the two-arrows button and click on the Join node to insert it onto the modeling canvas (Figure 7-66).



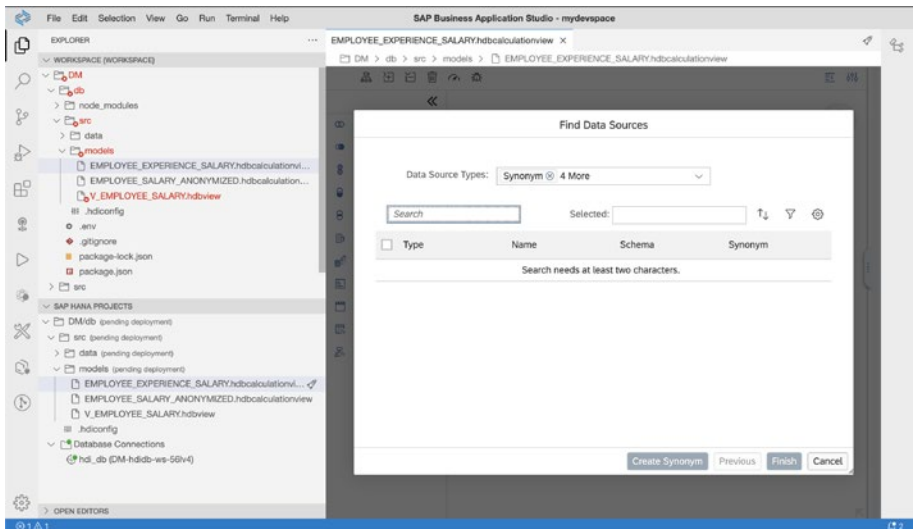
**Figure 7-66.** Calculation view on modeling canvas

3. With the Join node selected, click on the plus button to add the SQL view to the node (Figure 7-67).



**Figure 7-67.** Opening the Join node

4. Type in the name of the anonymized view `V_EMPLOYEE_SALARY` as the search criteria (Figure 7-68).



**Figure 7-68.** Finding data sources

5. Select it and click the Finish button (Figure 7-69).

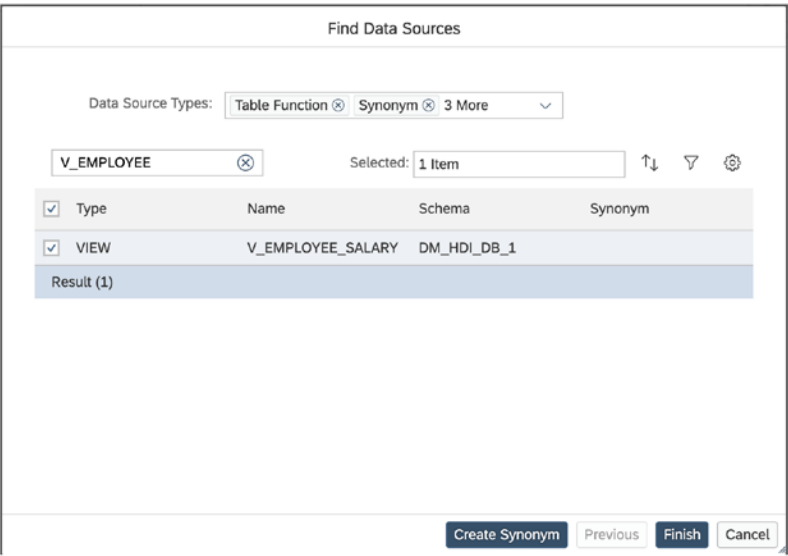


Figure 7-69. Finishing with V\_EMPLOYEE\_SALARY selected

6. The calculation view should look like Figure 7-70.

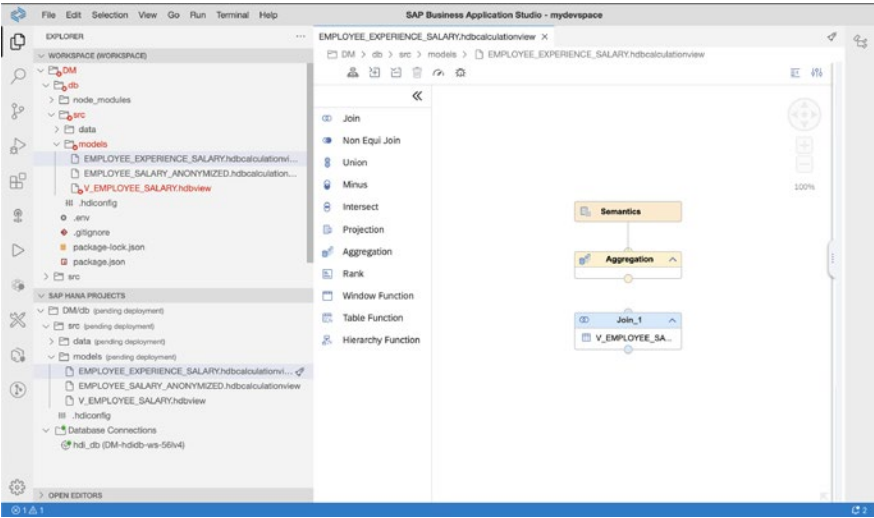
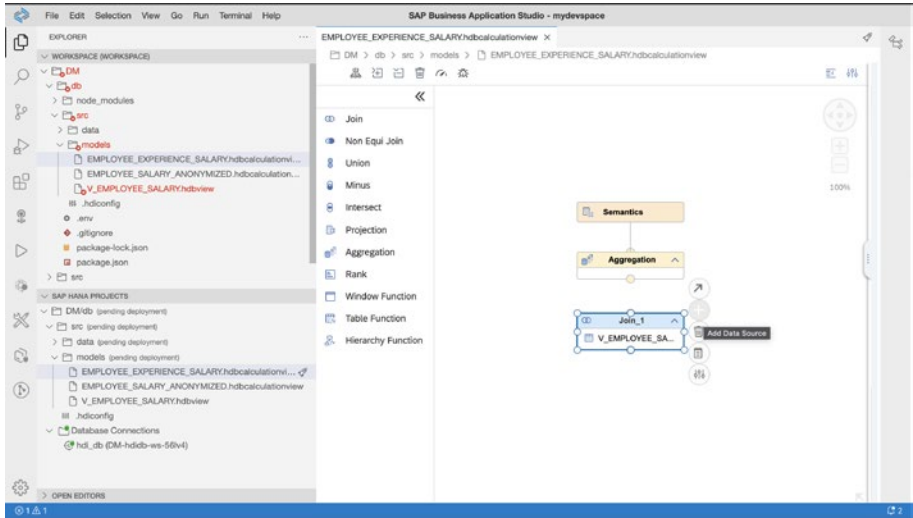


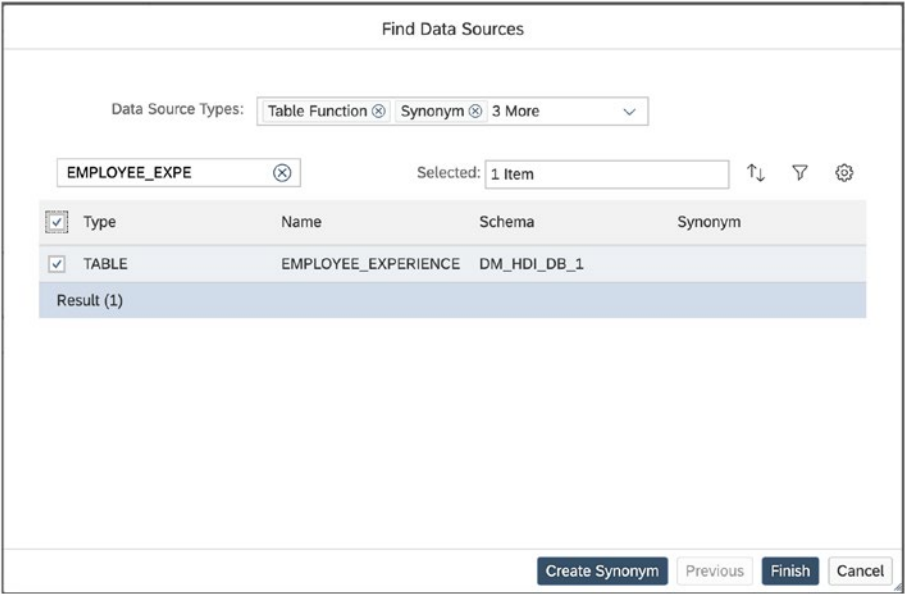
Figure 7-70. Calculation view with join definition partially configured

7. With the Join node selected, click on the plus button again and add the EMPLOYEE\_EXPERIENCE table (Figure 7-71).



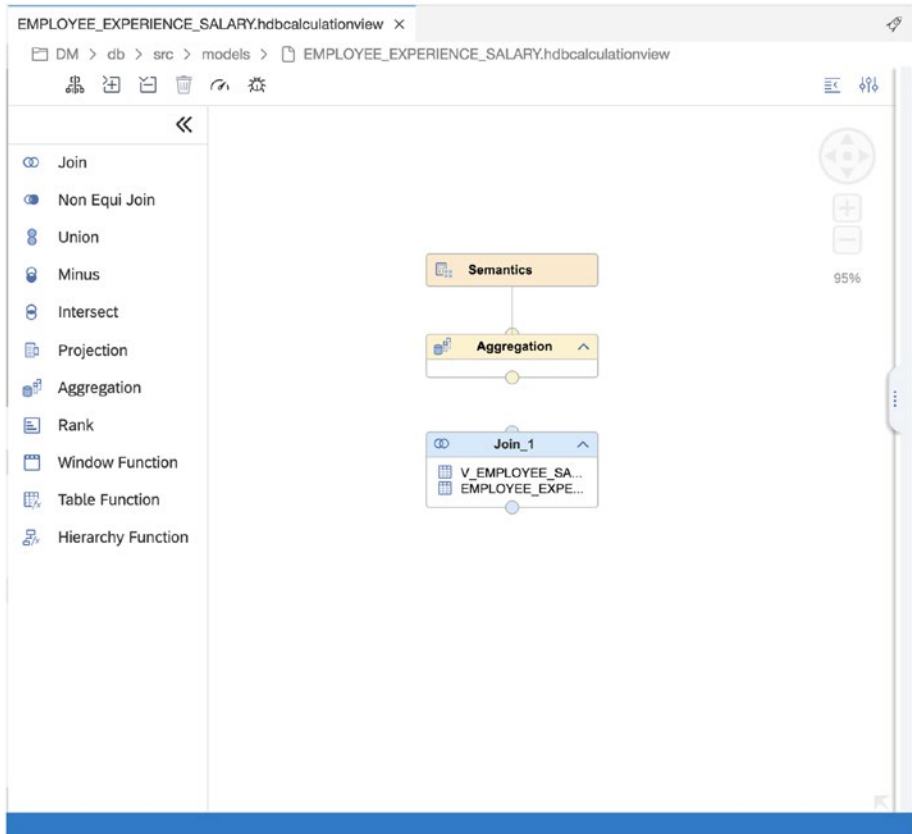
**Figure 7-71.** Opening the Join node for additional data source

8. Type in the name of the EMPLOYEE\_EXPERIENCE table as the search criteria. Select it and click the Finish button (Figure 7-72).



**Figure 7-72.** *Finishing with EMPLOYEE\_EXPERIENCE selected*

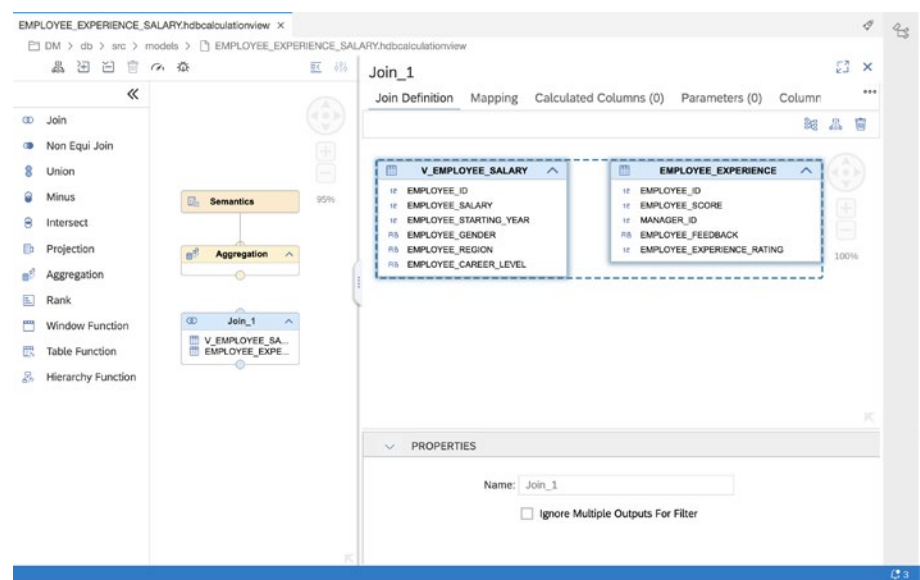
- 9. You should now see both artifacts in the Join node (Figure 7-73).



**Figure 7-73.** Calculation view with join definition fully configured

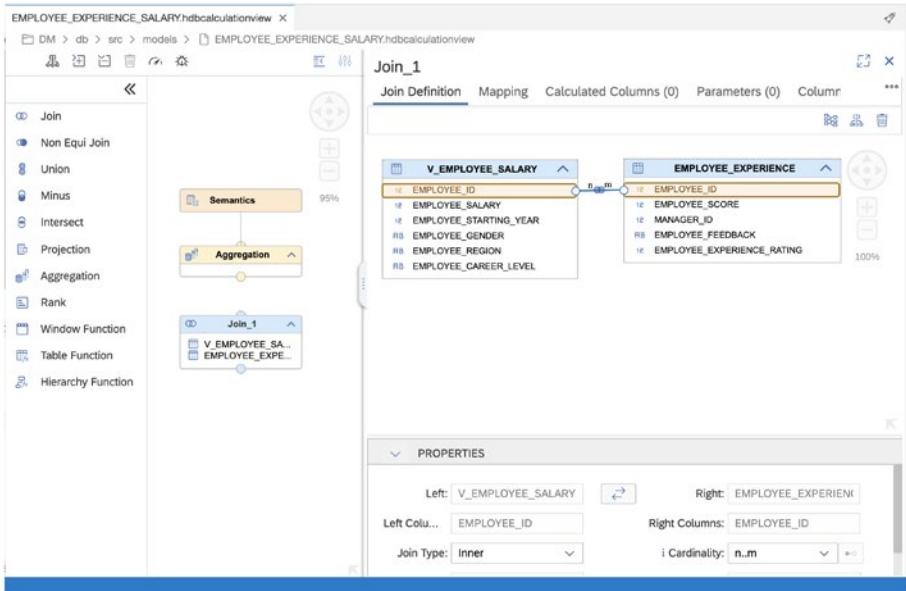
## Configure the Join

1. Double-click on the Join node. A panel will open on the right (Figure 7-74).



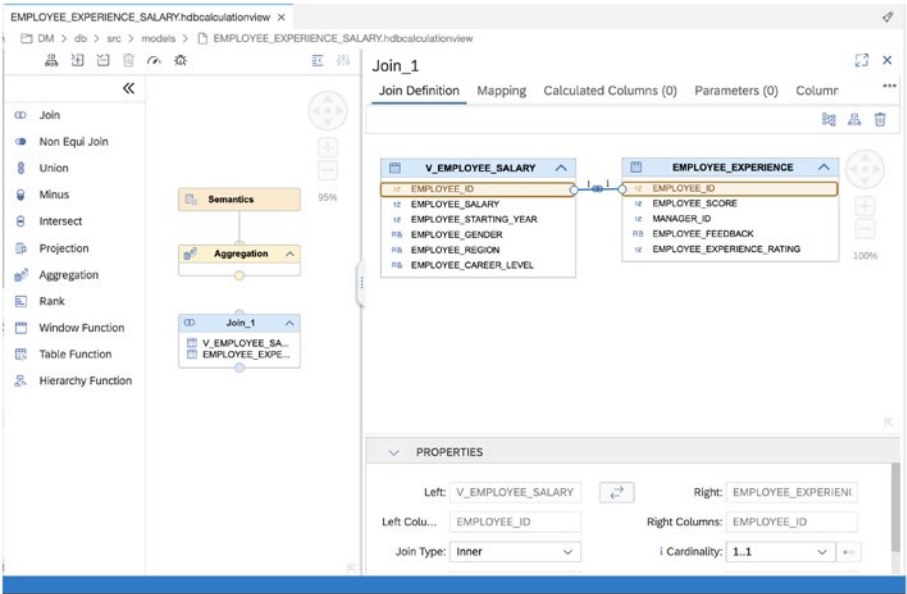
**Figure 7-74.** Calculation view join definition

2. Drag the EMPLOYEE\_ID field from the left side to on top of the same field on the right side to define the join criteria (Figure 7-75).



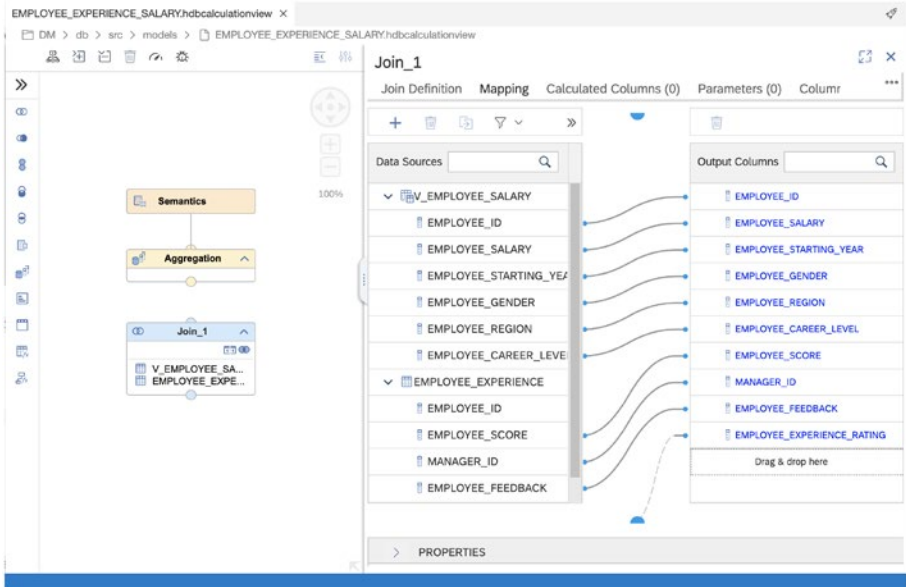
**Figure 7-75.** Calculation view join cardinality definition

3. Set the join cardinality to 1..1 in the PROPERTIES section at the bottom (Figure 7-76).



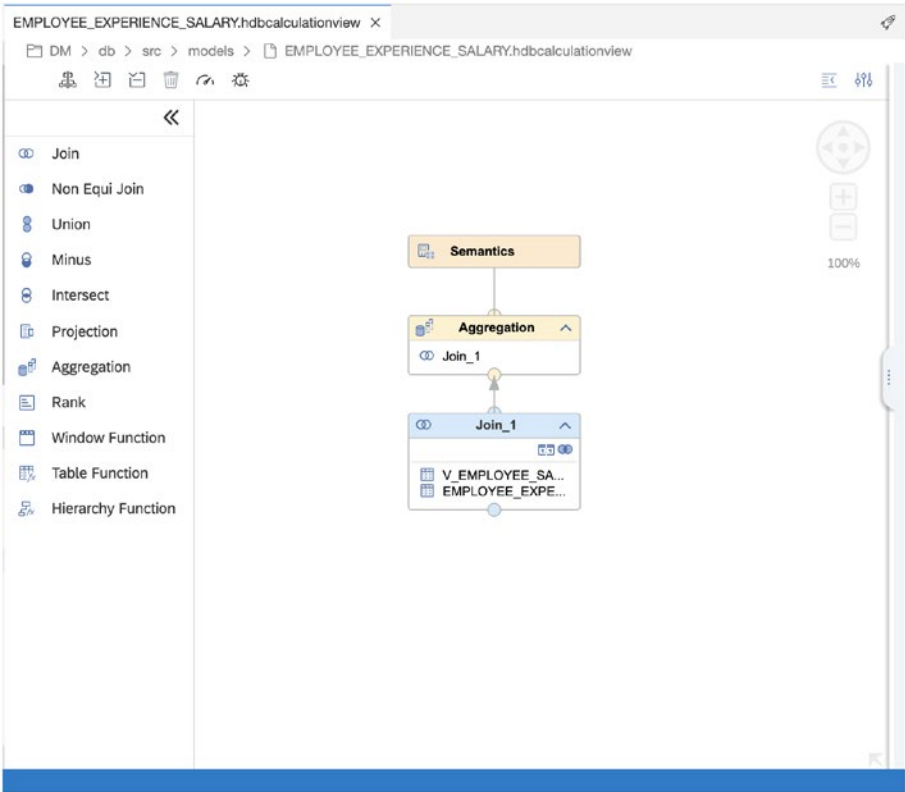
**Figure 7-76.** Calculation view join cardinality changed to 1..1

4. In the Mapping tab, add all the columns from both data sources as output columns (Figure 7-77). You can do this by double-clicking on both data sources on the left side. Make sure the EMPLOYEE\_ID column is only added once (remove the duplicate column if you used the double-click shortcut).



**Figure 7-77.** Calculation view join mapping definition

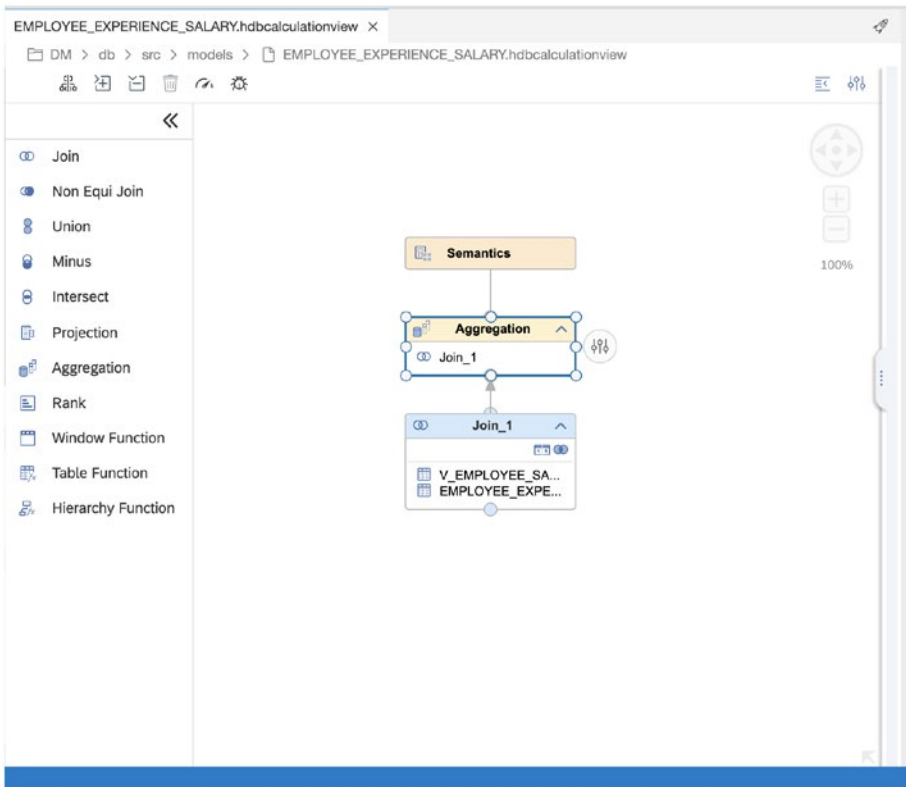
5. Connect the Join node with the Aggregation node (Figure 7-78).



**Figure 7-78.** Calculation view Join node connected to Aggregation node

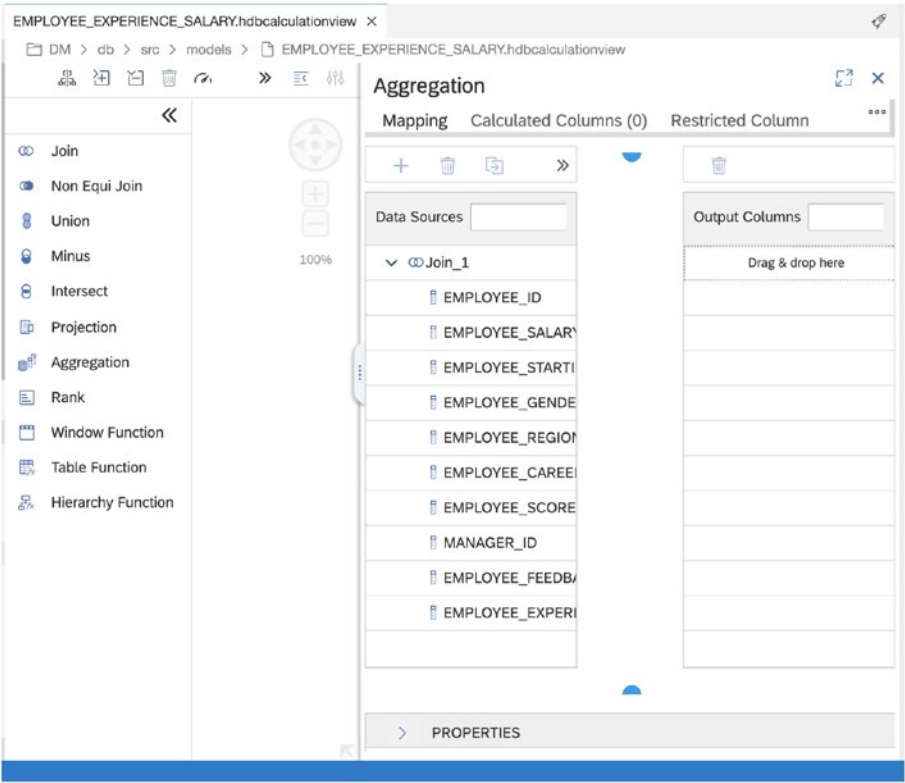
## Configure Semantics

1. Double-click on the Aggregation node to open the Aggregation definition pane (Figure 7-79).



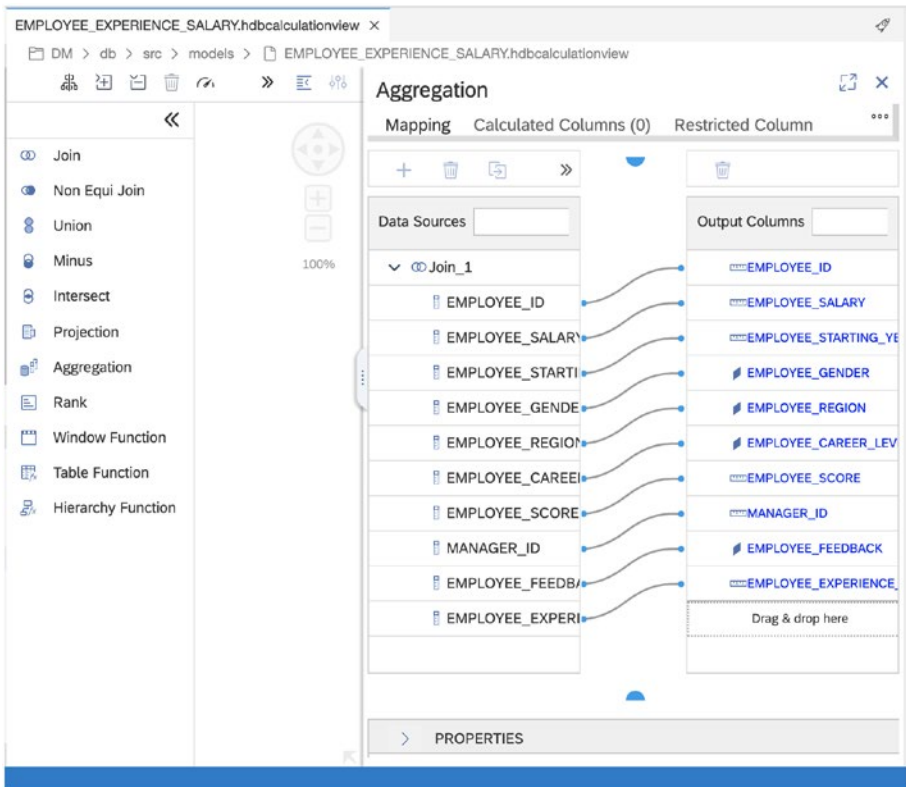
**Figure 7-79.** Calculation view Aggregation node

2. Double-click on the Join\_1 data source (root node) to add all columns to the output (Figure 7-80).



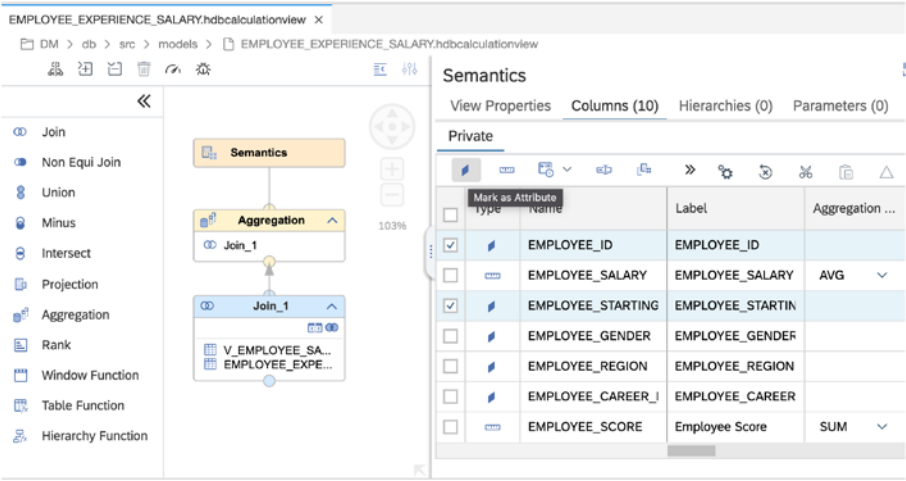
**Figure 7-80.** Mapping columns on Aggregation node

- 3. The mapping should look like the image in Figure 7-81.



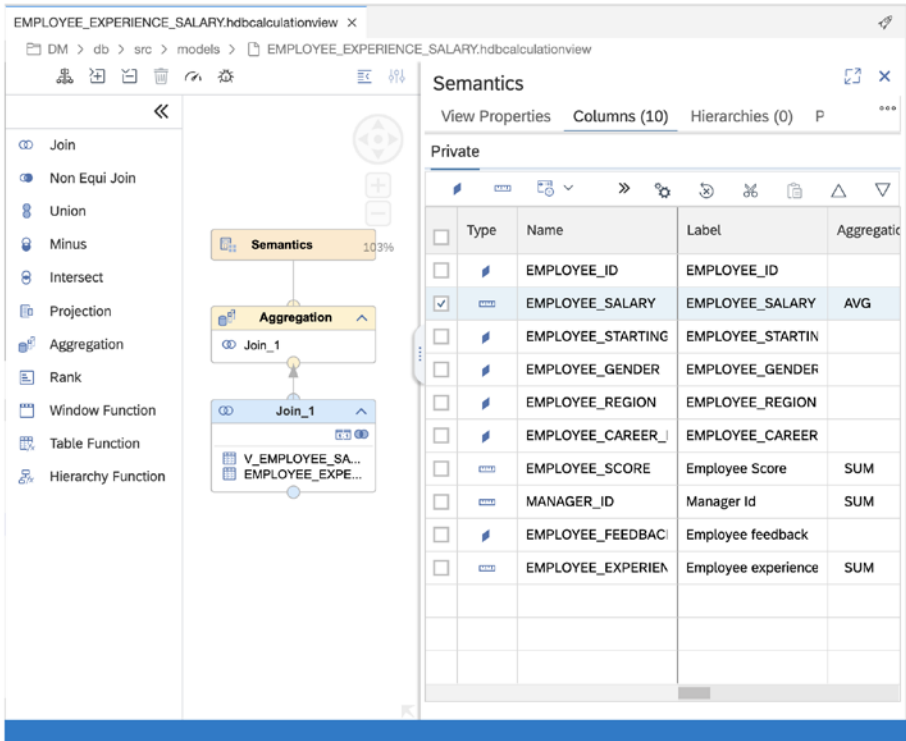
**Figure 7-81.** Calculation view Aggregation node with columns fully mapped

4. Click on the X to close the Aggregation definition pane.
5. Double-click on the Semantics node.
6. Select the EMPLOYEE\_ID and EMPLOYEE\_STARTING\_YEAR columns and click the Mark as Attribute button to change their type to ATTRIBUTE (Figure 7-82).



**Figure 7-82.** Calculation view Aggregation node

7. Set the aggregation of the EMPLOYEE\_SALARY as AVG (Figure 7-83).



**Figure 7-83.** Setting specific columns as AVG

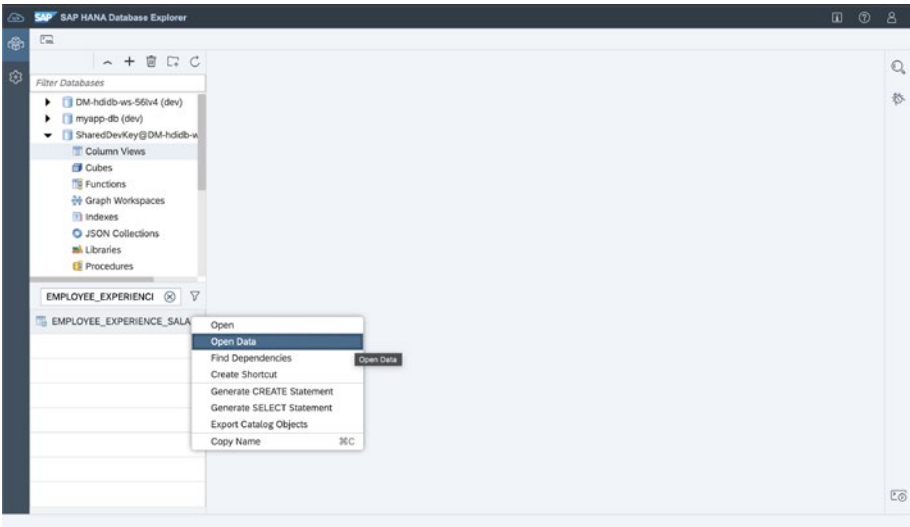
## Build and Test

1. Save and deploy the calculation view.
2. Open the HDI container in the Database Explorer (Figure 7-84).



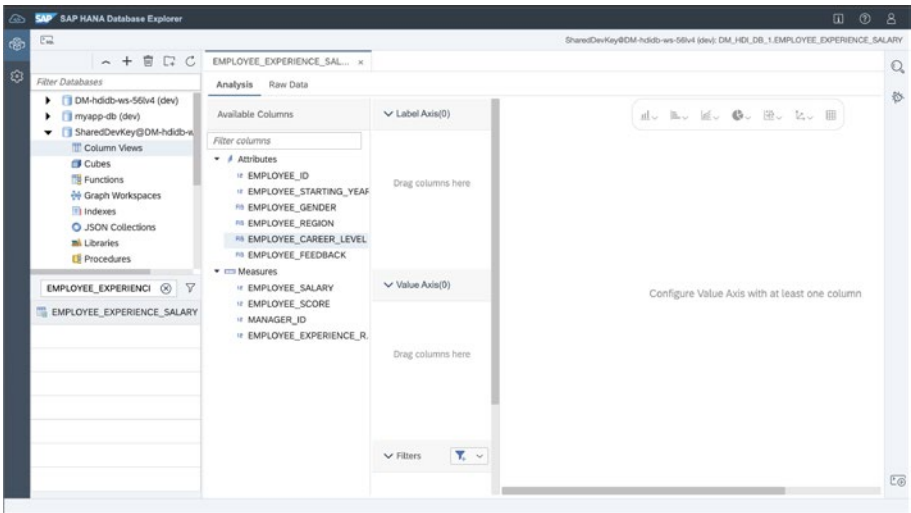
**Figure 7-84.** Opening the HDI container

- 3. Under Column Views you will find your calculation view. Choose Open Data from the context menu (Figure 7-85).



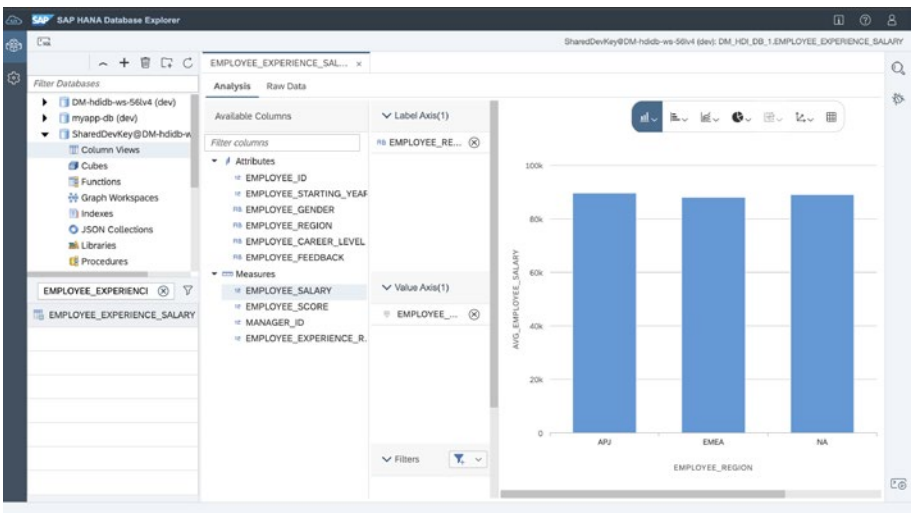
**Figure 7-85.** Opening data from context menu

- 4. On the Analysis tab, drag the EMPLOYEE\_REGION attribute to the Label Axis and the EMPLOYEE\_SALARY measure to the Values Axis (Figure 7-86).



**Figure 7-86.** Opening data from context menu

5. The generated chart should look like the image in [Figure 7-87](#).



**Figure 7-87.** Previewing calculation view data

## Summary

Congratulations! You have successfully completed our journey through the essentials of learning SAP HANA Cloud and understanding key aspects, including technology, architecture, and data modeling.

My suggestion is that you revisit this book to review concepts whenever you need. There are many business scenarios where SAP HANA Cloud can make the difference and provide the way to solve business challenges.

Hope to see you on the next journey.

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